

1. Clutch System**A: SPECIFICATIONS**

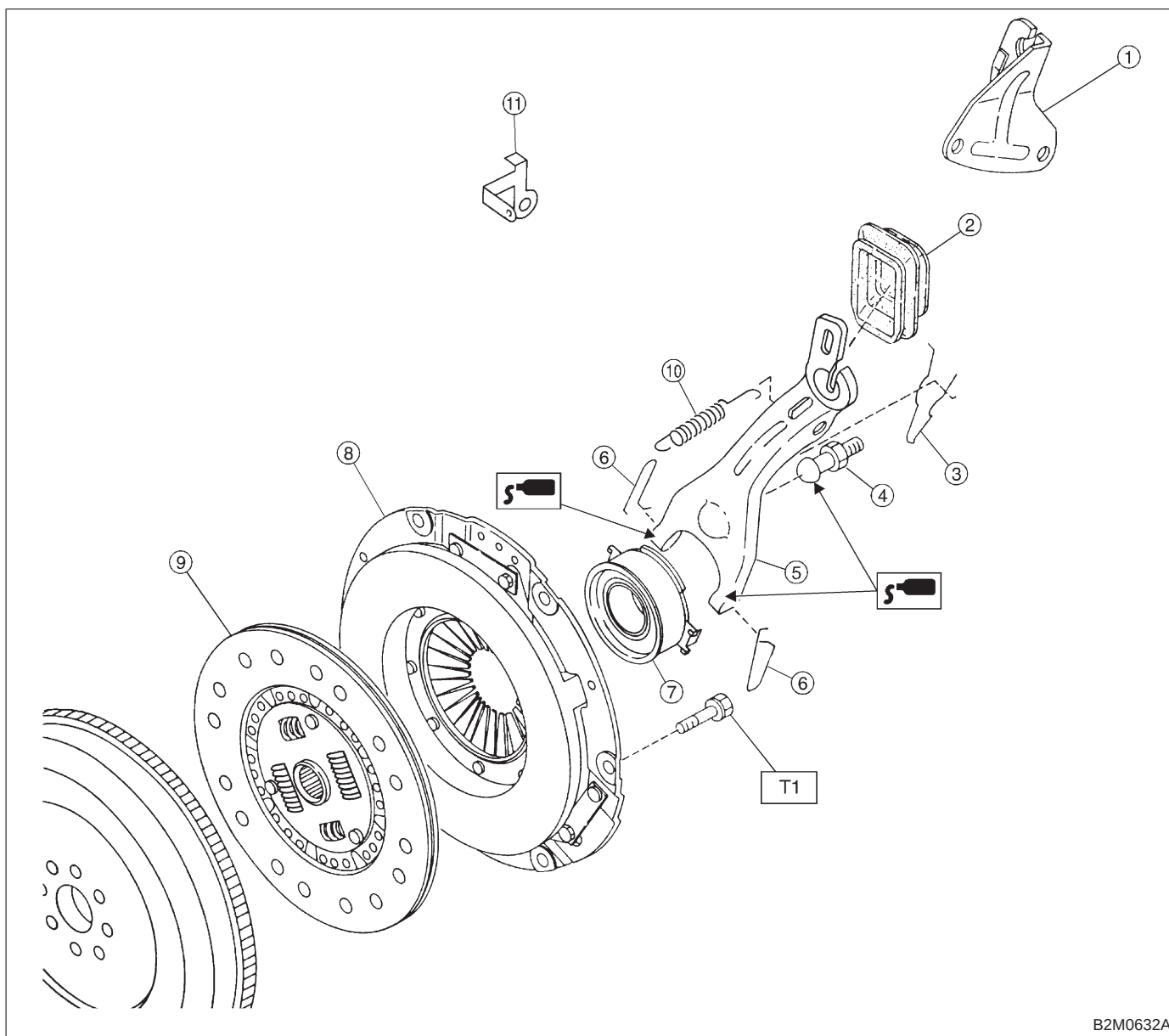
		2200 cc	2500 cc
Clutch cover	Diaphragm set load kg (lb)	450 (992)	550 (1,213)
Clutch disc	Facing material	Woven	
	O.D. x I.D. x thickness mm (in)	225 x 150 x 3.5 (8.86 x 5.91 x 0.138)	
	Spline O.D. (No. of teeth) mm (in)	25.2 (0.992) (24)	
Clutch release lever ratio		3.0	1.6
Release bearing		Grease-packed self-aligning	

B: SERVICE DATA

			2200 cc	2500 cc
Clutch pedal	Full stroke	mm (in)	140 — 150 (5.51 — 5.91)	
Release lever	Stroke	mm (in)	24 — 26 (0.94 — 1.02)	
	Play at release lever center	mm (in)	3 — 4 (0.12 — 0.16)	
Clutch disc	Depth of rivet head	Standard	1.3 — 1.9 (0.051 — 0.075)	
	mm (in)	Limit of sinking	0.3 (0.012)	
	Limit for runout	mm (in)	1.0 (0.039) at R = 107 (4.21)	

1. Clutch System

1. MECHANICAL APPLICATION TYPE



B2M0632A

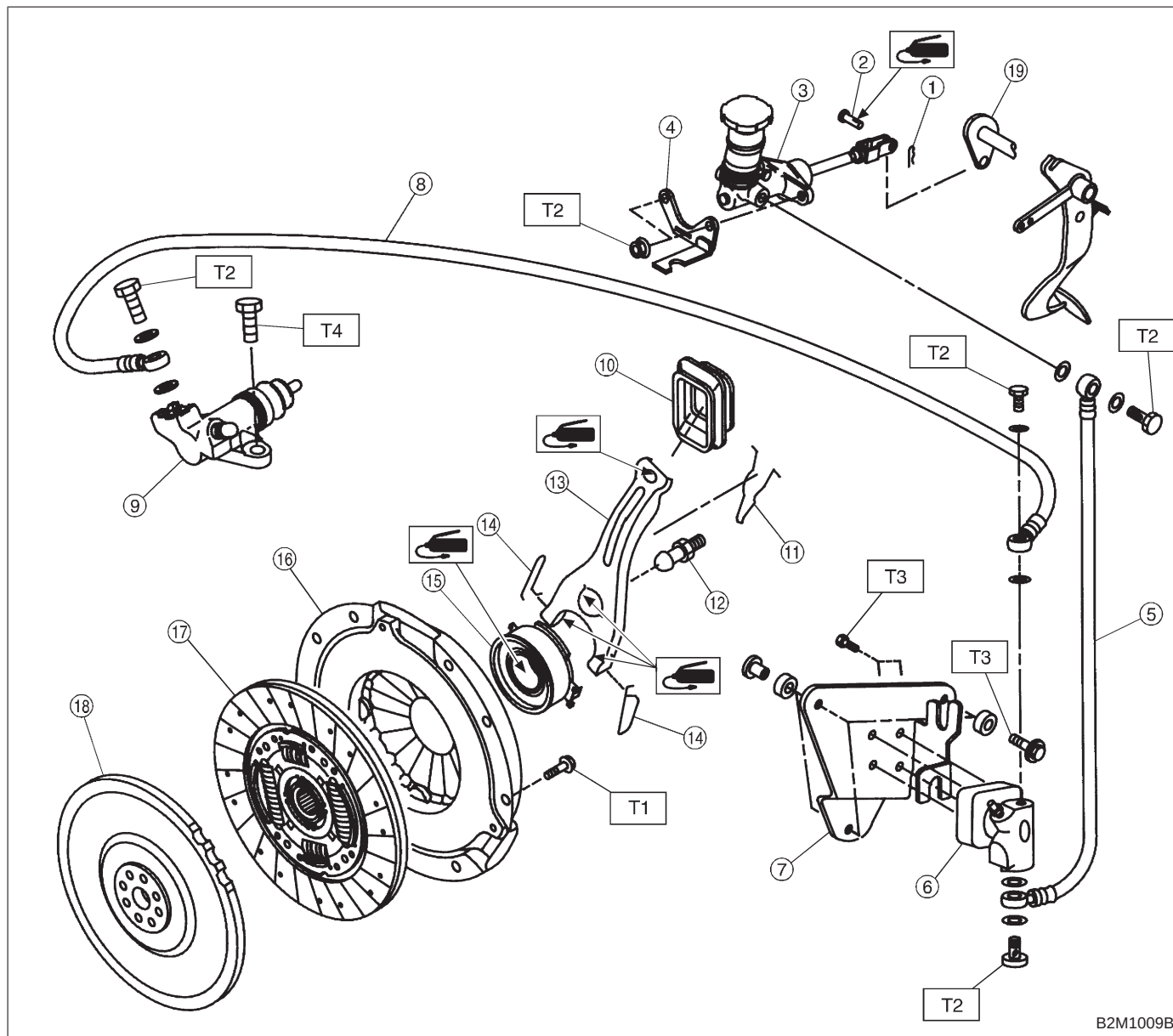
- ① Clutch cable bracket
- ② Clutch release lever sealing
- ③ Retainer spring
- ④ Pivot
- ⑤ Clutch release lever
- ⑥ Clip
- ⑦ Clutch release bearing
- ⑧ Clutch cover

- ⑨ Clutch disc
- ⑩ Return spring (Models without hill holder only)
- ⑪ Clutch return spring bracket

Tightening torque: N·m (kg·m, ft·lb)

T1: 15.7±1.5 (1.6±0.15, 11.6±1.1)

2. HYDRAULIC APPLICATION TYPE



B2M1009B

- ① Snap pin
- ② Clevis pin
- ③ Master cylinder ASSY
- ④ Bracket
- ⑤ Clutch hose B
- ⑥ Clutch damper
- ⑦ Damper bracket
- ⑧ Clutch hose A
- ⑨ Operating cylinder

- ⑩ Clutch release lever sealing
- ⑪ Retainer spring
- ⑫ Pivot
- ⑬ Release lever
- ⑭ Clip
- ⑮ Release bearing
- ⑯ Clutch cover
- ⑰ Clutch disc
- ⑱ Flywheel

- ⑲ Spacer

Tightening torque: N·m (kg-m, ft-lb)

T1: 15.7±1.5

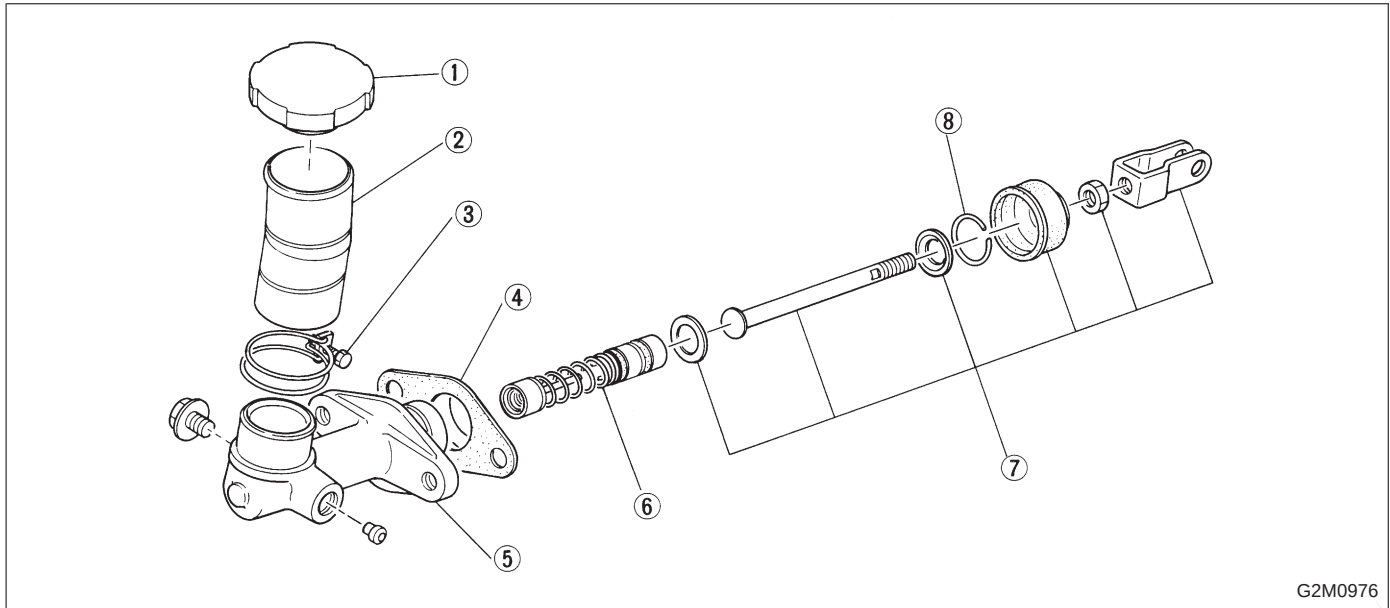
(1.6±0.15, 11.6±1.1)

T2: 18±3 (1.8±0.3, 13.0±2.2)

T3: 25±7 (2.5±0.7, 18.1±5.1)

T4: 37±3 (3.8±0.3, 27.5±2.2)

2. Master Cylinder and Reservoir Tank



- ① Reservoir cap
- ② Reservoir tank
- ③ Reservoir band
- ④ Seat

- ⑤ Master cylinder
- ⑥ Piston
- ⑦ Push rod
- ⑧ Piston stop ring

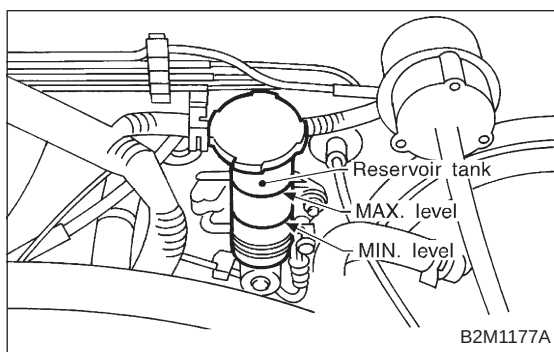
1. General

A: PRECAUTION

When servicing clutch system, pay attention to the following items.

1. MECHANICAL APPLICATION TYPE

- 1) Check the routing of clutch cable for smoothness.
- 2) Excessive tightness or looseness of clutch cable have a bad influence upon the cable durability.
- 3) Apply grease sufficiently to the connecting portion of clutch pedal.
- 4) Apply grease sufficiently to the release lever portion.
- 5) Position clutch cable through the center of toe board hole and route it smoothly. Adjustment is done by moving the outer cable.
- 6) Make sure not to let the clutch chatter when starting forward or rearward. If clutch chattering occurs, readjust so that the bend of clutch outer cable becomes flatter.



2. HYDRAULIC APPLICATION TYPE

- 1) Check fluid level using a scale on outside of reservoir tank. If the level is below "MIN", add clutch fluid to bring it up to "MAX".

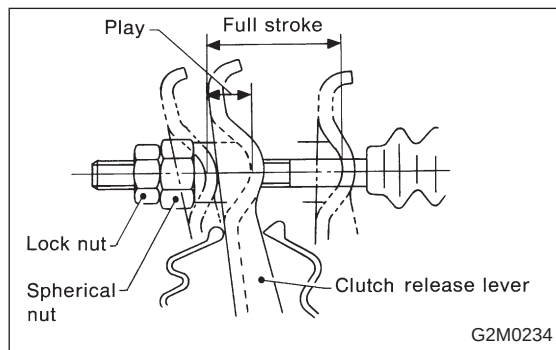
Recommended clutch fluid:

FMVSS No. 116, fresh DOT3 or DOT4 brake fluid

CAUTION:

- Avoid mixing different brands of brake fluid to prevent degradation of the fluid.
- Be careful not to allow dirt or dust to get into the reservoir tank.
- Use fresh DOT3 or DOT4 brake fluid when refilling fluid.

- 2) Make sure that clutch fluid does not leak from master cylinder, operating cylinder and piping.
- 3) Apply grease sufficiently to the release lever pinion.
- 4) Check for proper clutch disengagement and clutch pedal return ability.



2. On-Car Service

1. MECHANICAL APPLICATION TYPE

- 1) Remove release lever return spring from lever (Models without hill holder only).
- 2) Adjust spherical nut so that the play is within the specified value at the lever end (center of spherical nut).

CAUTION:

Take care not to twist the cable during adjustment

Play: 3 — 4 mm (0.12 — 0.16 in)

Full stroke: 24 — 26 mm (0.94 — 1.02 in)

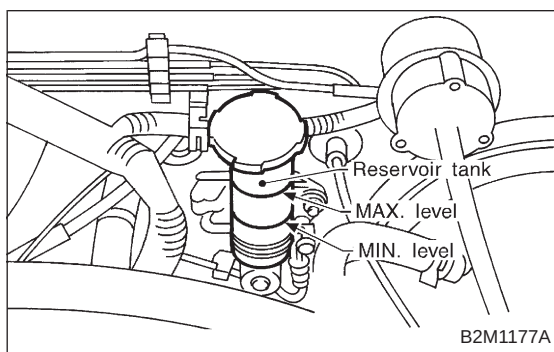
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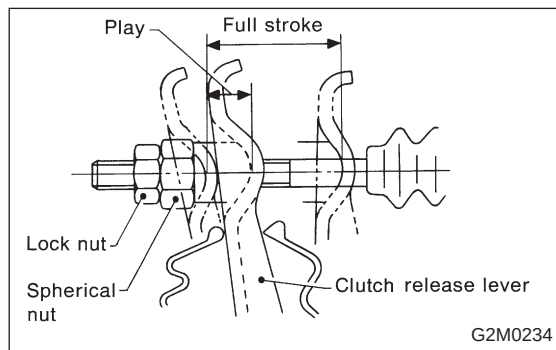
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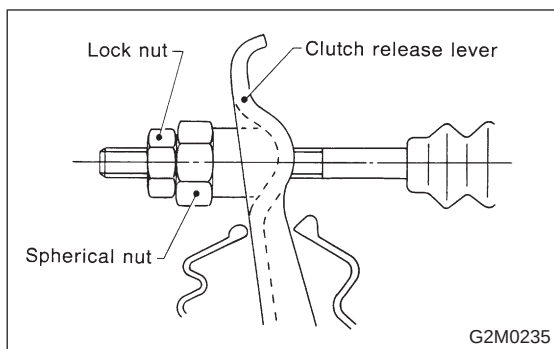
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Play: 3 — 4 mm (0.12 — 0.16 in)

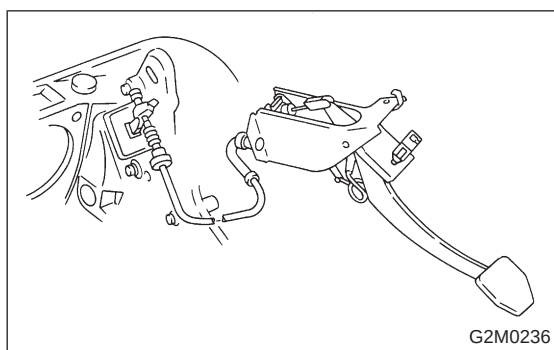
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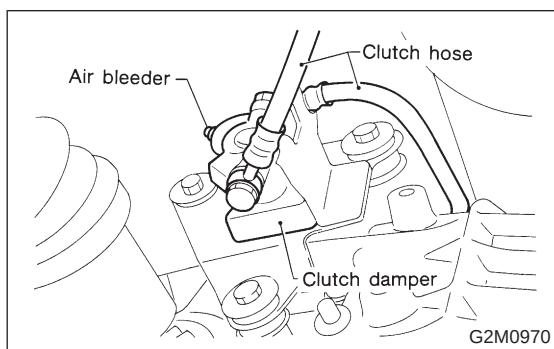
3) Upon completion of adjustment, securely lock spherical nut with lock nut.
Install return spring on lever (Models without hill holder only).

NOTE:

Hook the long hook side of the return spring with the lever (Models without hill holder only).



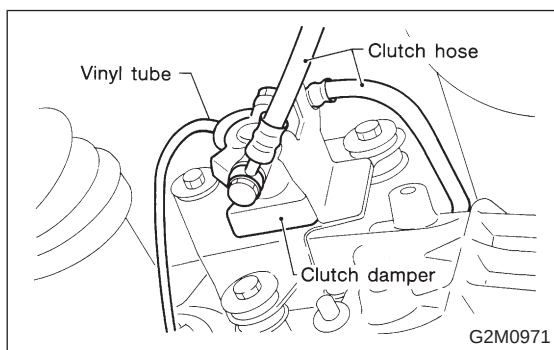
4) Depress clutch pedal to assure there is no abnormality in the clutch system.



2. HYDRAULIC APPLICATION TYPE

Bleed air from oil line with the help of a co-worker.

1) Fit one end of a vinyl tube into the air bleeder of clutch damper and put the other end into a clutch fluid container.



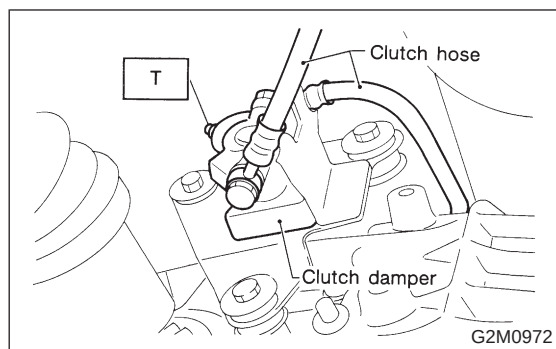
2) Slowly depress the clutch pedal and keep it depressed. Then open the air bleeder to discharge air together with the fluid.

Release the air bleeder for 1 or 2 seconds. Next, with the bleeder closed, slowly release the clutch pedal.

3) Repeat these steps until there are no more air bubbles in the vinyl tube.

CAUTION:

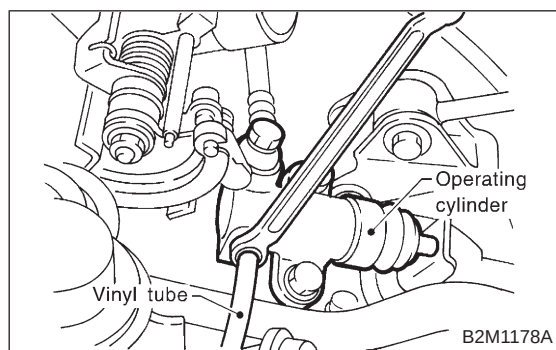
Cover bleeder with waste cloth when loosening it, to prevent clutch fluid from being splashed over surrounding parts.



4) Tighten air bleeder.

Tightening torque:

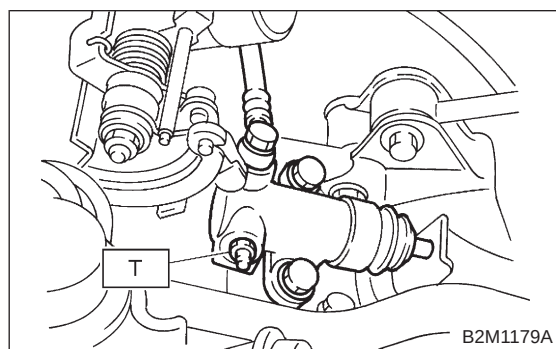
T: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)



5) Remove air chamber.

<Ref. to 2-7 [W18A0].>

6) Repeat steps 1) through 3) using air bleeder on operating cylinder.



7) Tighten air bleeder.

Tightening torque:

T: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)

8) After depressing the clutch pedal, make sure that there are no leaks evident in the entire system.

9) After bleeding air from system, ensure that clutch operates properly.

3. Release Bearing and Lever

A: REMOVAL

1. MECHANICAL APPLICATION TYPE

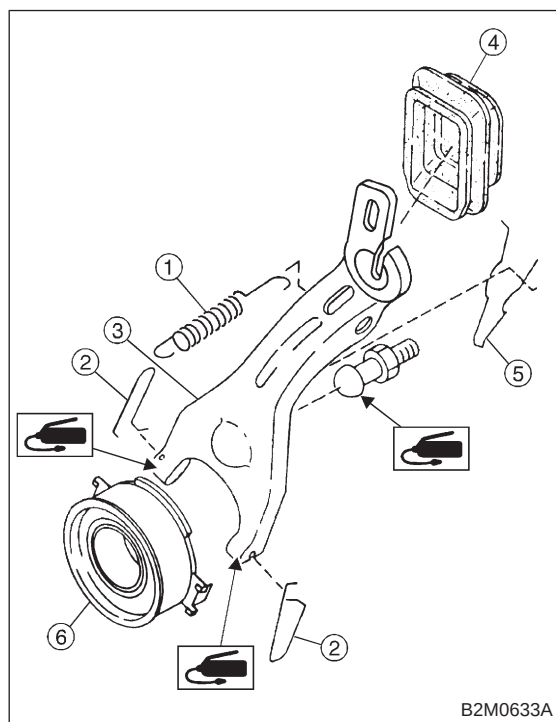
1) Remove release lever return spring ① (Models without hill holder only).

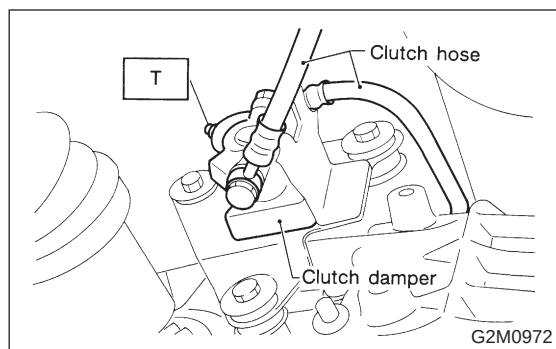
2) Remove the two clips ② from clutch release lever ③ and remove release bearing ⑥.

CAUTION:

Be careful not to deform clips.

3) Remove release lever seal ④.

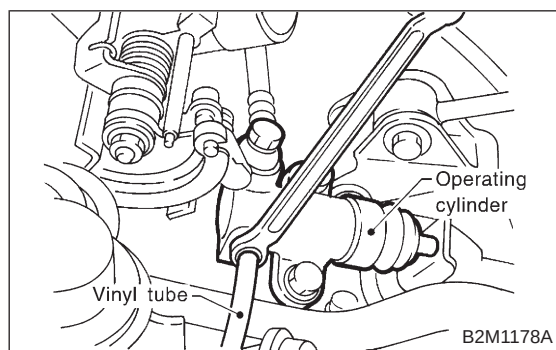




4) Tighten air bleeder.

Tightening torque:

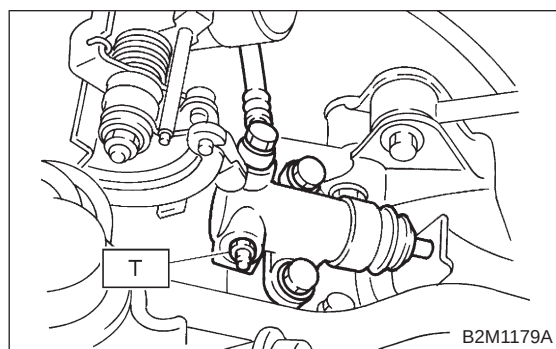
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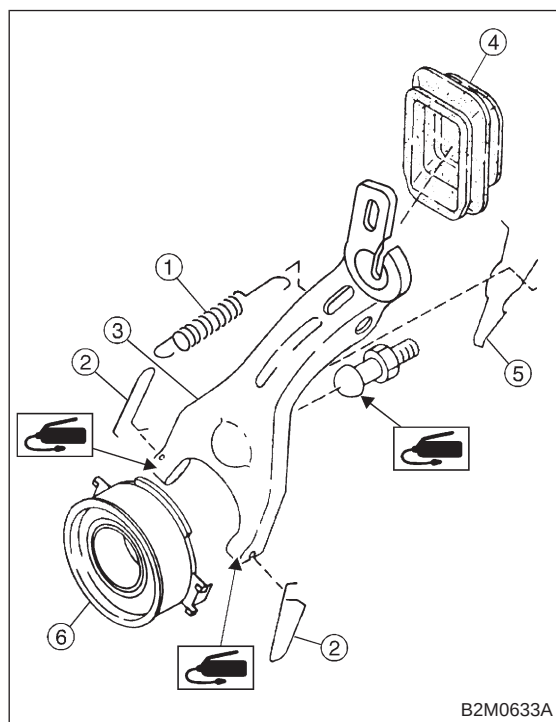
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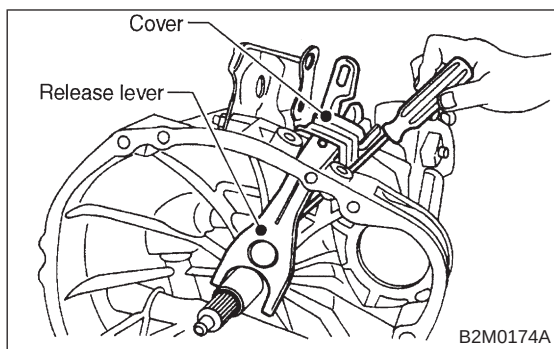
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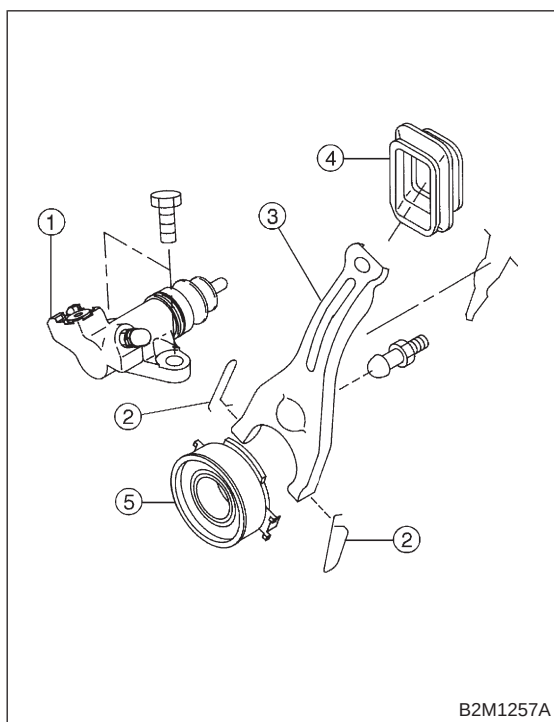
Be careful not to deform clips.

3) Remove release lever seal ④.





4) Remove release lever retainer spring from release lever pivot with a screwdriver by accessing it through clutch housing release lever hole. Then remove release lever.



2. HYDRAULIC APPLICATION TYPE

1) Remove transmission assembly from vehicle body.

<Ref. to 2-11 [W3A0].>

2) Remove operating cylinder ①.

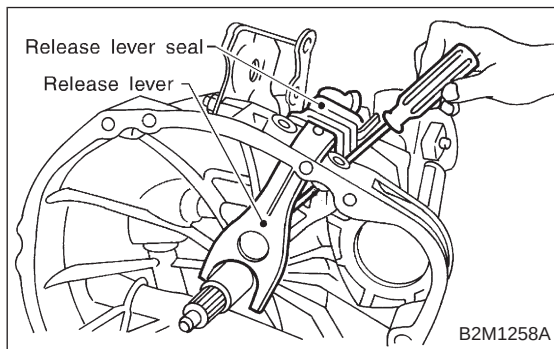
<Ref. to 2-10 [W5A0].>

3) Remove the two clips ② from clutch release lever ③ and remove release bearing ⑥.

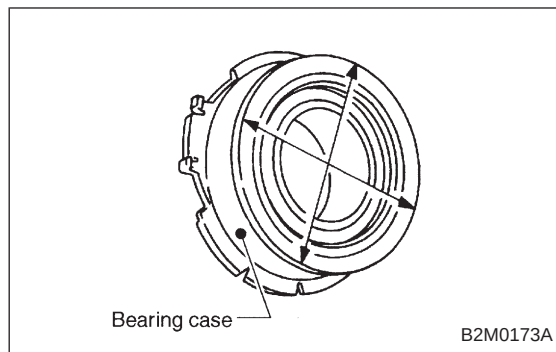
CAUTION:

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4) Remove release lever seal ④.



5) Remove release lever retainer spring from release lever pivot with a screwdriver by accessing it through clutch housing release lever hole. Then remove release lever.



B: INSPECTION

1. RELEASE BEARING

CAUTION:

Since this bearing is grease sealed and is of a nonlubrication type, do not wash with gasoline or any solvent when servicing the clutch.

1) Check the bearing for smooth movement by applying force in the radial direction.

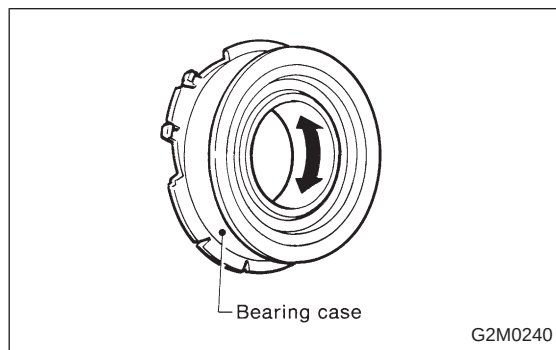
Radial direction stroke:

FWD; Approx.

1.0 mm (0.039 in)

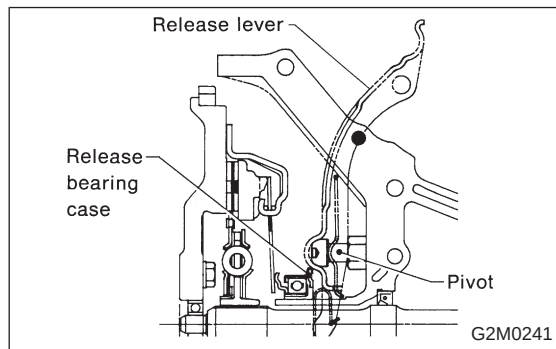
AWD; Approx.

1.4 mm (0.055 in)



2) Check the bearing for smooth rotation by applying pressure in the thrust direction.

3) Check wear and damage of bearing case surface contacting with lever.



2. RELEASE LEVER

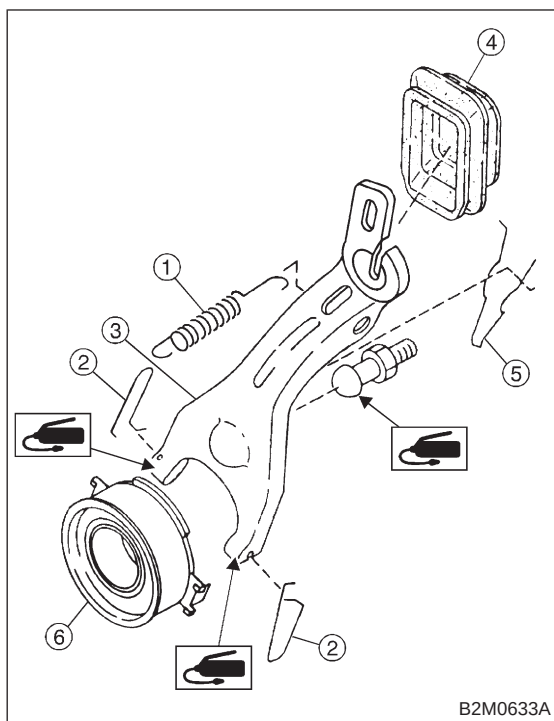
Check lever pivot portion and the point of contact with release bearing case for wear.

C: INSTALLATION

CAUTION:

Before or during assembling, lubricate the following points with a light coat of grease.

- Inner groove of release bearing
- Contact surface of lever and pivot
- Contact surface of lever and bearing
- Transmission main shaft spline (Use grease containing molybdenum disulphide.)



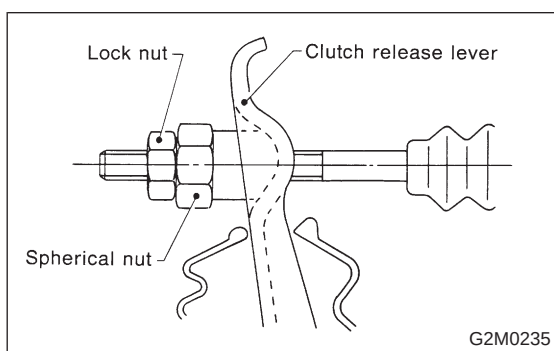
1. MECHANICAL APPLICATION TYPE

1) While pushing release lever (3) to pivot and twisting it to both sides, fit retainer spring (5) onto the constricted portion of pivot.

NOTE:

Confirm that retainer spring is securely fitted by observing it through the main case hole.

- 2) Install release bearing (6) and fasten it with two clips (2).
- 3) Install release lever seal (4).



4) After remounting engine and transmission on body, make adjustment of the clutch release lever end play.

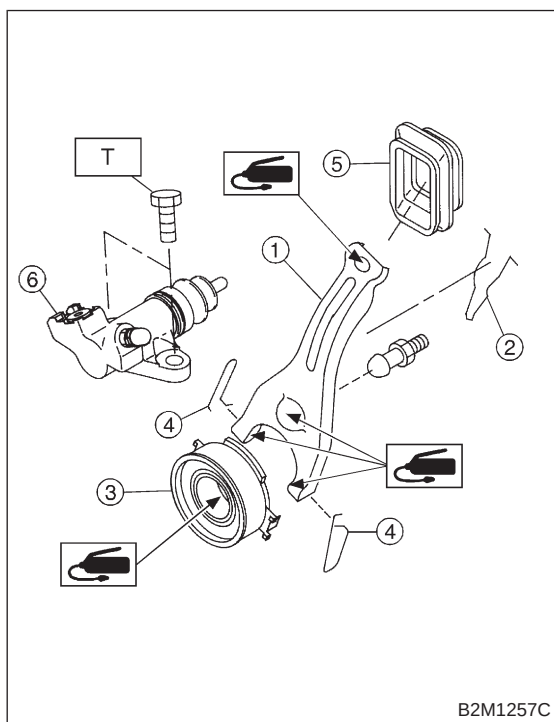
CAUTION:

Take care not to twist the cable during adjustment.

5) Install release lever return spring (Models without hill holder only).

NOTE:

Hook up the return spring to right side hole of the release lever.



2. HYDRAULIC APPLICATION TYPE

1) While pushing release lever (1) to pivot and twisting it to both sides, fit retainer spring (2) onto the constricted portion of pivot.

NOTE:

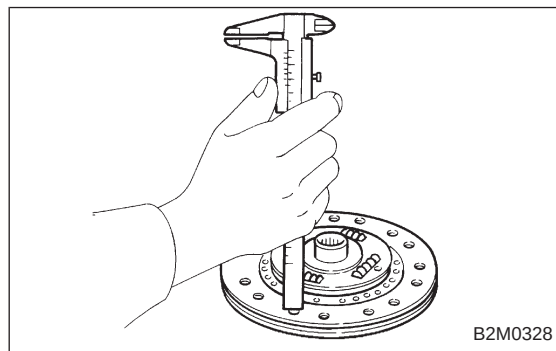
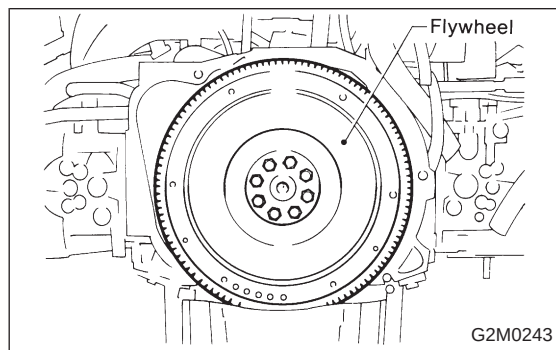
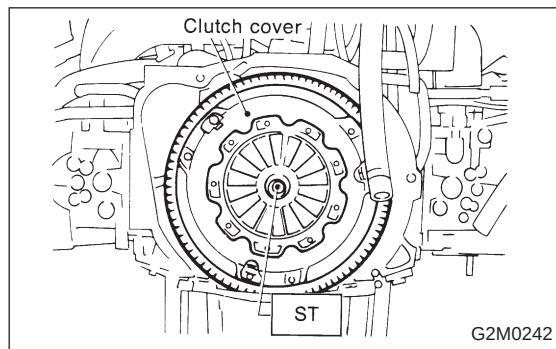
- Apply grease (SUNLIGHT 2: P/N 003602010) to contact point of release lever and operating cylinder.
- Confirm that retainer spring is securely fitted by observing it through the main case hole.

- 2) Install release bearing (3) and fasten it with two clips (4).
- 3) Install release lever seal (5).
- 4) Install operating cylinder (6).

Tightening torque:

T: 37±3 N·m (3.8±0.3 kg-m, 27.5±2.2 ft-lb)

- 5) After remounting engine and transmission on body;
<Ref. to 2-11 [W3B0].>
- 6) Bleed air from oil line with the help of a co-worker.
<Ref. to 2-10 [W202].>



4. Clutch Disc and Cover

A: REMOVAL

- 1) Install ST on flywheel.
ST 498497100 CRANKSHAFT STOPPER
- 2) Remove clutch cover and clutch disc.

CAUTION:

- Take care not to allow oil on the clutch disc facing.
- Do not disassemble either clutch cover or clutch disc.

- 3) Remove flywheel.

B: INSPECTION

1. CLUTCH DISC

- 1) Facing wear
Measure the depth of rivet head from the surface of facing.
Replace if facings are worn locally or worn down to less than the specified value.

Depth of rivet head:

Standard value

1.3 — 1.9 mm (0.051 — 0.075 in)

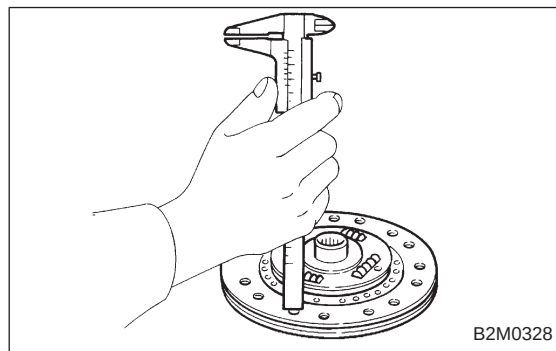
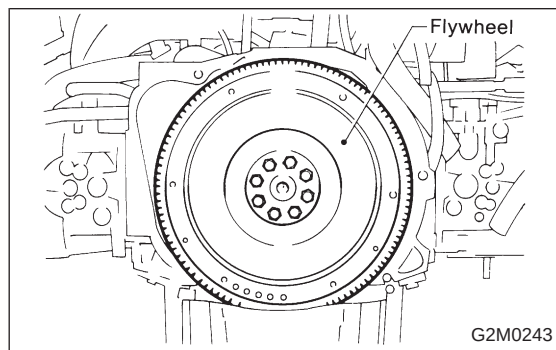
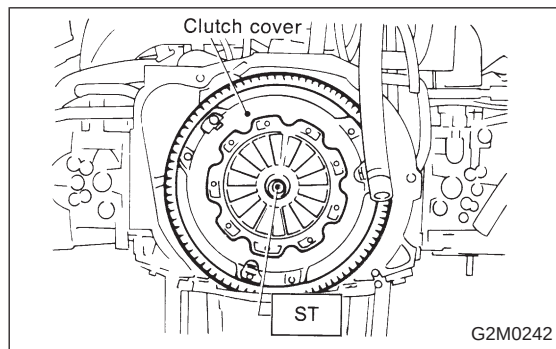
Limit of sinking

0.3 mm (0.012 in)

CAUTION:

Do not wash clutch disc with any cleaning fluid.

- 5) After remounting engine and transmission on body;
<Ref. to 2-11 [W3B0].>
- 6) Bleed air from oil line with the help of a co-worker.
<Ref. to 2-10 [W202].>



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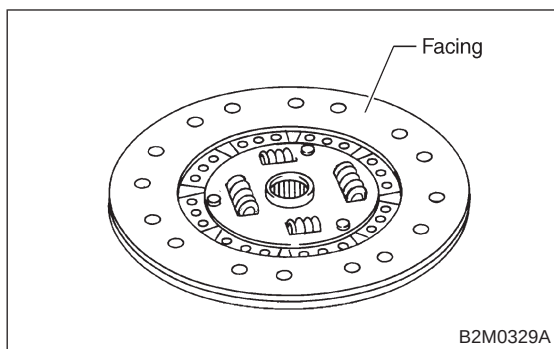
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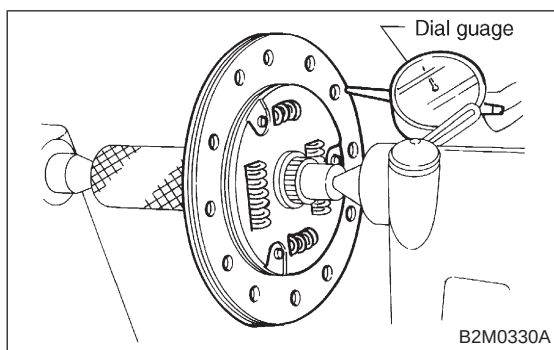


2) Hardened facing

Correct by using emery paper or replace.

3) Oil soakage on facing

Replace clutch disc and inspect transmission front oil seal, transmission case mating surface, engine rear oil seal and other points for oil leakage.

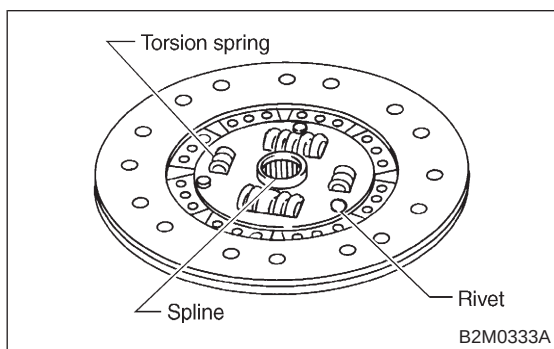


4) Deflection on facing

If deflection exceeds the specified value at the outer circumference of facing, repair or replace.

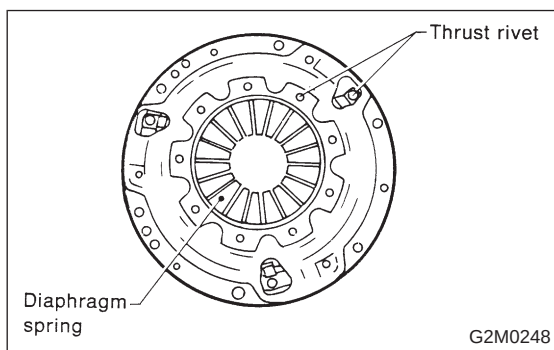
Limit for deflection:

1.0 mm (0.039 in) at R = 107 mm (4.21 in)



5) Worn spline, loose rivets and torsion spring failure

Replace defective parts.

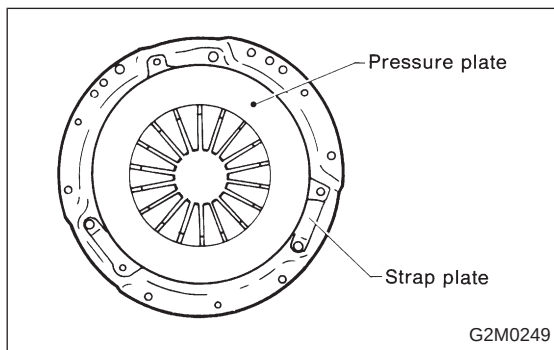


2. CLUTCH COVER

Visually check for the following items without disassembling, and replace or repair if defective.

1) Loose thrust rivet.

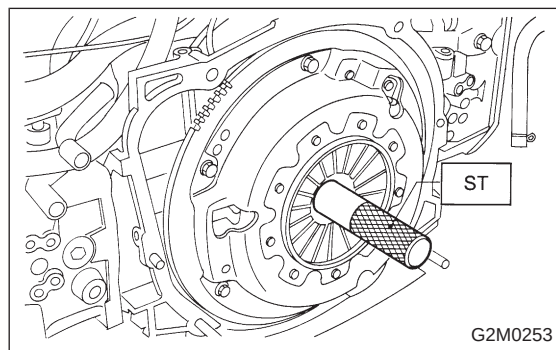
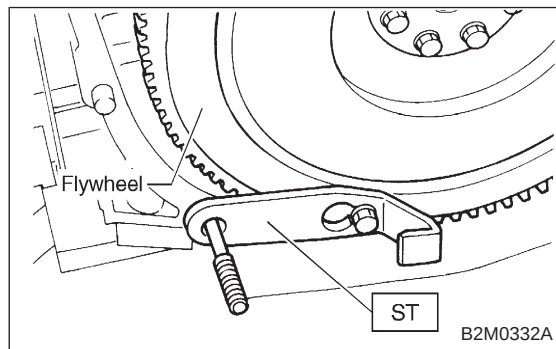
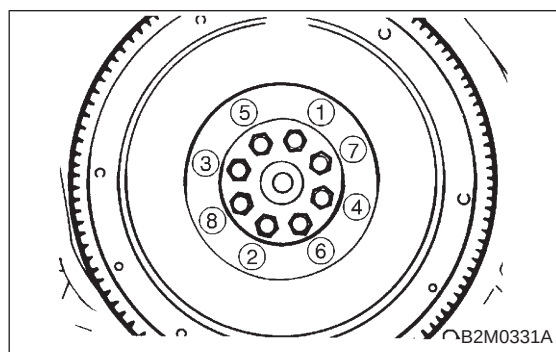
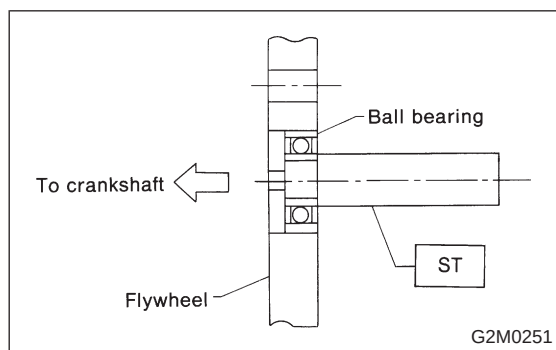
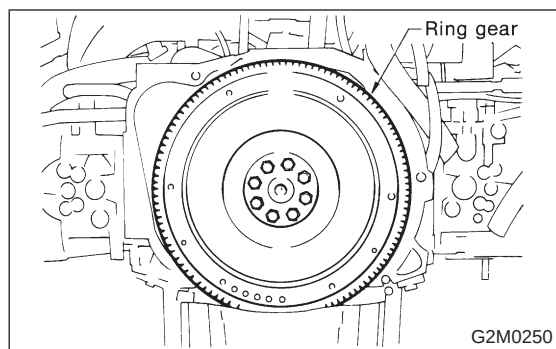
2) Damaged or worn bearing contact area at center of diaphragm spring.



3) Damaged or worn disc contact surface of pressure plate.

4) Loose strap plate setting bolt.

5) Worn diaphragm sliding surface.



3. FLYWHEEL

CAUTION:

Since this bearing is grease sealed and is of a nonlubrication type, do not wash with gasoline or any solvent.

- 1) Damage of facing and ring gear
If defective, replace flywheel.

- 2) Smoothness of rotation
Rotate ball bearing applying pressure in thrust direction.
If noise or excessive play is noted, replace ball bearing as follows:

- (1) Drive out ball bearing from flywheel.
- (2) Press bearing into flywheel until bearing end surface is flush with clutch disc contact surface of flywheel.
Do not press inner race.

ST 899754112 SNAP RING PRESS

C: INSTALLATION

- 1) Install flywheel.
- 2) Install ST, and tighten the flywheel attaching bolts to the specified torque.

ST 498497100 CRANKSHAFT STOPPER

Tightening torque:

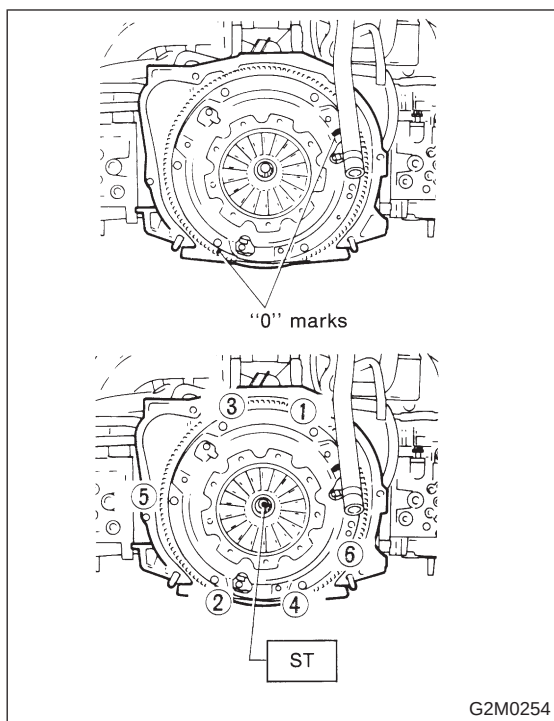
$72 \pm 3 \text{ N} \cdot \text{m}$ ($7.3 \pm 0.3 \text{ kg} \cdot \text{m}$, $52.8 \pm 2.2 \text{ ft} \cdot \text{lb}$)

NOTE:

Tighten flywheel installing bolts gradually. Each bolt should be tightened to the specified torque in a crisscross fashion.

- 3) Insert ST into the clutch disc and install them on the flywheel by inserting the ST end into the pilot bearing.

ST 499747100 CLUTCH DISC GUIDE



4) Install clutch cover on flywheel and tighten bolts to the specified torque.

NOTE:

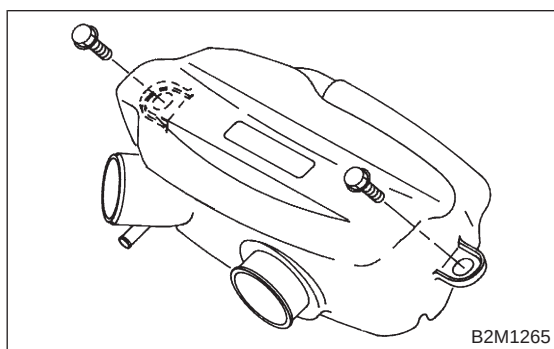
- When installing the clutch cover on the flywheel, position the clutch cover so that there is a gap of 120° or more between "0" marks on the flywheel and clutch cover. ("0" marks indicate the directions of residual unbalance.)
- Note the front and rear of the clutch disc when installing.
- Tighten clutch cover installing bolts gradually. Each bolt should be tightened to the specified torque in a crisscross fashion.

Tightening torque:

15.7±1.5 N·m (1.6±0.15 kg-m, 11.6±1.1 ft-lb)

5) Remove ST.

ST 499747100 CLUTCH DISC GUIDE

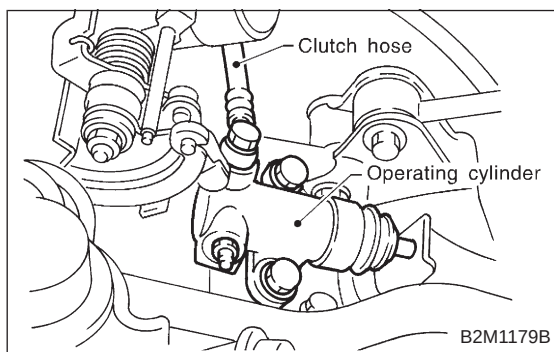


5. Operating Cylinder

A: REMOVAL AND INSTALLATION

1) Remove air chamber.

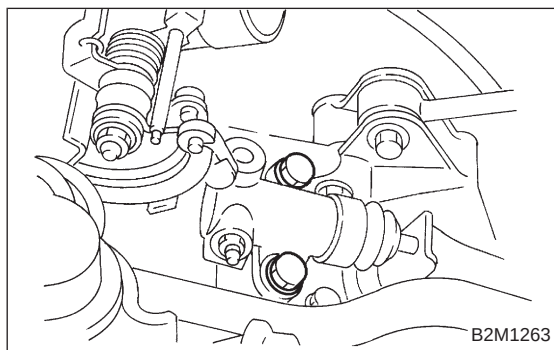
<Ref. to 2-7 [W18A0].>



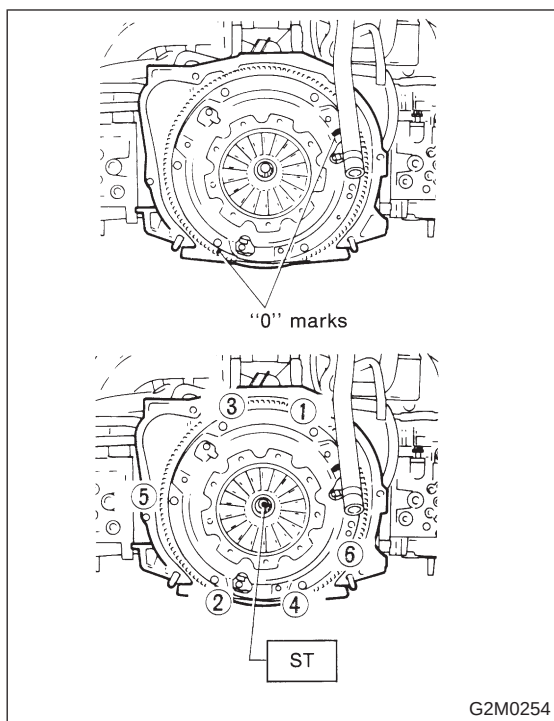
2) Remove clutch hose from operating cylinder.

CAUTION:

Cover hose joint to prevent clutch fluid from flowing out.



3) Remove operating cylinder from transmission.



4) Install clutch cover on flywheel and tighten bolts to the specified torque.

NOTE:

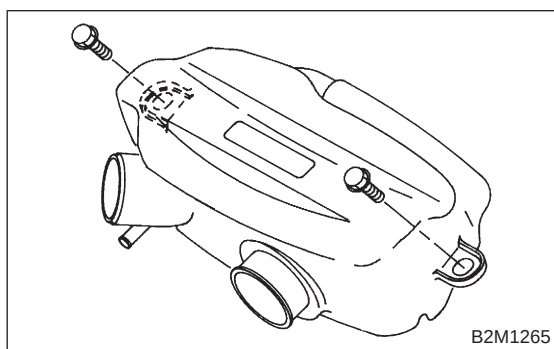
- When installing the clutch cover on the flywheel, position the clutch cover so that there is a gap of 120° or more between "0" marks on the flywheel and clutch cover. ("0" marks indicate the directions of residual unbalance.)
- Note the front and rear of the clutch disc when installing.
- Tighten clutch cover installing bolts gradually. Each bolt should be tightened to the specified torque in a crisscross fashion.

Tightening torque:

15.7±1.5 N·m (1.6±0.15 kg-m, 11.6±1.1 ft-lb)

5) Remove ST.

ST 499747100 CLUTCH DISC GUIDE

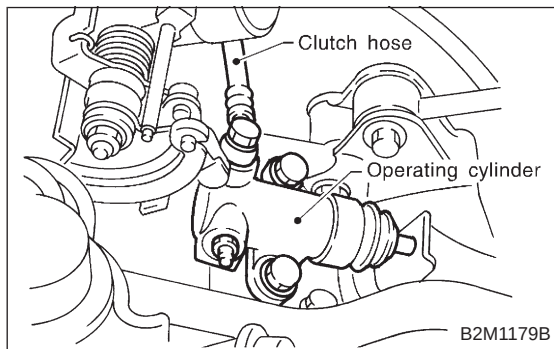


5. Operating Cylinder

A: REMOVAL AND INSTALLATION

1) Remove air chamber.

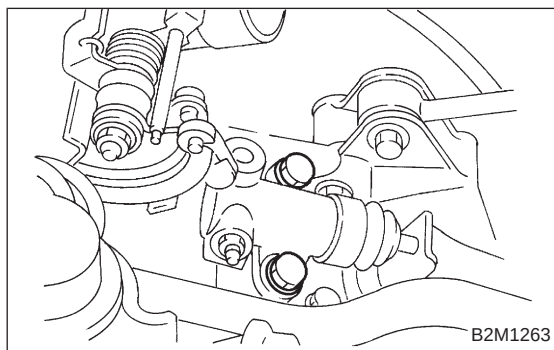
<Ref. to 2-7 [W18A0].>



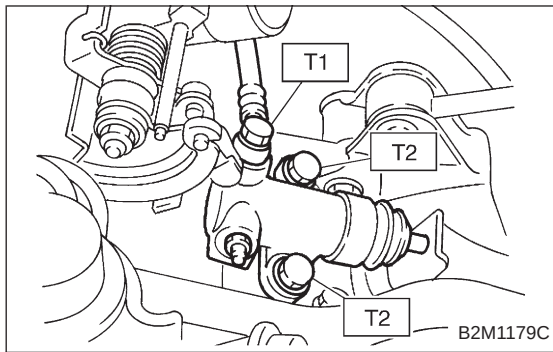
2) Remove clutch hose from operating cylinder.

CAUTION:

Cover hose joint to prevent clutch fluid from flowing out.



3) Remove operating cylinder from transmission.



4) Installation is in the reverse order of removal.

NOTE:

Before installing operating cylinder, apply grease (SUN-LIGHT 2: P/N 003602010) to contact point of release lever and operating cylinder.

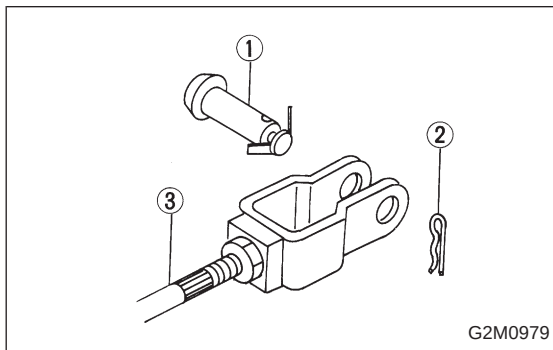
Tightening torque:

T1: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)

T2: 37 ± 3 N·m (3.8 ± 0.3 kg-m, 27.5 ± 2.2 ft-lb)

5) After bleeding air from operating cylinder, ensure that clutch operates properly.

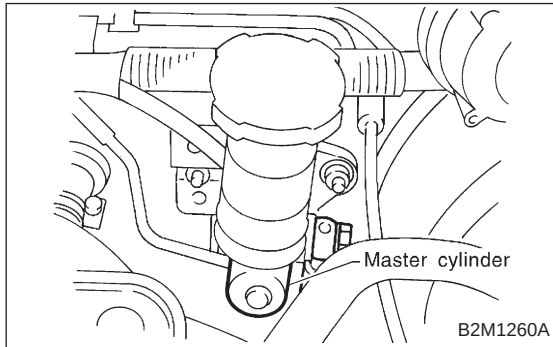
<Ref. to 2-10 [W202].>



6. Master Cylinder and Reservoir Tank

A: REMOVAL

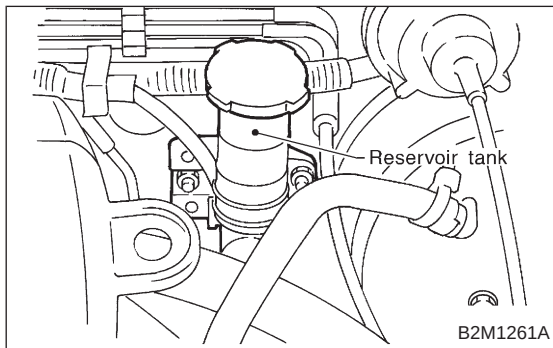
1) Remove snap pin (2), clevis pin (1) and separate push rod (3) of master cylinder from clutch pedal.



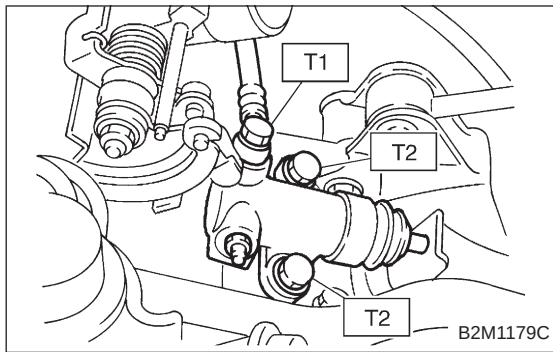
2) Remove clutch hose from master cylinder.

CAUTION:

Plug up hose connection to prevent clutch fluid from spilling out.



3) Remove master cylinder with reservoir tank.



4) Installation is in the reverse order of removal.

NOTE:

Before installing operating cylinder, apply grease (SUN-LIGHT 2: P/N 003602010) to contact point of release lever and operating cylinder.

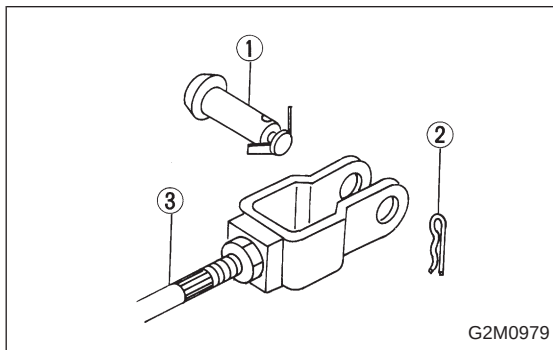
Tightening torque:

T1: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)

T2: 37 ± 3 N·m (3.8 ± 0.3 kg-m, 27.5 ± 2.2 ft-lb)

5) After bleeding air from operating cylinder, ensure that clutch operates properly.

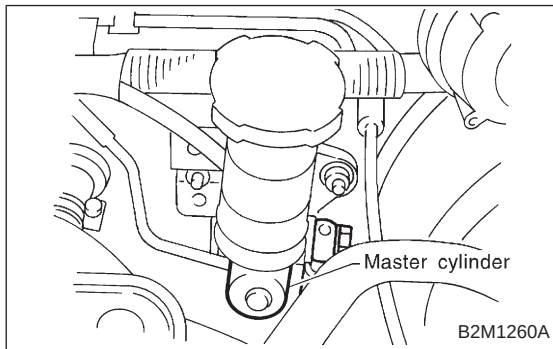
<Ref. to 2-10 [W202].>



6. Master Cylinder and Reservoir Tank

A: REMOVAL

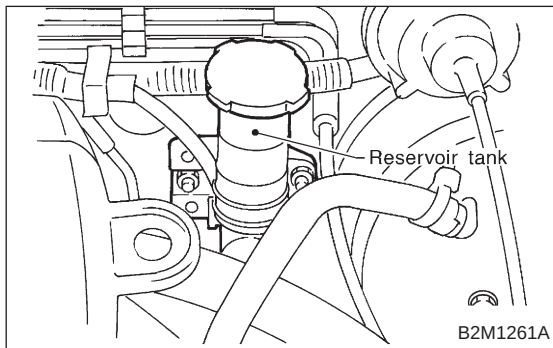
1) Remove snap pin ②, clevis pin ① and separate push rod ③ of master cylinder from clutch pedal.



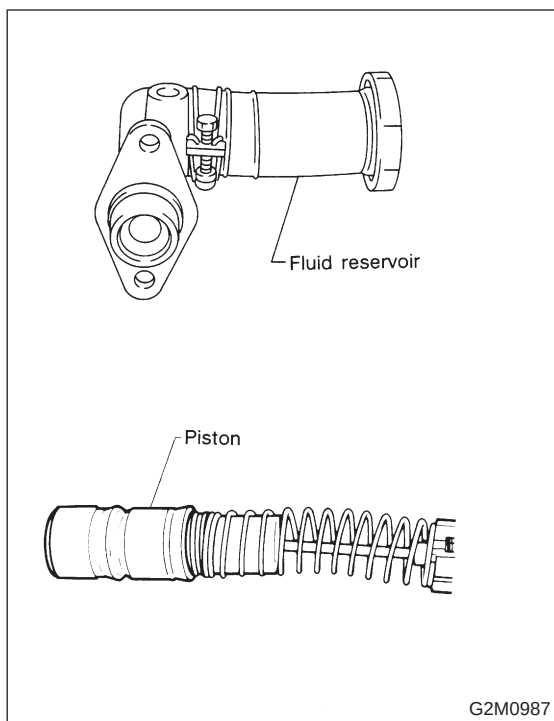
2) Remove clutch hose from master cylinder.

CAUTION:

Plug up hose connection to prevent clutch fluid from spilling out.

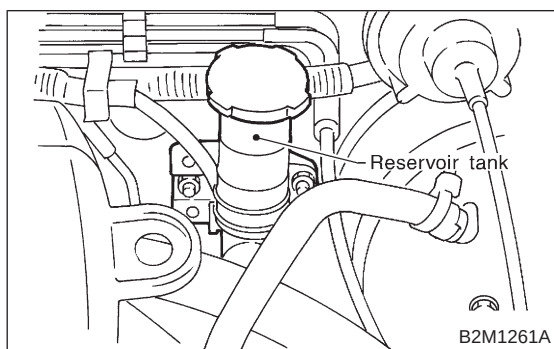


3) Remove master cylinder with reservoir tank.



B: INSPECTION

If any damage, deformation, wear, swelling, rust or other faults are found on the cylinder, piston, push rod, fluid reservoir, seat and gasket, replace the faulty part.

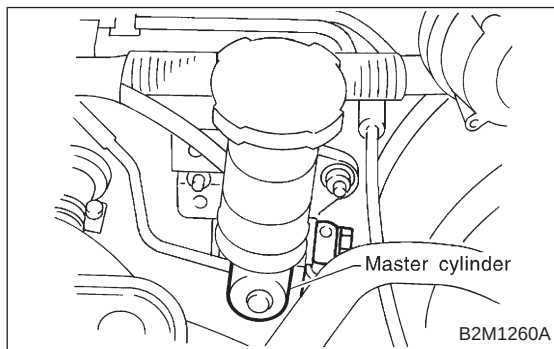


C: INSTALLATION

1) Install master cylinder to body.

Tightening torque:

$18 \pm 3 \text{ N}\cdot\text{m}$ ($1.8 \pm 0.3 \text{ kg}\cdot\text{m}$, $13.0 \pm 2.2 \text{ ft}\cdot\text{lb}$)



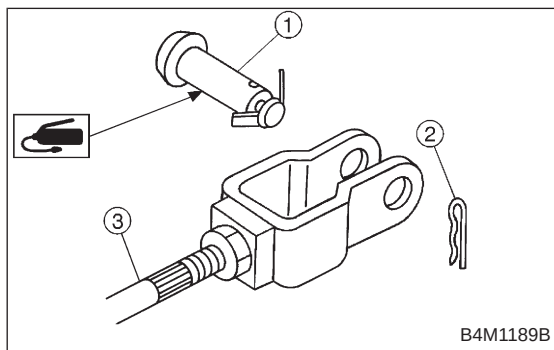
2) Install clutch hose to master cylinder.

CAUTION:

Check that hose is routed properly.

Tightening torque:

$18 \pm 3 \text{ N}\cdot\text{m}$ ($1.8 \pm 0.3 \text{ kg}\cdot\text{m}$, $13.0 \pm 2.2 \text{ ft}\cdot\text{lb}$)



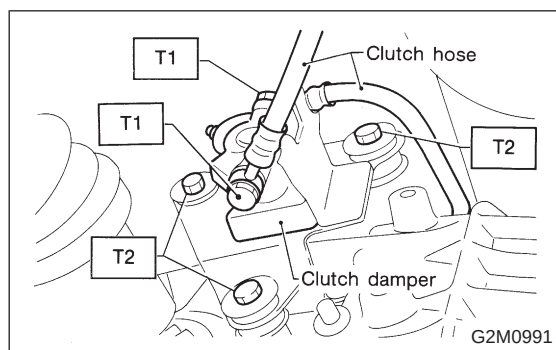
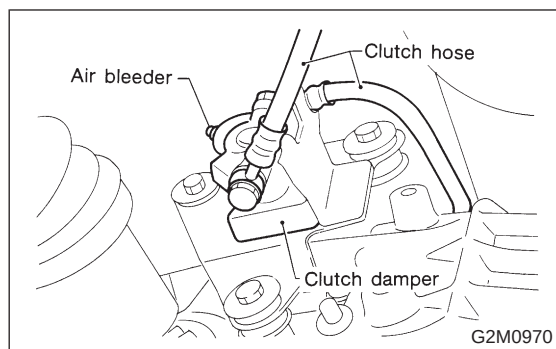
3) Connect push rod ③ of master cylinder to clutch pedal, and install clevis pin ① and snap pin ②.

NOTE:

Apply grease to clevis pin.

4) After bleeding air from system, ensure that clutch operates properly.

<Ref. to 2-10 [W202].>



7. Clutch Damper

A: REMOVAL AND INSTALLATION

- 1) Remove clutch hoses from clutch damper.

CAUTION:

Cover hose joint to prevent clutch fluid from flowing out.

- 2) Remove clutch damper with bracket.

- 3) Installation is in the reverse order of removal.

Tightening torque:

T1: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)

T2: 25 ± 7 N·m (2.5 ± 0.7 kg-m, 18.1 ± 5.1 ft-lb)

- 4) After bleeding air from system, ensure that clutch operates properly.

<Ref. to 2-10 [W202].>

8. Clutch Fluid

A: REPLACEMENT

CAUTION:

- The FMVSS No. 116, fresh DOT3 or 4 brake fluid must be used.
- Cover bleeder with waste cloth, when loosening it, to prevent clutch fluid from being splashed over surrounding parts.
- Avoid mixing different brands of clutch fluid to prevent degrading the quality of the fluid.
- Be careful not to allow dirt or dust to get into the reservoir tank.

NOTE:

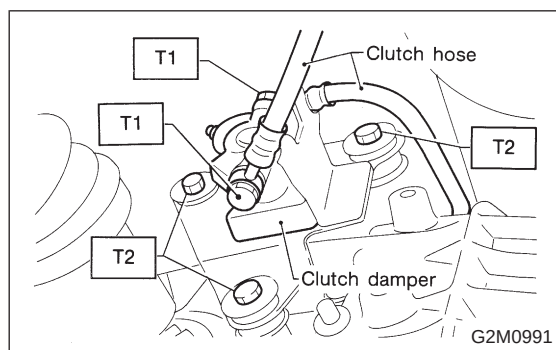
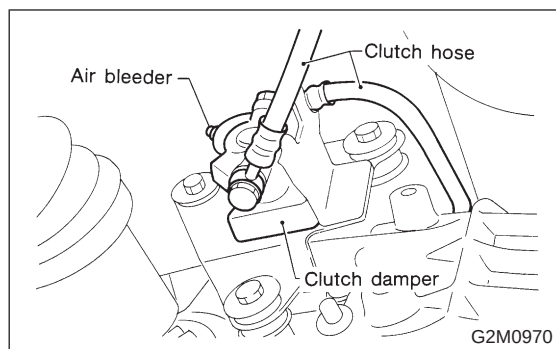
- During bleeding operation, keep the clutch reserve tank filled with clutch fluid to eliminate entry of air.
 - Clutch pedal operating must be very slow.
 - For convenience and safety, it is advisable to have two man working.
 - The amount of clutch fluid required is approximately 70 ml (2.4 US fl oz, 2.5 Imp fl oz) for total clutch system.
- 1) Either jack-up vehicle and place a safety stand under it, or lift-up vehicle.
 - 2) Remove both front and rear wheels.
 - 3) Draw out the clutch fluid from reserve tank with syringe.
 - 4) Refill reservoir tank with recommended clutch fluid.

Recommended clutch fluid:

FMVSS No. 116, fresh DOT3 or 4 brake fluid

- 5) Bleed air from oil line with the help of a co-worker.

<Ref. to 2-10 [W202].>



7. Clutch Damper

A: REMOVAL AND INSTALLATION

- 1) Remove clutch hoses from clutch damper.

CAUTION:

Cover hose joint to prevent clutch fluid from flowing out.

- 2) Remove clutch damper with bracket.

- 3) Installation is in the reverse order of removal.

Tightening torque:

T1: 18 ± 3 N·m (1.8 ± 0.3 kg-m, 13.0 ± 2.2 ft-lb)

T2: 25 ± 7 N·m (2.5 ± 0.7 kg-m, 18.1 ± 5.1 ft-lb)

- 4) After bleeding air from system, ensure that clutch operates properly.

<Ref. to 2-10 [W202].>

8. Clutch Fluid

A: REPLACEMENT

CAUTION:

- The FMVSS No. 116, fresh DOT3 or 4 brake fluid must be used.
- Cover bleeder with waste cloth, when loosening it, to prevent clutch fluid from being splashed over surrounding parts.
- Avoid mixing different brands of clutch fluid to prevent degrading the quality of the fluid.
- Be careful not to allow dirt or dust to get into the reservoir tank.

NOTE:

- During bleeding operation, keep the clutch reserve tank filled with clutch fluid to eliminate entry of air.
 - Clutch pedal operating must be very slow.
 - For convenience and safety, it is advisable to have two man working.
 - The amount of clutch fluid required is approximately 70 ml (2.4 US fl oz, 2.5 Imp fl oz) for total clutch system.
- 1) Either jack-up vehicle and place a safety stand under it, or lift-up vehicle.
 - 2) Remove both front and rear wheels.
 - 3) Draw out the clutch fluid from reserve tank with syringe.
 - 4) Refill reservoir tank with recommended clutch fluid.

Recommended clutch fluid:

FMVSS No. 116, fresh DOT3 or 4 brake fluid

- 5) Bleed air from oil line with the help of a co-worker.

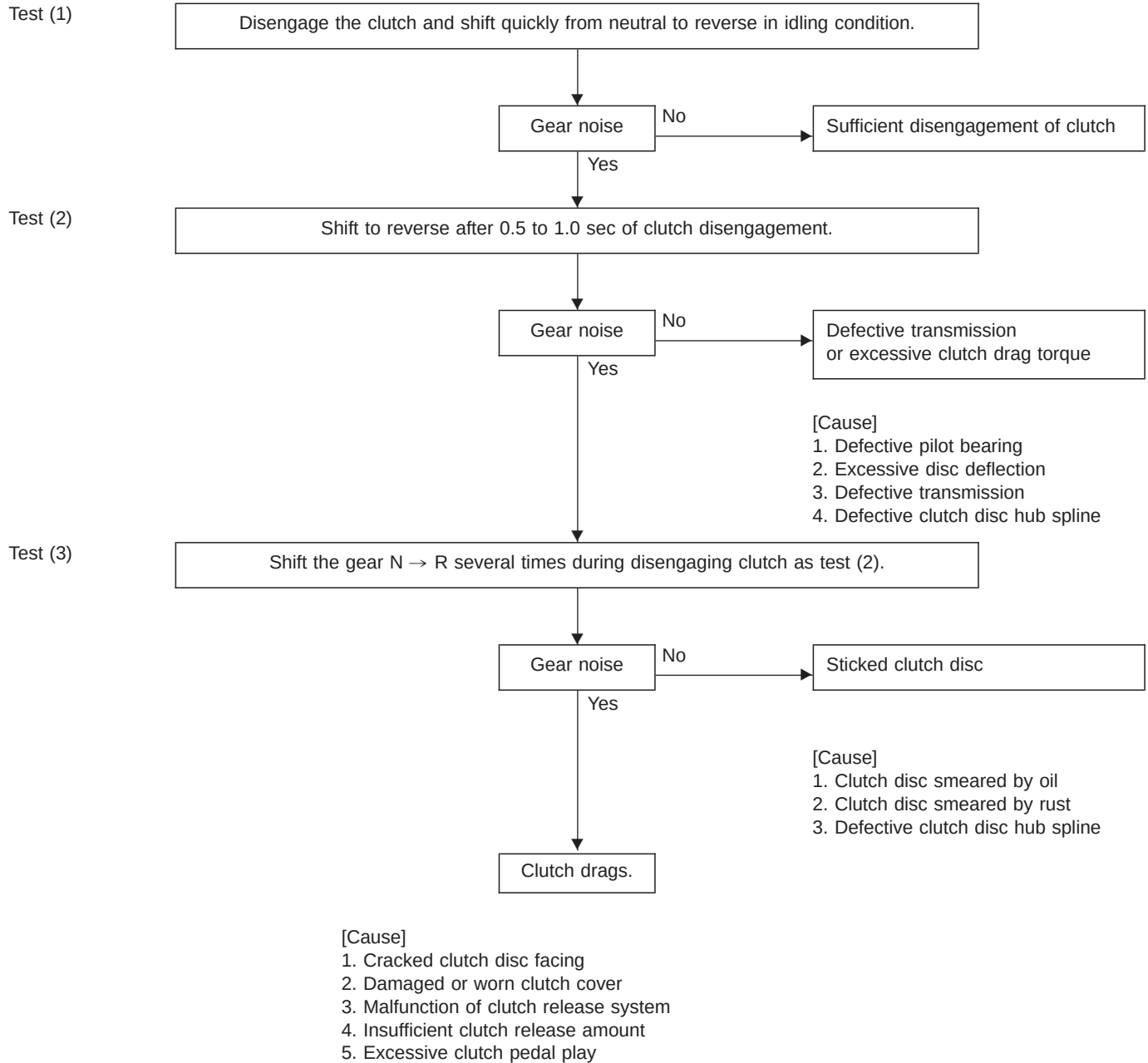
<Ref. to 2-10 [W202].>

1. Clutch System

Condition	Possible cause and testing	Corrective action
1. Clutch slip-page	It is hard to perceive clutch slippage in the early stage, but pay attention to the following symptoms.	
	(a) Engine revs up when shifting. (b) High speed driving is impossible; especially rapid acceleration impossible and vehicle speed does not increase in proportion to an increase in engine speed. (c) Power falls, particularly when ascending a slope, and there is a smell of burning of the clutch facing. ● Method of testing: Put the vehicle in stationary condition with parking brake fully applied. Disengage the clutch and shift the transmission gear into the first. Gradually allow the clutch to engage while gradually increasing the engine speed. The clutch function is satisfactory if the engine stalls. However, the clutch is slipping if the vehicle does not start off and the engine does not stall.	
2. Clutch drags.	(a) No clutch pedal play	Readjust.
	(b) No release lever end play	Readjust.
	(c) Clutch facing smeared by oil	Replace.
	(d) Worn clutch facing	Replace.
	(e) Deteriorated diaphragm spring	Replace.
	(f) Distorted pressure plate or flywheel	Correct or replace.
	(g) Defective release bearing holder	Correct or replace.
	(h) Defective pedal and cable system	Correct or replace.
2. Clutch drags.	As a symptom of this trouble, a harsh scratching noise develops and control becomes quite difficult when shifting gears. The symptom becomes more apparent when shifting into the first gear. However, because much trouble of the this sort is due to defective synchronization mechanism, carry out the test as described after.	
	● Method of testing: Refer to DIAGNOSTIC DIAGRAM on page after.	
	It may be judged as insufficient disengagement of clutch if any noise occurs during this test.	
	(a) Excessive clutch pedal play	Readjust.
	(b) Excessive clutch release lever play	Readjust.
	(c) Worn or rusty clutch disc hub spline	Replace clutch disc.
	(d) Excessive deflection of clutch disc facing	Correct or replace.
	(e) Seized crankshaft pilot needle bearing	Replace.
	(f) Malfunction of pedal and cable system	Correct or replace.
	(g) Cracked clutch disc facing	Replace.
	(h) Sticked clutch disc (smeared by oil or water)	Replace.
3. Clutch chatters.	Clutch chattering is an unpleasant vibration to the whole body when the vehicle is just started with clutch partially engaged.	
	(a) Improper clutch cable routing	Correct.
	(b) Adhesion of oil on the facing	Replace clutch disc.
	(c) Weak or broken torsion spring	Replace clutch disc.
	(d) Defective facing contact or excessive disc	Replace clutch disc defection.
	(e) Warped pressure plate or flywheel	Correct or replace.
	(f) Loose disc rivets	Replace clutch disc.
	(g) Loose engine mounting	Retighten or replace mounting.
	(h) Improper adjustment of pitching stopper	Adjustment.

Condition	Possible cause and testing	Corrective action
4. Noisy clutch	Examine whether the noise is generated when the clutch is disengaged, engaged, or partially engaged.	
	(a) Broken, worn or unlubricated release bearing	Replace release bearing.
	(b) Insufficient lubrication of pilot bearing	Apply grease.
	(c) Loose clutch disc hub	Replace clutch disc.
	(d) Loose torsion spring retainer	Replace clutch disc.
	(e) Deteriorated or broken torsion spring	Replace clutch disc.
5. Clutch grabs.	When starting the vehicle with the clutch partially engaged, the clutch engages suddenly and the vehicle jumps instead of making a smooth start.	
	(a) Grease or oil on facing	Replace clutch disc.
	(b) Deteriorated cushioning spring	Replace clutch disc.
	(c) Worn or rusted spline of clutch disc or main	Take off rust, apply grease or replace clutch shaft disc or mainshaft.
	(d) Deteriorated or broken torsion spring	Replace clutch disc.
	(e) Loose engine mounting	Retighten or replace mounting.
	(f) Deteriorated diaphragm spring	Replace.

1. DIAGNOSTIC DIAGRAM OF CLUTCH DRAG



1. AWD System

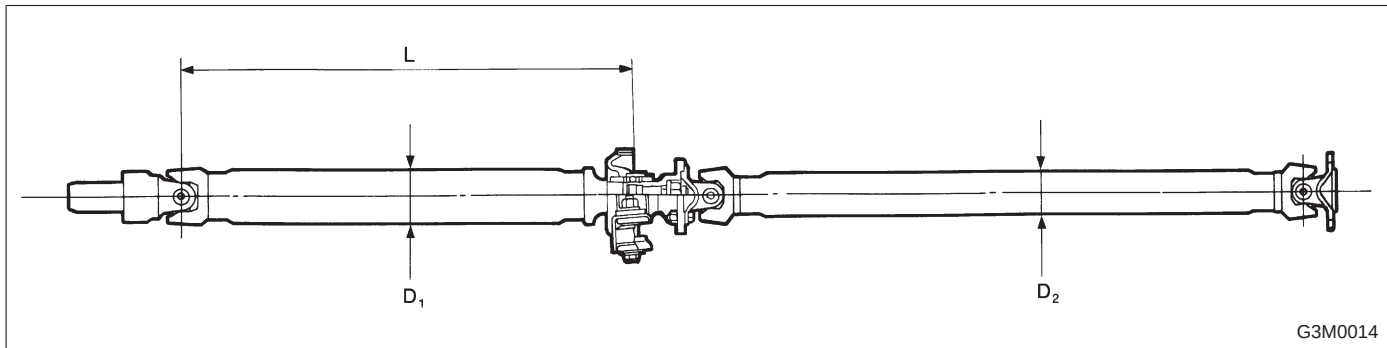
A. SPECIFICATIONS

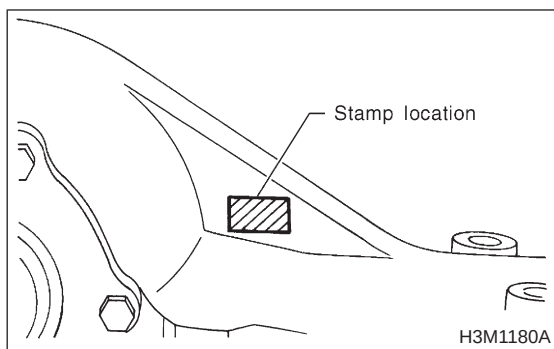
1. REAR FINAL REDUCTION GEAR RATIO

Type of gear	Hypoid			
	MT		AT	
	2200 cc	2500 cc	2200 cc	2500 cc
Gear ratio (Number of gear teeth)	3.900 (39/10)	4.111 (37/9)		4.444 (40/9)
Oil capacity	0.8 ℓ (0.8 US qt, 0.7 Imp qt)			
Rear differential gear oil	GL-5			

2. PROPELLER SHAFT

Front propeller shaft Joint-to-joint length: L mm (in)	AT	539 (21.22)
	MT	598 (23.54)
Outside dia. of tube mm (in)	D ₁	63.5 (2.500)
	D ₂	57.0 (2.244)



**B: IDENTIFICATION**

When replacing a rear differential assembly, select the correct one according to the following table.

CAUTION:

Using the different rear differential assembly causes the drive line and tires to “drag” or emit abnormal noise when AWD is selected.

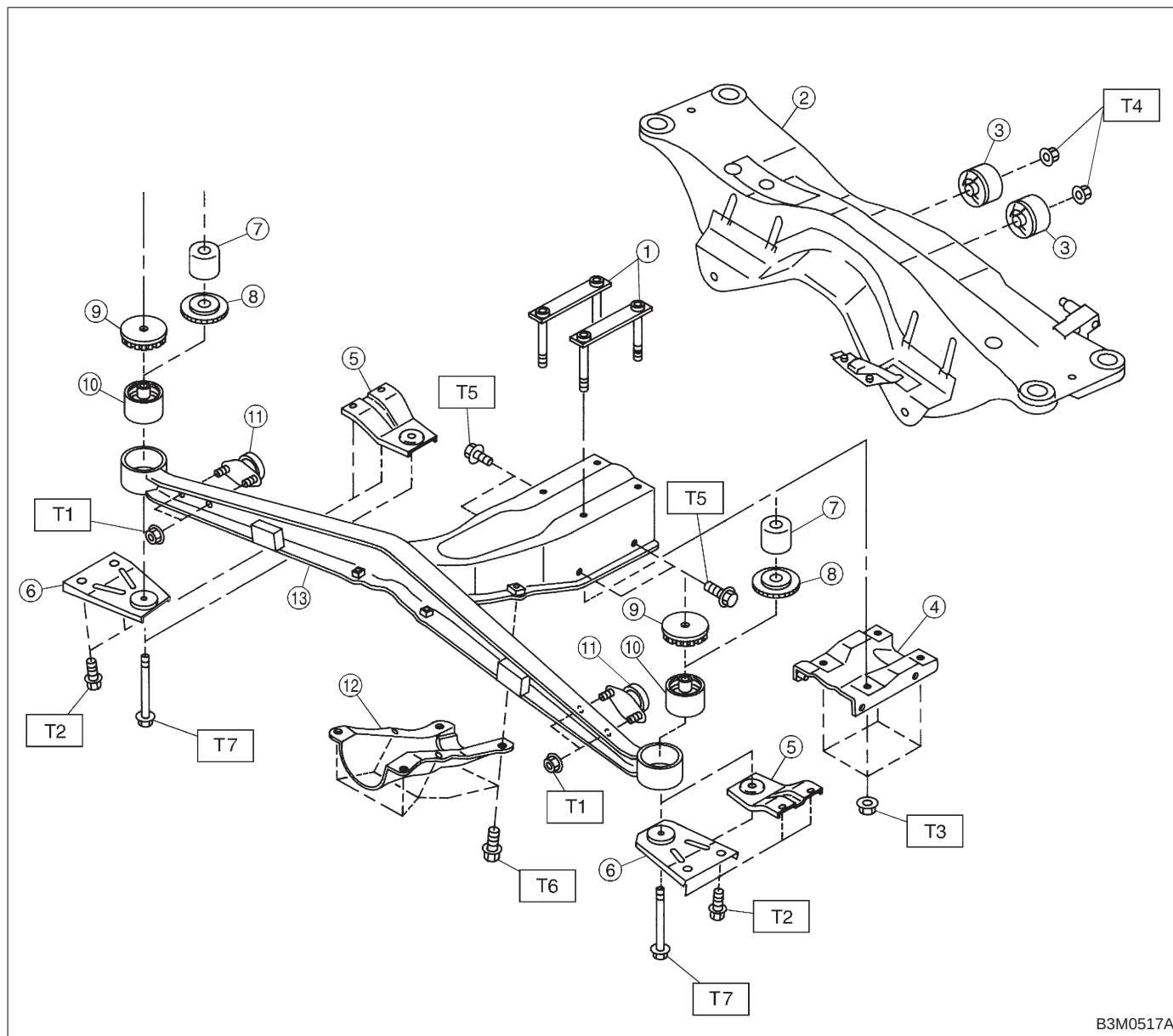
Gear ratio		Part number	Stamp on rear differential
2200 cc MT	3.900	27011AA330	T 1 B3M0124
2200 cc AT	4.111	27011AA340	T 2 B3M0127
2500 cc MT			
2500 cc AT	4.444	27011AA410	T P B3M0421

C: SERVICE DATA

Front and rear bearing preload at companion flange bolt hole	New bearing	19.6 — 28.4 N (2.0 — 2.9 kg, 4.4 — 6.4 lb)
	Used bearing	8.34 — 16.67 N (0.85 — 1.70 kg, 1.87 — 3.75 lb)
Preload adjusting washer length	Part No.	Length
	383705200	2.59 mm (0.1020 in)
	383715200	2.57 mm (0.1012 in)
	383725200	2.55 mm (0.1004 in)
	383735200	2.53 mm (0.0996 in)
	383745200	2.51 mm (0.0988 in)
	383755200	2.49 mm (0.0980 in)
	383765200	2.47 mm (0.0972 in)
	383775200	2.45 mm (0.0965 in)
	383785200	2.43 mm (0.0957 in)
	383795200	2.41 mm (0.0949 in)
	383805200	2.39 mm (0.0941 in)
	383815200	2.37 mm (0.0933 in)
	383825200	2.35 mm (0.0925 in)
	383835200	2.33 mm (0.0917 in)
	383845200	2.31 mm (0.0909 in)
Preload adjusting spacer length	Part No.	Length
	383695201	56.2 mm (2.213 in)
	383695202	56.4 mm (2.220 in)
	383695203	56.6 mm (2.228 in)
	383695204	56.8 mm (2.236 in)
	383695205	57.0 mm (2.244 in)
	383695206	57.2 mm (2.252 in)

Pinion height adjusting shim thickness	Part No.	Thickness
	383495200	3.09 mm (0.1217 in)
	383505200	3.12 mm (0.1228 in)
	383515200	3.15 mm (0.1240 in)
	383525200	3.18 mm (0.1252 in)
	383535200	3.21 mm (0.1264 in)
	383545200	3.24 mm (0.1276 in)
	383555200	3.27 mm (0.1287 in)
	383565200	3.30 mm (0.1299 in)
	383575200	3.33 mm (0.1311 in)
	383585200	3.36 mm (0.1323 in)
	383595200	3.39 mm (0.1335 in)
	383605200	3.42 mm (0.1346 in)
	383615200	3.45 mm (0.1358 in)
	383625200	3.48 mm (0.1370 in)
	383635200	3.51 mm (0.1382 in)
	383645200	3.54 mm (0.1394 in)
	383655200	3.57 mm (0.1406 in)
	383665200	3.60 mm (0.1417 in)
	383675200	3.63 mm (0.1429 in)
	383685200	3.66 mm (0.1441 in)
Side bearing standard width	—	20.00 mm (0.7874 in)
Side bearing retainer shim thickness	Part No.	Thickness
	383475201	0.20 mm (0.0079 in)
	383475202	0.25 mm (0.0098 in)
	383475203	0.30 mm (0.0118 in)
	383475204	0.40 mm (0.0157 in)
	383475205	0.50 mm (0.0197 in)
Crown gear to drive pinion backlash	Limit	0.10 — 0.20 mm (0.0039 — 0.0079 in)
Crown gear runout on its back surface		0.05 mm (0.0020 in)
Oil capacity		0.8 ℓ (0.8 US qt, 0.7 Imp qt)

1. Rear Differential Mounting System



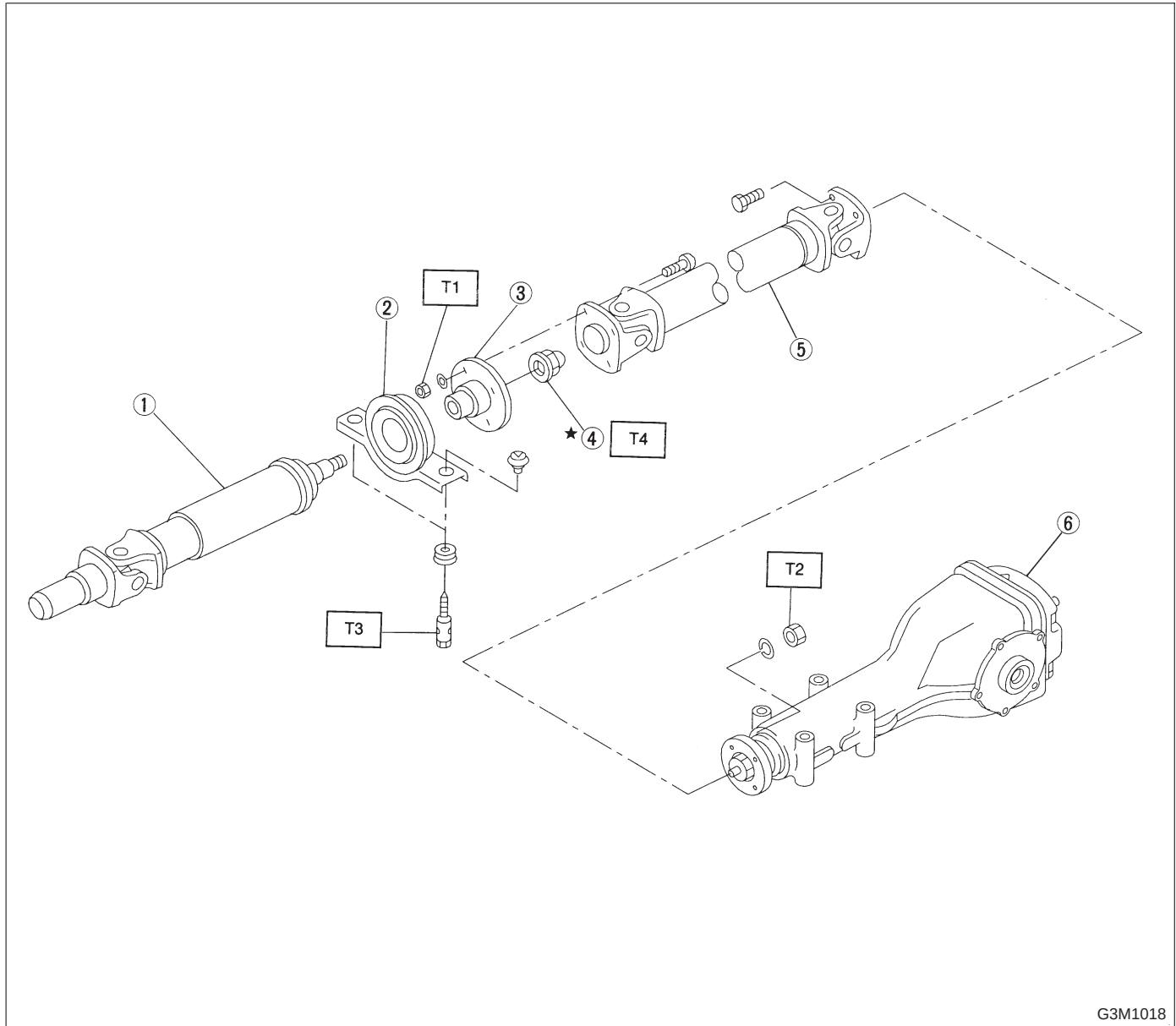
- ① Plate
- ② Crossmember
- ③ Rear bushing
- ④ Differential mount lower bracket
- ⑤ Differential mount bracket (OUTBACK model)
- ⑥ Differential mount bracket
- ⑦ Boss (OUTBACK model)
- ⑧ Stopper (OUTBACK model)
- ⑨ Stopper
- ⑩ Front bushing
- ⑪ Dynamic damper (2500 cc MT model)
- ⑫ Differential mount front cover

- ⑲ Differential front member

Tightening torque: N·m (kg·m, ft·lb)

- T1: 20±5 (2.0±0.5, 14.5±3.6)**
 - T2: 32±8 (3.3±0.8, 23.9±5.8)**
 - T3: 64±8 (6.5±0.8, 47.0±5.8)**
 - T4: 69±8 (7.0±0.8, 50.6±5.8)**
 - T5: 69±10 (7.0±1.0, 51.0±7.2)**
 - T6: 88±10 (9.0±1.0, 65.0±7.2)**
 - T7: 98±10 (10.0±1.0, 72.0±7.2)**
-

2. Propeller Shaft

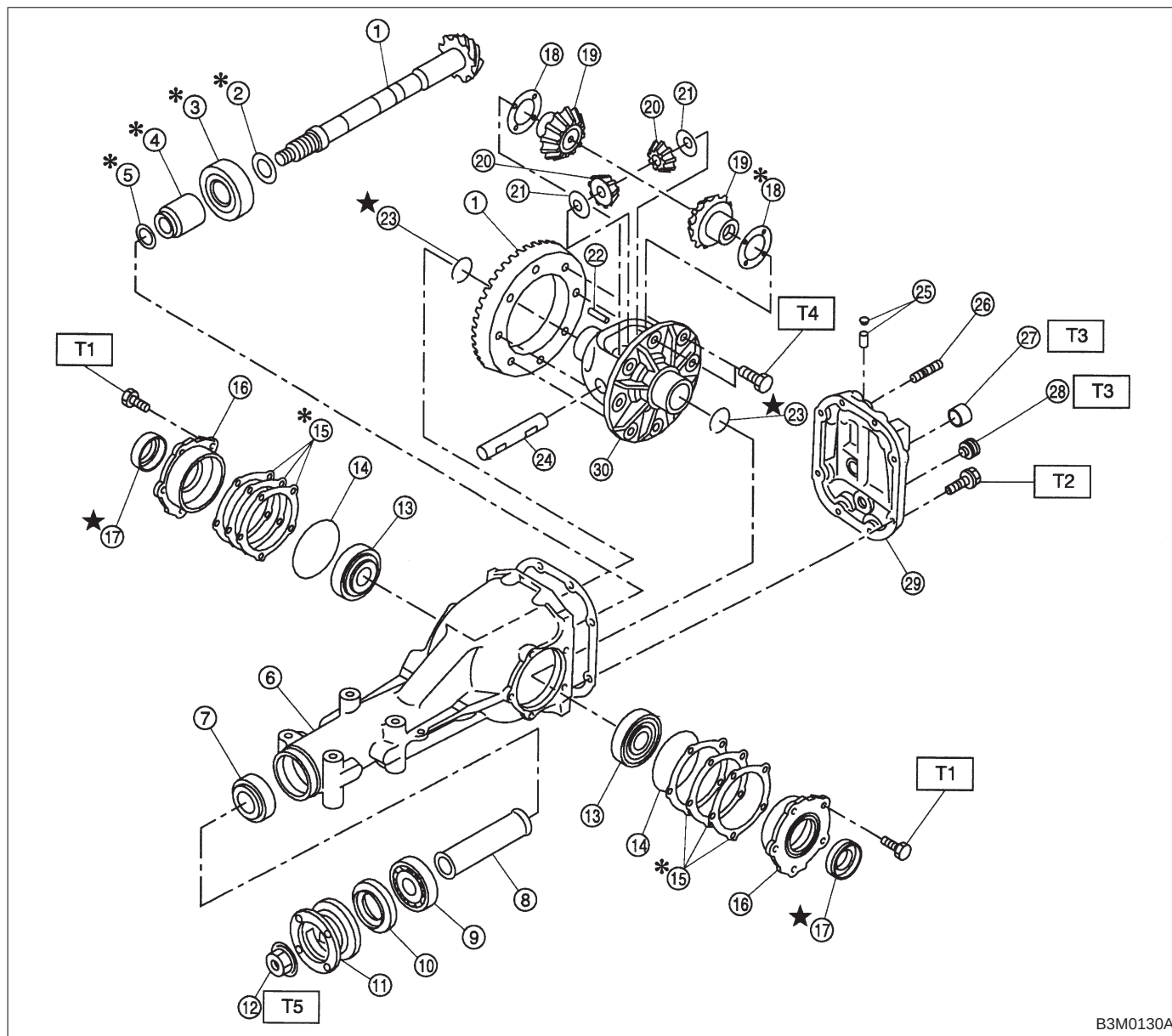


G3M1018

- ① Front propeller shaft
- ② Center bearing
- ③ Companion flange
- ④ Stake nut
- ⑤ Rear propeller shaft
- ⑥ Rear differential

Tightening torque: N·m (kg·m, ft·lb)
T1: 27.9±4.4 (2.85±0.45, 20.6±3.3)
T2: 31±8 (3.2±0.8, 23.1±5.8)
T3: 52±5 (5.3±0.5, 38.3±3.6)
T4: 270±25 (27.5±2.5, 199±18)

3. Rear Differential Assembly



B3M0130A

- ① Pinion crown gear set
- ② Pinion height adjusting washer
- ③ Rear bearing
- ④ Bearing preload adjusting spacer
- ⑤ Bearing preload adjusting washer
- ⑥ Differential carrier
- ⑦ Front bearing
- ⑧ Spacer
- ⑨ Pilot bearing
- ⑩ Front oil seal
- ⑪ Companion flange
- ⑫ Self-locking nut
- ⑬ Side bearing
- ⑭ O-ring

- ⑮ Side bearing retainer shim
- ⑯ Side bearing retainer
- ⑰ Side oil seal
- ⑱ Side gear thrust washer
- ⑲ Side gear
- ⑳ Pinion mate gear
- ㉑ Pinion mate gear washer
- ㉒ Pinion shaft lock pin
- ㉓ Circlip
- ㉔ Pinion mate shaft
- ㉕ Air breather cap
- ㉖ Stud bolt
- ㉗ Oil filler plug
- ㉘ Oil drain plug

- ㉙ Rear cover
- ㉚ Differential case

Tightening torque: N·m (kg·m, ft·lb)

- T1: 10.3±1.5**
(1.05±0.15, 7.6±1.1)
- T2: 29.4±4.9**
(3.00±0.50, 21.7±3.6)
- T3: 44.1±3.9**
(4.50±0.40, 32.5±2.9)
- T4: 103.0±9.8**
(10.50±1.00, 75.9±7.2)
- T5: 181.4±14.7**
(18.50±1.50, 133.8±10.8)

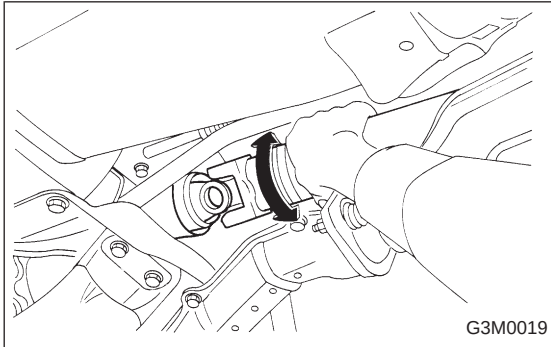
1. Propeller Shaft

A: ON-CAR SERVICE

Check the following points with propeller shaft installed in vehicle.

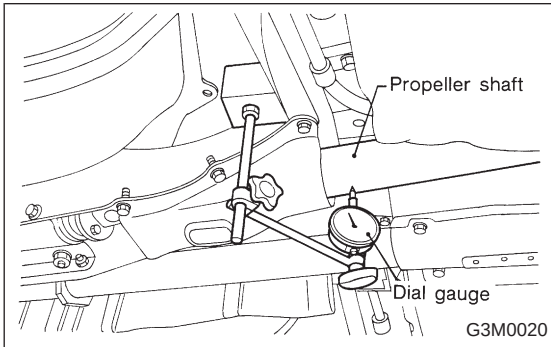
1) Joints and connections

Check for any looseness of yoke flange connecting bolts and center bearing retaining bolts.



2) Splines and bearing locations

Turn propeller shaft by hand to see if abnormal free play exists at splines. Also move yokes to see if abnormal free play exists at spiders and bearings.



3) Runout of propeller shaft

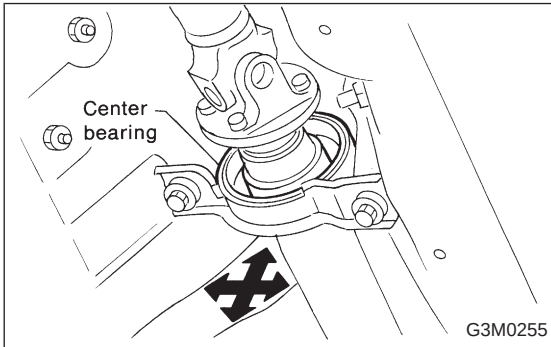
Turn rear wheels by hand to check for "runout" of propeller shaft.

Runout:

Limit 0.6 mm (0.024 in)

NOTE:

Measure runout with a dial gauge at the center of front and rear propeller shaft tubes.



4) Center bearing free play

While holding propeller shaft near center bearing with your hand, move it up and down, and left and right to check for any abnormal bearing free play.

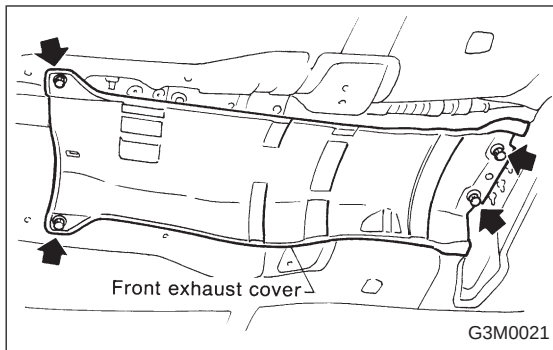
B: REMOVAL

NOTE:

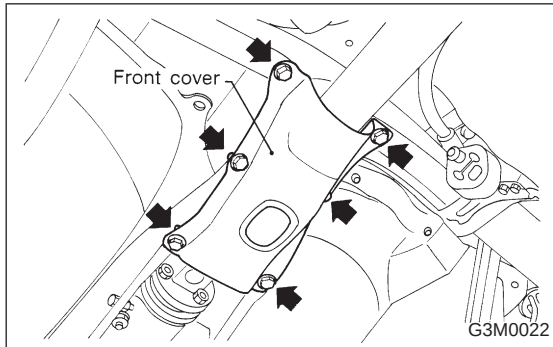
Before removing propeller shaft, wrap metal parts with a cloth or rubber material.

- 1) Disconnect ground cable from battery.
 - 2) Move selector lever or gear shift lever to "N".
 - 3) Release the parking brake.
 - 4) Jack-up vehicle and support it with sturdy racks.
 - 5) Remove rear exhaust pipe and muffler.
- <Ref. to 2-9 [W2A0], [W3A0].>

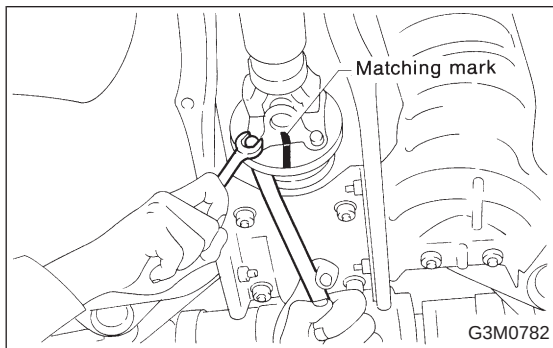
1. Propeller Shaft



6) Remove front exhaust cover.



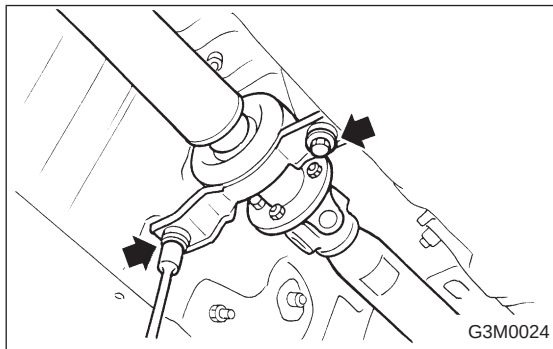
7) Remove differential mount front cover.



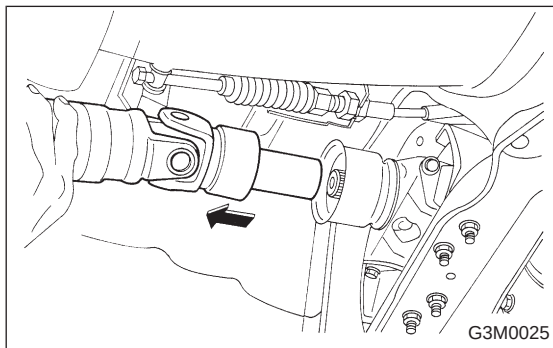
8) Remove the four bolts which hold propeller shaft to rear differential.

NOTE:

- Put matching mark on affected parts before removal.
- Remove all but one bolt.



9) Remove the two bolts which hold center bearing to vehicle body.



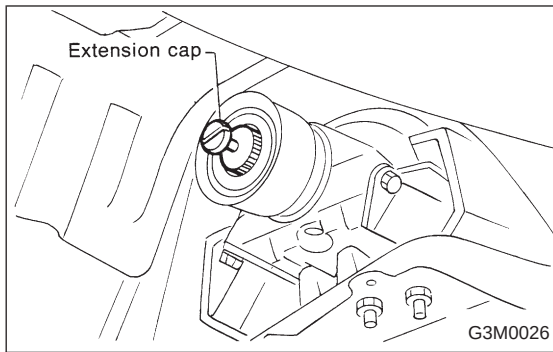
10) Remove propeller shaft from transmission.

CAUTION:

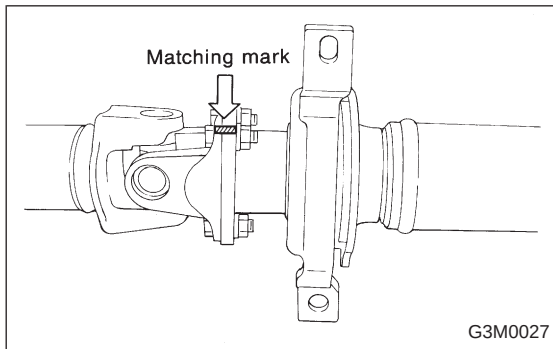
Be sure not to damage oil seals and the frictional surface of sleeve yoke.

NOTE:

- Be sure to use an empty oil can to catch oil flowing out when removing propeller shaft.
- Be sure to plug the opening in transmission after removal of propeller shaft.



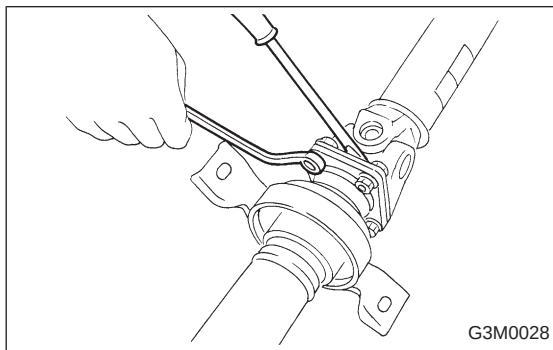
11) Install the extension cap to transmission.



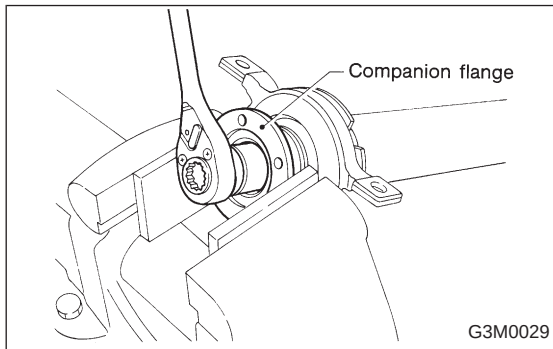
C: DISASSEMBLY

Before removing center bearing, check its condition. If it does not operate smoothly or if there is any free play or leakage, remove as follows:

1) Put matching marks on affected parts.



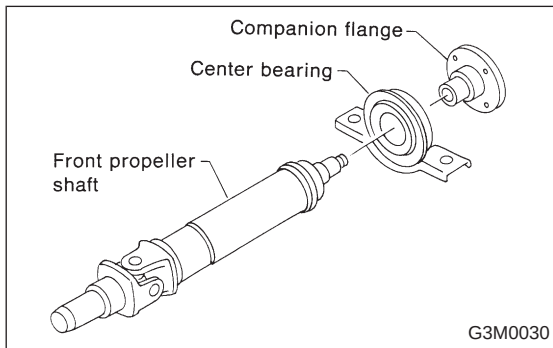
2) Remove bolts which hold front propeller shaft to rear propeller shaft.



3) Place companion flange in a vise and remove stake nut.

CAUTION:

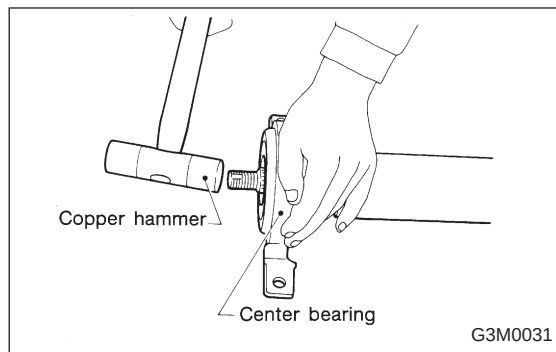
Be sure not to hold propeller shaft pipe portion in the vise.



4) Drive out companion flange with a puller or press.

NOTE:

Before disassembling, put matching mark on affected parts.



5) Lightly tap the head of front propeller shaft with a copper hammer until center bearing is removed.

CAUTION:

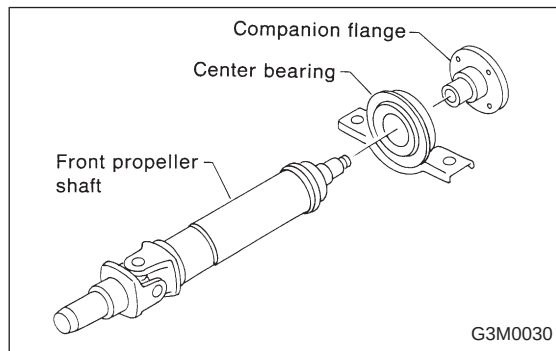
Be careful not to damage the thread portion.

D: INSPECTION

NOTE:

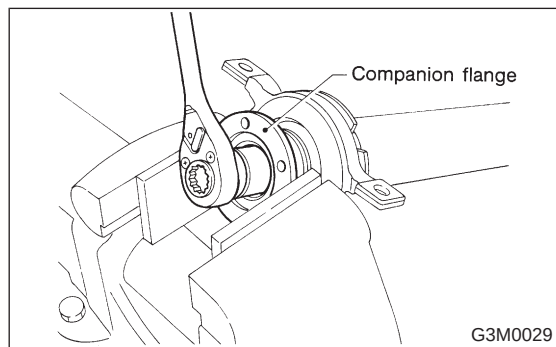
Do not disassemble propeller shaft. Check the following and replace if necessary.

- 1) Tube surfaces for dents or cracks
- 2) Splines for deformation or abnormal wear
- 3) Joints for non-smooth operation or abnormal noise
- 4) Center bearing for free play, noise or non-smooth operation
- 5) Oil seals for abnormal wear or damage
- 6) Center bearing for breakage



E: ASSEMBLY

- 1) Install center bearing onto front propeller shaft.
- 2) Align marks and install companion flange.



- 3) Tighten stake nut until center bearing is set in position.

CAUTION:

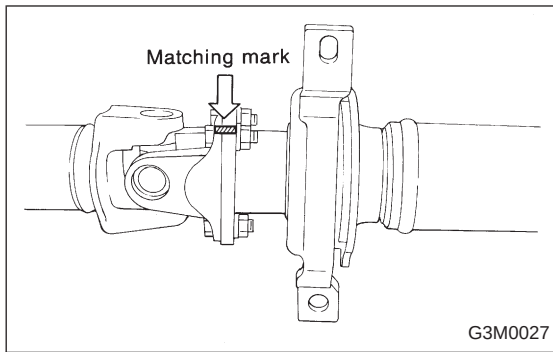
Be sure to install new stake nut.

Tightening torque:

$270 \pm 25 \text{ N} \cdot \text{m}$ ($27.5 \pm 2.5 \text{ kg} \cdot \text{m}$, $199 \pm 18 \text{ ft} \cdot \text{lb}$)

NOTE:

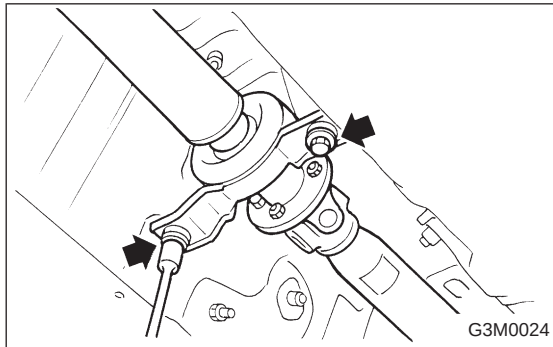
Stake the nut after tightening.



4) Align matching marks and connect front and rear propeller shafts.

Tightening torque:

$27.9 \pm 4.4 \text{ N}\cdot\text{m}$ ($2.85 \pm 0.45 \text{ kg}\cdot\text{m}$, $20.6 \pm 3.3 \text{ ft}\cdot\text{lb}$)

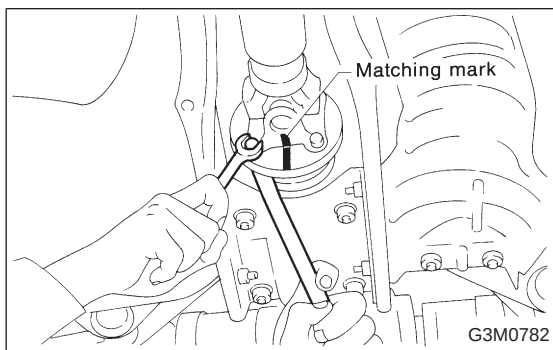


F: INSTALLATION

1) Insert sleeve yoke into transmission and attach center bearing to vehicle body.

Tightening torque:

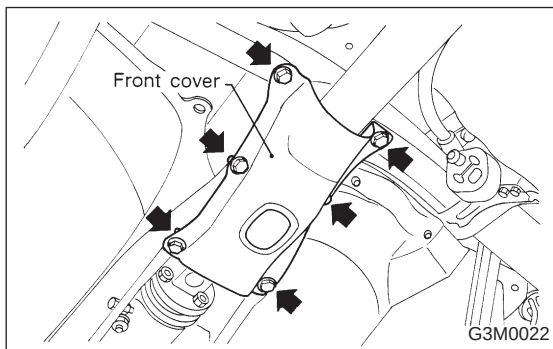
$52 \pm 5 \text{ N}\cdot\text{m}$ ($5.3 \pm 0.5 \text{ kg}\cdot\text{m}$, $38.3 \pm 3.6 \text{ ft}\cdot\text{lb}$)



2) Align matching marks and connect flange yoke and rear differential.

Tightening torque:

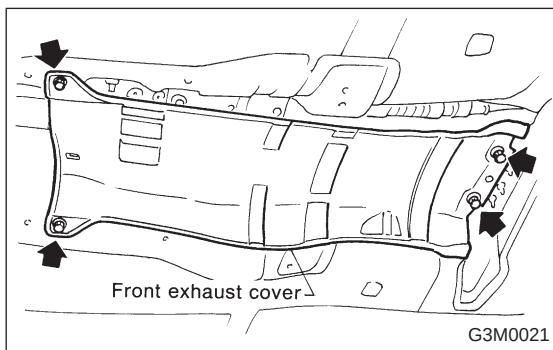
$31 \pm 8 \text{ N}\cdot\text{m}$ ($3.2 \pm 0.8 \text{ kg}\cdot\text{m}$, $23.1 \pm 5.8 \text{ ft}\cdot\text{lb}$)



3) Install differential mount front cover.

Tightening torque:

$88 \pm 10 \text{ N}\cdot\text{m}$ ($9.0 \pm 1.0 \text{ kg}\cdot\text{m}$, $65 \pm 7 \text{ ft}\cdot\text{lb}$)



4) Install front exhaust cover.

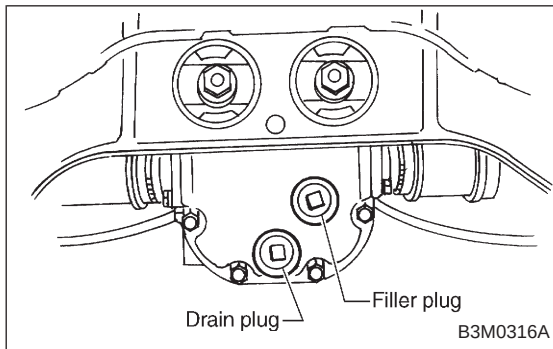
5) Install rear exhaust pipe and muffler.

2. Rear Differential

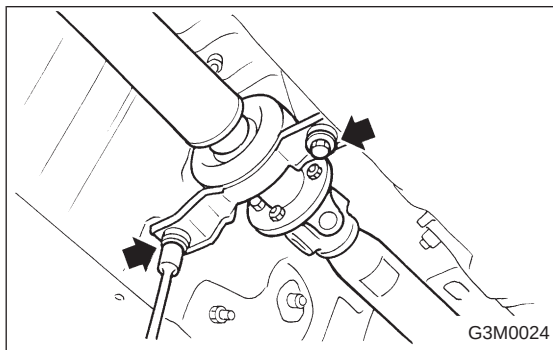
A: ON-CAR SERVICE

1. FRONT OIL SEAL

- 1) Disconnect ground cable from battery.
- 2) Move selector lever or gear shift lever to "N".
- 3) Release the parking brake.



- 4) Remove oil drain plug, and drain gear oil.

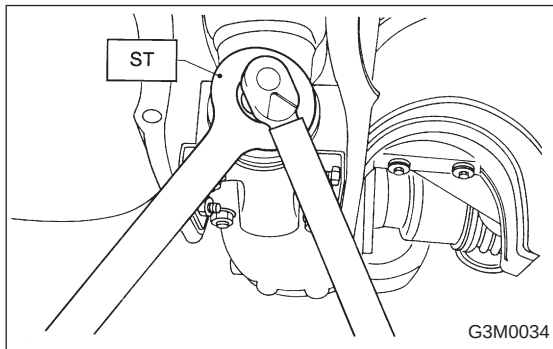


- 5) Jack-up rear wheels and support the vehicle body with sturdy racks.

- 6) Remove propeller shaft from body. <Ref. to 3-4 [W1B0].>

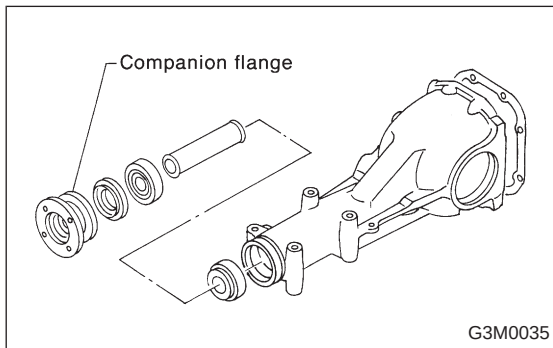
CAUTION:

Wrap metal parts with a cloth or rubber material to prevent damage from adjacent metal parts.

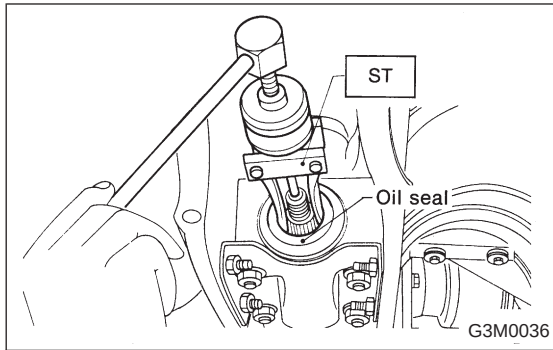


- 7) Remove self-locking nut while holding companion flange with ST.

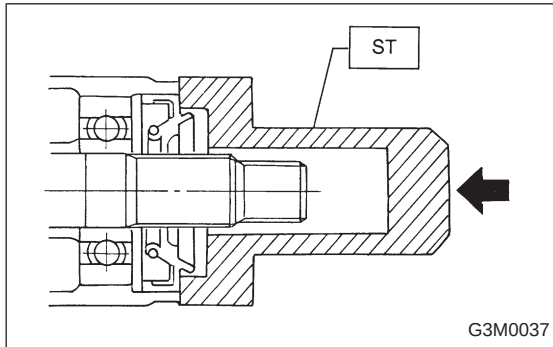
ST 498427200 FLANGE WRENCH



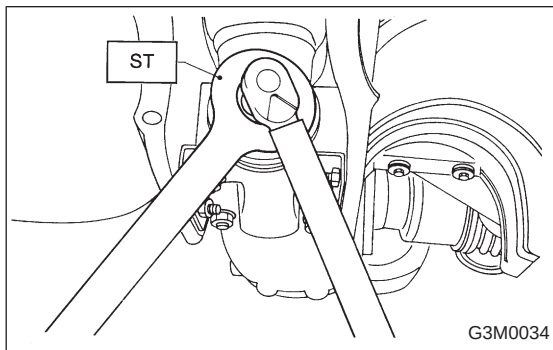
- 8) Extract companion flange with a puller.



- 9) Remove oil seal using ST.
ST 499705401 PULLER ASSY



- 10) Fit a new oil seal using ST.
ST 498447120 OIL SEAL INSTALLER



- 11) Install companion flange.
12) Tighten self-locking nut within the specified torque range so that the turning resistance of companion flange becomes the same as that before replacing oil seal.

ST 498427200 FLANGE WRENCH

CAUTION:

Use a new self-locking nut.

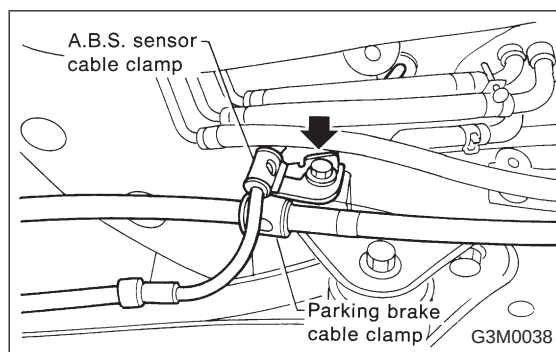
Tightening torque:

$181.4 \pm 14.7 \text{ N}\cdot\text{m}$ ($18.50 \pm 1.50 \text{ kg}\cdot\text{m}$, $133.8 \pm 10.8 \text{ ft}\cdot\text{lb}$)

- 13) Reassembling procedure hereafter is the reverse of the disassembling.

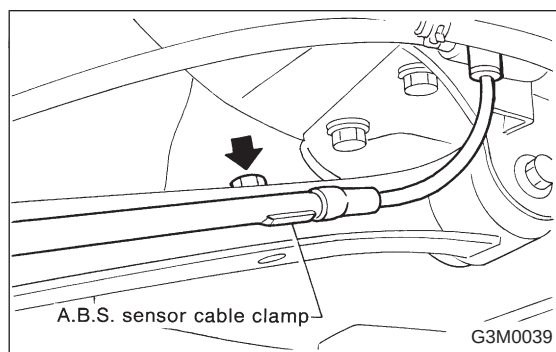
2. SIDE OIL SEAL

- 1) Disconnect ground cable from battery.
 - 2) Move selector lever or gear shift lever to "N".
 - 3) Release the parking brake.
 - 4) Loosen both wheel nuts.
 - 5) Jack-up the vehicle and support it with rigid racks.
 - 6) Remove wheels.
 - 7) Remove rear exhaust pipe and muffler.
- <Ref. to 2-9 [W2A0], [W3A0].>

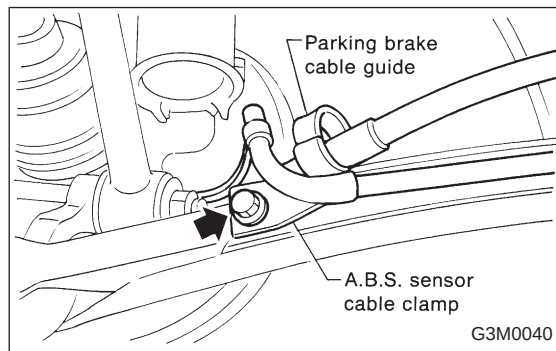


- 8) Remove the DOJ of rear drive shaft from rear differential.

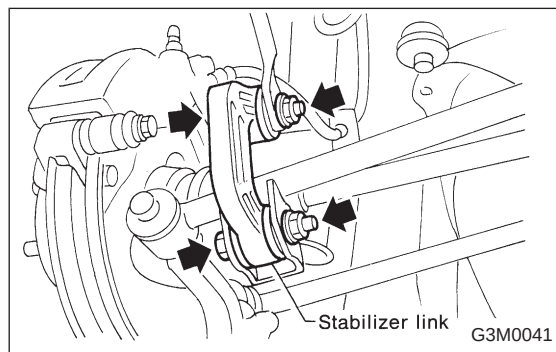
- (1) Remove the A.B.S. sensor cable clamp and parking brake cable clamp from bracket.



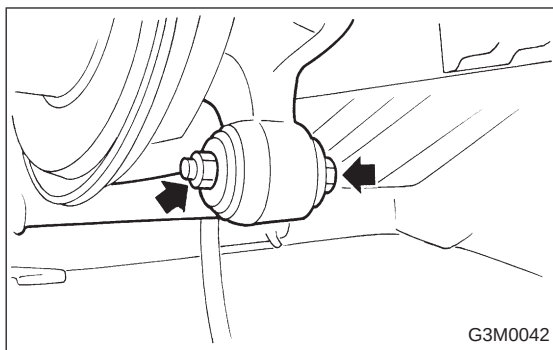
- (2) Remove the A.B.S. sensor cable clamp from the trailing link.



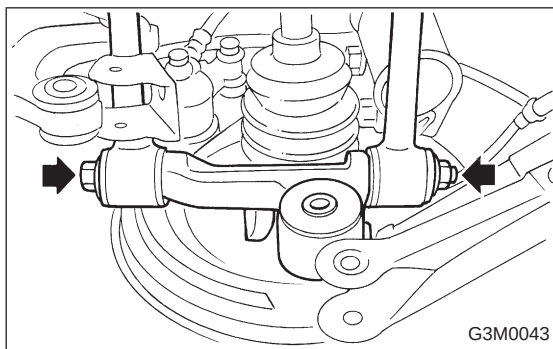
- (3) Remove the A.B.S. sensor cable clamp and parking brake cable guide from the trailing link.



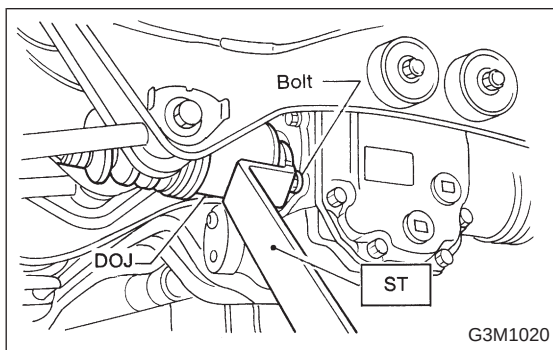
- (4) Remove the rear stabilizer link.



(5) Remove the bolts which secure the trailing link to the rear housing.



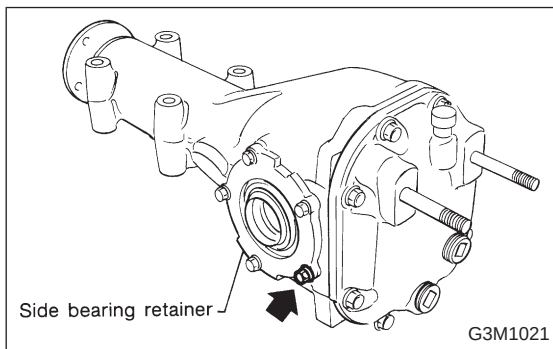
(6) Remove the bolts which secure the front and rear lateral link to the rear housing.



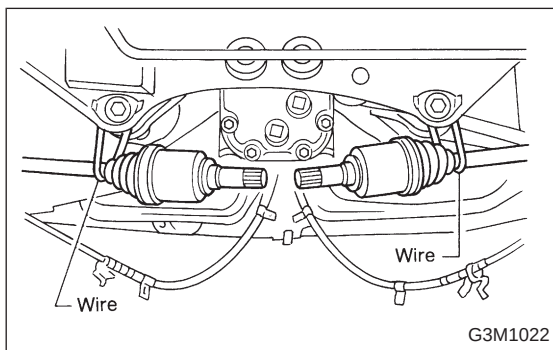
(7) Remove crossmember reinforcement lower (AWD Sedan only).

(8) Remove the DOJ from the rear differential by using ST.

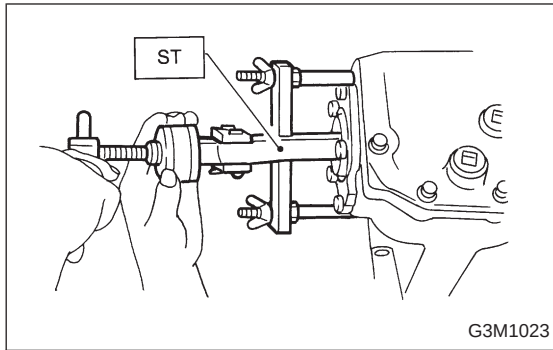
ST 208099PA100 DRIVE SHAFT REMOVER



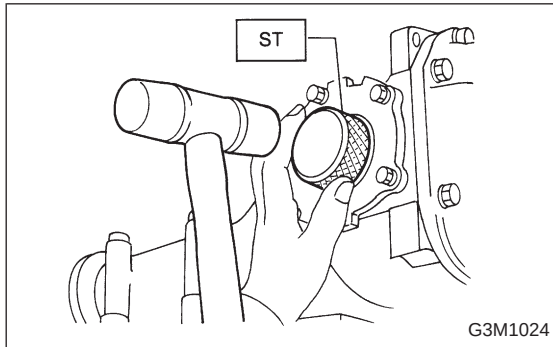
CAUTION:
When removing the DOJ from the rear differential, fit tire lever to the bolt as shown in figure so as not to damage the side bearing retainer.



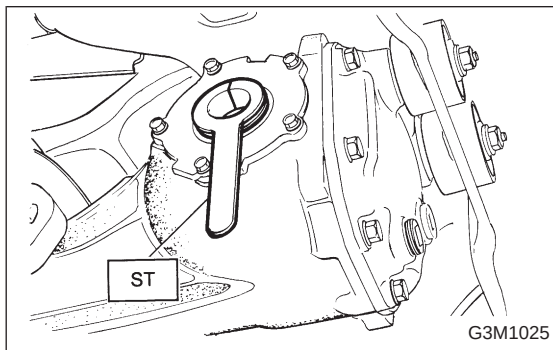
9) Secure rear drive shaft to rear crossmember using wire.



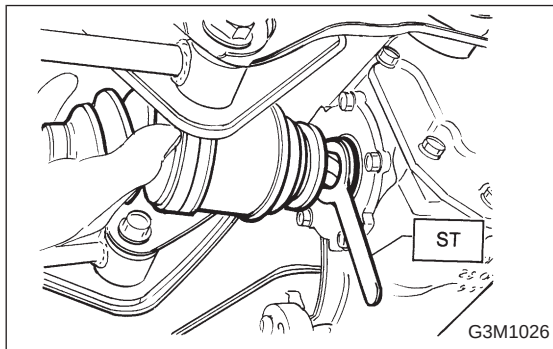
- 10) Remove side oil seal with ST.
ST 398527700 PULLER ASSY



- 11) Drive in a new side oil seal with ST.
CAUTION:
Apply chassis grease between the oil seal lips.
ST 398437700 DRIFT



- 12) Insert the DOJ into rear differential.
(1) Install ST to rear differential.
ST 28099PA090 SIDE OIL SEAL PROTECTOR



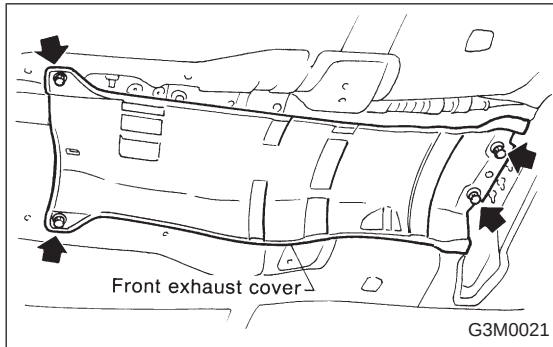
- (2) Insert the spline shaft until the spline portion is inside the side oil seal.
(3) Remove ST.
ST 28099PA090 SIDE OIL SEAL PROTECTOR

- 13) Hereafter, re-assemble in reverse order of disassembly.

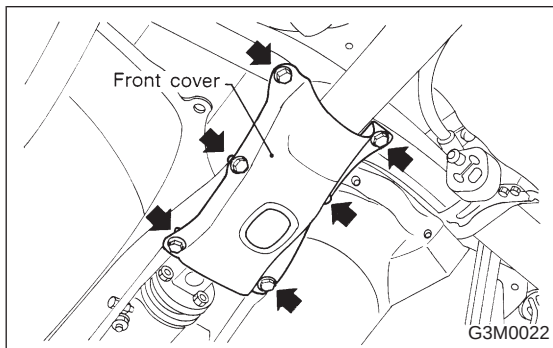
B: REMOVAL

- 1) Disconnect ground cable from battery.
- 2) Move selector lever or gear shift lever to "N".
- 3) Release the parking brake.
- 4) Loosen wheel nuts.
- 5) Jack-up vehicle and support it with sturdy racks.
- 6) Remove wheels.
- 7) Remove rear exhaust pipe and muffler.
<Ref. to 2-9 [W2A0], [W3A0].>

- 8) Remove front exhaust cover.



- 9) Remove front cover of rear differential mount.



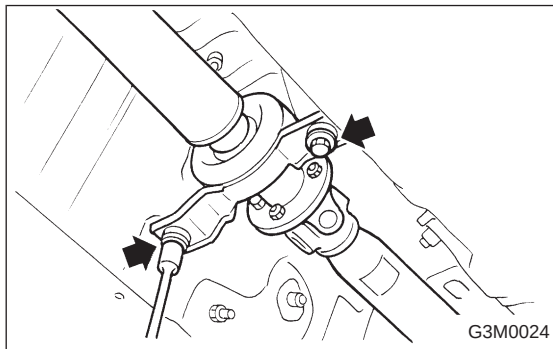
- 10) Remove propeller shaft.

CAUTION:

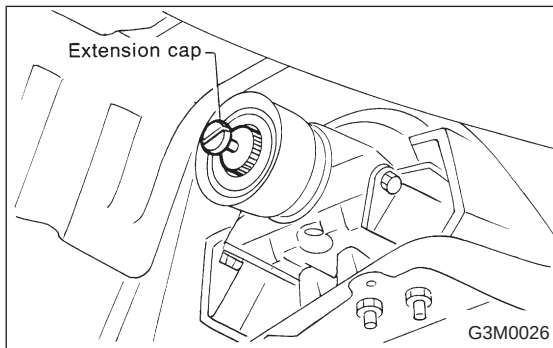
When removing propeller shaft, pay attention not to damage the sliding surfaces of rear drive shaft (extension) spline, oil seal and sleeve yoke.

NOTE:

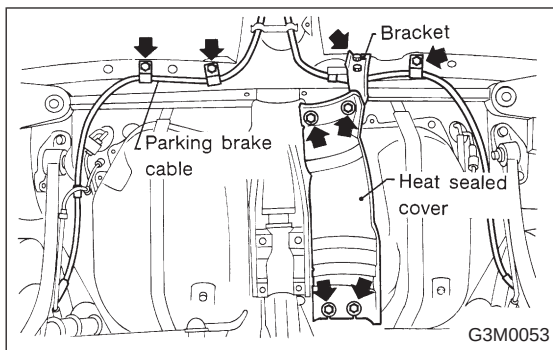
Prepare an oil can and cap since the transmission oil flows out from the extension at removing propeller shaft.

**NOTE:**

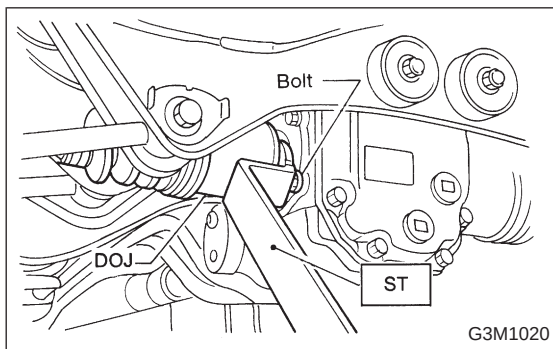
Insert the cap into the extension to prevent transmission oil from flowing out immediately after removing the propeller shaft.



2. Rear Differential

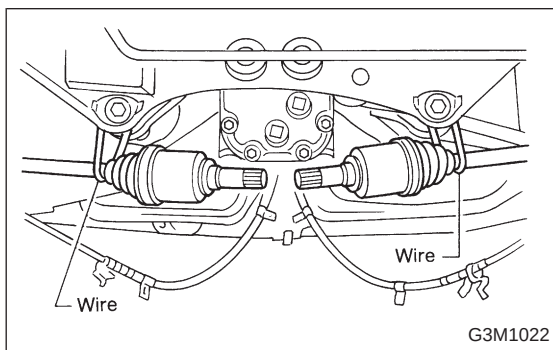


- 11) Remove heat sealed cover.
- 12) Remove clamps and bracket of parking brake cable.
- 13) Remove crossmember reinforcement lower (AWD Sedan only).

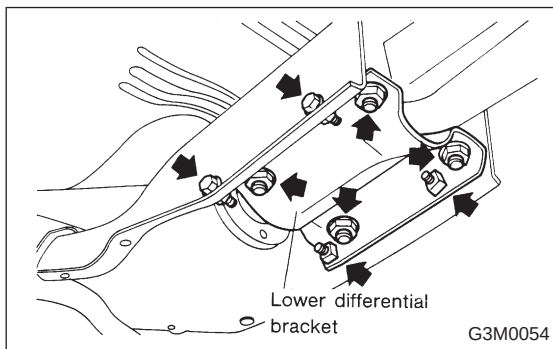


- 14) Remove DOJ of rear drive shaft from rear differential using ST. <Ref. to 3-4 [W2A2].>

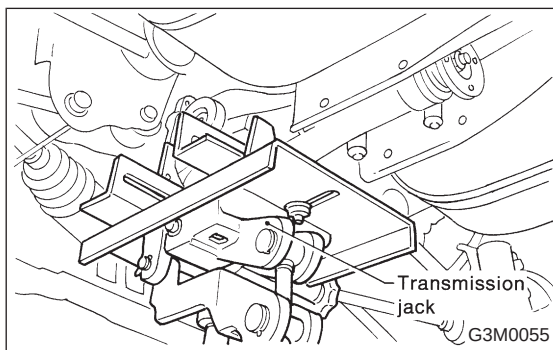
ST 28099PA100 DRIVE SHAFT REMOVER



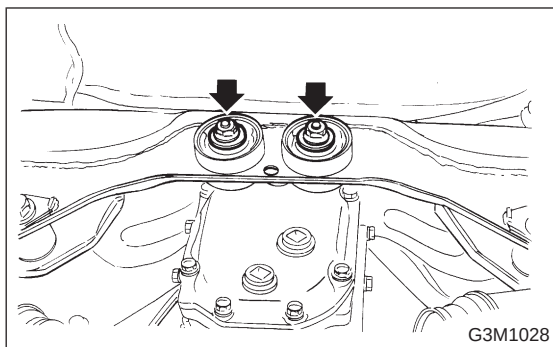
- 15) Secure rear drive shaft to rear crossmember using wire.



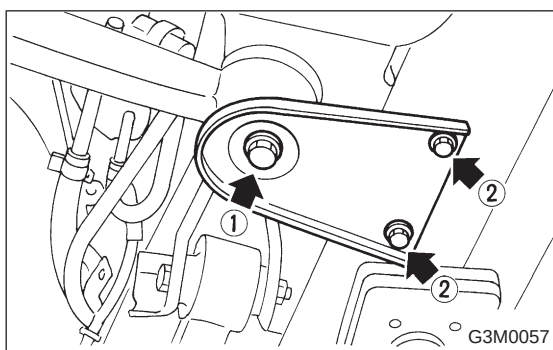
- 16) Remove lower differential bracket.



- 17) Support rear differential with transmission jack.



18) Remove self-locking nuts connecting rear differential to rear crossmember.



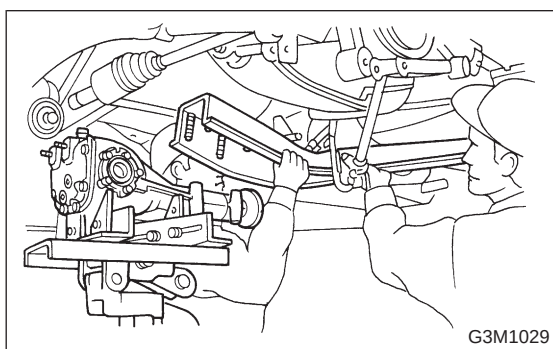
19) Remove bolts which secure rear differential front member to body.

Loosen bolt ① first, then remove bolts ②.

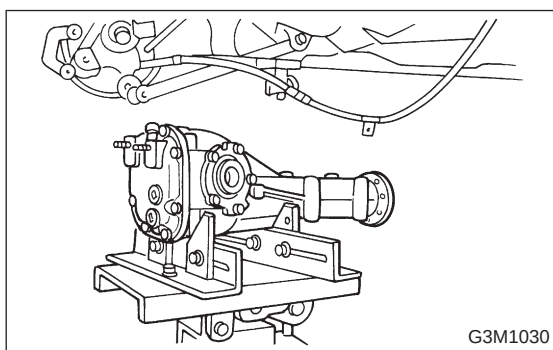
NOTE:

Support front member with the use of a helper to prevent it from dropping.

20) Remove bolt ①.



21) While slowly lowering transmission jack, move rear differential forward and remove front member and rear differential from body.

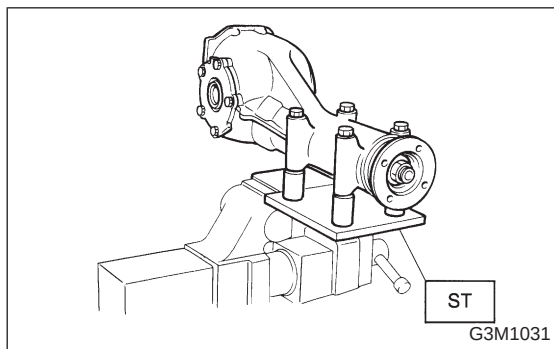


22) Remove rear differential from front member.

C: DISASSEMBLY

To detect real cause of trouble, inspect the following items before disassembling. <Ref. to 3-4 [W2E0].>

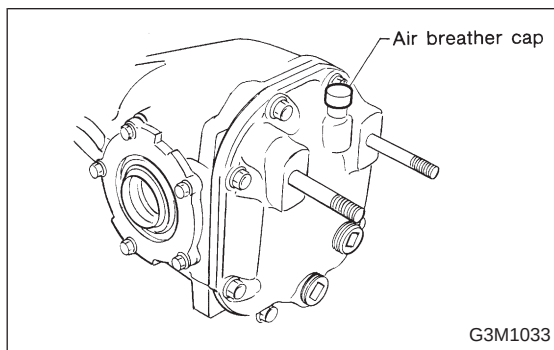
- Tooth contact of crown gear and pinion, and backlash
- Runout of crown gear at its back surface
- Turning resistance of drive pinion



1) Set ST on vise and install the differential assembly to ST.

ST 398217700 ATTACHMENT

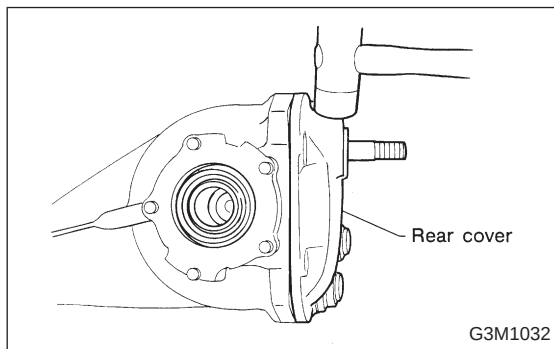
2) Drain gear oil by removing plug.



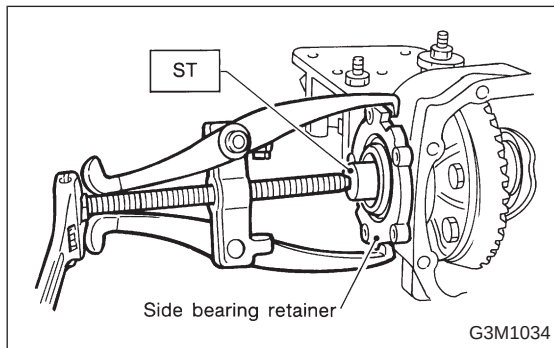
3) Remove the air breather cap.

NOTE:

Do not attempt to replace the air breather cap unless necessary.



4) Remove rear cover by loosening retaining bolts.

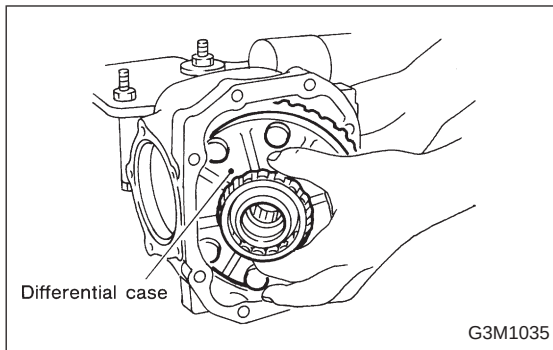


5) Make right and left side bearing retainers in order to identify them at reassembly. Remove side bearing retainer attaching bolts, set ST to differential case, and extract right and left side bearing retainers with a puller.

ST 398457700 ATTACHMENT

CAUTION:

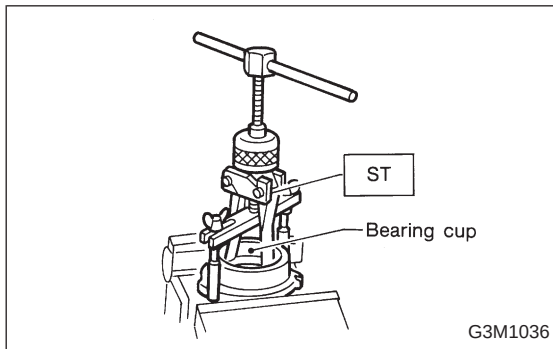
Each shim, which is installed to adjust the side bearing preload, should be kept together with its mating retainer.



6) Pull out differential assembly from differential carrier.

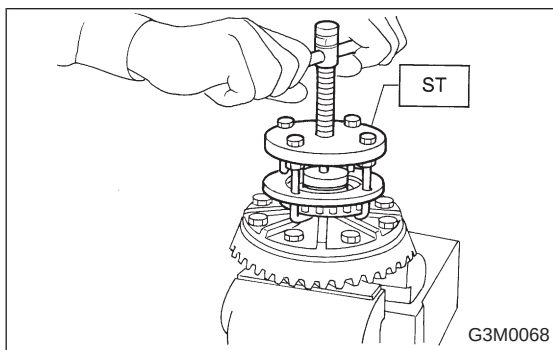
CAUTION:

Be careful not to hit the teeth against the case.



7) When replacing side bearing, pull bearing cup from side bearing retainer using ST.

ST 398527700 PULLER ASSY



8) Extract bearing cone with ST.

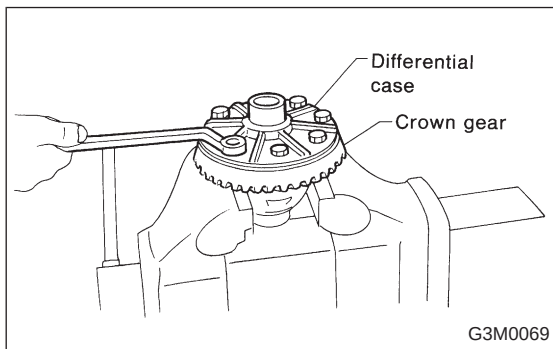
CAUTION:

Do not attempt to disassemble the parts unless necessary.

NOTE:

- Set puller so that its claw catch the edge of the bearing cone.
- Never mix up the right and left hand bearing cups and cones.

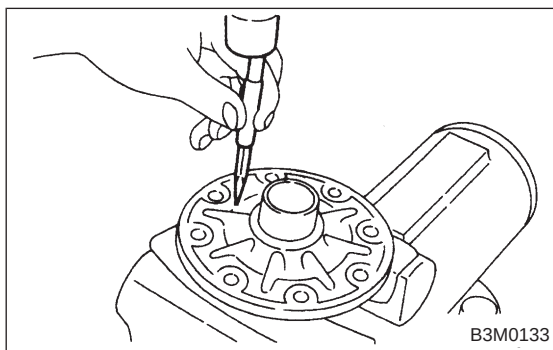
ST 399527700 PULLER SET



9) Remove crown gear by loosening crown gear bolts.

CAUTION:

Further disassembling is not allowed.

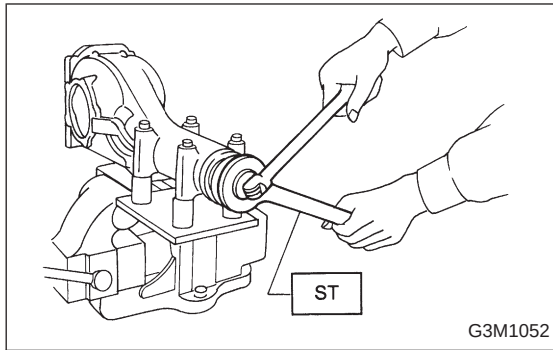


10) Drive out pinion shaft lock pin from crown gear side.

NOTE:

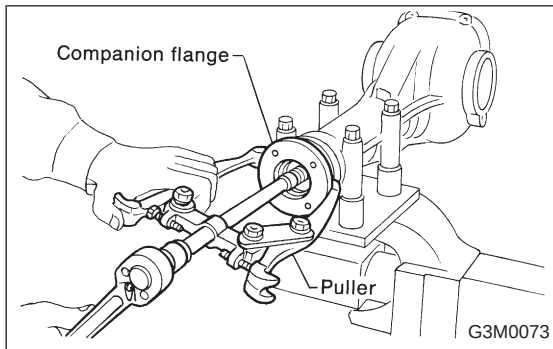
The lock pin is staked at the pin hole end on the differential carrier; do not drive it out forcibly before unstaking it.

ST 899904100 STRAIGHT PIN REMOVER

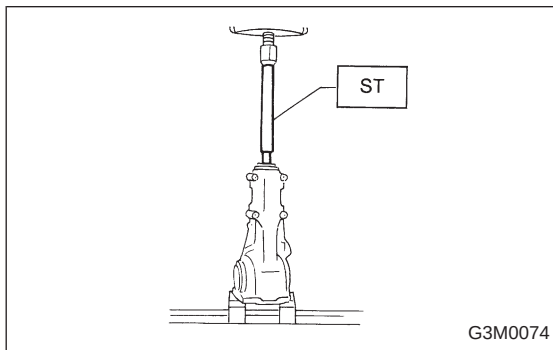


11) Hold companion flange with ST and remove drive pinion nut.

ST 498427200 FLANGE WRENCH



12) Extract the companion flange with a puller.

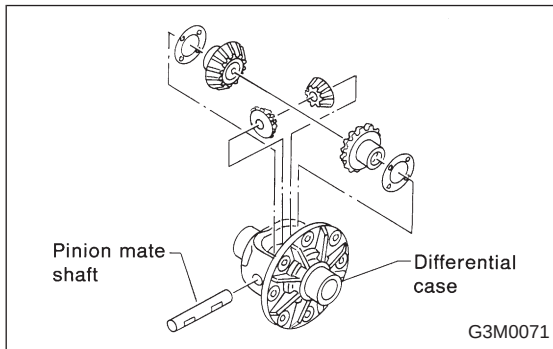


13) Press the end of drive pinion shaft and extract it together with rear bearing cone, preload adjusting spacer and washer.

NOTE:

Hold the drive pinion so as not to drop it.

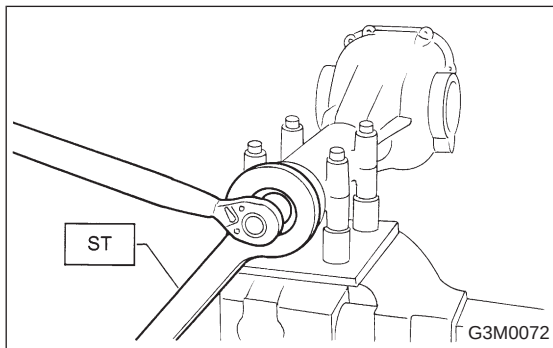
ST 398467700 DRIFT



14) Draw out pinion mate shaft and remove pinion mate gears, side gears and thrust washers.

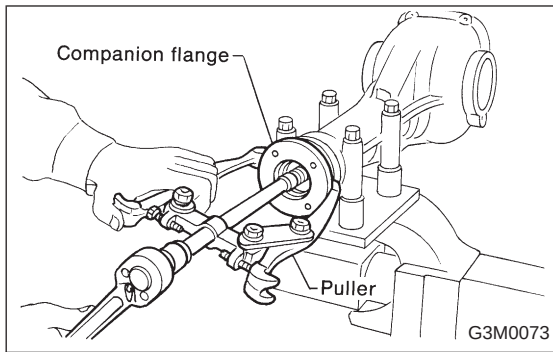
NOTE:

The gears as well as thrust washers should be marked or kept separated left and right, and front and rear.

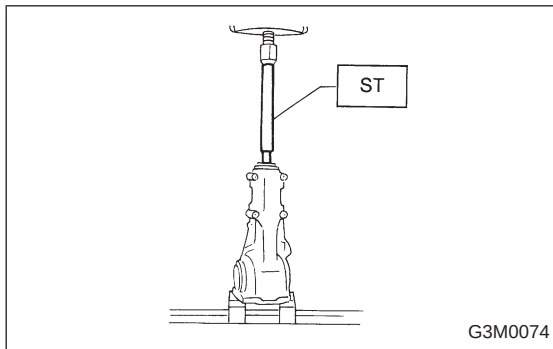


15) Hold companion flange with ST and remove drive pinion nut.

ST 498427200 FLANGE WRENCH



16) Extract the companion flange with a puller.

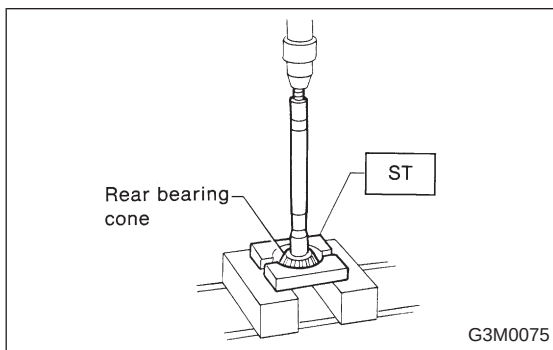


17) Press the end of drive pinion shaft and extract it together with rear bearing cone, preload adjusting spacer and washer.

NOTE:

Hold the drive pinion so as not to drop it.

ST 398467700 DRIFT

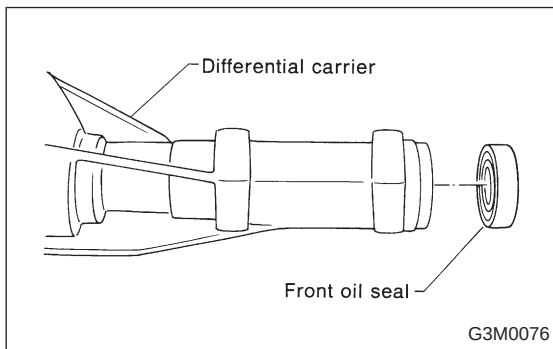


18) Remove rear bearing cone from drive pinion by supporting cone with ST.

NOTE:

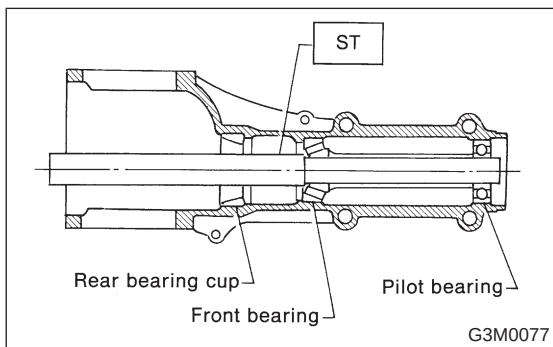
Place the replacer so that its center-recessed side faces the pinion gear.

ST 498515500 REPLACER



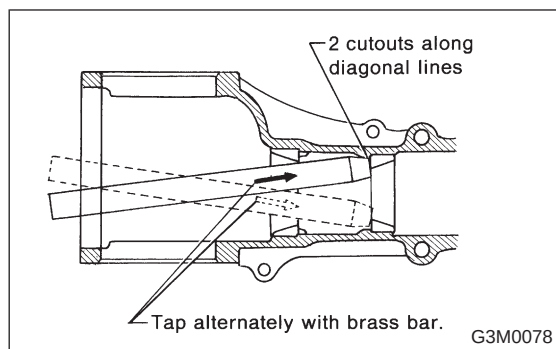
19) Remove front oil seal from differential carrier using ST.

ST 398527700 PULLER ASSY



20) Remove pilot bearing together with front bearing cone using ST.

ST 398467700 DRIFT



21) When replacing bearings, tap front bearing cup and rear bearing cup in this order out of case by using a brass bar.

D: INSPECTION

Wash all the disassembled parts clean, and examine them for wear, damage, or other defects. Repair or replace defective parts as necessary.

1) Crown gear and drive pinion

(1) If abnormal tooth contact is evident, find out the cause and adjust to give correct tooth contact at assembly. Replace the gear if excessively worn or incapable of adjustment.

(2) If crack, score, or seizure is evident, replace as a set. Slight damage of tooth can be corrected by oil stone or the like.

2) Side gear and pinion mate gear

(1) Replace if crack, score, or other defects are evident on tooth surface.

(2) Replace if thrust washer contacting surface is worn or seized. Slight damage of the surface can be corrected by oil stone or the like.

3) Bearing

Replace if seizure, peeling, wear, rust, dragging during rotation, abnormal noise or other defect is evident.

4) Thrust washers of side gear and pinion mate gear

Replace if seizure, flaw, abnormal wear or other defect is evident.

5) Oil seal

Replace if deformed or damaged, and at every disassembling.

6) Differential carrier

Replace if the bearing bores are worn or damaged.

7) Differential case

Replace if its sliding surfaces are worn or cracked.

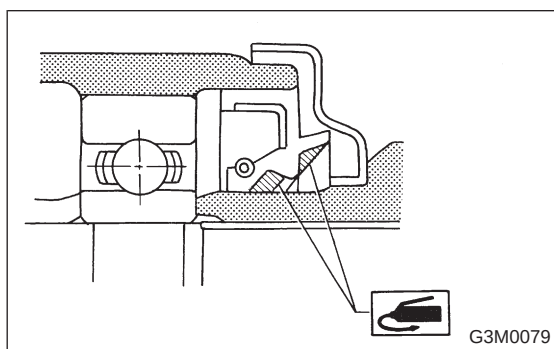
8) Companion flange

Replace if the oil seal lip contacting surfaces have flaws.

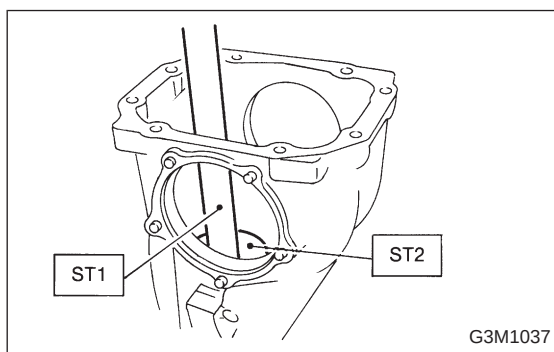
E: ASSEMBLY

1) Precautions for assembling

- (1) Assemble in the reverse order of disassembling.
- (2) Check and adjust each part during assembly.
- (3) Keep the shims and washers in order, so that they are not misinstalled.
- (4) Thoroughly clean the surfaces on which the shims, washers and bearings are to be installed.
- (5) Apply gear oil when installing the bearings and thrust washers.
- (6) Be careful not to mix up the right and left hand cups of the bearings.



- (7) Replace the oil seal with new one at every disassembly. Apply chassis grease between the lips when installing the oil seal.

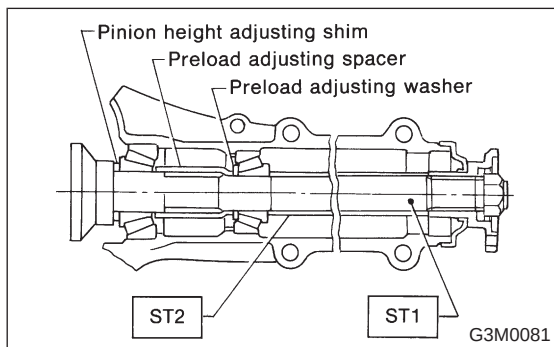


- 2) Adjusting preload for front and rear bearings
Adjust the bearing preload with spacer and washer between front and rear bearings. Pinion height adjusting washer are not affected by this adjustment. The adjustment must be carried out without oil seal inserted.

- (1) Press rear bearing race into differential carrier with ST1 and ST2.

ST1 398477701 HANDLE

ST2 398427703 DRIFT 2



- (2) Insert ST1 into case with pinion height adjusting washer and rear bearing cone fitted onto it.

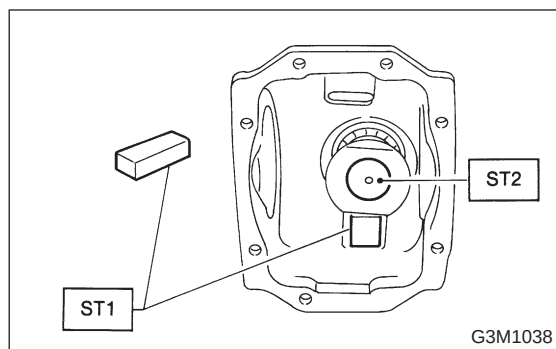
CAUTION:

- Re-use the used washer if not deformed.
- Use a new rear bearing cone.

- (3) Then install preload adjusting spacer and washer, front bearing cone, ST2, companion flange, and washer and drive pinion nut.

ST1 398507702 DUMMY SHAFT

ST2 398507703 DUMMY COLLAR



(4) Turn ST1 with hand to make it seated, and tighten drive pinion nut while measuring the preload with spring balance. Select preload adjusting washer and spacer so that the specified preload is obtained when nut is tightened to the specified torque with ST2.

CAUTION:

Use a new lock nut.

NOTE:

- Be careful not to give excessive preload.

- When tightening the drive pinion nut, lock ST1 with ST2 as shown in the figure.

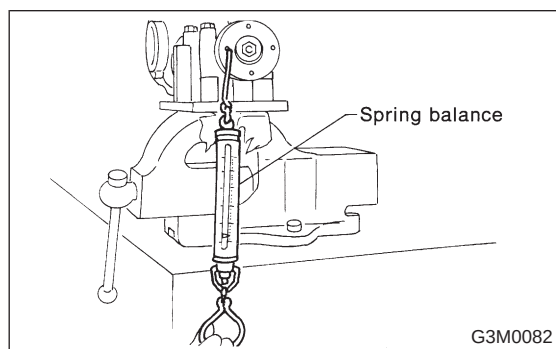
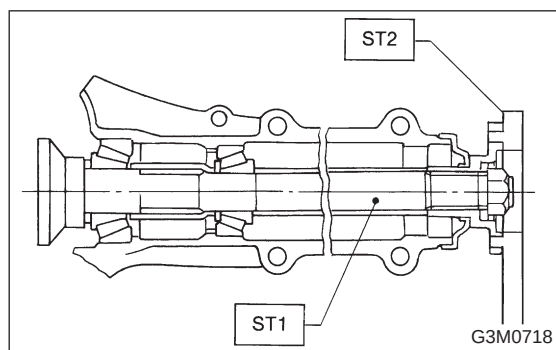
ST1 398507704 BLOCK

ST2 398507702 DUMMY SHAFT

ST3 498427200 FLANGE WRENCH

Tightening Torque:

181±15 N·m (18.5±1.5 kg·m, 134±11 ft·lb)



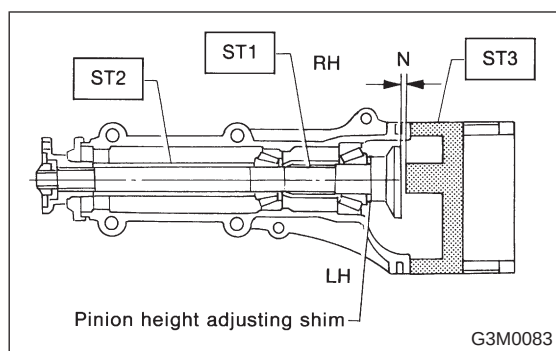
Front and rear bearing preload

For new bearing:

17.7 — 25.5 N (1.8 — 2.6 kg, 4.0 — 5.7 lb)

at companion flange bolt hole

	Part No.	Length mm (in)
● Preload adjusting washer length	383705200	2.59 (0.1020)
	383715200	2.57 (0.1012)
	383725200	2.55 (0.1004)
	383735200	2.53 (0.0996)
	383745200	2.51 (0.0988)
	383755200	2.49 (0.0980)
	383765200	2.47 (0.0972)
	383775200	2.45 (0.0965)
	383785200	2.43 (0.0957)
	383795200	2.41 (0.0949)
	383805200	2.39 (0.0941)
	383815200	2.37 (0.0933)
	383825200	2.35 (0.0925)
	383835200	2.33 (0.0917)
	383845200	2.31 (0.0909)
● Preload adjusting spacer length	383695201	56.2 (2.213)
	383695202	56.4 (2.220)
	383695203	56.6 (2.228)
	383695204	56.8 (2.236)
	383695205	57.0 (2.244)
	383695206	57.2 (2.252)



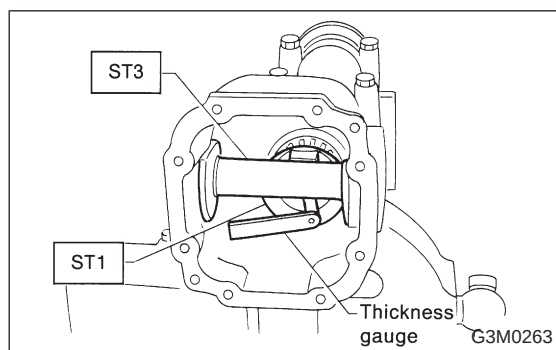
3) Adjusting drive pinion height

Adjust drive pinion height with shim installed between rear bearing cone and the back of pinion gear.

- (1) Install ST1, ST2 and ST3, as shown in the figure, and apply the specified preload on the bearings. <Ref. to 2).> Adjusting preload for front and rear bearings

NOTE:

At this time, install a pinion height adjusting shim which is temporarily selected or the same as that used before.



- (2) Measure the clearance N between the end of ST3 and the end surface of ST1 by using a thickness gauge.

NOTE:

Make sure there is no clearance between the case and ST3.

- ST1 398507702 DUMMY SHAFT
 ST2 398507703 DUMMY COLLAR
 ST3 398507701 DIFFERENTIAL CARRIER GAUGE

- (3) Obtain the thickness of pinion height adjusting shim to be inserted from the following formula, and replace the temporarily installed shim with this one.

$$T = T_o + N - (H \times 0.01) - 0.20 \text{ (mm)}$$

Where:

T = Thickness of pinion height adjusting shim (mm)

T_o = Thickness of shim temporarily inserted (mm)

N = Reading of thickness gauge (mm)

H = Figure marked on drive pinion head

(Example of calculation)

$$T_o = 2.20 + 1.20 = 3.40 \text{ mm}$$

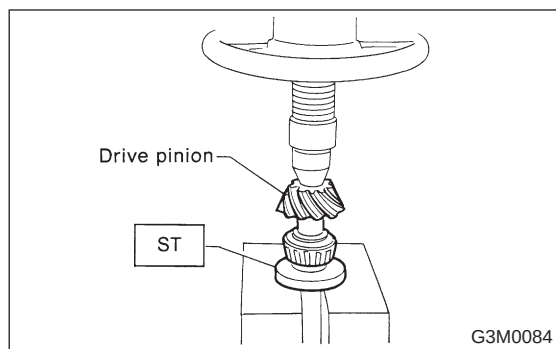
$$N = 0.23 \text{ mm } H = + 1,$$

$$T = 3.40 + 0.23 - 0.01 - 0.20 = 3.42$$

Result: Thickness = 3.42 mm

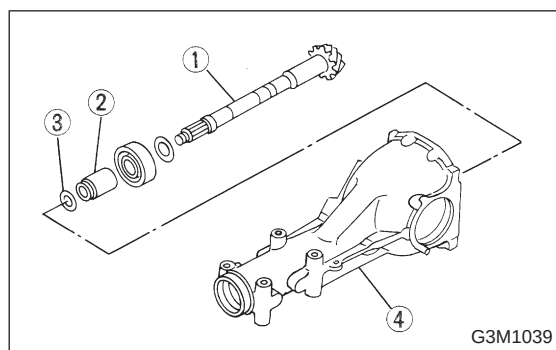
Therefore use the shim 383605200.

	Part No.	Thickness mm (in)
	383495200	3.09 (0.1217)
● Pinion height adjusting shim thickness	383505200	3.12 (0.1228)
	383515200	3.15 (0.1240)
	383525200	3.18 (0.1252)
	383535200	3.21 (0.1264)
	383545200	3.24 (0.1276)
	383555200	3.27 (0.1287)
	383565200	3.30 (0.1299)
	383575200	3.33 (0.1311)
	383585200	3.36 (0.1323)
	383595200	3.39 (0.1335)
	383605200	3.42 (0.1346)
	383615200	3.45 (0.1358)
	383625200	3.48 (0.1370)
	383635200	3.51 (0.1382)
	383645200	3.54 (0.1394)
	383655200	3.57 (0.1406)
	383665200	3.60 (0.1417)
	383675200	3.63 (0.1429)
	383685200	3.66 (0.1441)

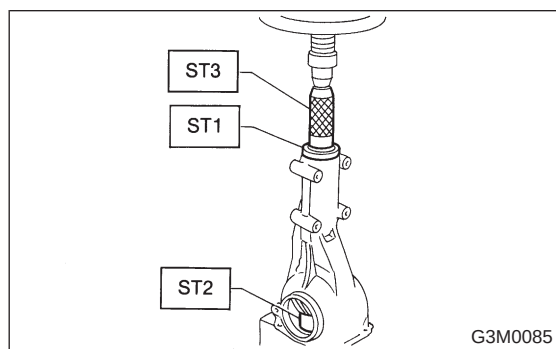


4) Install the selected pinion height adjusting shim on drive pinion, and press the rear bearing cone into position with ST.

ST 398177700 INSTALLER



5) Insert drive pinion ① into differential carrier ④, install the previously selected bearing preload adjusting spacer ② and washer ③.

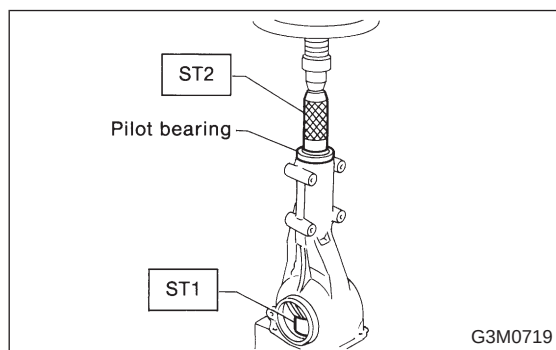


6) Press-fit front bearing cone into case with ST1, ST2 and ST3.

ST1 398507703 DUMMY COLLAR

ST2 399780104 WEIGHT

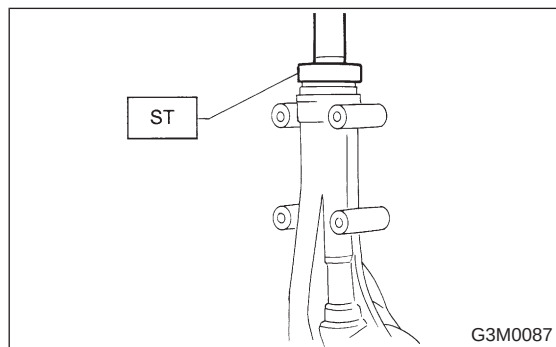
ST3 899580100 INSTALLER



7) Insert spacer, then press-fit pilot bearing with ST1 and ST2.

ST1 399780104 WEIGHT

ST2 899580100 INSTALLER

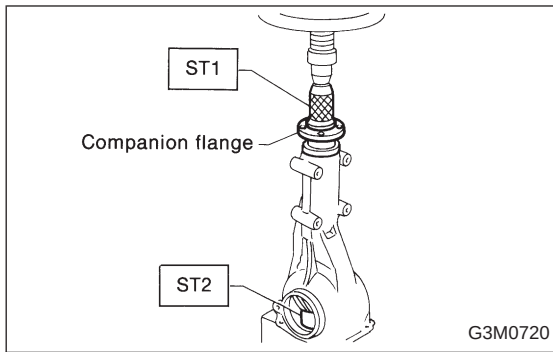


8) Fit a new oil seal with ST.

NOTE:

- Press-fit until end of oil seal is 1 mm (0.04 in) inward from end of carrier.
- Apply grease between the oil seal lips. <Ref. to 3-4 [W2E0] 1).>

ST 498447120 OIL SEAL INSTALLER



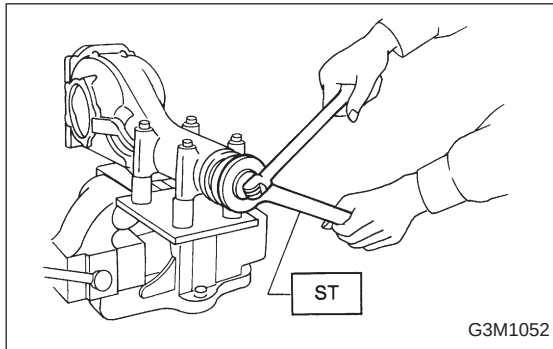
9) Press-fit companion flange with ST1 and ST2.

CAUTION:

Be careful not to damage bearing.

ST1 899874100 INSTALLER

ST2 399780104 WEIGHT

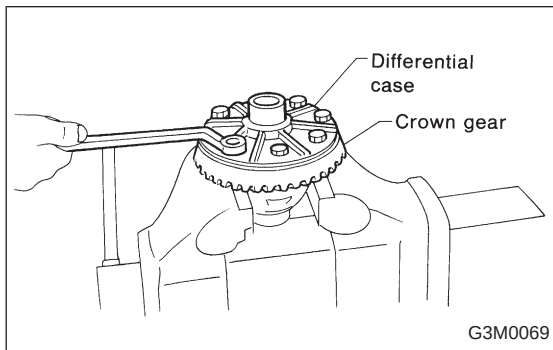


10) Install self-locking nut. Then tighten it with ST.

ST 498427200 FLANGE WRENCH

Tightening torque:

$181 \pm 15 \text{ N}\cdot\text{m}$ ($18.5 \pm 1.5 \text{ kg}\cdot\text{m}$, $134 \pm 11 \text{ ft}\cdot\text{lb}$)



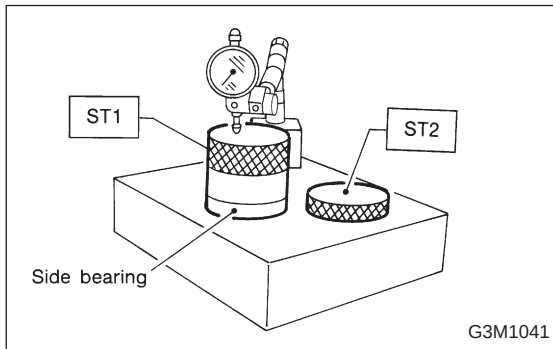
11) Install crown gear on differential case.

Tightening Torque:

$103 \pm 10 \text{ N}\cdot\text{m}$ ($10.5 \pm 1.0 \text{ kg}\cdot\text{m}$, $76 \pm 7 \text{ ft}\cdot\text{lb}$)

NOTE:

Tighten diagonally while tapping the bolt heads.



12) Before installing side bearing, measure the bearing width by using a dial gauge, ST1 and ST2.

Standard bearing width:

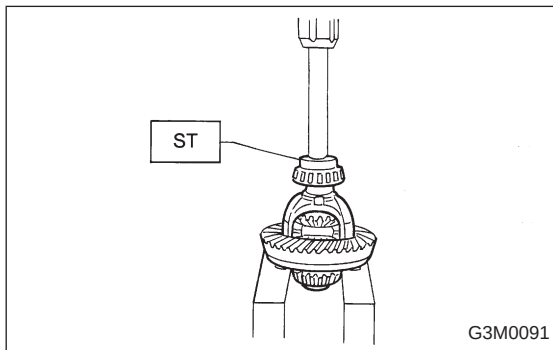
20.00 mm (0.7874 in)

NOTE:

Set the dial gauge needle to zero, using a standard bearing or block of specified height in advance.

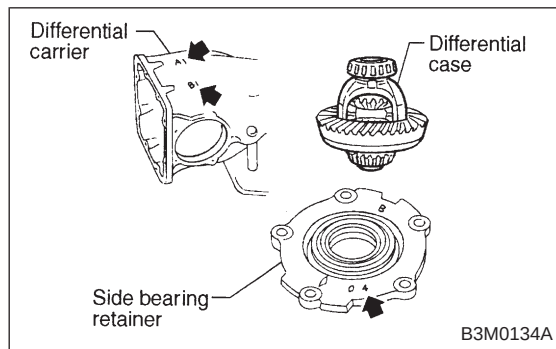
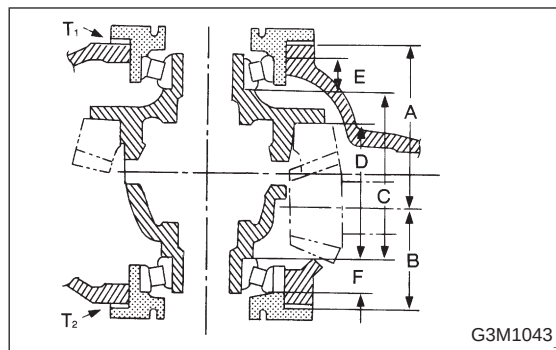
ST1 398227700 WEIGHT

ST2 398237700 GAUGE



13) Press side bearing cone onto differential case with ST1.

ST1 398487700 DRIFT



14) Adjusting side bearing retainer shims

(1) The drive gear backlash and side bearing preload can be determined by the side bearing retainer shim thickness.

(2) When replacing differential case, differential carrier, side bearing and side bearing retainer, obtain the right and left retainer shim thickness from the following formulas.

$$T_1 \text{ (Left)} = (A + C + G_1 - D) \times 0.01 + 0.76 - E \text{ (mm)}$$

$$T_2 \text{ (Right)} = (B + D + G_2) \times 0.01 + 0.76 - F \text{ (mm)}$$

T₁ & T₂ : Thickness of left and right side bearing retainer shim (mm)

A & B : Number marked on differential carrier

C & D : Number marked on differential case

E & F : Difference of width of left and right side bearing from standard width 20.0 mm, expressed in a unit of 0.01 mm. For example, if the bearing measured width is 19.89 mm, value of E or F is as follows.
20.00 - 19.89 = 0.11 (E or F)

G₁ & G₂ : Number marked on side bearing retainer

If a number is not marked, regard it as zero.

NOTE:

Use several shims to obtain the calculated thickness.

	Part No.	Thickness mm (in)
	383475201	0.20 (0.0079)
● Side bearing retainer shim thickness	383475202	0.25 (0.0098)
	383475203	0.30 (0.0118)
	383475204	0.40 (0.0157)
	383475205	0.50 (0.0197)

Example of calculation

Ex. 1

A = 5, B = 5, C = 3, D = 3, G₁ = 4, G₂ = 1,
E = 0.10 mm, F = 0.15 mm

Left side

$$\begin{aligned} T_1 &= (A + C + G_1 - D) \times 0.01 + 0.76 - E \\ &= (5 + 3 + 4 - 3) \times 0.01 + 0.76 - 0.10 \\ &= 0.09 + 0.76 - 0.10 = 0.75 \text{ mm} \end{aligned}$$

The correct shims are as follows:

Thickness	Q'ty	
0.25	x 1	= 0.25
0.50	x 1	= 0.50
Total shim thickness = 0.75 mm		

Right side

$$\begin{aligned} T_2 &= (B + D + G_2) \times 0.01 + 0.76 - F \\ &= (5 + 3 + 1) \times 0.01 + 0.76 - 0.15 \\ &= 0.09 + 0.76 - 0.15 \\ &= 0.70 \text{ mm} \end{aligned}$$

The correct shims are as follows:

Thickness	Q'ty	
0.20	x 1	= 0.20
0.50	x 1	= 0.50
Total shim thickness = 0.70 mm		

Ex. 2

A = 2, B = 3, C = 0, D = 3, G₁ = 2, G₂ = 3,
E = 0.22 mm, F = 0.10 mm

Left side

$$\begin{aligned} T_1 &= (A + C + G_1 - D) \times 0.01 + 0.76 - E \\ &= (2 + 0 + 2 - 3) \times 0.01 + 0.76 - 0.22 \\ &= 0.01 + 0.76 - 0.22 \\ &= 0.55 \text{ mm} \end{aligned}$$

The correct shims are as follows:

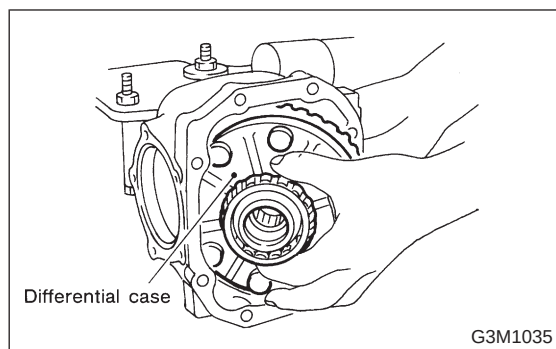
Thickness	Q'ty	
0.25	x 1	= 0.25
0.30	x 1	= 0.30
Total shim thickness = 0.55 mm		

Right side

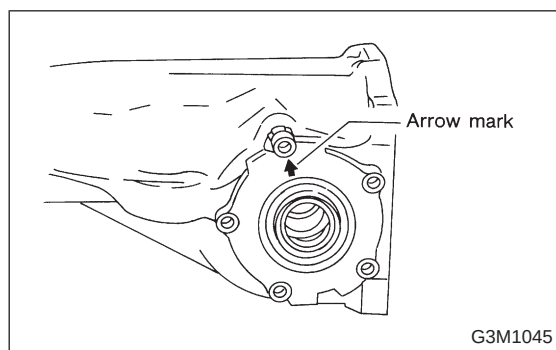
$$\begin{aligned} T_2 &= (B + D + G_2) \times 0.01 + 0.76 - F \\ &= (3 + 3 + 3) \times 0.01 + 0.76 - 0.10 \\ &= 0.09 + 0.76 - 0.10 \\ &= 0.75 \text{ mm} \end{aligned}$$

The correct shims are as follows:

Thickness	Q'ty	
0.25	x 1	= 0.25
0.50	x 1	= 0.50
Total shim thickness = 0.75 mm		



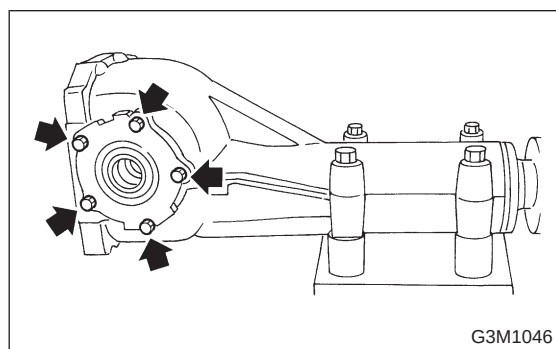
(3) Install the differential case assembly into differential carrier in the reverse order of disassembly.



(4) Fit the selected shims and O-ring on side bearing retainer and install them on differential carrier with the arrow mark on the retainer directed as shown in figure.

CAUTION:

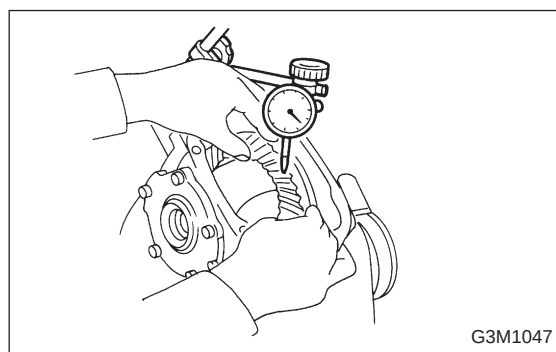
Be careful that side bearing cup is not damaged by bearing roller.



(5) Tighten side bearing retainer bolts.

Tightening Torque:

$10.3 \pm 1.5 \text{ N} \cdot \text{m}$ ($1.05 \pm 0.15 \text{ kg} \cdot \text{m}$, $7.6 \pm 1.1 \text{ ft} \cdot \text{lb}$)



(6) Measure the crown gear-to-drive pinion backlash. Set magnet base on differential carrier. Align contact point of dial gauge with tooth face of crown gear, and move crown gear while holding drive pinion still. Read value indicated on dial gauge.

Backlash:

$0.10 - 0.20 \text{ mm}$ ($0.0039 - 0.0079 \text{ in}$)

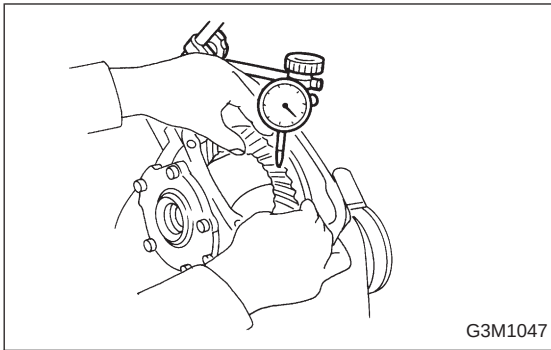
(7) At the same time, measure the turning resistance of drive pinion. Compared with the resistance when differential case is not installed, if the increase of the resistance is not within the specified range, readjust side bearing retainer shims.

Turning resistance increase:

$0.1 - 0.6 \text{ N} \cdot \text{m}$ ($1 - 6 \text{ kg} \cdot \text{cm}$, $0.9 - 5.2 \text{ in} \cdot \text{lb}$)

NOTE:

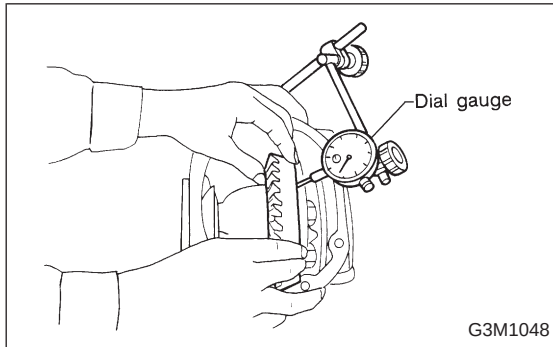
If measured backlash is not within specified range, repeat steps (1) thru (7).



15) Re-check crown gear-to-pinion backlash.

Backlash:

0.10 — 0.20 mm (0.0039 — 0.0079 in)



16) Check the crown gear runout on its back surface, and make sure pinion and crown gear rotate smoothly.

Limit of runout:

0.05 mm (0.0020 in)

17) Checking and adjusting tooth contact of crown gear

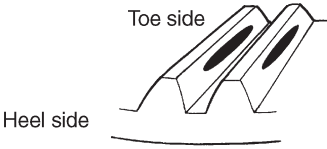
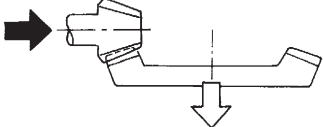
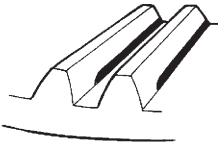
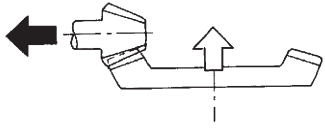
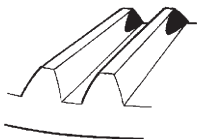
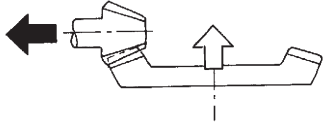
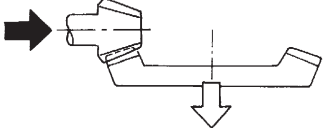
(1) Apply an even coat of red lead on both sides of three or four teeth on the crown gear. Check the contact pattern after rotating crown gear several revolutions back and forth until a definite contact pattern appears on the crown gear.

(2) When the contact pattern is incorrect, readjust according to the instructions given in "TOOTH CONTACT PATTERN".

NOTE:

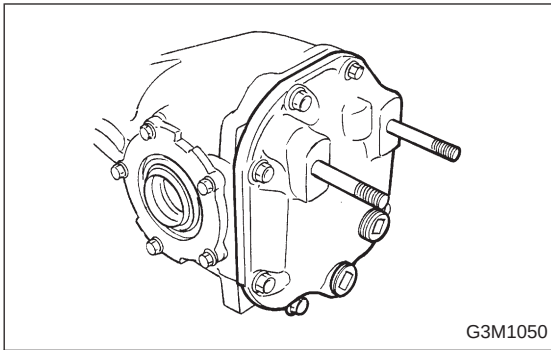
Be sure to wipe off red lead completely after adjustment is completed.

18) If proper tooth contact is not obtained, once again adjust the drive pinion height and the differential side bearing preload (mentioned above) and the hypoid gear backlash. <Ref. to 2).>

TOOTH CONTACT PATTERN		
Condition	Contact pattern	Adjustment
Correct tooth contact Tooth contact pattern slightly shifted towards toe under no load rotation. (When loaded, contact pattern moves toward heel.)	 B3M0317A	
Face contact Backlash is too large.	 B3M0319	Increase thickness of drive pinion height adjusting shim in order to bring drive pinion closer to crown gear center.  B3M0323
Flank contact Backlash is too small.	 B3M0320	Reduce thickness of drive pinion height adjusting shim in order to move drive pinion away from crown gear.  B3M0324
Toe contact Contact area is small.	 B3M0321	Adjust as for flank contact.  B3M0324
Heel contact Contact area is small.	 B3M0322	Adjust as for face contact.  B3M0323

➡ : Adjusting direction of drive pinion

⇨ : Adjusting direction of crown gear



19) Install rear cover and tighten bolts to specified torque.

Tightening torque:

29 ± 5 N·m (3.0 ± 0.5 kg·m, 21.7 ± 3.6 ft·lb)

F: INSTALLATION

To install, reverse the removal sequence.

1) Install the air breather cap tapping with a plastic hammer.

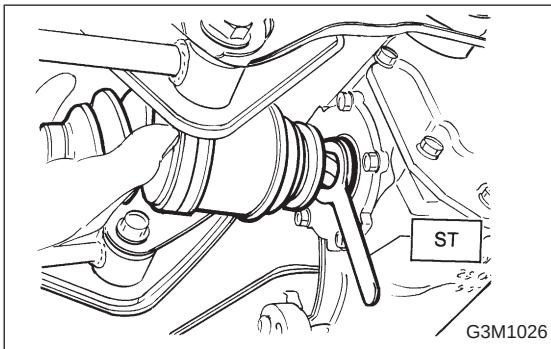
CAUTION:

Be sure to install new air breather cap.

2) Position front member on body by passing it under parking brake cable and securing to rear differential.

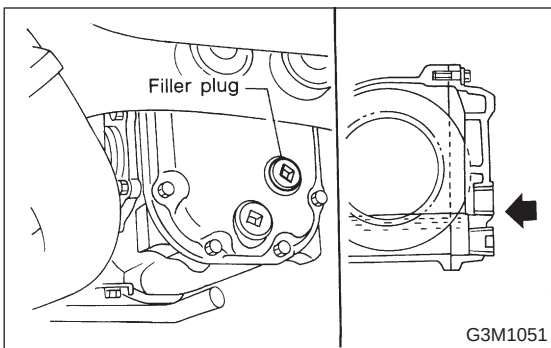
NOTE:

When installing rear differential front member, do not confuse the installation sequence of the upper and lower stoppers.



3) Install DOJ of rear drive shaft into rear differential. <Ref. to 3-4 [W2A2].>

ST 28099PA090 SIDE OIL SEAL PROTECTOR



4) Installing procedure hereafter is in the reverse order of removal.

5) After installation, fill differential carrier with gear oil to the upper plug level.

CAUTION:

Apply fluid packing to plug.

Fluid packing:

THREE BOND 1205 or equivalent

Oil capacity:

0.8 ℓ (0.8 US qt, 0.7 Imp qt)

Tightening torque:

44 ± 4 N·m (4.5 ± 0.4 kg·m, 32.5 ± 2.9 ft·lb)

3. Rear Differential Front Member

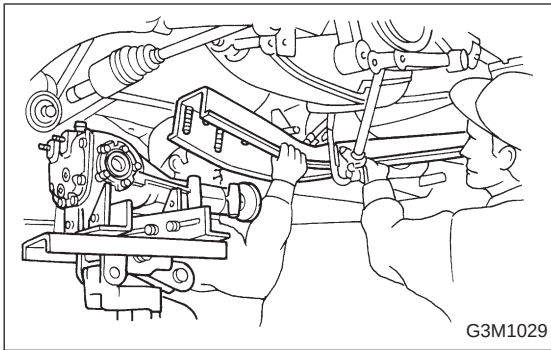
A: REMOVAL

- 1) Disconnect ground cable from battery.
- 2) Move selector lever or gear shift lever to "N".
- 3) Release the parking brake.
- 4) Loosen wheel nuts.
- 5) Jack-up vehicle and support it with sturdy racks.
- 6) Remove wheels.
- 7) Remove rear exhaust pipe and muffler.
<Ref. to 2-9 [W2A0], [W3A0].>
- 8) Remove rear differential front member.

NOTE:

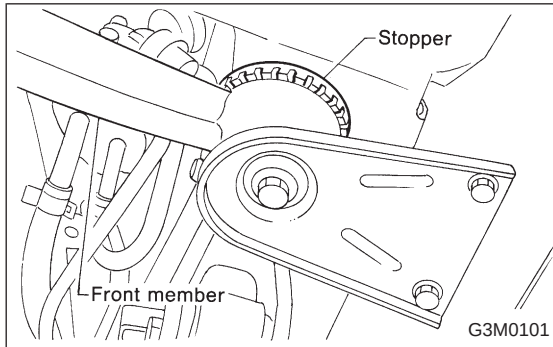
When removing rear differential front member, work the removal procedure as rear differential.

<Ref. to 3-4 [W2B0].>

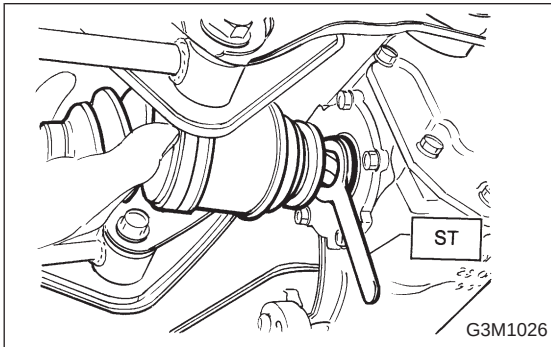
**B: INSTALLATION**

To install, reverse the removal sequence.

- 1) Position front member on body by passing it under parking brake cable and securing to rear differential.

**NOTE:**

When installing rear differential front member, do not confuse the installation sequence of the stopper.



- 2) Insert DOJ of rear drive shaft into rear differential.

ST 28099PA090 SIDE OIL SEAL PROTECTOR

CAUTION:

Before inserting, replace the differential side oil seal and the circlip at the end of the spline shaft with a new one.

- 3) Installing procedure hereafter is in the reverse order of removal.

1. Rear Differential

Symptom and possible cause	Remedy
1. Oil leakage	
① Worn, scratched, or incorrectly seated front or side oil seal. Scored, battered, or excessively worn sliding surface of companion flange.	Repair or replace.
② Clogged or damaged air breather.	Clean, repair or replace.
③ Loose bolts on differential spindle or side retainer, or incorrectly fitted O-ring.	Tighten bolts to specified torque. Replace O-ring.
④ Loose rear cover attaching bolts or damaged gasket.	Tighten bolts to specified torque. Replace gasket and apply liquid packing.
⑤ Loose oil filler or drain plug.	Retighten and apply liquid packing.
⑥ Wear, damage or incorrectly fitting for spindle, side retainer and oil seal.	Repair or replace.
2. Seizure Seized or damaged parts should be replaced, and also other parts should be thoroughly checked for any defect and should be repaired or replaced as required.	
① Insufficient backlash for hypoid gear.	Readjust or replace.
② Excessive preload for side, rear, or front bearing.	Readjust or replace.
③ Insufficient or improper oil used.	Replace seized part and fill with specified oil to specified level.
3. Damage Damaged parts should be replaced, and also other parts should be thoroughly checked for any defect and should be repaired or replaced as required.	
① Improper backlash for hypoid gear.	Replace.
② Insufficient or excessive preload for side, rear, or front bearing.	Readjust or replace.
③ Excessive backlash for differential gear.	Replace gear or thrust washer.
④ Loose bolts and nuts such as crown gear bolt.	Retighten.
⑤ Damage due to overloading.	Replace.
4. Noises when starting or shifting gears Noises may be caused by differential assembly, universal joint, wheel bearing, etc. Find out what is actually making noise before disassembly.	
① Excessive backlash for hypoid gear.	Readjust.
② Excessive backlash for differential gear.	Replace gear or thrust washer.
③ Insufficient preload for front or rear bearing.	Readjust.
④ Loose drive pinion nut.	Tighten to specified torque.
⑤ Loose bolts and nuts such as side bearing retainer attaching bolt.	Tighten to specified torque.

Symptom and possible cause	Remedy
5. Noises when cornering	
① Damaged differential gear.	Replace.
② Excessive wear or damage of thrust washer.	Replace.
③ Broken pinion mate shaft.	Replace.
④ Seized or damaged side bearing.	Replace.
6. Gear noises	
Since noises from engine, muffler, transmission, propeller shaft, wheel bearings, tires, and body are sometimes mistaken for noises from differential assembly, be careful in checking them. Inspection methods to locate noises include coasting, accelerating, cruising, and jacking-up all four wheels. Perform these inspections according to condition of trouble. When listening to noises, shift gears into four wheel drive and fourth speed position, trying to pick up only differential noise.	
① Improper tooth contact of hypoid gear.	Readjust or replace hypoid gear set.
② Improper backlash for hypoid gear.	Readjust.
③ Scored or chipped teeth of hypoid gear.	Replace hypoid gear set.
④ Seized hypoid gear.	Replace hypoid gear set.
⑤ Improper preload for front or rear bearings.	Readjust.
⑥ Seized, scored, or chipped front or rear bearing.	Replace.
⑦ Seized, scored, or chipped side bearing.	Replace.
⑧ Vibrating differential carrier.	Replace.

2. Propeller Shaft

Symptom and possible cause	Remedy
1. Vibration of propeller shaft	
Vibration is caused by propeller shaft during operation and is transferred to vehicle body. Generally vibration increase in proportion to vehicle speed.	
① Worn or damaged universal joint.	Replace.
② Unbalanced propeller shaft due to bend or dent.	Replace.
③ Loose installation of propeller shaft.	Retighten.
④ Worn or damaged center bearing and damaged center mounting rubber.	Replace.
2. Tapping when starting and noise while cruising, caused by propeller shaft.	
① Worn or damaged universal joint.	Replace.
② Worn spline of sleeve yoke.	Replace.
③ Loose installation of propeller shaft.	Retighten.
④ Loose installation of joint.	Replace.
⑤ Worn or damaged center bearing and damaged center mounting rubber.	Replace.

NOTE:

Vibration while cruising may be caused by an unbalanced tire, improper tire inflation pressure, improper wheel alignment, etc.

Symptom and possible cause	Remedy
5. Noises when cornering	
① Damaged differential gear.	Replace.
② Excessive wear or damage of thrust washer.	Replace.
③ Broken pinion mate shaft.	Replace.
④ Seized or damaged side bearing.	Replace.
6. Gear noises	
Since noises from engine, muffler, transmission, propeller shaft, wheel bearings, tires, and body are sometimes mistaken for noises from differential assembly, be careful in checking them. Inspection methods to locate noises include coasting, accelerating, cruising, and jacking-up all four wheels. Perform these inspections according to condition of trouble. When listening to noises, shift gears into four wheel drive and fourth speed position, trying to pick up only differential noise.	
① Improper tooth contact of hypoid gear.	Readjust or replace hypoid gear set.
② Improper backlash for hypoid gear.	Readjust.
③ Scored or chipped teeth of hypoid gear.	Replace hypoid gear set.
④ Seized hypoid gear.	Replace hypoid gear set.
⑤ Improper preload for front or rear bearings.	Readjust.
⑥ Seized, scored, or chipped front or rear bearing.	Replace.
⑦ Seized, scored, or chipped side bearing.	Replace.
⑧ Vibrating differential carrier.	Replace.

2. Propeller Shaft

Symptom and possible cause	Remedy
1. Vibration of propeller shaft	
Vibration is caused by propeller shaft during operation and is transferred to vehicle body. Generally vibration increase in proportion to vehicle speed.	
① Worn or damaged universal joint.	Replace.
② Unbalanced propeller shaft due to bend or dent.	Replace.
③ Loose installation of propeller shaft.	Retighten.
④ Worn or damaged center bearing and damaged center mounting rubber.	Replace.
2. Tapping when starting and noise while cruising, caused by propeller shaft.	
① Worn or damaged universal joint.	Replace.
② Worn spline of sleeve yoke.	Replace.
③ Loose installation of propeller shaft.	Retighten.
④ Loose installation of joint.	Replace.
⑤ Worn or damaged center bearing and damaged center mounting rubber.	Replace.

NOTE:

Vibration while cruising may be caused by an unbalanced tire, improper tire inflation pressure, improper wheel alignment, etc.

1. Manual Transmission and Differential

A: SPECIFICATIONS

Item			Model		
			2200 cc	2500 cc	OUTBACK
Type			5-forward speeds with synchromesh and 1-reverse		
Transmission gear ratio		1st	3.545		
		2nd	2.111		
		3rd	1.448		
		4th	1.088		
		5th	0.780	0.871	
		Reverse	3.416		
Front reduction gear	Final	Type of gear	Hypoid		
		Gear ratio	3.900	4.111	
Rear reduction gear	Transfer	Type of gear	Helical		
		Gear ratio	1.000		
	Final	Type of gear	Hypoid		
		Gear ratio	3.900	4.111	
Front differential	Type and number of gear		Straight bevel gear (Bevel pinion: 2, Bevel gear: 2)		
Center differential	Type and number of gear		Straight bevel gear (Bevel pinion: 2, Bevel gear: 2 and viscous coupling)		
Transmission gear oil			GL-5		
Transmission oil capacity			3.5 ℓ (3.7 US qt, 3.1 Imp qt)		

B: SERVICE DATA**1. EXTENSION (AWD Model)**

Snap ring (Inner-72) to ball bearing side clearance
0 — 0.15 mm (0 — 0.0059 in)

Snap ring (Inner-72)	
Part No.	Thickness mm (in)
805172071	1.78 (0.0701)
805172072	1.90 (0.0748)
805172073	2.02 (0.0795)

Snap ring (Outer-30) to ball bearing side clearance
0 — 0.15 mm (0 — 0.0059 in)

Snap ring (Outer-30)	
Part No.	Thickness mm (in)
805030041	1.53 (0.0602)
805030042	1.65 (0.0650)
805030043	1.77 (0.0697)

2. EXTENSION ASSEMBLY (AWD Model)

Thrust washer (52 x 61 x t) to ball bearing side clearance
0.05 — 0.30 mm (0.0020 — 0.0118 in)

Thrust washer (52 x 61 x t)	
Part No.	Thickness mm (in)
803052021	0.50 (0.0197)
803052022	0.75 (0.0295)
803052023	1.00 (0.0394)

3. TRANSFER CASE OR REAR CASE

Neutral position adjustment

Adjustment shim	
Part No.	Thickness mm (in)
32190AA000	0.15 (0.0059)
32190AA010	0.30 (0.0118)

Reverse accent shaft		
Part No.	Mark	Remarks
32188AA040	1	Neutral position is closer to 1st.
32188AA011	No mark or 2	Standard
32188AA050	3	Neutral position is closer to reverse gear.

Reverse check plate adjustment

Reverse check plate			
Part No.	Mark	Angle θ	Remarks
32189AA000	0	28°	Arm stops closer to 5th gear.
32189AA010	1	31°	Arm stops closer to 5th gear.
33189AA020	2	34°	Arm stops in the center.
32189AA030	3	37°	Arm stops closer to reverse gear.
32189AA040	4	40°	Arm stops closer to reverse gear.

4. REVERSE IDLER GEAR

Adjustment of reverse idler gear position

Reverse idler gear to transmission case (LH) wall clearance
6.0 — 7.5 mm (0.236 — 0.295 in)

Reverse shifter lever		
Part No.	Mark	Remarks
32820AA000	0	Further from case wall
32820AA010	No mark	Standard
32820AA020	2	Closer to the case wall

After installing a suitable reverse shifter lever, adjust reverse idler gear to transmission case wall clearance to within 0 to 0.5 mm (0 to 0.020 in) using washers.

Washer (20.5 x 26 x t)			
Part No.	Thickness mm (in)	Part No.	Thickness mm (in)
803020151	0.4 (0.016)	803020154	1.9 (0.075)
803020152	1.1 (0.043)	803020155	2.3 (0.091)
803020153	1.5 (0.059)		

5. SHIFTER FORK AND ROD

Select suitable shifter forks so that both coupling sleeve and reverse driven gear are positioned in the center of their synchromesh mechanisms.

1st-2nd shifter fork		
Part No.	Mark	Remarks
32804AA060	1	Approach to 1st gear by 0.2 mm (0.008 in)
32804AA070	No mark	Standard
32804AA080	3	Approach to 2nd gear by 0.2 mm (0.008 in)

3rd-4th shifter fork		
Part No.	Mark	Remarks
32810AA060	1	Approach to 4th gear by 0.2 mm (0.008 in)
32810AA070	No mark	Standard
32810AA100	3	Approach to 3rd gear by 0.2 mm (0.008 in)

5th shifter fork		
Part No.	Mark	Remarks
32812AA200	4	Approach to 5th gear by 0.2 mm (0.008 in)
32812AA210	No mark	Standard
32812AA220	6	Become distant from 5th gear by 0.2 mm (0.008 in)

Rod end clearance

A: 1st-2nd — 3rd-4th

0.5 — 1.5 mm (0.020 — 0.059 in)

B: 3rd-4th — 5th

0.6 — 1.4 mm (0.024 — 0.055 in)

6. TRANSMISSION CASE ASSEMBLY

Drive pinion shim adjustment

Drive pinion shim			
Part No.	Thickness mm (in)	Part No.	Thickness mm (in)
32295AA031	0.150 (0.0059)	32295AA071	0.250 (0.0098)
32295AA041	0.175 (0.0069)	32295AA081	0.275 (0.0108)
32295AA051	0.200 (0.0079)	32295AA091	0.300 (0.0118)
32295AA061	0.225 (0.0089)	32295AA101	0.500 (0.0197)

Hypoid gear backlash

0.13 — 0.18 mm (0.0051 — 0.0071 in)

Selection of main shaft rear plate

Main shaft rear plate		
Dimension "A" mm (in)	Part No.	Mark
4.00 — 4.13 (0.1575 — 0.1626)	32294AA041	1
3.87 — 3.99 (0.1524 — 0.1571)	32294AA051	2

7. DRIVE PINION ASSEMBLY

Preload adjustment of thrust bearing

Starting torque

0.3 — 0.8 N·m (3 — 8 kg-cm, 2.6 — 6.9 in-lb)

Adjusting washer No. 1	
Part No.	Thickness mm (in)
803025051	3.925 (0.1545)
803025052	3.950 (0.1555)
803025053	3.975 (0.1565)
803025054	4.000 (0.1575)
803025055	4.025 (0.1585)
803025056	4.050 (0.1594)
803025057	4.075 (0.1604)

Adjusting washer No. 2	
Part No.	Thickness mm (in)
803025059	3.850 (0.1516)
803025054	4.000 (0.1575)
803025058	4.150 (0.1634)

Assemble a driven shaft and 1st driven gear that are selected for the proper radial clearance adjustment.

Driven shaft		1st driven gear
Part No.	Diameter A mm (in)	Part No.
32229AA150	49.959 — 49.966 (1.9669 — 1.9672)	32231AA290
32229AA140	49.967 — 49.975 (1.9672 — 1.9675)	32231AA280

8. DRIVE PINION ASSEMBLY (FWD Model)

Selection of 1st driven gear:

1st driven gear	
Outer diameter of bushing mm (in)	Part No.
41.983 — 41.996 (1.6529 — 1.6534)	32231AA320
41.968 — 41.982 (1.6523 — 1.6528)	32231AA330
41.954 — 41.967 (1.6517 — 1.6522)	32231AA340

9. CENTER DIFFERENTIAL (AWD Model)

Snap ring (Inner-110) to center differential case clearance
0 — 0.15 mm (0 — 0.0059 in)

Snap ring (Inner-110)	
Part No.	Thickness mm (in)
805100061	2.10 (0.0827)
805100062	2.21 (0.0870)
805100063	2.32 (0.0913)

Backlash adjustment axial movement
0.62 — 0.86 mm (0.0244 — 0.0339 in)

Adjusting washer (45 x 62 x t)	
Part No.	Thickness mm (in)
803045041	1.60 (0.0630)
803045042	1.80 (0.0709)
803045043	2.00 (0.0787)
803045044	2.20 (0.0866)
803045045	2.40 (0.0945)

10. FRONT DIFFERENTIAL

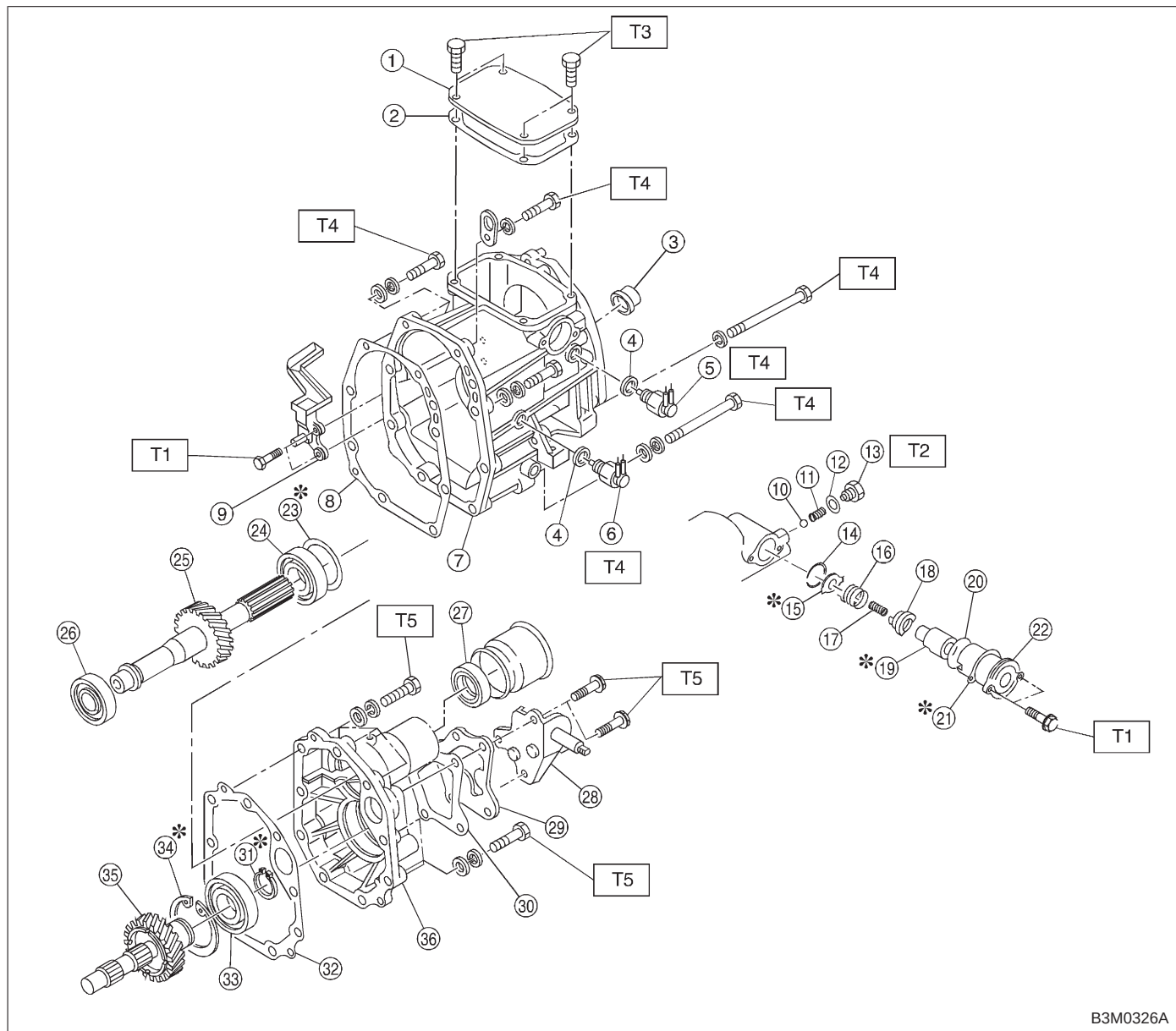
Bevel gear to pinion backlash
0.13 — 0.18 mm (0.0051 — 0.0071 in)

Washer (38.1 x 50 x t)			
Part No.	Thickness mm (in)	Part No.	Thickness mm (in)
803038021	0.925 — 0.950 (0.0364 — 0.0374)	803038023	1.025 — 1.050 (0.0404 — 0.0413)
803038022	0.975 — 1.000 (0.0384 — 0.0394)		

Pinion shaft to axle drive shaft clearance
0 — 0.2 mm (0 — 0.008 in)

Snap ring (Outer-28)			
Part No.	Thickness mm (in)	Part No.	Thickness mm (in)
805028011	1.05 (0.0413)	805028012	1.20 (0.0472)

1. Transfer Case and Extension (AWD Model)



- ① Transfer cover
- ② Cover gasket
- ③ Oil seal
- ④ Gasket
- ⑤ Neutral switch
- ⑥ Back-up light switch
- ⑦ Transfer case
- ⑧ Gasket
- ⑨ Oil guide
- ⑩ Ball
- ⑪ Reverse accent spring
- ⑫ Gasket
- ⑬ Plug
- ⑭ Snap ring (Inner)
- ⑮ Reverse check plate

- ⑯ Reverse check spring
- ⑰ Reverse return spring
- ⑱ Reverse check cam
- ⑲ Reverse accent shaft
- ⑳ O-ring
- ㉑ Adjusting select shim
- ㉒ Reverse check sleeve
- ㉓ Adjusting washer
- ㉔ Ball bearing
- ㉕ Transfer driven gear
- ㉖ Ball bearing
- ㉗ Oil seal
- ㉘ Shift bracket
- ㉙ Extension cover
- ㉚ Gasket

- ⑳ Snap ring (Outer-30)
- ㉑ Gasket
- ㉒ Ball bearing
- ㉓ Snap ring (Inner-72)
- ㉔ Transfer drive gear
- ㉕ Extension

Tightening torque: N·m (kg·m, ft·lb)

T1: 5±1 (0.5±0.1, 3.6±0.7)

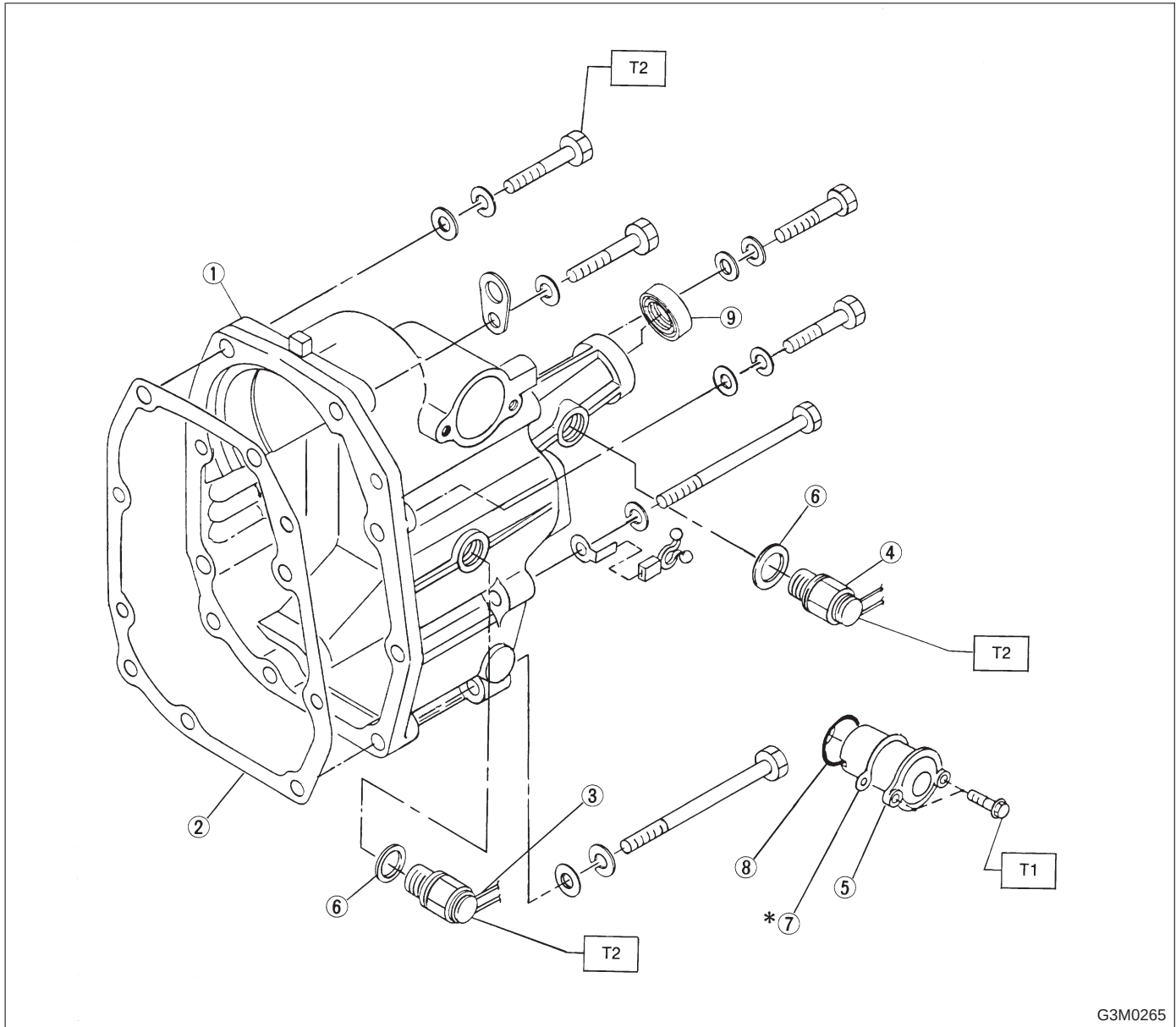
T2: 10±1 (1.0±0.1, 7.2±0.7)

**T3: 15.7±1.5
(1.6±0.15, 11.6±1.1)**

T4: 25±2 (2.5±0.2, 18.1±1.4)

T5: 37±3 (3.8±0.3, 27.5±2.2)

2. Rear Case (FWD Model)



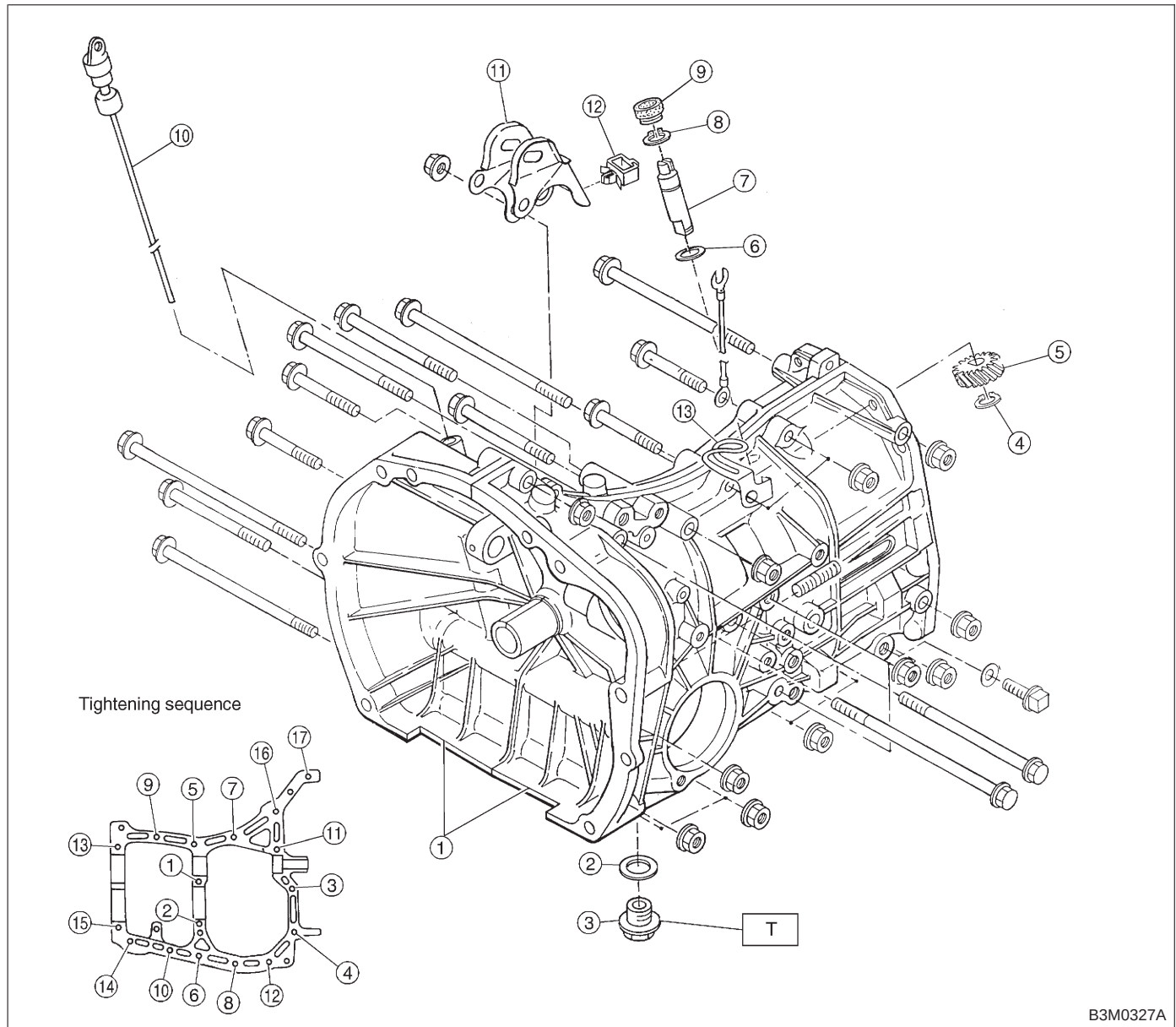
G3M0265

- ① Rear case
- ② Case gasket
- ③ Back-up light switch
- ④ Neutral switch
- ⑤ Reverse check sleeve ASSY
- ⑥ Gasket
- ⑦ Adjusting shim

- ⑧ O-ring
- ⑨ Oil seal

Tightening torque: N·m (kg-m, ft-lb)
T1: 6.4±0.5 (0.65±0.05, 4.7±0.4)
T2: 25±2 (2.5±0.2, 18.1±1.4)

3. Transmission Case

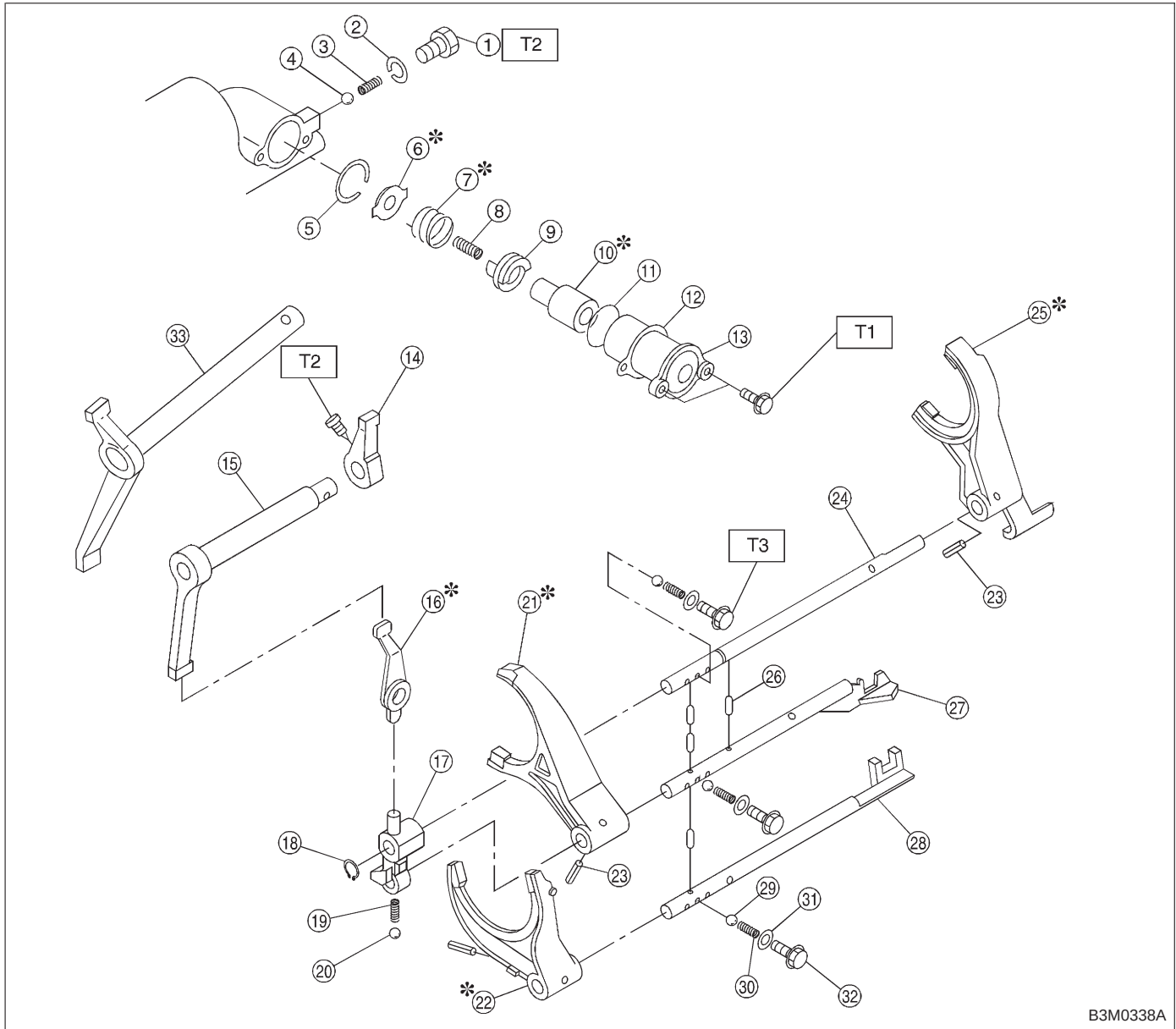


- ① Transmission case ASSY
- ② Gasket
- ③ Drain plug
- ④ Snap ring (Outer)
- ⑤ Speedometer driven gear
- ⑥ Washer
- ⑦ Speedometer shaft
- ⑧ Snap ring (Outer)
- ⑨ Oil seal
- ⑩ Oil level gauge
- ⑪ Pitching stopper bracket
- ⑫ Clamp
- ⑬ Clip

Tightening torque: N·m (kg-m, ft-lb)
T: 44±3 (4.5±0.3, 32.5±2.2)

Size	All models	Torque
8 mm bolt	⑤ — ⑮	25±2 N·m (2.5±0.2 kg-m, 18.1±1.4 ft-lb)
10 mm bolt	① — ④ ⑯ — ⑰	39±2 N·m (4.0±0.2 kg-m, 28.9±1.4 ft-lb)

4. Shifter Fork and Shifter Rod



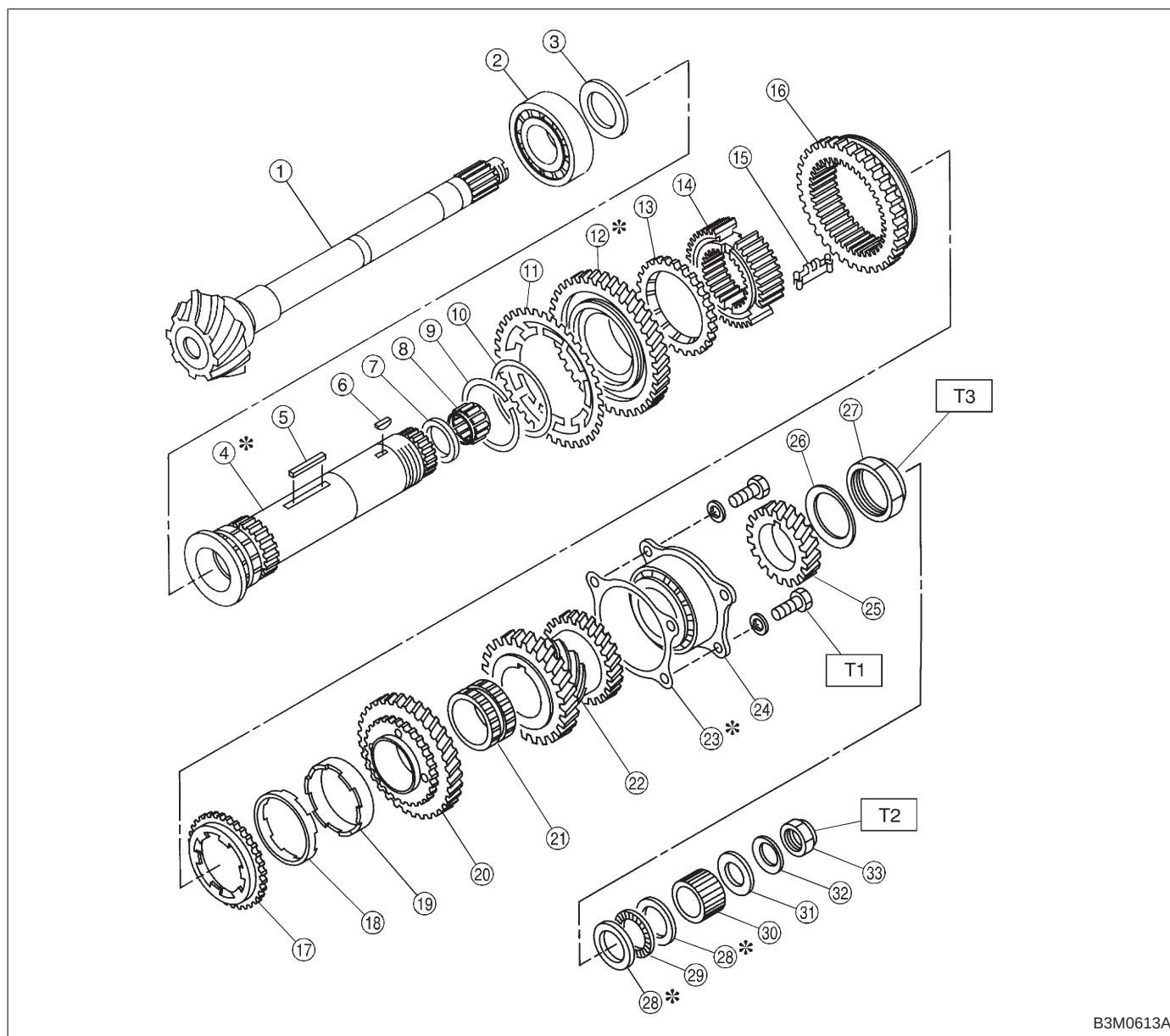
B3M0338A

- | | | |
|-------------------------|----------------------------|---------------------------|
| ① Plug | ⑭ Selector arm (AWD model) | ⑳ 3rd-4th fork rod |
| ② Gasket | ⑮ Shifter arm (AWD model) | ㉑ 1st-2nd fork rod |
| ③ Reverse accent spring | ⑯ Reverse shifter lever | ㉒ Ball |
| ④ Ball | ⑰ Reverse fork rod arm | ㉓ Checking ball spring |
| ⑤ Snap ring (Inner) | ⑱ Snap ring (Outer) | ㉔ Gasket |
| ⑥ Reverse check plate | ⑲ Spring | ㉕ Checking ball plug |
| ⑦ Reverse check spring | ⑳ Ball | ㉖ Shifter arm (FWD model) |
| ⑧ Reverse return spring | ㉑ 3rd-4th shifter fork | |
| ⑨ Reverse check cam | ㉒ 1st-2nd shifter fork | |
| ⑩ Reverse accent shaft | ㉓ Straight pin | |
| ⑪ O-ring | ㉔ Reverse fork rod | |
| ⑫ Adjusting select shim | ㉕ 5th shifter fork | |
| ⑬ Reverse check sleeve | ㉖ Interlock plunger | |

Tightening torque: N·m (kg-m, ft-lb)
T1: 6.4±0.5 (0.65±0.05, 4.7±0.4)
T2: 10±1 (1.0±0.1, 7.2±0.7)
T3: 19.6±1.5
(2.00±0.15, 14.5±1.1)

5. Drive Pinion Assembly

1. AWD MODEL



B3M0613A

- ① Drive pinion shaft
- ② Roller bearing
- ③ Washer
- ④ Driven shaft
- ⑤ Key
- ⑥ Woodruff key
- ⑦ Drive pinion collar
- ⑧ Needle bearing
- ⑨ Snap ring (Outer)
- ⑩ Washer
- ⑪ Sub gear
- ⑫ 1st driven gear
- ⑬ Baulk ring

- ⑭ 1st-2nd synchronizer hub
- ⑮ Insert key
- ⑯ Reverse driven gear
- ⑰ Outer baulk ring
- ⑱ Synchro cone
- ⑲ Inner baulk ring
- ⑳ 2nd driven gear
- ㉑ 2nd driven gear bush
- ㉒ 3rd-4th driven gear
- ㉓ Driven pinion shim
- ㉔ Roller bearing
- ㉕ 5th driven gear
- ㉖ Lock washer

- ㉗ Lock nut
- ㉘ Washer
- ㉙ Thrust bearing
- ㉚ Differential bevel gear sleeve
- ㉛ Washer
- ㉜ Lock washer
- ㉝ Lock nut

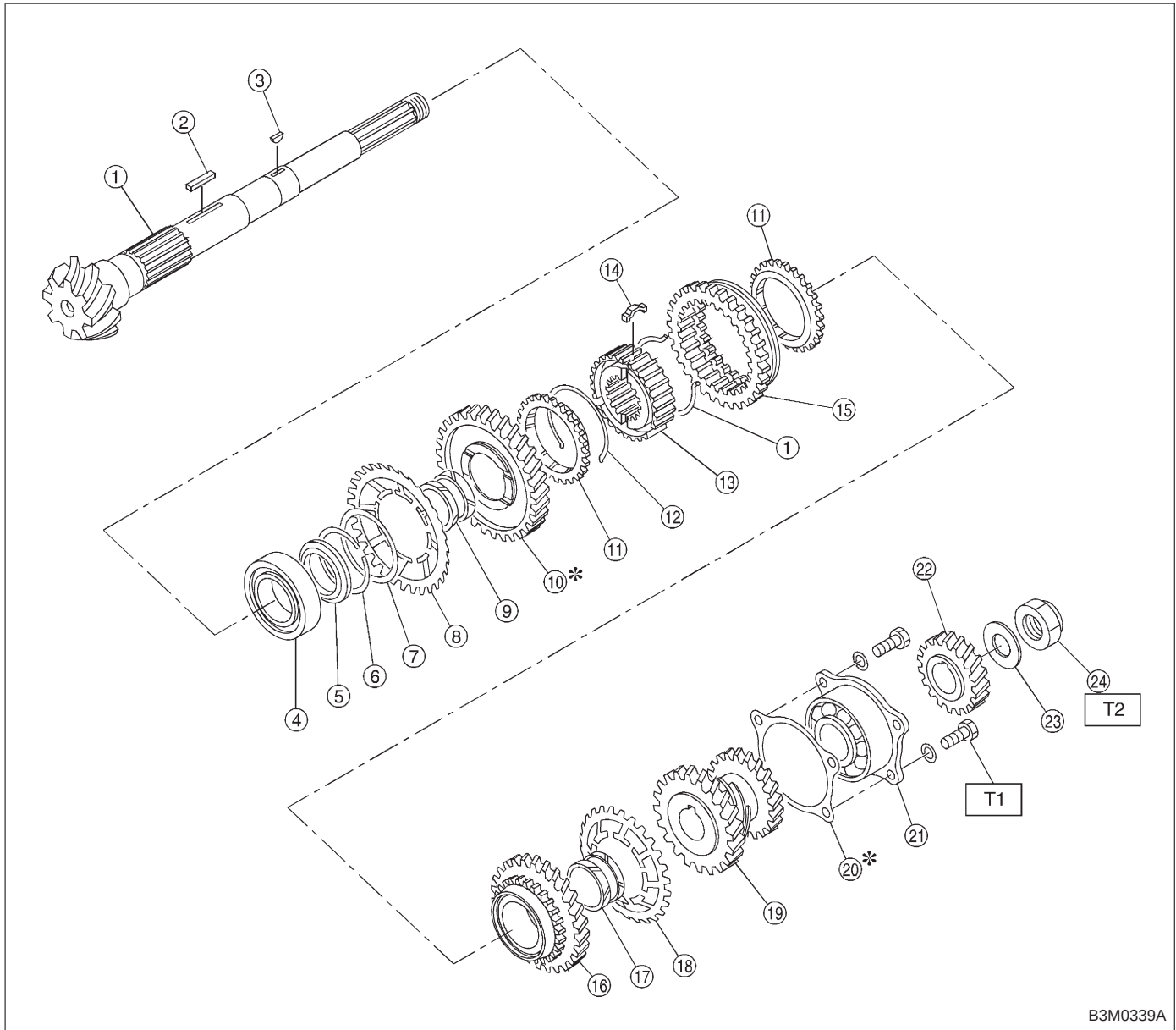
Tightening torque: N·m (kg·m, ft·lb)

T1: 29±3 (3.0±0.3, 21.7±2.2)

T2: 118±8 (12.0±0.8, 86.8±5.8)

T3: 265±10 (27±1, 195±7)

2. FWD MODEL



- ① Drive pinion shaft
- ② Key
- ③ Woodruff key
- ④ Roller bearing
- ⑤ 1st gear thrust plate
- ⑥ Snap ring (Outer)
- ⑦ Washer
- ⑧ 1st sub gear
- ⑨ 1st gear bushing
- ⑩ 1st driven gear

- ⑪ 1st-2nd baulk ring
- ⑫ 1st-2nd synchronizer spring
- ⑬ 1st-2nd synchronizer hub
- ⑭ 1st-2nd shifting insert
- ⑮ Reverse driven gear
- ⑯ 2nd driven gear
- ⑰ 2nd gear bushing
- ⑱ 2nd sub gear
- ⑲ 3rd-4th driven gear
- ⑳ Drive pinion shim

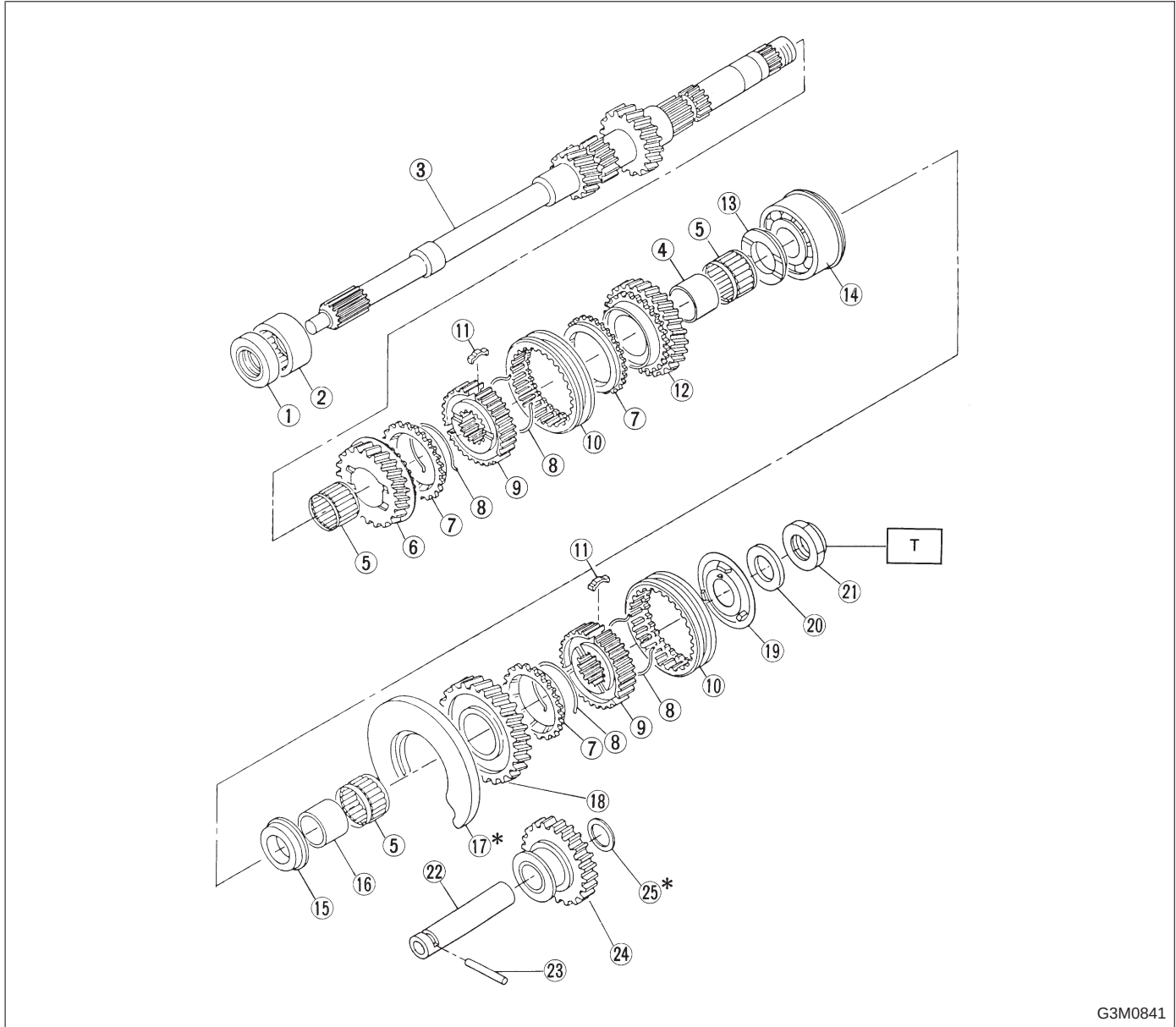
- ㉑ Ball bearing
- ㉒ 5th driven gear
- ㉓ Lock washer
- ㉔ Lock nut

Tightening torque: N·m (kg·m, ft·lb)

T1: 29±3 (3.0±0.3, 22±2)

T2: 118±6 (12.0±0.6, 86.8±4.3)

6. Main Shaft Assembly



G3M0841

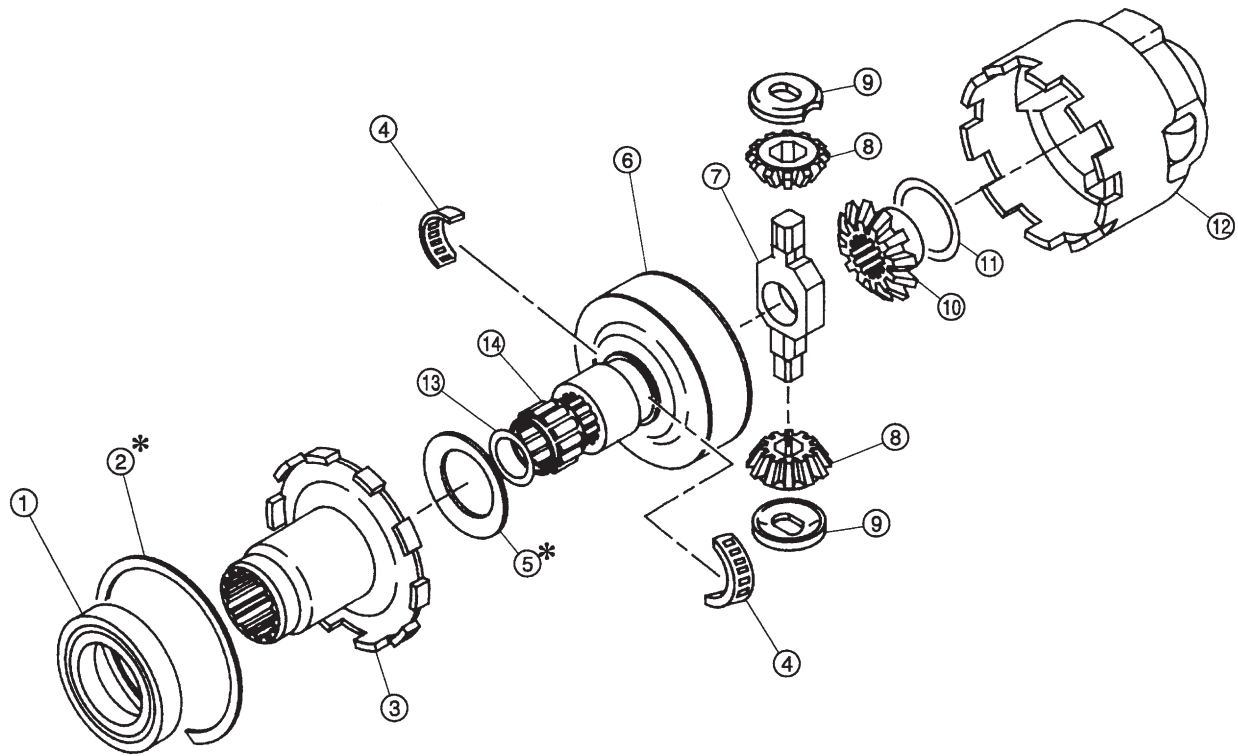
- ① Oil seal
- ② Needle bearing
- ③ Transmission main shaft
- ④ 4th needle bearing race
- ⑤ Needle bearing
- ⑥ 3rd drive gear
- ⑦ Baulk ring
- ⑧ Synchronizer spring
- ⑨ Synchronizer hub
- ⑩ Coupling sleeve

- ⑪ Shifting insert
- ⑫ 4th drive gear
- ⑬ 4th gear thrust washer
- ⑭ Ball bearing
- ⑮ 5th gear thrust washer
- ⑯ 5th needle bearing race
- ⑰ Main shaft rear plate
- ⑱ 5th drive gear
- ⑲ Insert stopper plate
- ⑳ Lock washer

- ㉑ Lock nut
- ㉒ Reverse idler gear shaft
- ㉓ Straight pin
- ㉔ Reverse idler gear
- ㉕ Washer

Tightening torque: N·m (kg-m, ft-lb)
T: 118±6 (12.0±0.6, 86.8±4.3)

7. Center Differential (AWD Model)

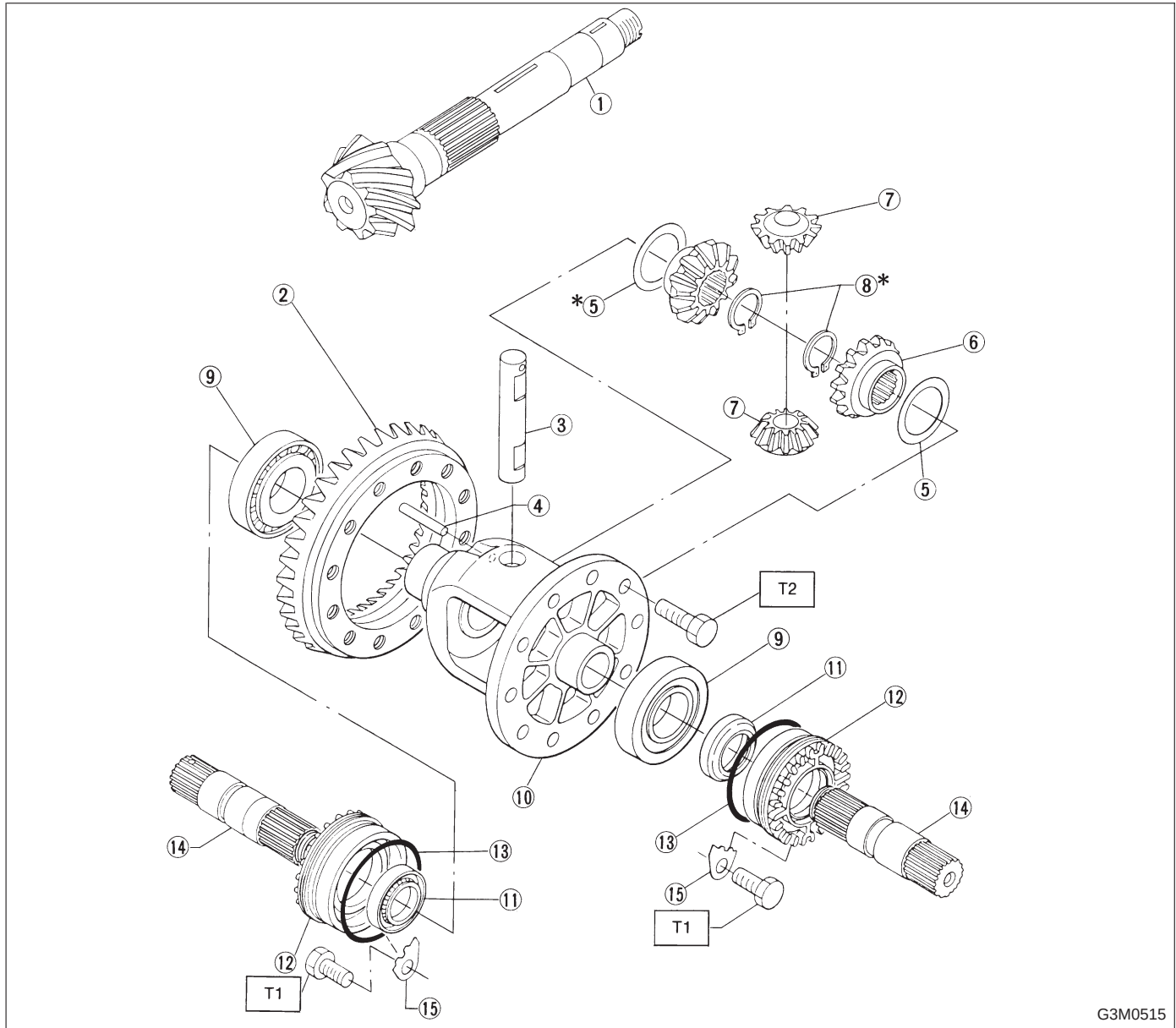


B3M0036A

- ① Ball bearing
- ② Snap ring (Inner-110)
- ③ Center differential cover
- ④ Needle bearing
- ⑤ Adjusting washer
- ⑥ Viscous coupling
- ⑦ Pinion shaft

- ⑧ Differential bevel pinion
- ⑨ Retainer
- ⑩ Differential bevel gear
- ⑪ Washer
- ⑫ Center differential case
- ⑬ Snap ring
- ⑭ Roller bearing

8. Front Differential



G3M0515

- ① Drive pinion shaft
- ② Hypoid driven gear
- ③ Pinion shaft
- ④ Straight pin
- ⑤ Washer
- ⑥ Differential bevel gear
- ⑦ Differential bevel pinion
- ⑧ Snap ring (Outer)
- ⑨ Roller bearing
- ⑩ Differential case

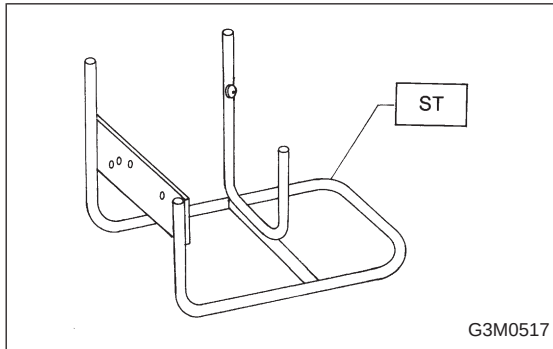
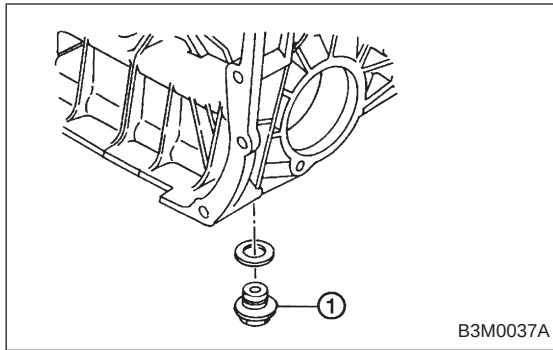
- ⑪ Oil seal
- ⑫ Differential side retainer
- ⑬ O-ring
- ⑭ Axle drive shaft
- ⑮ Retainer lock plate

Tightening torque: N·m (kg·m, ft·lb)**T1: 25±5 (2.5±0.5, 18.1±3.6)****T2: 62±5 (6.3±0.5, 45.6±3.6)**

1. General**A: APPLICATION**

Large title		Model	
		AWD	FWD
1	General	○	○
2	Transfer Case and Extension (AWD Model)	○	
3	Rear Case (FWD Model)		○
4	Transmission Case	○	○
5	Drive Pinion Assembly (AWD Model)	○	
6	Drive Pinion Assembly (FWD Model)		○
7	Main Shaft Assembly	○	○
8	Center Differential (AWD Model)	○	
9	Front Differential	○	○

The table above shows the titles of the main sections in Service Procedures and the applicable vehicle models. Carry out service operations by referring to the sections applicable to the vehicle to be serviced.



B: PRECAUTIONS

1) The following job should be followed before disassembly:

- (1) Clean oil, grease, dirt and dust from transmission.
- (2) Remove drain plug ① to drain oil. After draining, retighten it as before.

CAUTION:

Replace gasket with a new one.

Tightening torque:

44 ± 3 N·m (4.5 ± 0.3 kg-m, 32.5 ± 2.2 ft-lb)

- (3) Attach transmission to ST.

ST 499937100 TRANSMISSION STAND SET

2) Rotating parts should be coated with oil prior to assembly.

3) All disassembled parts, if to be reused, should be reinstalled in the original positions and directions.

4) Gaskets and lock washers must be replaced with new ones.

5) Liquid gasket should be used where specified to prevent leakage.

6) Fill transmission gear oil through the oil level gauge hole up to upper point level gauge. <Ref. to 1-5 [09A1].>

C: INSPECTION

Disassembled parts should be washed clean first and then inspected carefully.

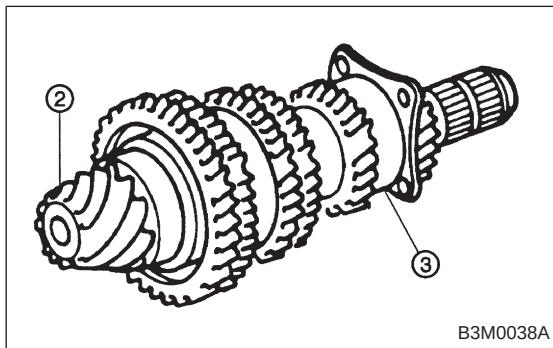
1) Bearings

Replace bearings in the following cases:

- Bearings whose balls, outer races and inner races are broken or rusty.
- Worn bearings
- Bearings that fail to turn smoothly or make abnormal noise when turned after gear oil lubrication.

The ball bearing ③ on the rear side of the drive pinion shaft ② should be checked for smooth rotation before the drive pinion assembly is disassembled. In this case, because a preload is working on the bearing, its rotation feels like it is slightly dragging unlike the other bearings.

- Bearings having other defects



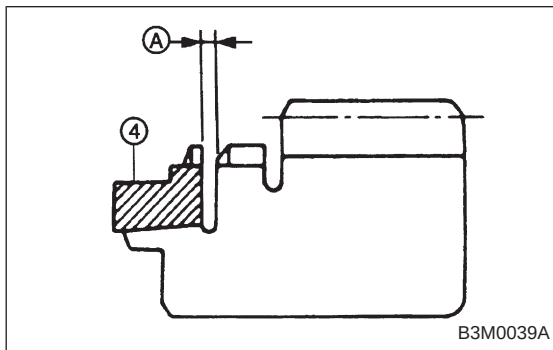
2) Bushing (each gear)

Replace the bushing in the following cases:

- (1) When the sliding surface is damaged or abnormally worn.
- (2) When the inner wall is abnormally worn.

3) Gears

- (1) Replace gears with new ones if their tooth surfaces are broken, damaged, or excessively worn.
- (2) Correct or replace if the cone that contacts the baulk ring is rough or damaged.
- (3) Correct or replace if the inner surface or end face is damaged.



4) Baulk ring

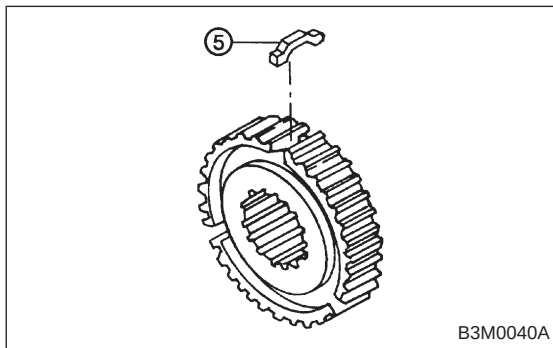
Replace the ring in the following cases:

- When the inner surface and end face are damaged.
- When the ring inner surface is abnormally or partially worn down.
- If the gap between the end faces of the ring and the gear splined part is excessively small when the ring is pressed against the cone.

Clearance (A) :

0.5 — 1.0 mm (0.020 — 0.040 in)

- When the contact surface of the synchronizer ring (4) insert is scored or abnormally worn down.



5) Insert (5) (shifting)

Replace the insert if deformed, excessively worn, or defective in any way.

6) Oil seal

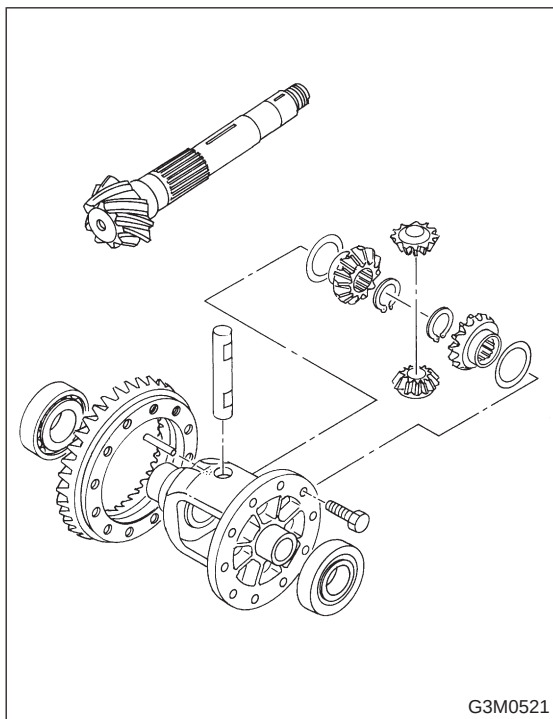
Replace the oil seal if the lip is deformed, hardened, damaged, worn, or defective in any way.

7) O-ring

Replace the O-ring if the sealing face is deformed, hardened, damaged, worn, or defective in any way.

8) Gearshift mechanism

Repair or replace the gearshift mechanism if excessively worn, bent, or defective in any way.



G3M0521

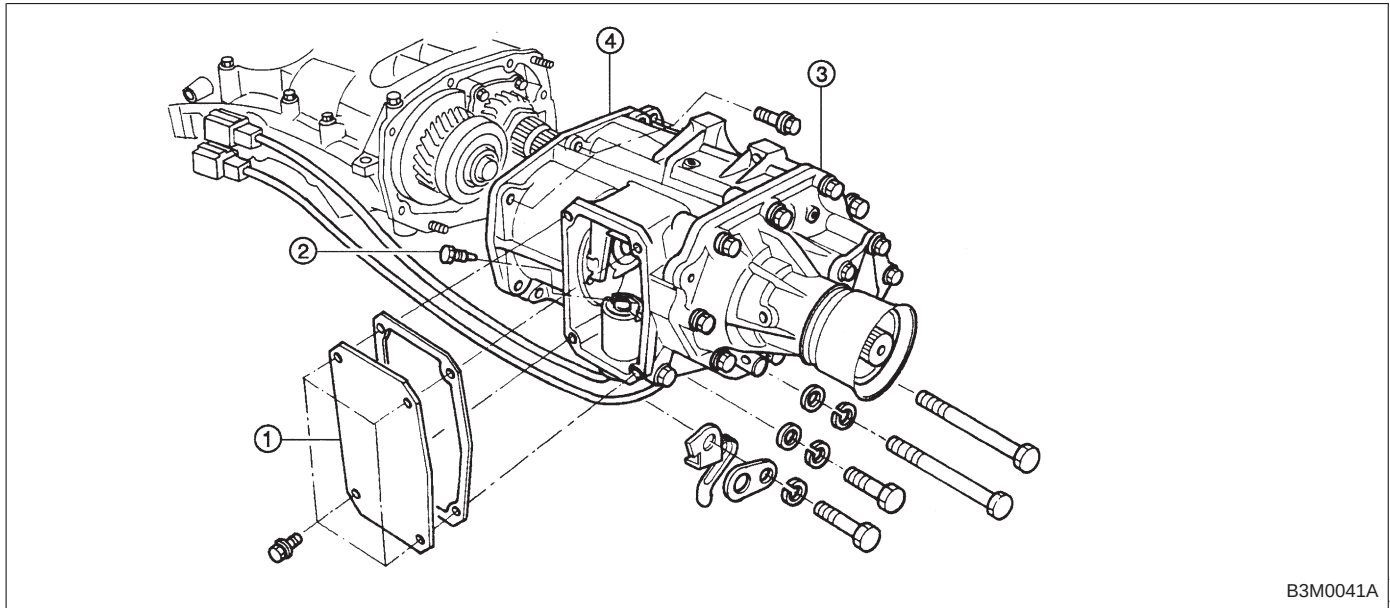
9) Differential gear

Repair or replace the differential gear in the following cases:

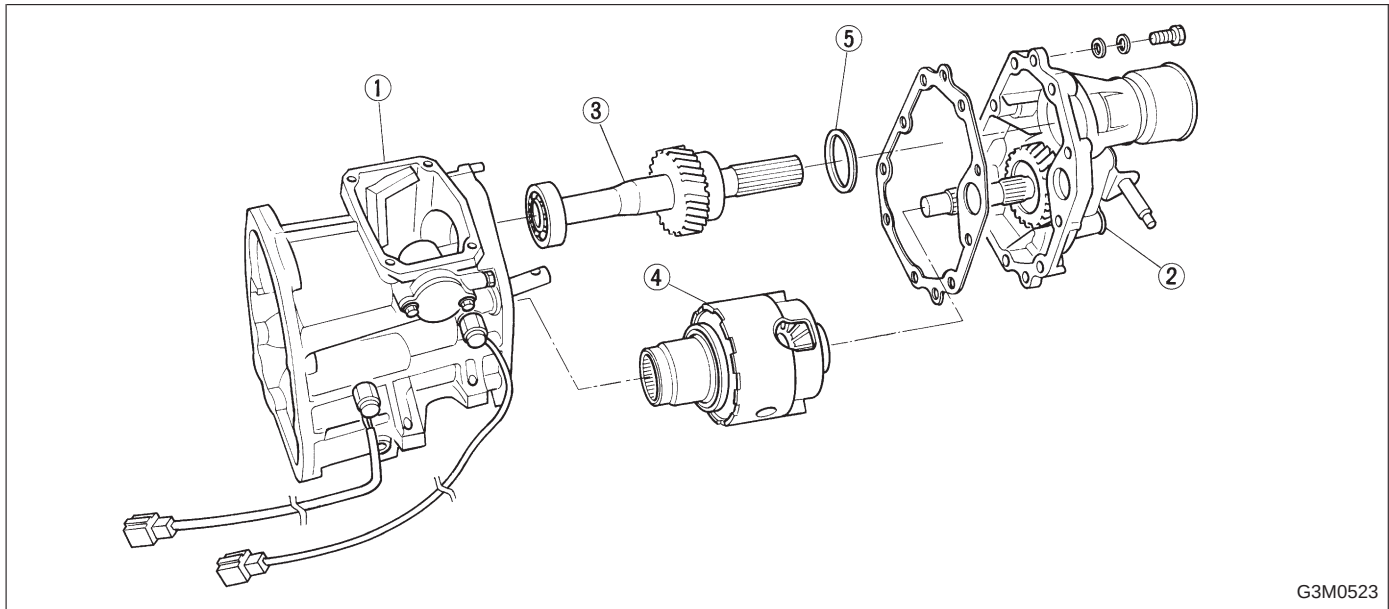
- (1) The hypoid drive gear and drive pinion shaft tooth surface are damaged, excessively worn, or seized.
- (2) The roller bearing on the drive pinion shaft has a worn or damaged roller path.
- (3) There is damage, wear, or seizure of the differential bevel pinion, differential bevel gear, washer, pinion shaft, and straight pin.
- (4) The differential case has worn or damaged sliding surfaces.

2. Transfer Case and Extension (AWD Model)

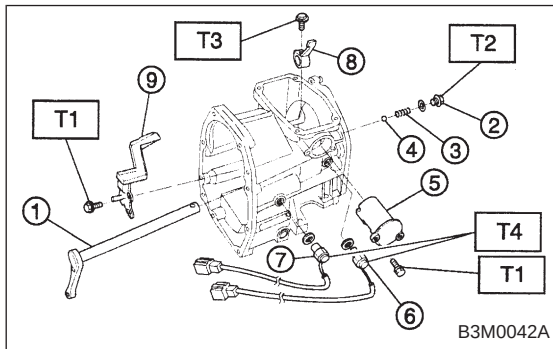
A: REMOVAL



- 1) Remove transfer cover ①.
- 2) Remove shifter fork screw ② which secures selector arm to shifter arm.
- 3) Remove transfer case ④ with extension assembly ③.

B: DISASSEMBLY**1. SEPARATION OF TRANSFER CASE AND EXTENSION ASSEMBLY**

- 1) Separate transfer case ① and extension assembly ②.
- 2) Remove transfer driven gear ③ and center differential ④ as a set.
- 3) Remove thrust washer ⑤ (52 x 61 x t).

**2. TRANSFER CASE**

- 1) Remove neutral switch ⑥.

NOTE:

Before removing shifter arm, disconnect neutral switch.

- 2) Draw out shifter arm ① and remove selector arm ⑧.
- 3) Remove plug ②, spring ③ and reverse check ball ④.

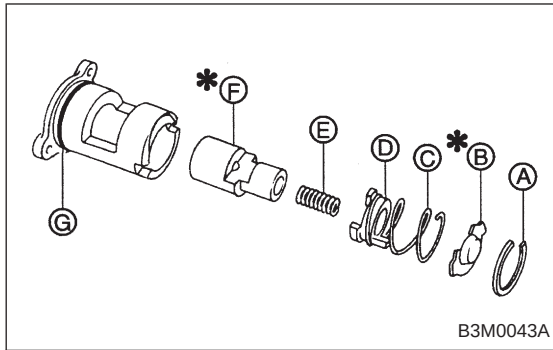
Tightening torque:

T1: 6.4±0.5 N·m (0.65±0.05 kg-m, 4.7±0.4 ft-lb)

T2: 10±1 N·m (1.0±0.1 kg-m, 7.2±0.7 ft-lb)

T3: 19.6±1.5 N·m (2.00±0.15 kg-m, 14.5±1.1 ft-lb)

T4: 25±2 N·m (2.5±0.2 kg-m, 18.1±1.4 ft-lb)



4) Remove reverse check sleeve (5).

Disassembly procedure is as follows:

- (1) Using a standard screwdriver, remove snap ring (A) (Inner-28).

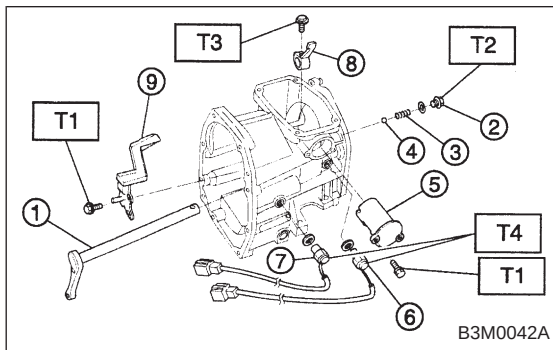
NOTE:

Replace snap ring with a new one if deformed or weakened.

- (2) Remove reverse check plate (B).
- (3) Remove reverse check spring (C) with cam (D).
- (4) Remove reverse return spring (E).
- (5) Remove reverse accent shaft (F).
- (6) Remove O-ring (G).

NOTE:

- Reverse check sleeve assembly uses an O-ring which should not be scratched.
- Be careful not to break adjustment shim placed between reverse check sleeve assembly and case.



5) Remove back-up light switch (7).

6) Remove oil guide (9).

Tightening torque:

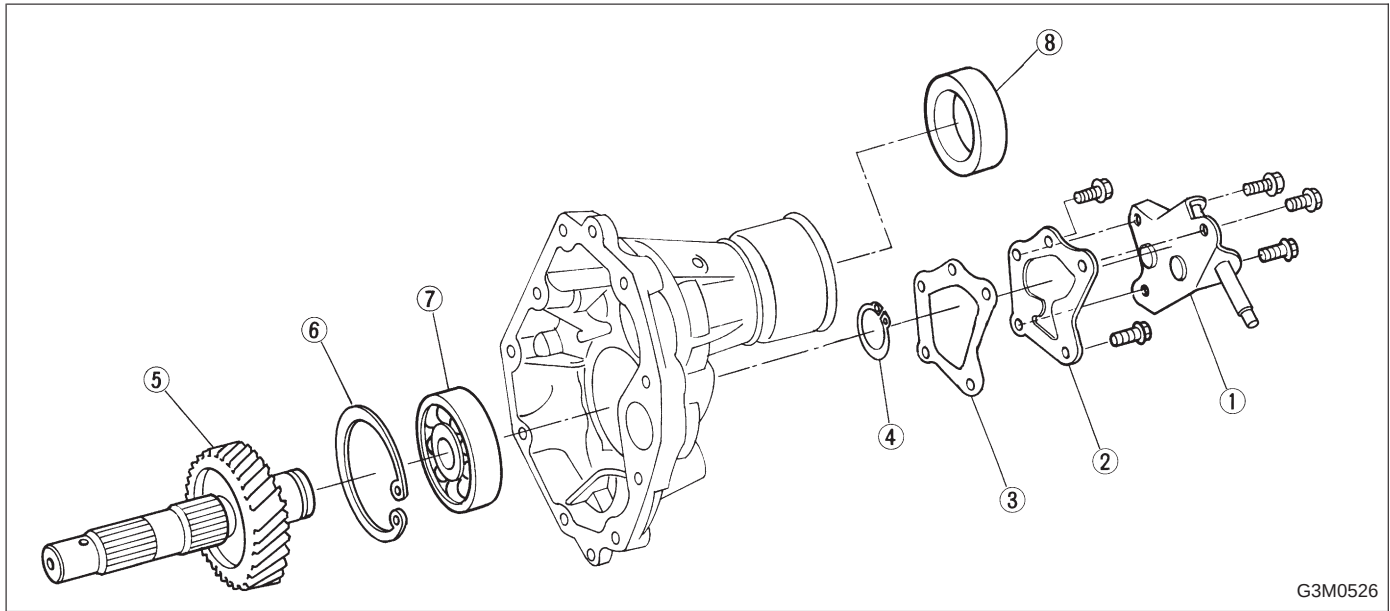
T1: 6.4 ± 0.5 N·m (0.65 ± 0.05 kg-m, 4.7 ± 0.4 ft-lb)

T2: 10 ± 1 N·m (1.0 ± 0.1 kg-m, 7.2 ± 0.7 ft-lb)

T3: 19.6 ± 1.5 N·m (2.00 ± 0.15 kg-m, 14.5 ± 1.1 ft-lb)

T4: 25 ± 2 N·m (2.5 ± 0.2 kg-m, 18.1 ± 1.4 ft-lb)

3. EXTENSION



G3M0526

- 1) Remove extension cover ② and shift bracket ①.
- 2) Remove snap ring ④ (Outer-30).
- 3) Remove transfer drive gear ⑤.

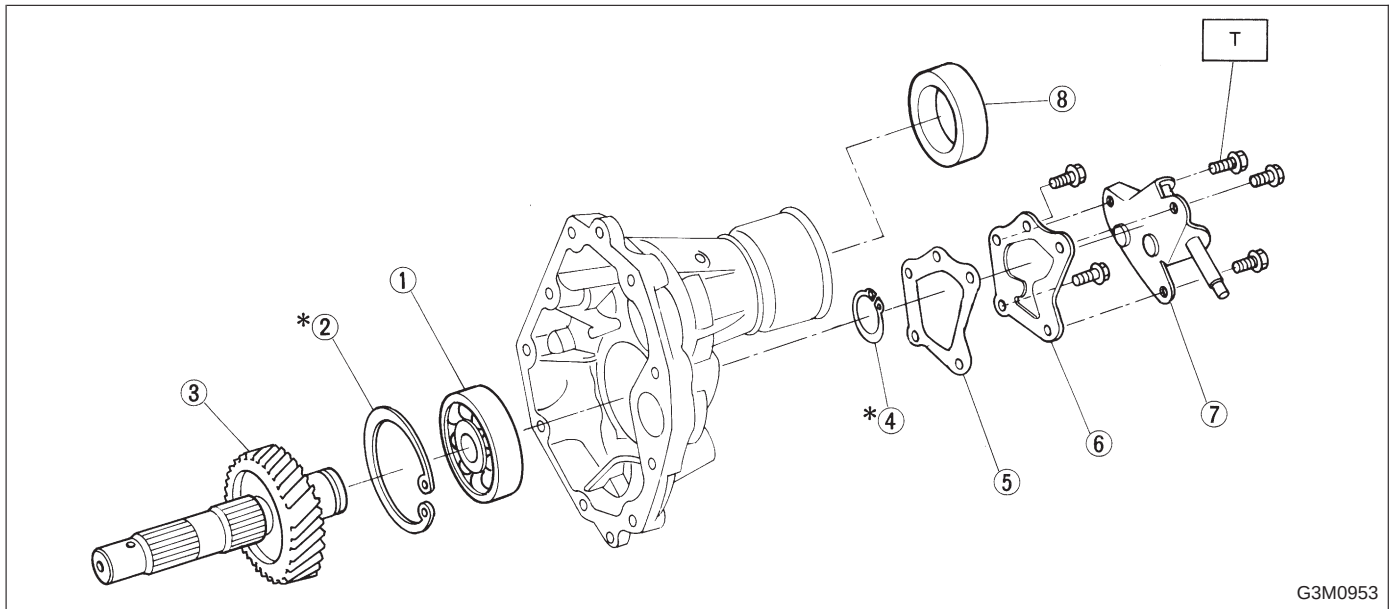
CAUTION:**Do not remove ball bearing unless replacing.**

- 4) Remove snap ring ⑥ (Inner-72).
- 5) Remove ball bearing ⑦.

CAUTION:**Do not reuse ball bearing.**

- 6) Remove oil seal ⑧.

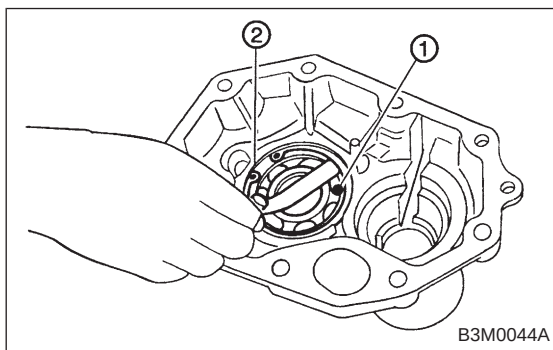
CAUTION:**Do not reuse oil seal.**

C: ASSEMBLY**1. EXTENSION**

Tightening torque: N·m (kg-m, ft-lb)
T: 25±2 (2.5±0.2, 18.1±1.4)

- 1) Installation of ball bearing ① and selection of snap ring ② (Inner-72)

(1) Attach ball bearing ① (30 x 72 x 17) to extension and install snap ring ②.



- (2) Measure clearance between snap ring ② and outer race of ball bearing ① with a thickness gauge.

CAUTION:

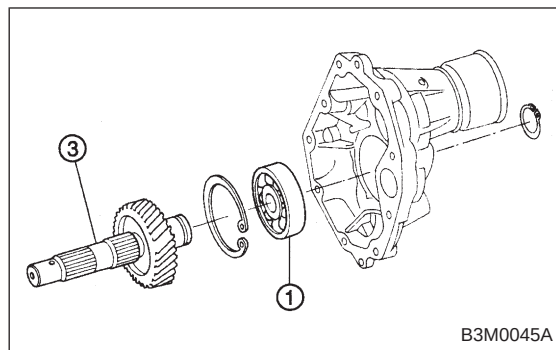
Replace ball bearing with a new one.

Clearance:

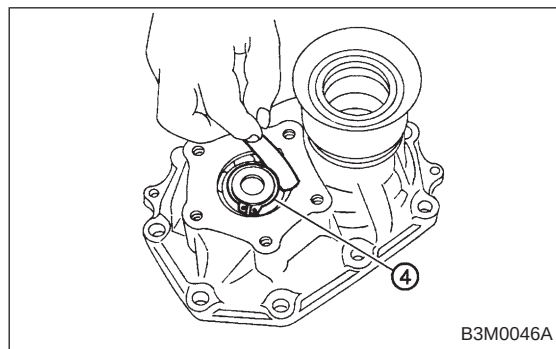
0 — 0.15 mm (0 — 0.0059 in)

- (3) If the measurement is not within the specification, select suitable snap ring ②.

Snap ring (Inner-72)	
Part No.	Thickness mm (in)
805172071	1.78 (0.0701)
805172072	1.90 (0.0748)
805172073	2.02 (0.0795)



- 2) Installation of transfer drive gear ③
Press transfer drive gear into inner race of ball bearing.



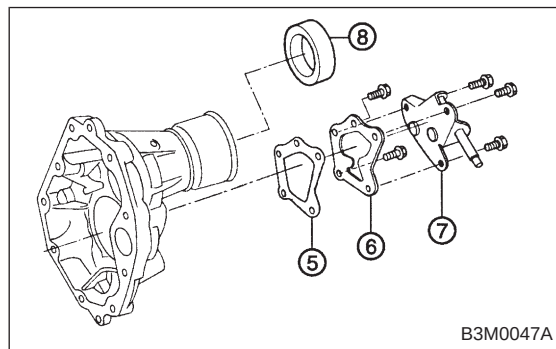
- 3) Selection of snap ring ④ (Outer-30)
(1) Install snap ring ④ on transfer drive shaft.
(2) Measure clearance between snap ring and inner race of ball bearing with a thickness gauge.

Clearance:

0 — 0.15 mm (0 — 0.0059 in)

- (3) If the measurement is not within the specification, select suitable snap ring.

Snap Ring (Outer-30)	
Part No.	Thickness mm (in)
805030041	1.53 (0.0602)
805030042	1.65 (0.0650)
805030043	1.77 (0.0697)

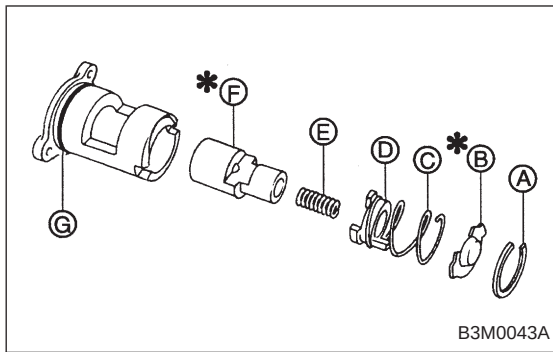


- 4) Install extension cover ⑥, gasket ⑤ and shift bracket ⑦.

CAUTION:
Use new gasket.

- 5) Install oil seal ⑧ with ST.
ST 498057300 INSTALLER

CAUTION:
Use new oil seal.



2. TRANSFER CASE

Assembly of transfer case is in the reverse order of disassembly. Do the following:

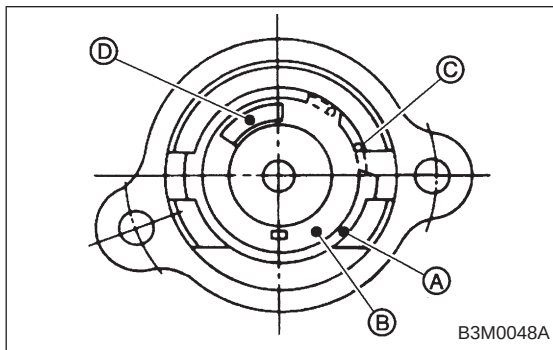
1) Assembly of reverse check sleeve

- (1) Install reverse accent shaft **F**, check cam **D**, return spring **E** and check spring **C** onto reverse check sleeve.

NOTE:

Be sure the bent section of reverse check spring is positioned in the groove in check cam.

- (2) Hook the bent section of reverse check spring over reverse check plate **B**.
- (3) Rotate cam so that the protrusion of reverse check cam is at the opening in plate.
- (4) With cam held in that position, install plate onto reverse check sleeve and hold with snap ring **A** (Inner-28).
- (5) Position O-ring **G** (35.4 x 1.5) in groove in sleeve.



CAUTION:

- Make sure the cutout section of reverse accent shaft is aligned with the opening in reverse check sleeve.

- Spin cam by hand for smooth rotation.

If it does not return properly, replace reverse check spring.

- Move cam and shaft all the way toward plate and release.

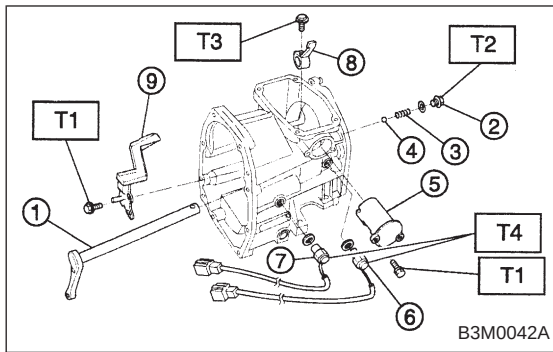
If cam does not return properly, replace reverse check spring; if shaft does not, check for scratches on the inner surface of sleeve. If sleeve is in good order, replace spring.

- Select a suitable reverse accent shaft and reverse check plate by referring to "Neutral Position Adjustment."

3-1

SERVICE PROCEDURE

2. Transfer Case and Extension (AWD Model)



2) Installation of shifter arm ① and selector arm ⑧
Install shifter arm into the partition from the front while inserting selector arm into the opening in reverse check sleeve. Pass shaft through hole in selector arm until its end comes out of the rear of transfer case.

NOTE:

Apply a coat of gear oil to shifter arm. Also make sure oil seal is positioned properly.

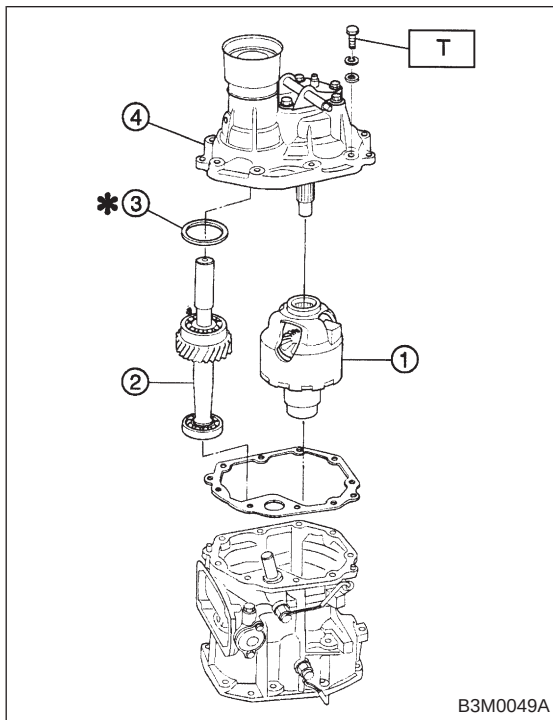
Tightening torque:

T1: $6.4 \pm 0.5 \text{ N}\cdot\text{m}$ ($0.65 \pm 0.05 \text{ kg}\cdot\text{m}$, $4.7 \pm 0.4 \text{ ft}\cdot\text{lb}$)

T2: $10 \pm 1 \text{ N}\cdot\text{m}$ ($1.0 \pm 0.1 \text{ kg}\cdot\text{m}$, $7.2 \pm 0.7 \text{ ft}\cdot\text{lb}$)

T3: $19.6 \pm 1.5 \text{ N}\cdot\text{m}$ ($2.00 \pm 0.15 \text{ kg}\cdot\text{m}$, $14.5 \pm 1.1 \text{ ft}\cdot\text{lb}$)

T4: $25 \pm 2 \text{ N}\cdot\text{m}$ ($2.5 \pm 0.2 \text{ kg}\cdot\text{m}$, $18.1 \pm 1.4 \text{ ft}\cdot\text{lb}$)

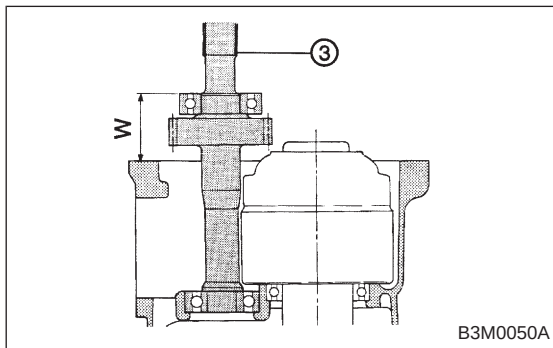


3. COMBINATION OF TRANSFER CASE AND EXTENSION ASSEMBLY

1) Install center differential ① and transfer driven gear ② into transfer case.

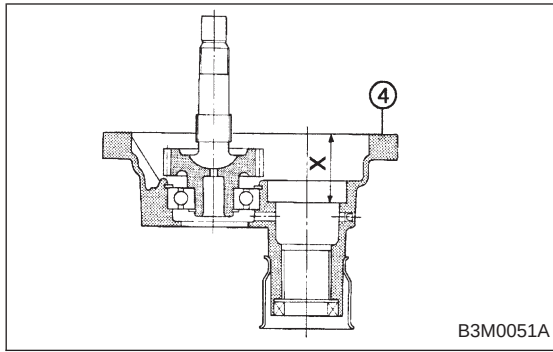
Tightening torque:

T: $37 \pm 3 \text{ N}\cdot\text{m}$ ($3.8 \pm 0.3 \text{ kg}\cdot\text{m}$, $27.5 \pm 2.2 \text{ ft}\cdot\text{lb}$)



2) Selection of thrust washer (52 x 61 x t)

(1) Measure height "W" between transfer case and ball bearing on the transfer driven gear ③.



(2) Measure depth "X" as shown in figure.

④ Extension

(3) Calculate space "Y" using the following equation: $Y = X - W + 0.24 \text{ mm (0.0094 in)}$ [Thickness of gasket]

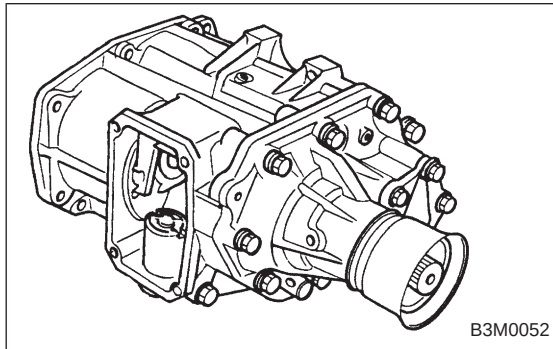
(4) Select suitable washer in the following table:

Space "Y" mm (in)	Thrust washer (52 x 61 x t)	
	Part No.	Thickness mm (in)
0.55 — 0.79 (0.0217 — 0.0311)	803052021	0.50 (0.0197)
0.80 — 1.04 (0.0315 — 0.0409)	803052022	0.75 (0.0295)
1.05 — 1.30 (0.0413 — 0.0512)	803052023	1.00 (0.0394)

Standard clearance between thrust washer and ball bearing:

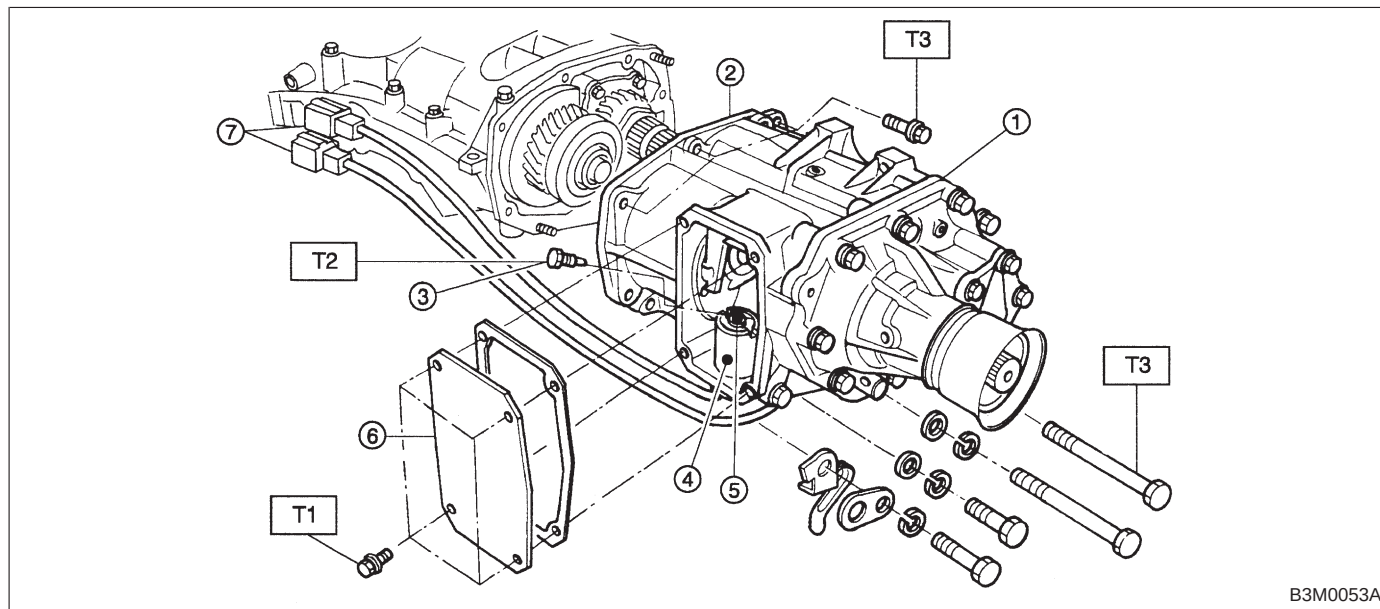
0.05 — 0.30 mm (0.0020 — 0.0118 in)

(5) Fit thrust washers on transfer drive shaft.



3) Install extension assembly into transfer case.

D: INSTALLATION



Tightening torque: N·m (kg-m, ft-lb)

T1: 15.7±1.5 (1.6±0.15, 11.6±1.1)

T2: 19.6±1.5 (2.00±0.15, 14.5±1.1)

T3: 24.5±2.0 (2.50±0.20, 18.1±1.4)

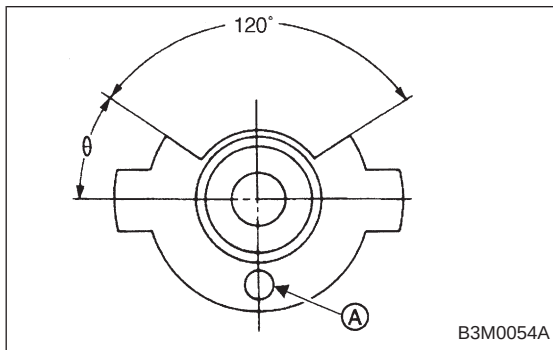
- 1) Install transfer case ② with extension assembly ①.
- 2) Secure selector arm to shifter arm with shifter fork screw ③. Shifter arm should be caught by pawl of rod. Selector arm must be engaged with reverse check sleeve assembly.
- 3) Adjustment of neutral position
 - (1) Shift gear into 3rd gear position.
 - (2) Shifter arm turns lightly toward the 1st/2nd gear side but heavily toward the reverse gear side because of the function of the return spring, until arm contacts the stopper.
 - (3) Make adjustment so that the heavy stroke (reverse side) is a little more than the light stroke (1st/2nd side).
 - (4) To adjust, remove bolts holding reverse check sleeve assembly ④ to the case, move sleeve assembly outward, and place adjustment shim (0 to 1 ea.) between sleeve assembly and case to adjust the clearance.

CAUTION:**Be careful not to break O-ring when placing shim(s).**

Adjustment shim	
Part No.	Thickness mm (in)
32190AA000	0.15 (0.0059)
32190AA010	0.30 (0.0118)

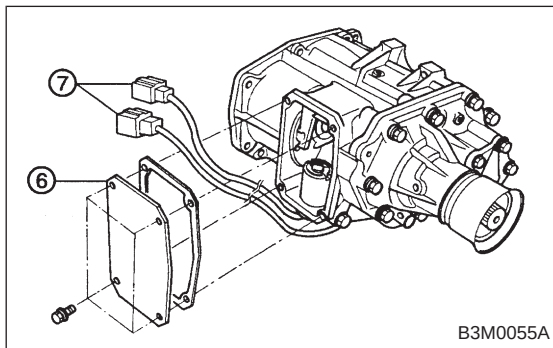
- When shim is removed, the neutral position will move closer to reverse; when shim is added, the neutral position will move closer to 1st gear.
- If shims alone cannot adjust the clearance, replace reverse accent shaft and re-adjust.

Reverse accent shaft		
Part No.	Mark	Remarks
32188AA040	1	Neutral position is closer to 1st gear.
32188AA011	No mark or 2	Standard
32188AA050	3	Neutral position is closer to reverse gear.

**4) Reverse check plate ⑤ adjustment**

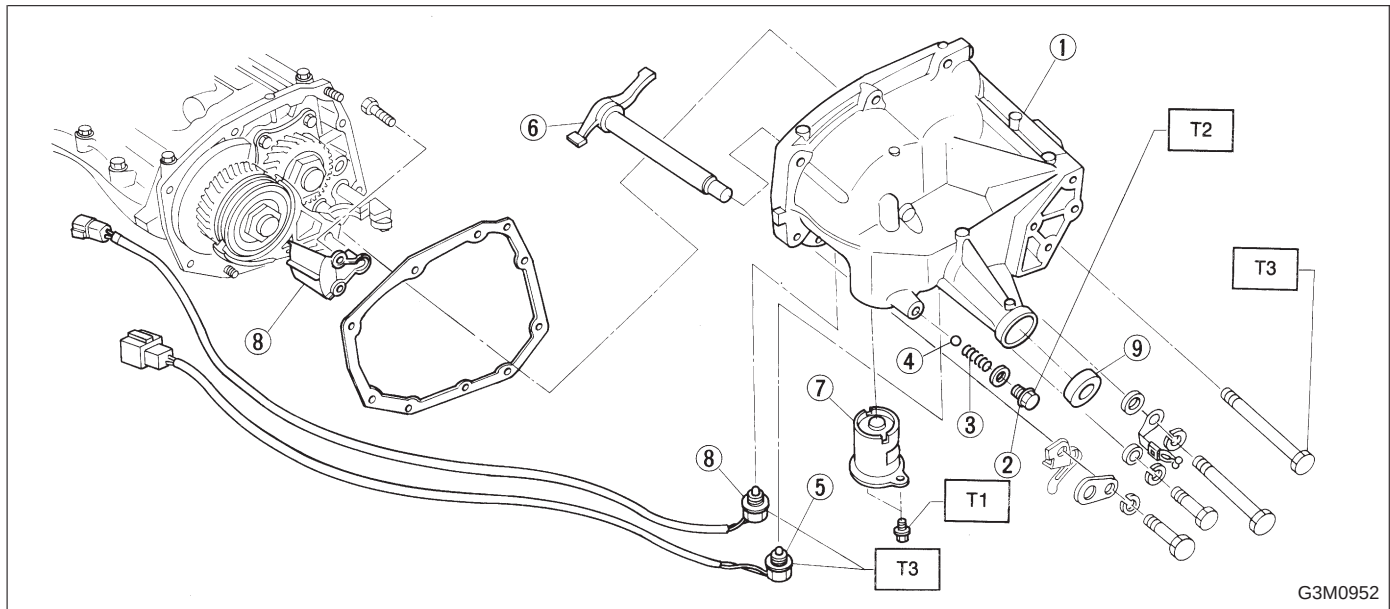
Shift shifter arm to "5th" and then to reverse to see if reverse check mechanism operates properly. Also check to see if arm returns to neutral when released from the reverse position. If arm does not return properly, replace reverse check plate.

Reverse check plate			
Part No.	⑤: No.	Angle θ	Remarks
32189AA000	0	28°	Arm stops closer to 5th gear.
32189AA010	1	31°	Arm stops closer to 5th gear.
32189AA020	2	34°	Arm stops in the center.
32189AA030	3	37°	Arm stops closer to reverse gear.
32189AA040	4	40°	Arm stops closer to reverse gear.

**5) Install transfer ⑥ cover and gasket.****6) Connect each connector ⑦.**

3. Rear Case (FWD Model)

A: DISASSEMBLY



Tightening torque: N·m (kg-m, ft-lb)

T1: 6.4 ± 0.5 (0.65 ± 0.05 , 4.7 ± 0.4)

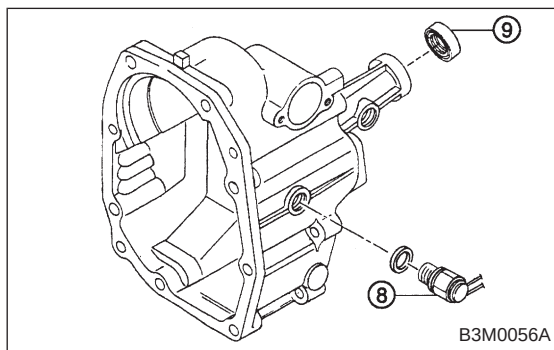
T2: 10 ± 1 (1.0 ± 0.1 , 7.2 ± 0.7)

T3: 25 ± 2 (2.5 ± 0.2 , 18.1 ± 1.4)

- 1) Remove rear case ①.
- 2) Remove plug ②, spring ③ and reverse check ball ④.
- 3) Remove neutral switch ⑤.
- 4) Pull out shifter arm ⑥.
- 5) Remove reverse check sleeve ⑦.

NOTE:

The disassembly procedure is the same as for AWD model.



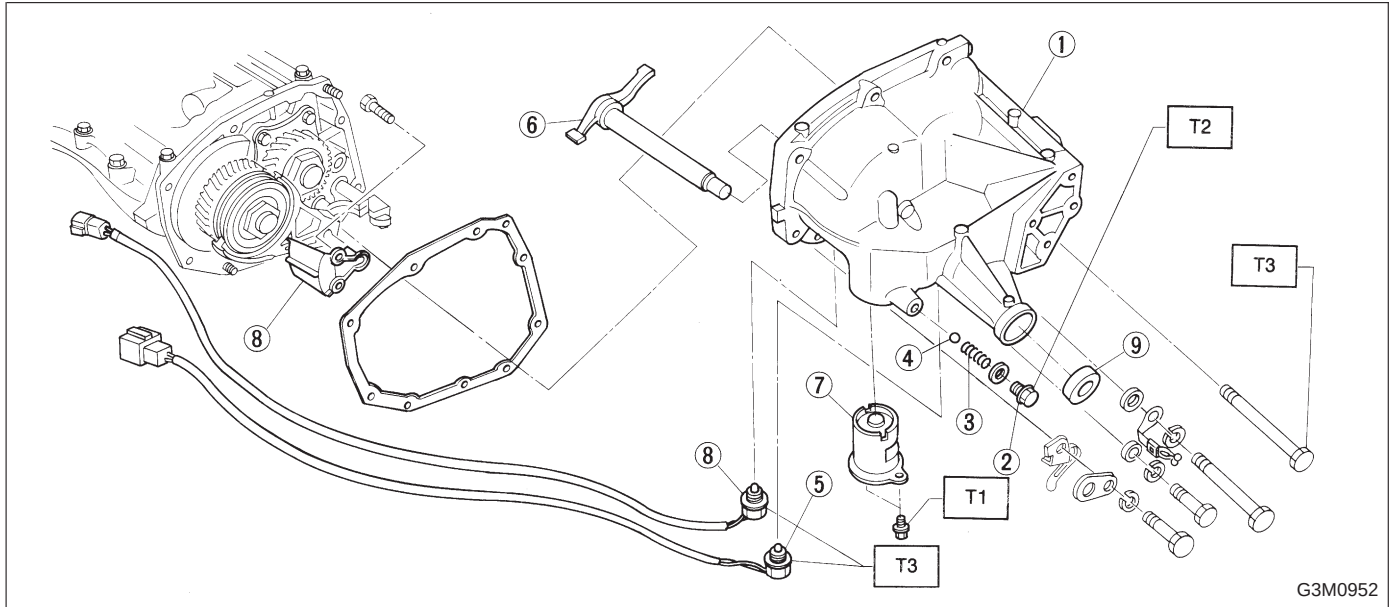
- 6) Remove back-up light switch ⑧.
- 7) Remove oil seal ⑨.

CAUTION:

Do not reuse oil seal.

B: ASSEMBLY

1) Assembly of rear case is in the reverse order of disassembly.



Tightening torque: N·m (kg-m, ft-lb)

T1: 6.4±0.5 (0.65±0.05, 4.7±0.4)

T2: 10±1 (1.0±0.1, 7.2±0.7)

T3: 25±5 (2.5±0.5, 18.1±3.6)

2) Installation of shifter arm ⑥

Install shifter arm into the partition from the front while inserting selector arm into the opening in reverse check sleeve. Pass shaft through hole in selector arm until its end comes out of the rear of transmission case assembly.

CAUTION:

Apply a coat of gear oil to shifter arm. Also make sure oil seal is positioned properly.

3) Adjustment of neutral position

NOTE:

After assembling and installing rear case, adjust neutral position.

(1) Shift gear into 3rd gear position.

(2) Shifter arm turns lightly toward the 1st/2nd gear side but heavily toward the reverse gear side because of the function of the return spring, until arm contacts the stopper.

(3) Make adjustment so that the heavy stroke (reverse side) is a little more than the light stroke (1st/2nd side).

(4) To adjust, remove bolts holding reverse check sleeve assembly to the case, move sleeve assembly outward, and place adjustment shim (0 to 1 ea.) between sleeve assembly and case to adjust the clearance.

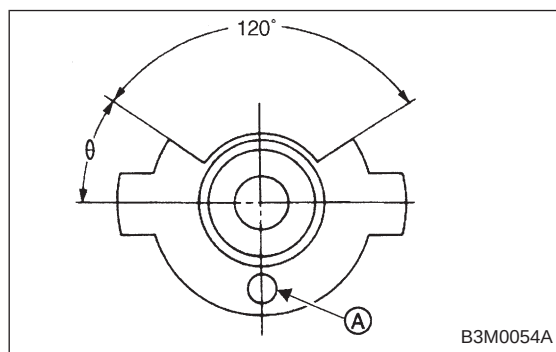
CAUTION:**Be careful not to break O-ring when placing shim(s).**

Adjustment shim	
Part No.	Thickness mm (in)
32190AA000	0.15 (0.0059)
32190AA010	0.30 (0.0118)

NOTE:

- When shim is removed, the neutral position will move closer to reverse; when shim is added, the neutral position will move closer to 1st gear.
- If shims alone cannot adjust the clearance, replace reverse accent shaft and re-adjust.

Reverse accent shaft		
Part No.	Mark	Remarks
32188AA040	1	Neutral position is closer to 1st gear.
32188AA011	No mark or 2	Standard
32188AA050	3	Neutral position is closer to reverse gear.



B3M0054A

4) Reverse check plate adjustment**NOTE:**

After assembling and installing rear case, adjust reverse check plate.

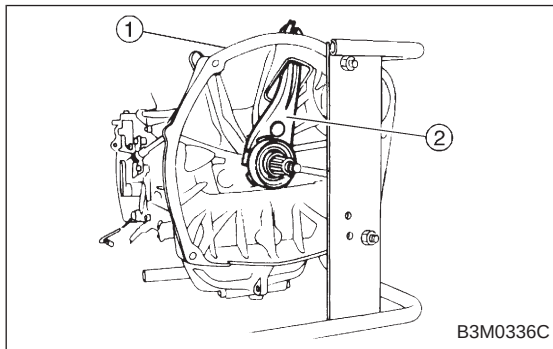
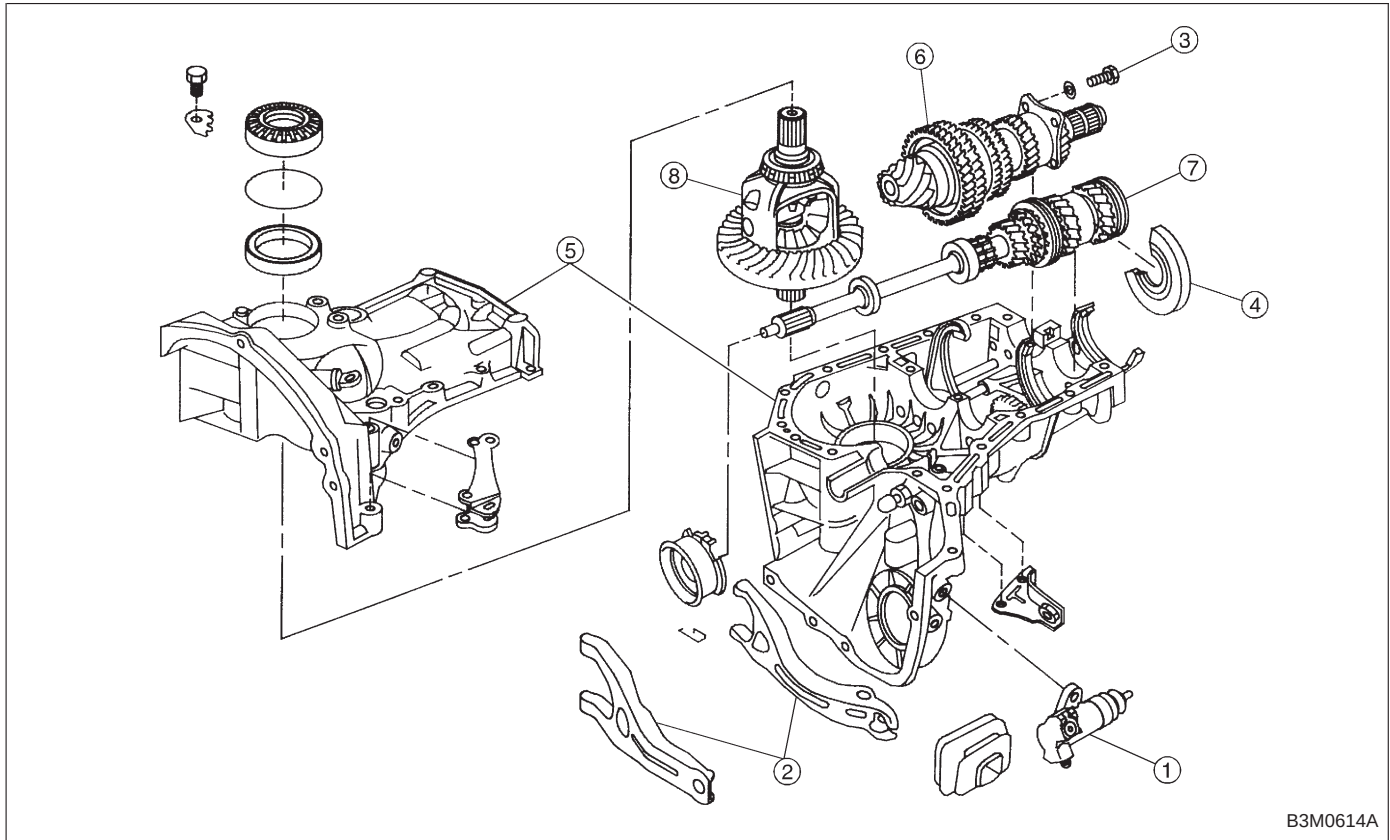
Shift shifter arm to "5th" and then to reverse to see if reverse checking mechanism operates properly. Also check to see if arm returns to Neutral when released from the reverse position. If arm does not return properly, replace reverse check plate.

Reverse check plate			
Part No.	Ⓐ: No.	Angle θ	Remarks
32189AA000	0	28°	Arm stops closer to 5th gear.
32189AA010	1	31°	Arm stops closer to 5th gear.
32189AA020	2	34°	Arm stops in the center.
32189AA030	3	37°	Arm stops closer to reverse gear.
32189AA040	4	40°	Arm stops closer to reverse gear.

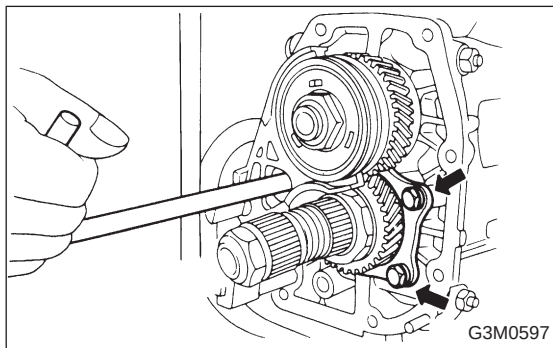
4. Transmission Case

A: DISASSEMBLY

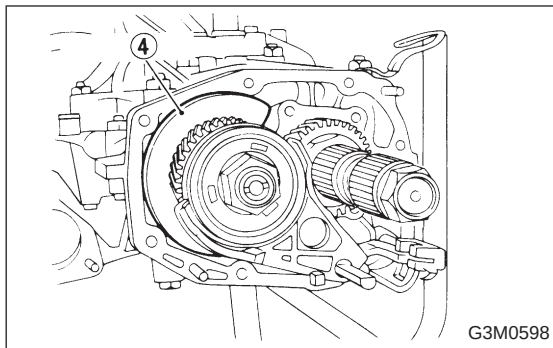
1. SEPARATION OF TRANSMISSION



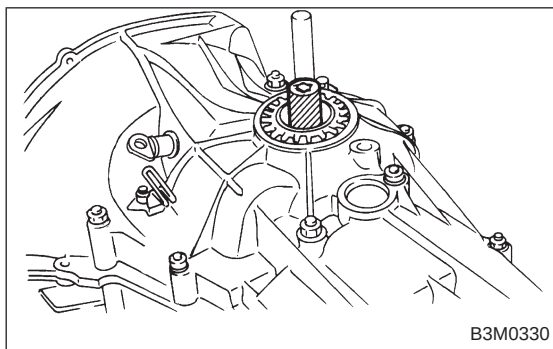
1) Remove operating cylinder (hydraulic application type) ① and clutch release lever ②. <Ref. to 2-10 [W3A1], [W3A2], [W5A0].>



2) Remove bearing mounting bolts.

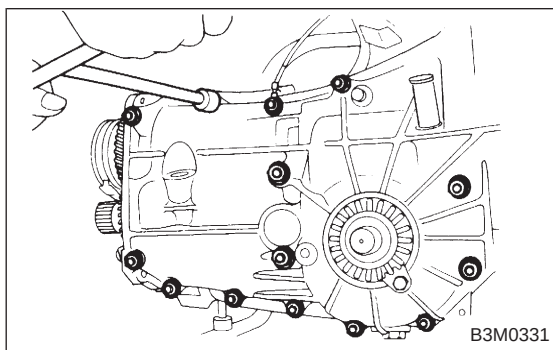


3) Remove main shaft rear plate ④.

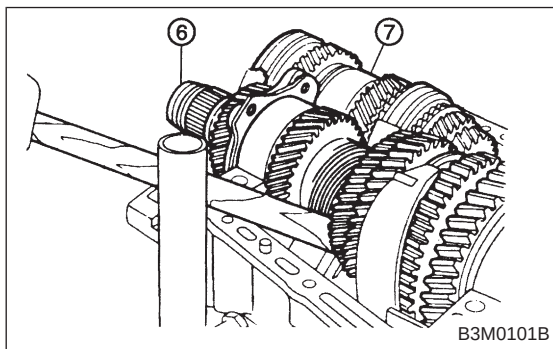


4) Separating transmission case

(1) Put vinyl tape around splines of right and left axle drive shafts to prevent damage to oil seals.



(2) Separate transmission case into right and left cases by loosening seventeen coupling bolts and nuts.

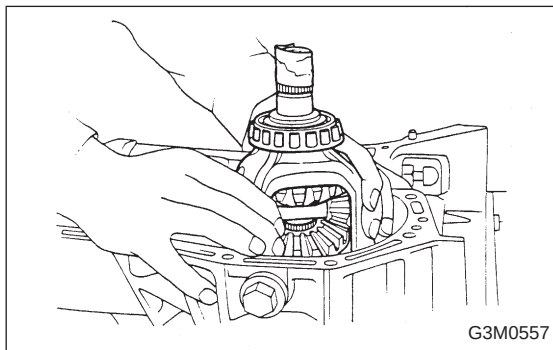


5) Remove drive pinion shaft assembly ⑥ from left side transmission case.

NOTE:

Use a hammer handle, etc. to remove if too tight.

6) Remove main shaft assembly ⑦.

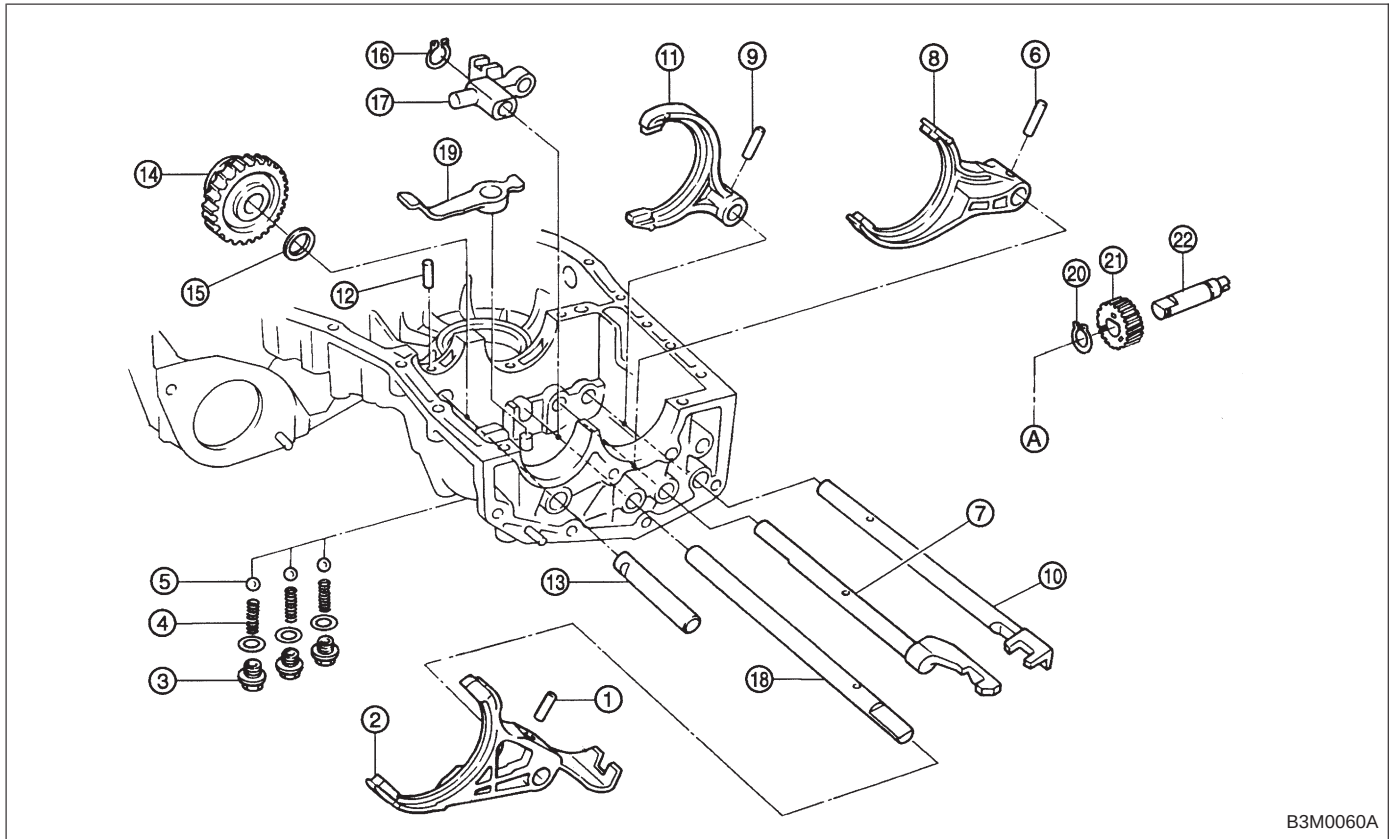


7) Remove differential assembly.

CAUTION:

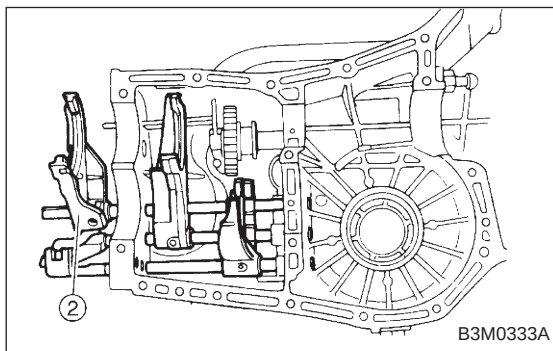
- Be careful not to confuse right and left roller bearing outer races.
- Be careful not to damage retainer oil seal.

2. TRANSMISSION CASE



B3M0060A

Ⓐ: Front right hand transmission case

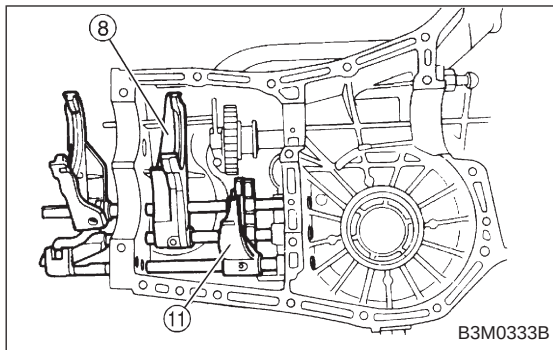


B3M0333A

1) Drive out spring pin ① with ST, and remove 5th shifter fork ②.

ST 398791700 STRAIGHT PIN REMOVER

2) Remove plugs ③, springs ④ and checking balls ⑤.

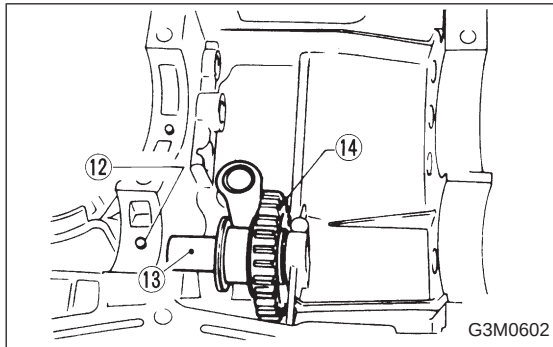


3) Drive out spring pin ⑥, and pull out 3-4 fork rod ⑦ and shifter fork ⑧.

NOTE:

When removing rod, keep other rods in neutral. Also, when pulling out straight pin, remove it toward inside of case so that it may not hit against case.

4) Drive out straight pin ⑨, and pull out 1-2 fork rod ⑩ and shifter fork ⑪.



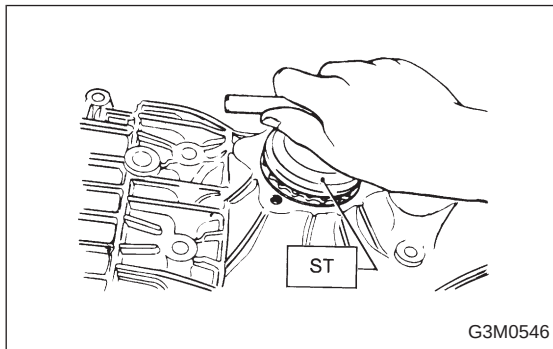
5) Pull out straight pin ⑫, and remove idler gear shaft ⑬, reverse idler gear ⑭ and washer ⑮.

6) Remove outer snap ring ⑯, and pull out reverse shifter rod arm ⑰ from reverse fork rod ⑱. Then take out ball, spring and interlock plunger from rod. And then remove rod.

NOTE:

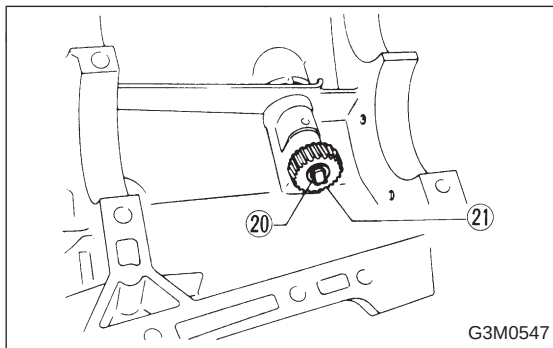
When pulling out reverse shifter rod arm, be careful not to let ball pop out of arm.

7) Remove reverse shifter lever ⑲.



8) Remove differential side retainers using ST.

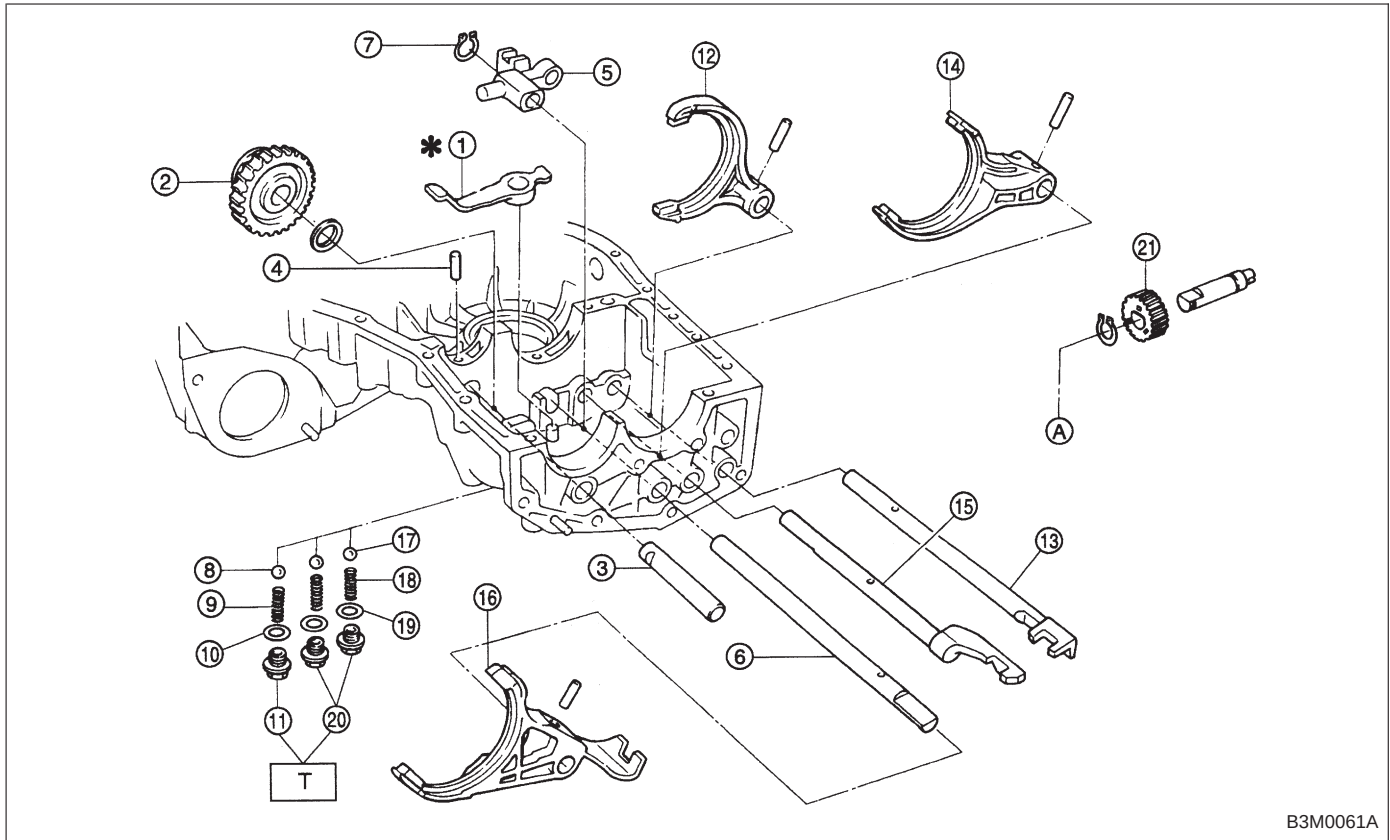
ST 499787000 WRENCH ASSY



9) Remove outer snap ring ⑳ and pull out speedometer driven gear ㉑. Next, remove vehicle speed sensor 2, oil seal, speedometer shaft ㉒ and washer.

B: ASSEMBLY

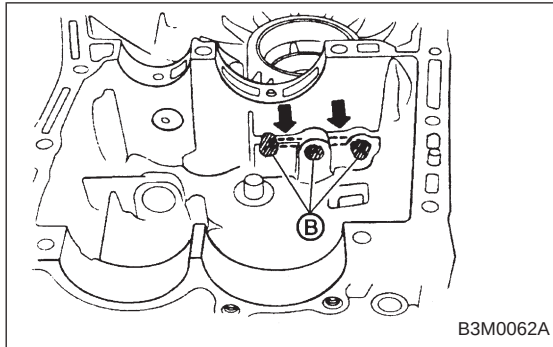
1. TRANSMISSION CASE



B3M0061A

Ⓐ: To right hand transmission case

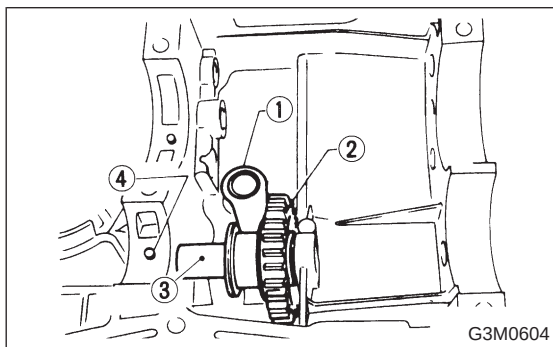
Tightening torque: N·m (kg·m, ft·lb)
T: 19.6±0.1 (2.00±0.015, 14.5±0.1)



B3M0062A

1) Position interlock plungers (5.56 x 19.6), one plunger in hole between 1-2 and 3-4 fork rod holes, and one plunger in hole between 3-4 and reverse fork rod holes.

Ⓑ: Rod holes



G3M0604

2) Install reverse shifter lever ①, reverse idler gear ② and reverse idler gear shaft ③, and secure with straight pin ④.

NOTE:

Be sure to install reverse idler shaft from the rear side.

3) Install reverse arm fork spring, ball and interlock plunger (5.56 x 19.6) to reverse fork rod arm ⑤. Insert reverse fork rod ⑥ into hole in reverse fork rod arm ⑤, and hold it with outer snap ring ⑦ using ST.

ST 399411700 ACCENT BALL INSTALLER

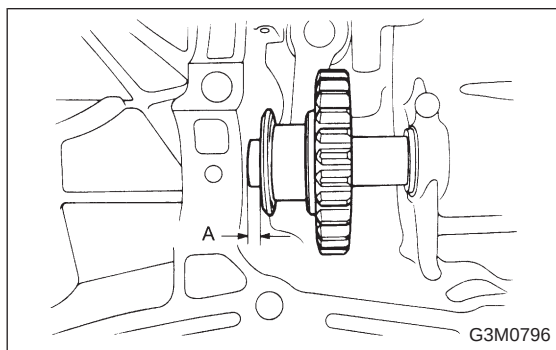
CAUTION:

Apply grease to plunger to prevent it from falling.

4) Position ball ⑧ (7.1438), spring ⑨ and gasket ⑩ in reverse shifter rod hole, on left side transmission case, and tighten checking ball plug ⑪.

CAUTION:

Replace gasket with a new one.



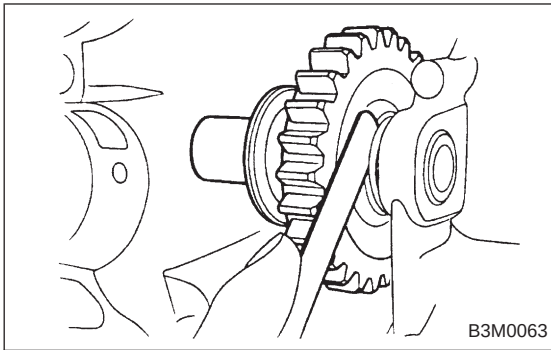
5) Adjustment of reverse idler gear position

(1) Move reverse shifter rod toward REV side. Adjust clearance between reverse idler gear and transmission case wall, using reverse shifter lever ①.

Clearance A:

6.0 — 7.5 mm (0.236 — 0.295 in)

Reverse shifter lever		
Part No.	No.	Remarks
32820AA000	0	Further from case wall
32820AA010	No mark	Standard
32820AA020	2	Closer to case wall

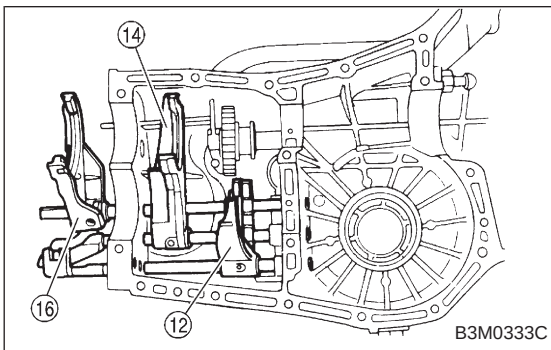


(2) After installing a suitable reverse shifter lever, shift into neutral. Using a thickness gauge, measure clearance between reverse idler gear and transmission case wall and adjust with washer(s).

Clearance:

0 — 0.5 mm (0 — 0.020 in)

Washer (20.5 x 26 x t)	
Part No.	Thickness mm (in)
803020151	0.4 (0.016)
803020152	1.1 (0.043)
803020153	1.5 (0.059)
803020154	1.9 (0.075)
803020155	2.3 (0.091)



6) Installation of 1-2 shifter fork (12) and rod (13)

(1) Install 1-2 fork rod into 1-2 shifter fork via the hole on the rear of transmission case.

(2) Align the holes in rod and fork, and drive straight pin (6 x 22) into these holes using ST.

ST 398791700 STRAIGHT PIN REMOVER

NOTE:

- Set other rods to neutral.
- Make sure interlock plunger (5.56 x 19.6) is on the 3-4 fork rod side.

7) Installation of 3-4 shifter fork (14) and rod (15)

(1) Install interlock plunger (3 x 11.9) onto 3-4 fork rod.

CAUTION:

Apply a coat of grease to plunger to prevent it from falling.

(2) Install 3-4 fork rod into 3-4 shifter fork via the hole on the rear of transmission case.

(3) Align the holes in rod and fork, and drive straight pin (6 x 22) into these holes.

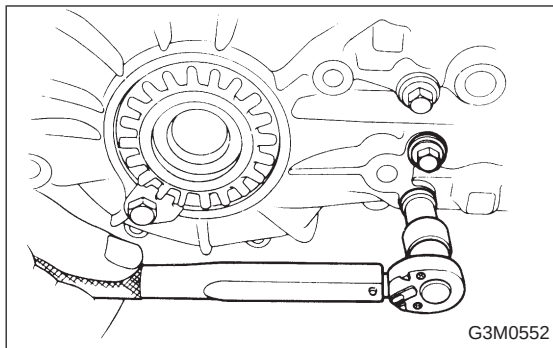
ST 398791700 STRAIGHT PIN REMOVER

NOTE:

- Set reverse fork rod to neutral.
- Make sure interlock plunger (installed before) is on the reverse fork rod side.

8) Install 5th shifter fork (16) onto the rear of reverse fork rod (6). Align holes in the two parts and drive straight pin into place.

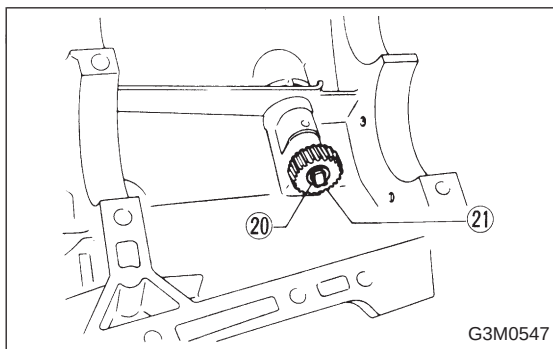
ST 398791700 STRAIGHT PIN REMOVER



9) Position balls ⑰, checking ball springs ⑱ and gaskets ⑲ into 3-4 and 1-2 rod holes, and install plugs ⑳.

CAUTION:

Replace gasket with a new one.



10) Installation of speedometer driven gear ⑳

(1) Install washer and speedometer shaft, and press fit oil seal with ST.

ST 899824100 or 499827000 PRESS

CAUTION:

Use new oil seal, if it has been removed.

(2) Install vehicle speed sensor 2.

CAUTION:

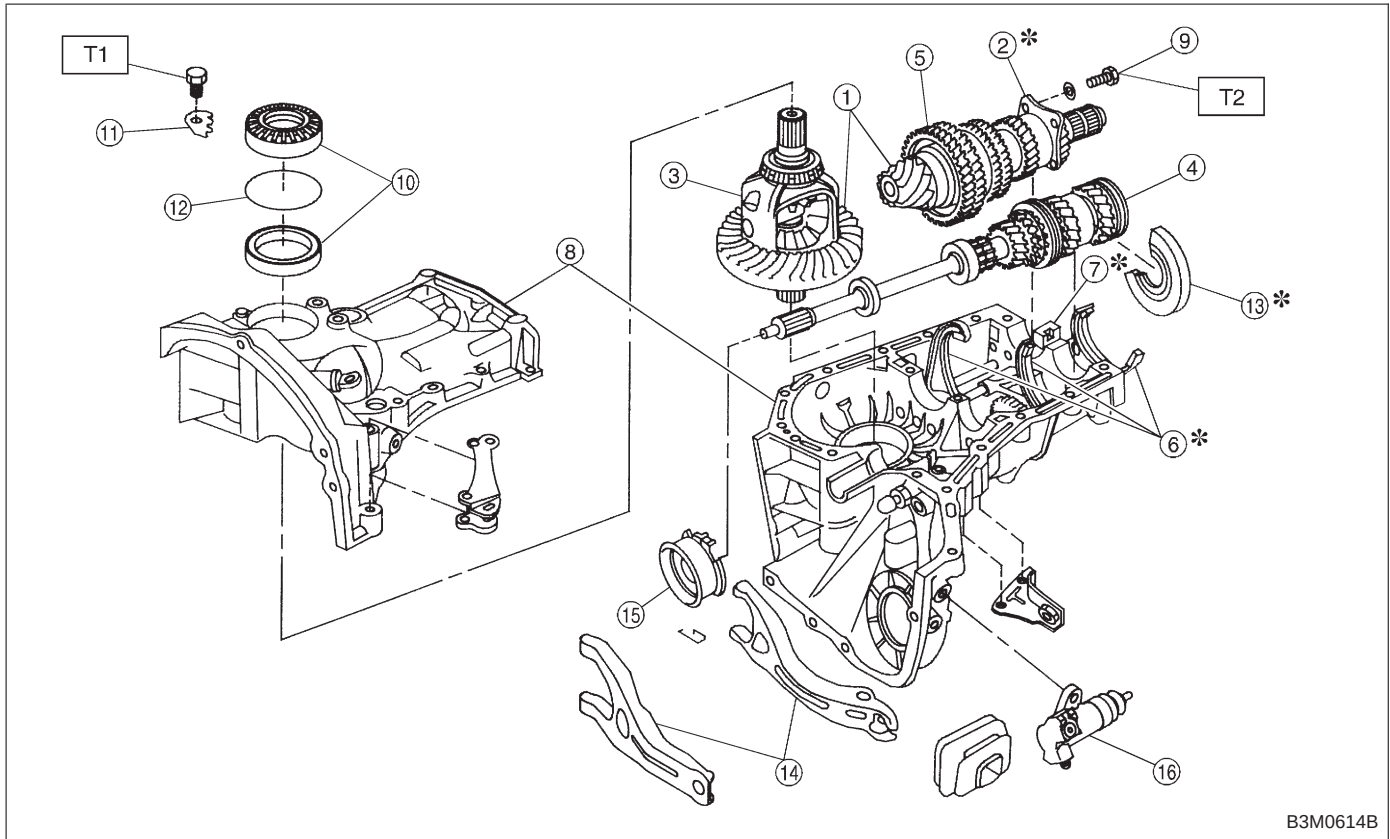
Use new vehicle speed sensor 2, if it has been removed.

Tightening torque:

5.9±1.5 N·m (60±15 kg-cm, 52±13 in-lb)

(3) Install speedometer driven gear and snap ring.

2. COMBINATION OF TRANSMISSION CASE

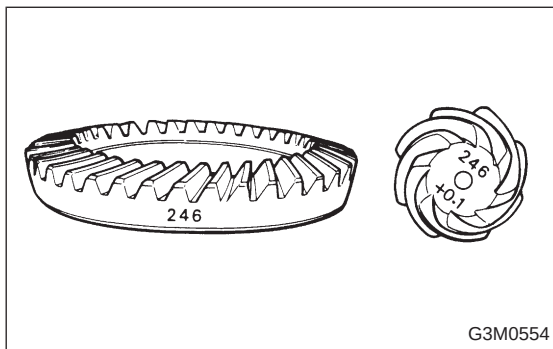


B3M0614B

Tightening torque: N·m (kg·m, ft·lb)

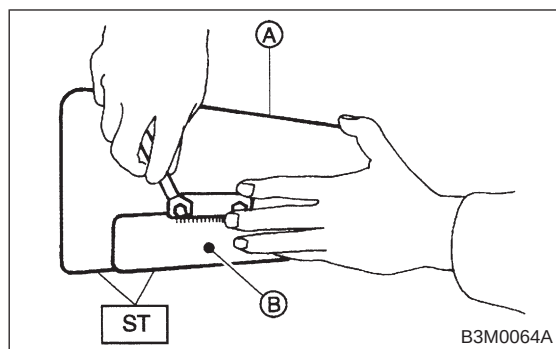
T1: 25 (2.5, 18)

T2: 29±3 (3.0±0.3, 21.7±2.2)



G3M0554

1) Alignment marks/numbers on hypoid gear set ①
The upper number on driven pinion is the match number for combining it with hypoid driven gear. The lower number is for shim adjustment. If no lower number is shown, the value is zero. The number on hypoid driven gear indicates a number for combination with drive pinion.



2) Adjustment of drive pinion shim ②

(1) Place drive pinion shaft assembly on right hand transmission main case without shim and tighten bearing mounting bolts.

(2) Inspection and adjustment of ST

NOTE:

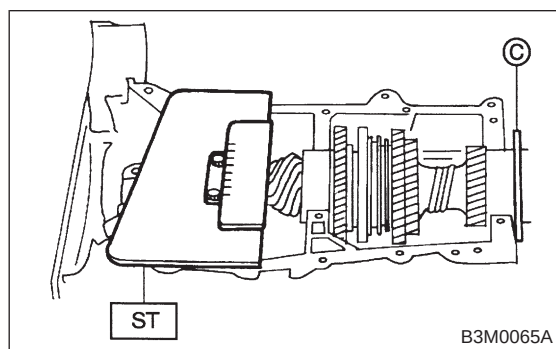
● Loosen the two bolts and adjust so that the scale indicates 0.5 correctly when the plate end and the scale end are on the same level.

● Tighten the two bolts.

ST 499917500 DRIVE PINION GAUGE ASSY

Ⓐ: Plate

Ⓑ: Scale



(3) Position the ST by inserting the knock pin of ST into the knock hole in the transmission case.

(4) Slide the drive pinion gauge scale with finger tip and read the value at the point where it matches with the end face of drive pinion.

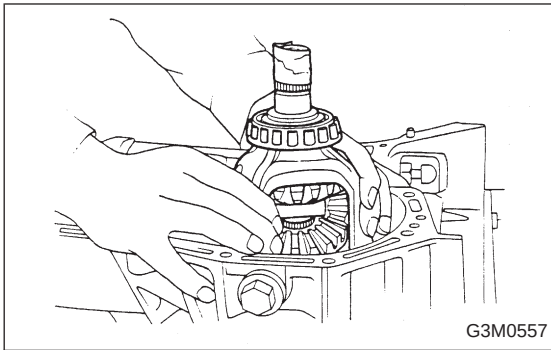
Ⓒ: Adjust clearance to zero without shim.

(5) The thickness of shim shall be determined by adding the value indicated on drive pinion to the value indicated on the ST. (Add if the number on drive pinion is prefixed by + and subtract if the number is prefixed by -.)

ST 499917500 DRIVE PINION GAUGE ASSY

Select one to three shims from the next table for the value determined as described above and take a shim thickness which is closest to the said value.

Drive pinion shim	
Part No.	Thickness mm (in)
32295AA031	0.150 (0.0059)
32295AA041	0.175 (0.0069)
32295AA051	0.200 (0.0079)
32295AA061	0.225 (0.0089)
32295AA071	0.250 (0.0098)
32295AA081	0.275 (0.0108)
32295AA091	0.300 (0.0118)
32295AA101	0.500 (0.0197)



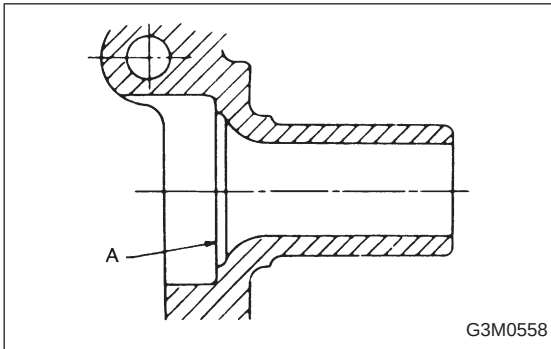
3) Install differential assembly ③ on left hand transmission case.

CAUTION:

Be careful not to fold the sealing lip of oil seal.

NOTE:

Wrap the left and right splined sections of axle shaft with vinyl tape to prevent scratches.



4) Install needle bearing and oil seal onto the front of transmission main shaft assembly ④, and position in left side transmission case.

CAUTION:

● Wrap clutch splined section with vinyl tape to prevent damage to oil seal.

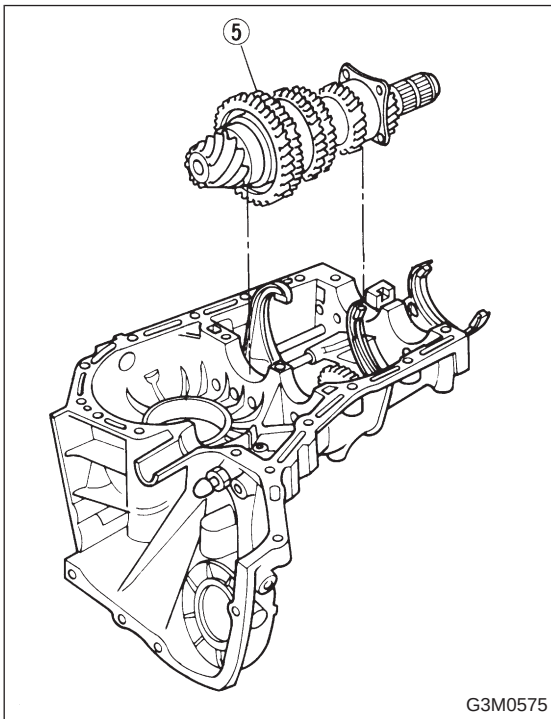
● Apply grease (Unilube #2 or equivalent) to the sealing lip of oil seal.

NOTE:

● Align the end face of seal with surface A of left side transmission main case when installing oil seal.

● Be careful not to drop oil seal when installing right side transmission main case.

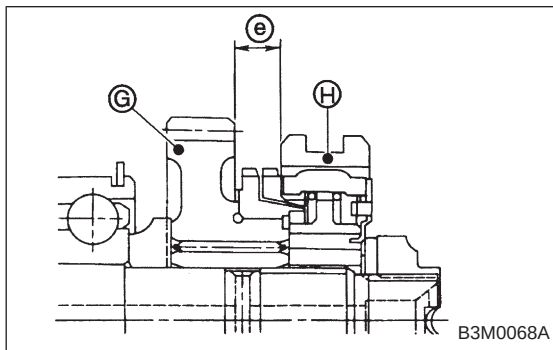
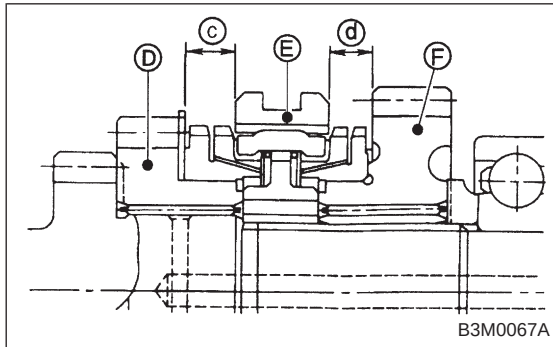
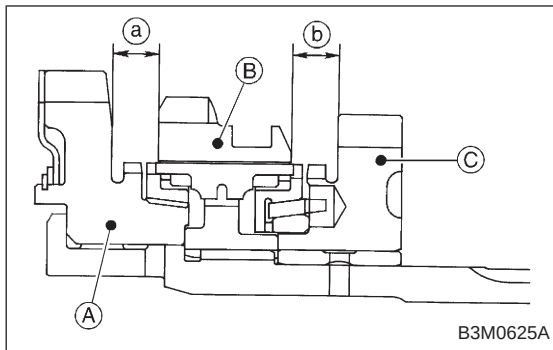
● Make sure straight pin is positioned in hole in needle bearing's outer race.



5) Install drive pinion shaft assembly ⑤ with shims selected before into transmission case.

NOTE:

Ensure that the knock pin of the case is fitted into the hole in the bearing outer race.

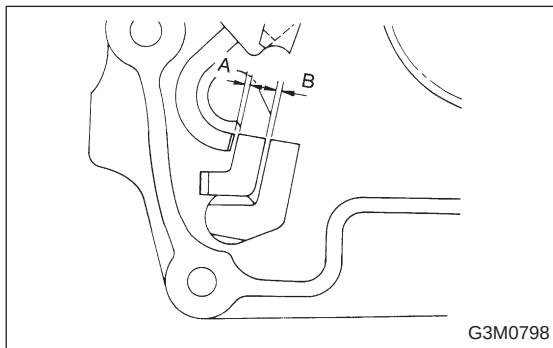


6) Selection of suitable 1st-2nd, 3rd-4th and 5th shifter fork

Set transmission main shaft assembly and drive pinion shaft assembly in position (so there is no clearance between the two when moved all the way to the front). Select suitable 1st-2nd, 3rd-4th and 5th shifter fork so that coupling sleeve and reverse driven gear are positioned in the center of their synchronizing mechanisms.

	Clearance mm (in)
1st driven gear (A) to reverse driven gear (B)	(a): 9.5 (0.374)
2nd driven gear (C) to reverse driven gear (B)	(b): 9.5 (0.374)
3rd drive gear (D) to coupling sleeve (E)	(c): 9.3 (0.366)
4th drive gear (F) to coupling sleeve (E)	(d): 9.3 (0.366)
5th drive gear (G) to coupling sleeve (H)	(e): 9.3 (0.366)

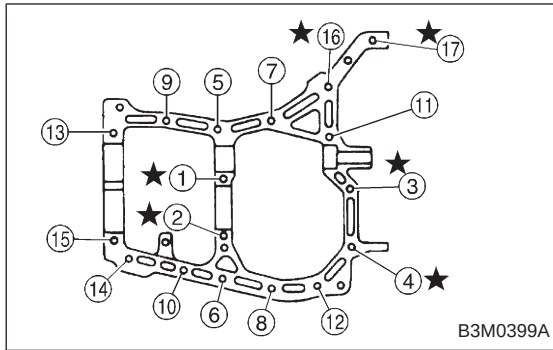
1st-2nd shifter fork		
Part No.	No.	Remarks
32804AA060	1	Approach to 1st gear by 0.2 mm (0.008 in)
32804AA070	No mark	Standard
32804AA080	3	Approach to 2nd gear by 0.2 mm (0.008 in)
3rd-4th shifter fork		
Part No.	No.	Remarks
32810AA060	1	Approach to 4th gear by 0.2 mm (0.008 in)
32810AA070	No mark	Standard
32810AA100	3	Approach to 3rd gear by 0.2 mm (0.008 in)
5th shifter fork		
Part No.	No.	Remarks
32812AA200	4	Approach to 5th gear by 0.2 mm (0.008 in)
32812AA210	No mark	Standard
32812AA220	5	Become distant from 5th gear by 0.2 mm (0.008 in)



7) Inspection of rod end clearance

Measure rod end clearances A and B. If any clearance is not within specifications, replace rod or fork as required.

A: 1st-2nd to 3rd-4th	0.4 — 1.4 mm (0.016 — 0.055 in)
B: 3rd-4th to 5th	0.5 — 1.3 mm (0.020 — 0.051 in)



8) Combination of transmission case

(1) Wipe off grease, oil and dust on the mating surfaces of transmission cases with white gasoline, and apply liquid gasket, and then put case right side and left side together.

Liquid gasket:

THREE BOND 1215 or equivalent

(2) Tighten 17 bolts with bracket, clip, etc. as shown in the figure.

Tightening torque:

8 mm bolt

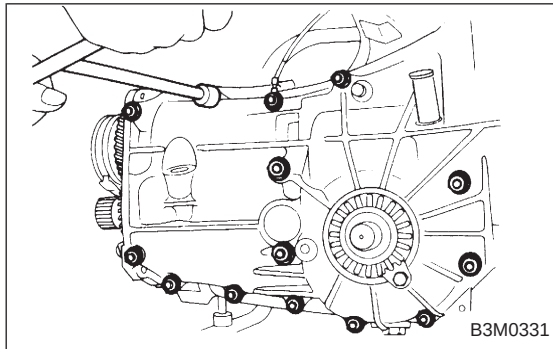
$25 \pm 2 \text{ N}\cdot\text{m}$ ($2.5 \pm 0.2 \text{ kg}\cdot\text{m}$, $18.1 \pm 1.4 \text{ ft}\cdot\text{lb}$)

★ 10 mm bolt

$39 \pm 3 \text{ N}\cdot\text{m}$ ($4.0 \pm 0.3 \text{ kg}\cdot\text{m}$, $28.9 \pm 2.2 \text{ ft}\cdot\text{lb}$)

NOTE:

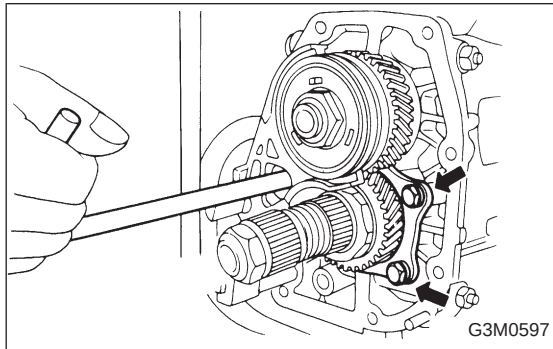
- Insert bolts from the bottom and tighten nuts at the top.
- Put cases together so that drive pinion shim and input shaft holder shim are not caught up in between.
- Confirm that counter gear and speedometer gear are meshed.



9) Tighten ball bearing attachment bolts.

Tightening torque:

$29 \pm 3 \text{ N}\cdot\text{m}$ ($3.0 \pm 0.3 \text{ kg}\cdot\text{m}$, $21.7 \pm 2.2 \text{ ft}\cdot\text{lb}$)

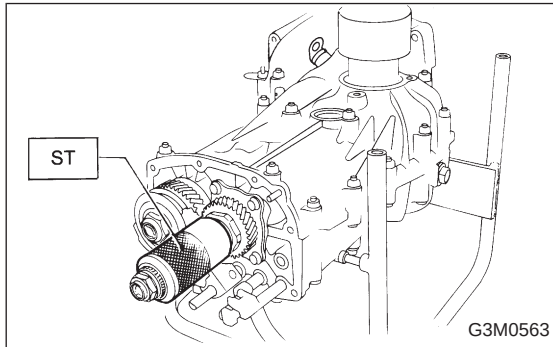


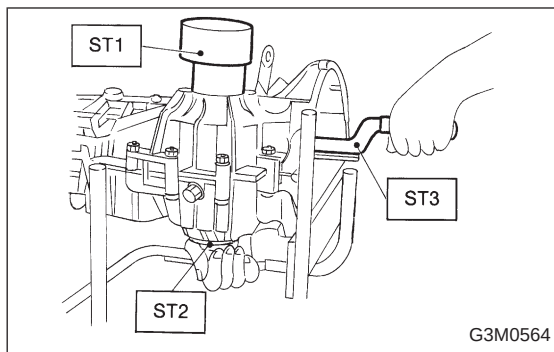
10) Backlash adjustment of hypoid gear and preload adjustment of roller bearing

NOTE:

Support drive pinion assembly with ST.

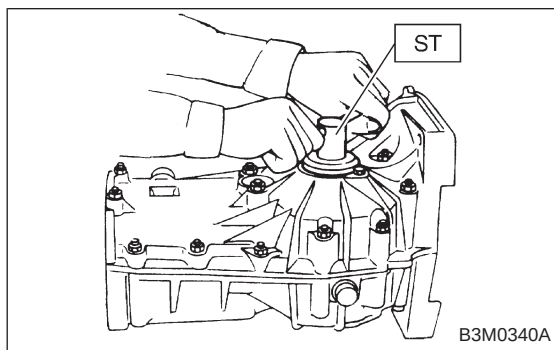
ST 498427100 STOPPER





- (1) Place the transmission with case left side facing downward and put ST1 on bearing cup.
- (2) Screw retainer assembly into left case from the bottom with ST2. Fit ST3 on the transmission main shaft. Shift gear into 4th or 5th and turn the shaft several times. Screw in the retainer while turning ST3 until a slight resistance is felt on ST2. This is the contact point of hypoid gear and drive pinion shaft. Repeat the above sequence several times to ensure the contact point.

ST1	399780104	WEIGHT
ST2	499787000	WRENCH ASSY
ST3	499927100	HANDLE

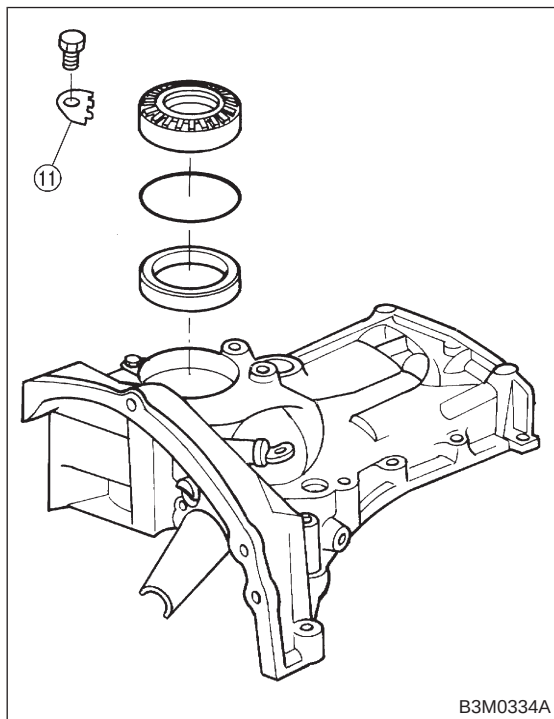


- (3) Remove weight and screw in retainer without O-ring on the upper side and stop at the point where slight resistance is felt.

NOTE:

At this point, the backlash between the hypoid gear and drive pinion shaft is zero.

ST	499787000	WRENCH ASSY
----	-----------	-------------

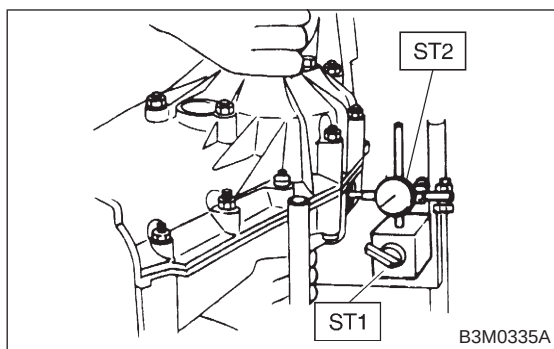


- (4) Fit lock plate ⑪. Loosen the retainer on the lower side by 1-1/2 notches of lock plate and turn in the retainer on the upper side by the same amount in order to obtain the backlash.

NOTE:

The notch on the lock plate moves by 1/2 notch if the plate is turned upside down.

- (5) Turn in the retainer on the upper side additionally by 1 notch in order to apply preload on taper roller bearing.
- (6) Tighten temporarily both the upper and lower lock plates and mark both holder and lock plate for later readjustment.
- (7) Turn transmission main shaft several times while tapping around retainer lightly with plastic hammer.
- (8) Set ST1 and ST2. Insert the needle through transmission oil drain plug hole so that the needle comes in contact with the tooth surface at a right angle and check the backlash.



ST1 498247001 MAGNET BASE

ST2 498247100 DIAL GAUGE

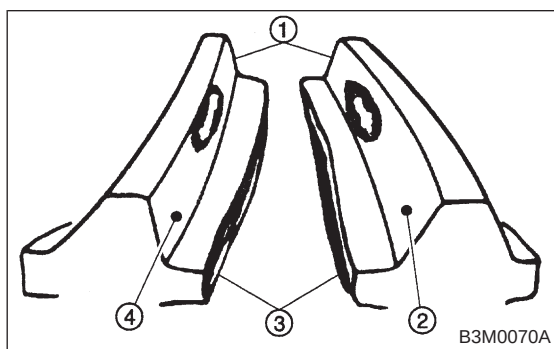
Backlash:

0.13 — 0.18 mm (0.0051 — 0.0071 in)

NOTE:

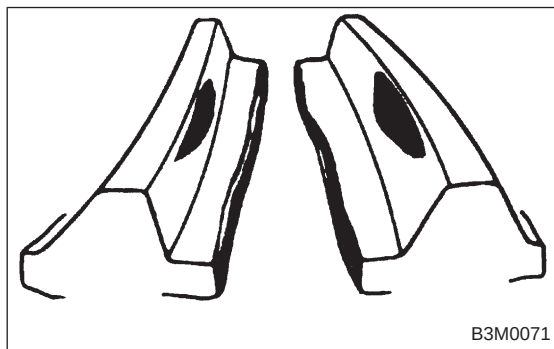
- If backlash is outside specified range, adjust it by turning holder in right side case.
- Each time holder rotates one tooth, backlash changes by 0.05 mm (0.0020 in).

(9) Check tooth contact of hypoid gear as follows:
Apply a uniform thin coat of red lead on both tooth surfaces of 3 or 4 teeth of the hypoid gear. Move the hypoid gear back and forth by turning the transmission main shaft until a definite contact pattern is developed on hypoid gear, and judge whether face contact is correct. If it is incorrect, make the following correction.



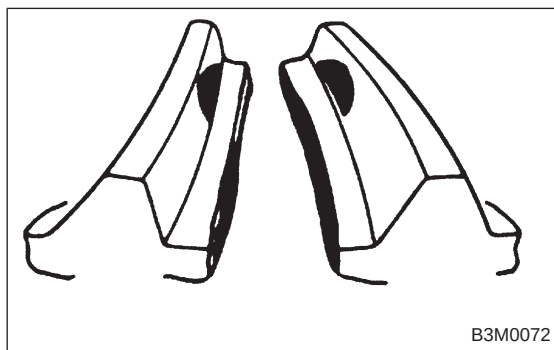
- Tooth contact is correct.

- ① Toe
- ② Coast side
- ③ Heel
- ④ Drive side



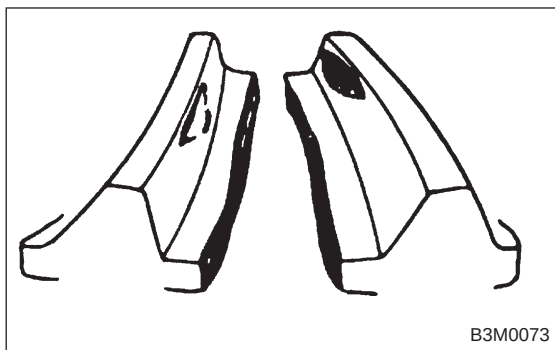
- Backlash is excessive.

To reduce backlash, loosen holder on the upper side (case right side) and turn in the holder on the lower side (case left side) by the same amount.

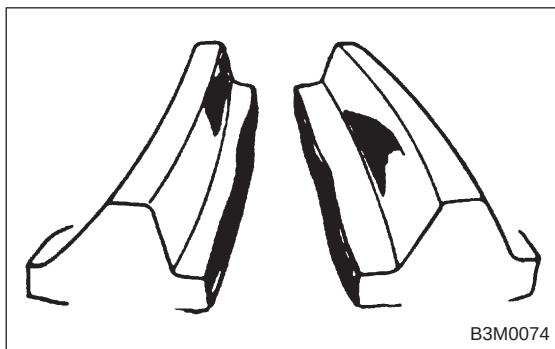


- Backlash is insufficient.

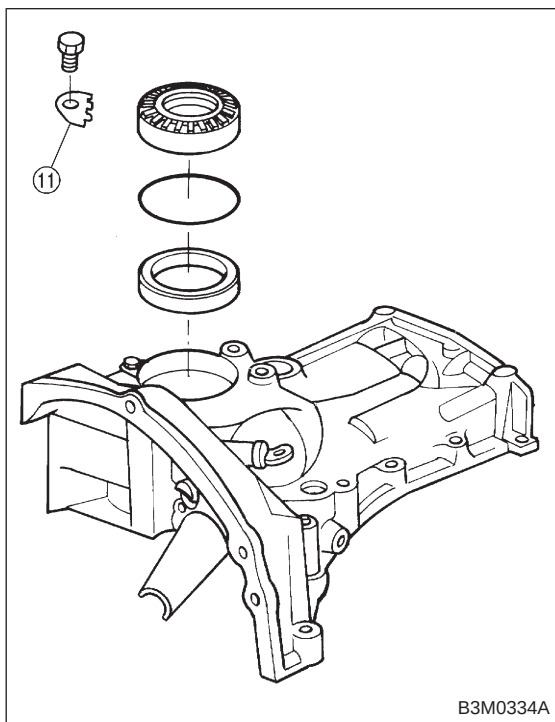
To increase backlash, loosen holder on the lower side (case left side) and turn in the holder on the upper side (case right side) by the same amount.



- The drive pinion shim selected before is too thick. Reduce its thickness.



- The drive pinion shim selected before is too thin. Increase its thickness.



11) After checking the tooth contact of hypoid gears, remove the lock plate ⑪. Then loosen retainer until the O-ring groove appears. Fit O-ring into the groove and tighten retainer into the position where retainer has been tightened in.

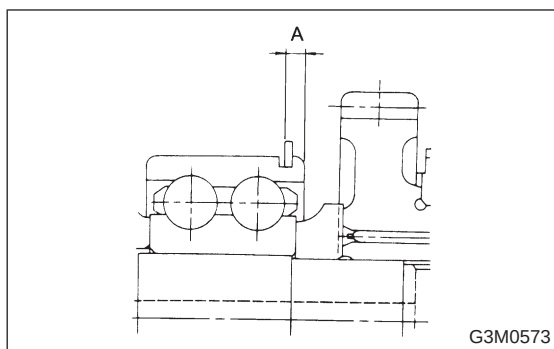
Tighten lock plate ⑪.

Tightening torque:

$25 \pm 3 \text{ N}\cdot\text{m}$ ($2.5 \pm 0.3 \text{ kg}\cdot\text{m}$, $18.1 \pm 2.2 \text{ ft}\cdot\text{lb}$)

NOTE:

Carry out this job on both upper and lower retainers.



12) Selecting of main shaft rear plate

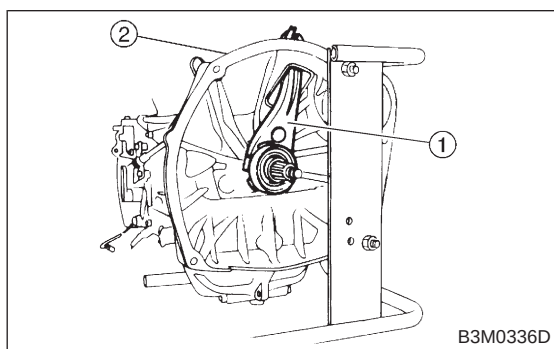
Using ST, measure the amount A of ball bearing protrusion from transmission main case surface and select the proper plate in the following table:

ST 498147000 DEPTH GAUGE

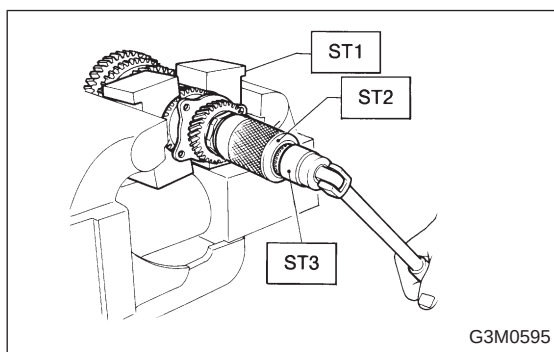
Dimension "A" mm (in)	Part No.	Mark
4.00 — 4.13 (0.1575 — 0.1626)	32294AA041	1
3.87 — 3.99 (0.1524 — 0.1571)	32294AA051	2

NOTE:

Before measuring, tap the end of main shaft with a plastic hammer lightly in order to make the clearance zero between the main case surface and the moving flange of bearing.



13) Install clutch release lever ① and operating cylinder (hydraulic application type) ②. <Ref. to 2-10 [W3C1], [W3C2], [W5A0].>



5. Drive Pinion Assembly (AWD Model)

A: DISASSEMBLY

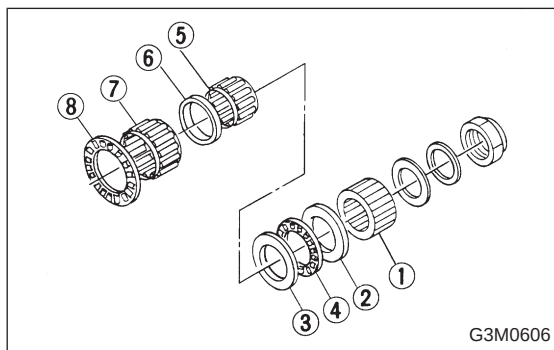
1. DRIVE PINION SHAFT

1) Straighten lock nut at staked portion. Remove the lock nut using ST1, ST2 and ST3.

ST1 899884100 HOLDER

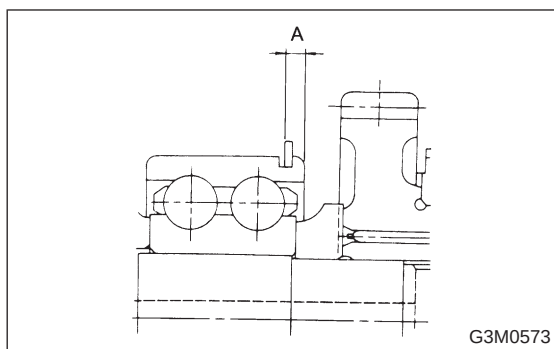
ST2 498427100 STOPPER

ST3 899988608 SOCKET WRENCH



2) Withdraw drive pinion from driven shaft.

Remove differential bevel gear sleeve ①, adjusting washer No. 1 ② (25 x 37.5 x t), adjusting washer No. 2 ③ (25 x 37.5 x 4), thrust bearing ④ (25 x 37.5 x 3), needle bearing ⑤ (25 x 30 x 20), drive pinion collar ⑥, needle bearing ⑦ (30 x 37 x 23) and thrust bearing ⑧ (33 x 50 x 3).



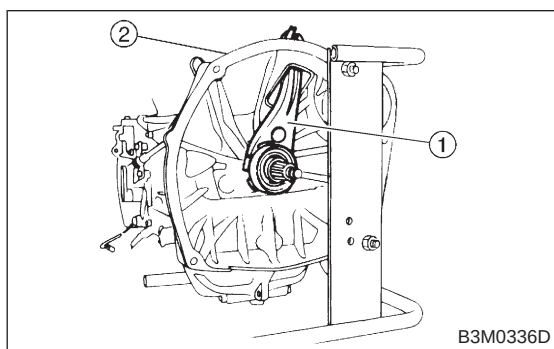
12) Selecting of main shaft rear plate
Using ST, measure the amount A of ball bearing protrusion from transmission main case surface and select the proper plate in the following table:

ST 498147000 DEPTH GAUGE

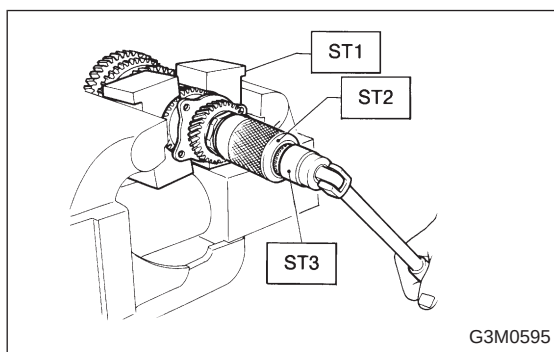
Dimension "A" mm (in)	Part No.	Mark
4.00 — 4.13 (0.1575 — 0.1626)	32294AA041	1
3.87 — 3.99 (0.1524 — 0.1571)	32294AA051	2

NOTE:

Before measuring, tap the end of main shaft with a plastic hammer lightly in order to make the clearance zero between the main case surface and the moving flange of bearing.



13) Install clutch release lever ① and operating cylinder (hydraulic application type) ②. <Ref. to 2-10 [W3C1], [W3C2], [W5A0].>



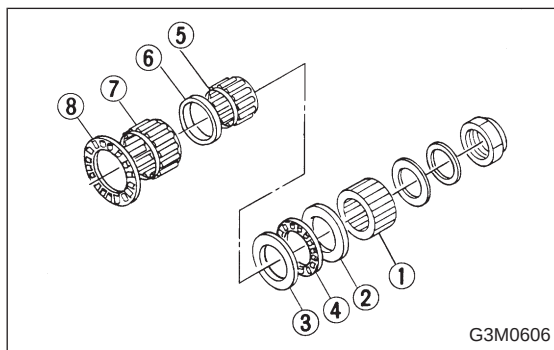
5. Drive Pinion Assembly (AWD Model)

A: DISASSEMBLY

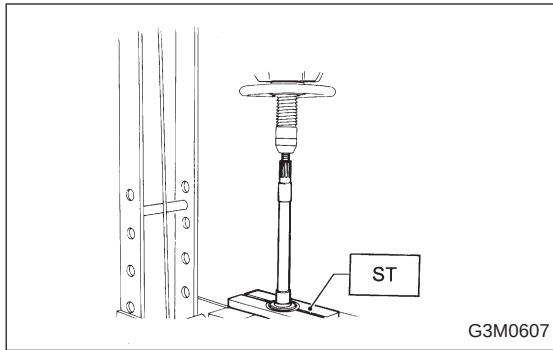
1. DRIVE PINION SHAFT

1) Straighten lock nut at staked portion. Remove the lock nut using ST1, ST2 and ST3.

ST1 899884100 HOLDER
ST2 498427100 STOPPER
ST3 899988608 SOCKET WRENCH



2) Withdraw drive pinion from driven shaft.
Remove differential bevel gear sleeve ①, adjusting washer No. 1 ② (25 x 37.5 x t), adjusting washer No. 2 ③ (25 x 37.5 x 4), thrust bearing ④ (25 x 37.5 x 3), needle bearing ⑤ (25 x 30 x 20), drive pinion collar ⑥, needle bearing ⑦ (30 x 37 x 23) and thrust bearing ⑧ (33 x 50 x 3).

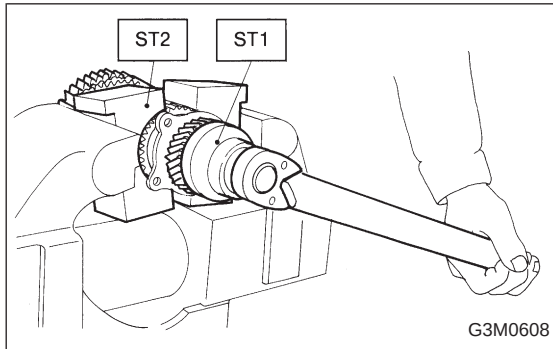


3) Remove roller bearing and washer (33 x 50 x 5) using ST and press.

ST 498077000 REMOVER

CAUTION:

Do not reuse roller bearing.



2. DRIVEN GEAR ASSEMBLY (2200 cc MODEL)

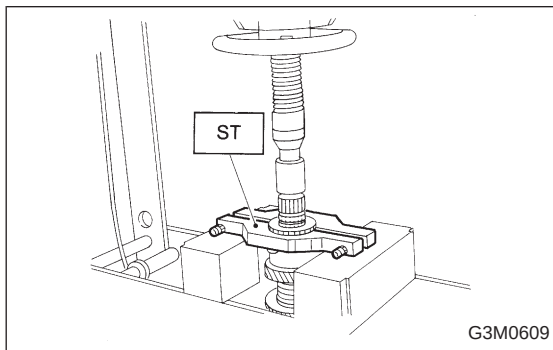
CAUTION:

Attach a cloth to the end of driven shaft (on the frictional side of thrust needle bearing) during disassembly or reassembly to prevent damage.

1) Straighten lock nut at staked portion. Remove the lock nut using ST1 and ST2.

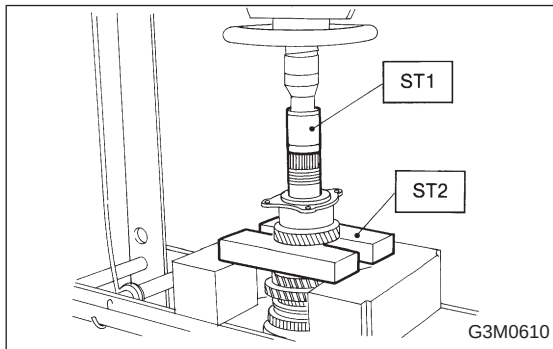
ST1 499987300 SOCKET WRENCH (50)

ST2 899884100 HOLDER



2) Remove 5th driven gear using ST.

ST 499857000 5TH DRIVEN GEAR REMOVER

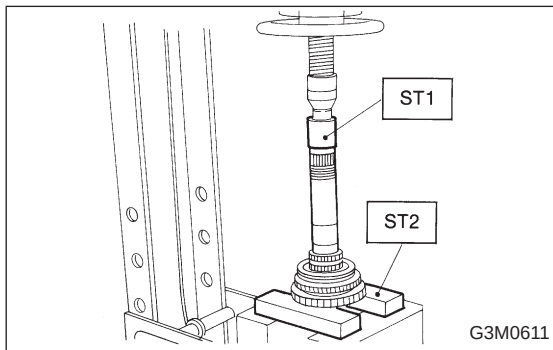


3) Remove woodruff key.

4) Remove roller bearing (42 x 74 x 40), 3rd and 4th driven gear using ST1 and ST2.

ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER



5) Remove the key.

6) Remove 2nd driven gear assembly.

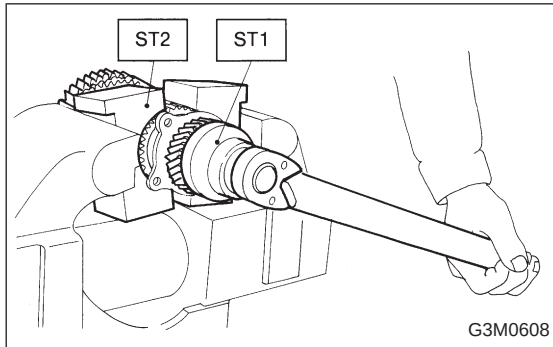
7) Remove 1st driven gear, 2nd gear bushing, gear and hub using ST1 and ST2.

Replace gear and hub if necessary. Do not attempt to disassemble if at all possible because they must engage at a specified point. If they have to be disassembled, mark the engaging point beforehand.

ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER

8) Remove sub gears for 1st and 2nd driven gear.



3. DRIVEN GEAR ASSEMBLY (2500 cc MODEL)

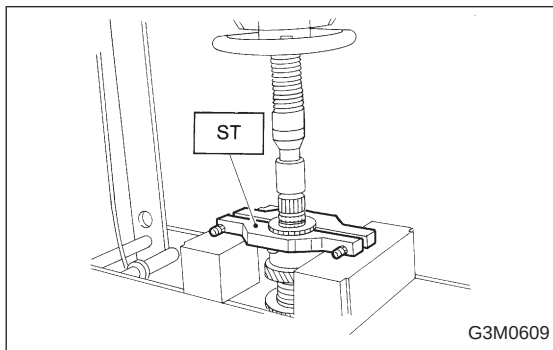
CAUTION:

Attach a cloth to the end of driven shaft (on the frictional side of thrust needle bearing) during disassembly or reassembly to prevent damage.

1) Straighten lock nut at staked portion. Remove the lock nut using ST1 and ST2.

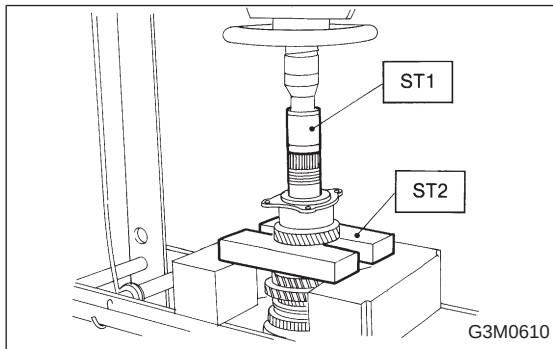
ST1 499987300 SOCKET WRENCH (50)

ST2 899884100 HOLDER



2) Remove 5th driven gear using ST.

ST 499857000 5TH DRIVEN GEAR REMOVER

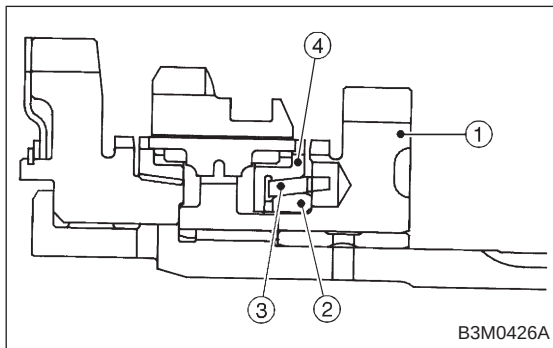


3) Remove woodruff key.

4) Remove roller bearing (42 x 74 x 40), 3rd-4th driven gear using ST1 and ST2.

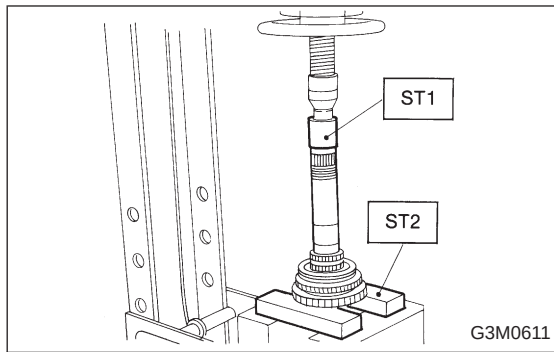
ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER



5) Remove the key.

6) Remove 2nd driven gear ①, inner baulk ring ②, synchro cone ③ and outer baulk ring ④.



7) Remove 1st driven gear, 2nd gear bushing, gear and hub using ST1 and ST2.

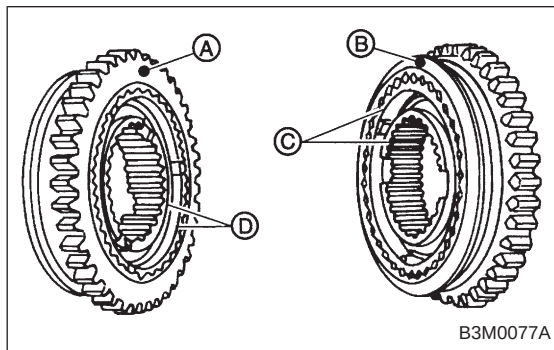
NOTE:

Replace gear and hub if necessary. Do not attempt to disassemble if at all possible because they must engage at a specified point. If they have to be disassembled, mark the engaging point beforehand.

ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER

8) Remove sub gear for 1st driven gear.



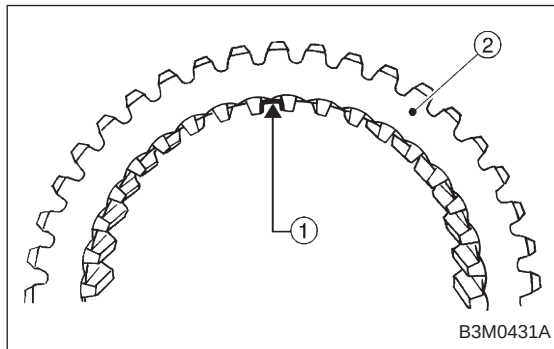
B: ASSEMBLY

1. GEAR AND HUB ASSEMBLY (2200 cc MODEL)

NOTE:

Position open ends of springs 120° apart.

- Ⓐ: 1st gear side
- Ⓑ: 2nd gear side
- Ⓒ: Flush surface
- Ⓓ: Stepped surface

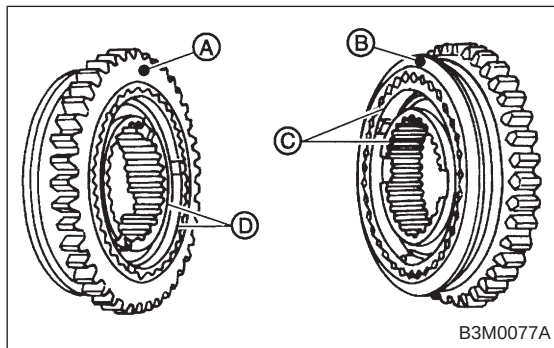


2. GEAR AND HUB ASSEMBLY (2500 cc MODEL)

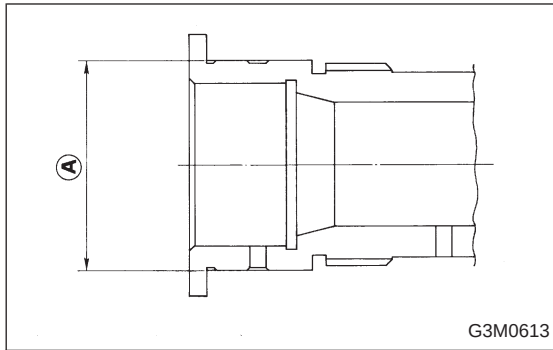
Assemble gear and hub assembly.

NOTE:

- Use new gear and hub assembly, if gear or hub have been replaced.
- Be sure the insert keys are correctly located in the insert key grooves ① inside the reverse driven gear ②.



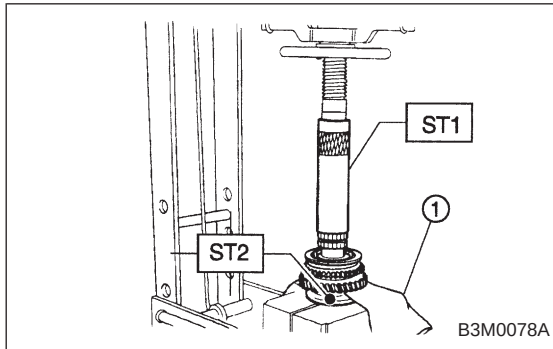
- Ⓐ: 1st gear side
- Ⓑ: 2nd gear side
- Ⓒ: Flush surface
- Ⓓ: Stepped surface



3. DRIVEN GEAR ASSEMBLY (2200 cc MODEL)

Assemble a driven shaft and 1st driven gear that select for adjustment the proper radial clearance.

Driven shaft		1st driven gear
Part No.	Diameter A mm (in)	Part No.
32229AA150	49.959 — 49.966 (1.9669 — 1.9672)	32231AA290
32229AA140	49.967 — 49.975 (1.9672 — 1.9675)	32231AA280



1) Install 1st driven gear, 1st-2nd baulk ring and gear and hub assembly onto driven shaft.

NOTE:

Take care to install gear hub in proper direction.

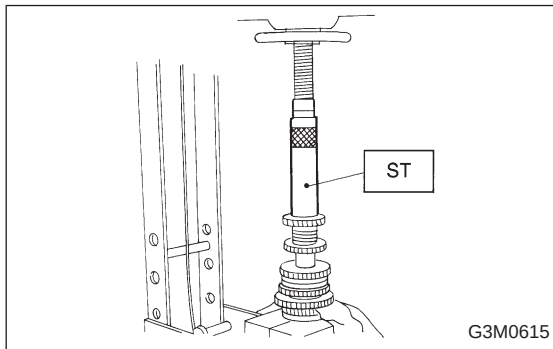
2) Install 2nd driven gear bushing onto driven shaft using ST1, ST2 and press.

ST1 499277200 INSTALLER

ST2 499587000 INSTALLER

CAUTION:

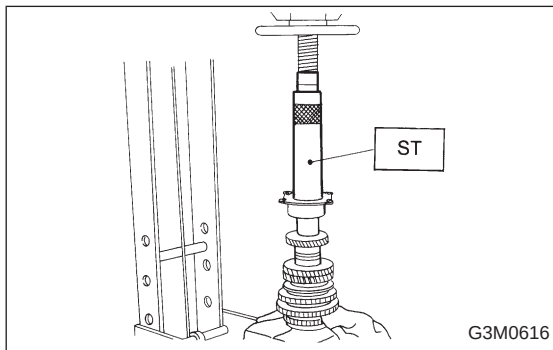
Attach a cloth ① to the end of driven shaft to prevent damage.



3) Install 2nd driven gear, 1st-2nd baulk ring and insert onto driven shaft. After installing key on driven shaft, install 3rd-4th driven gear using ST and press.

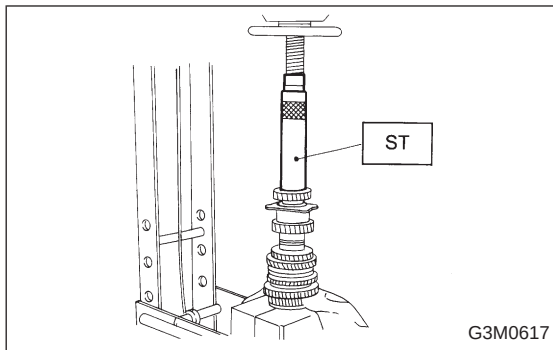
ST 499277200 INSTALLER

Align groove in baulk ring with insert.



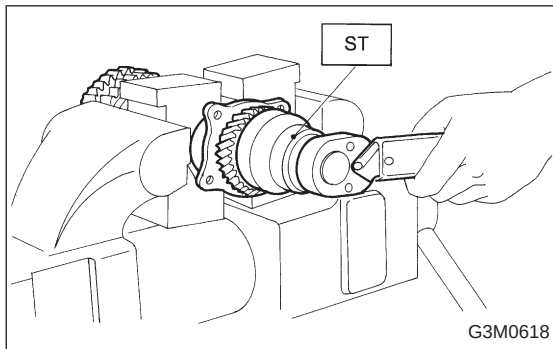
4) Install a set of roller bearings (42 x 74 x 40) onto the driven shaft using ST and press.

ST 499277200 INSTALLER



5) Position woodruff key in groove on the rear of driven shaft. Install 5th driven gear onto drive shaft using ST and press.

ST 499277200 INSTALLER

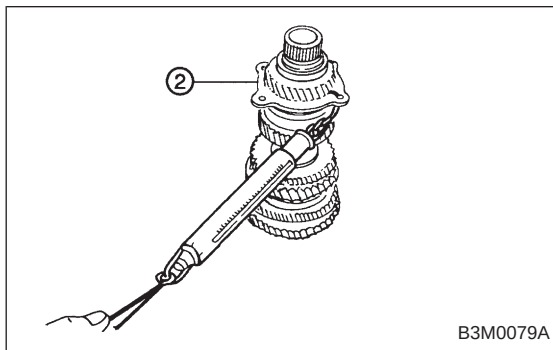


6) Install lock washer (42 x 53 x 2). Install lock nut (42 x 13) and tighten to the specified torque using ST.

ST 499987300 SOCKET WRENCH (50)

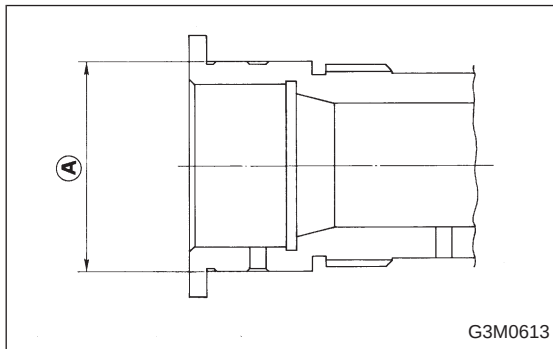
Tightening torque:

265±10 N·m (27±1 kg-m, 195±7 ft-lb)



NOTE:

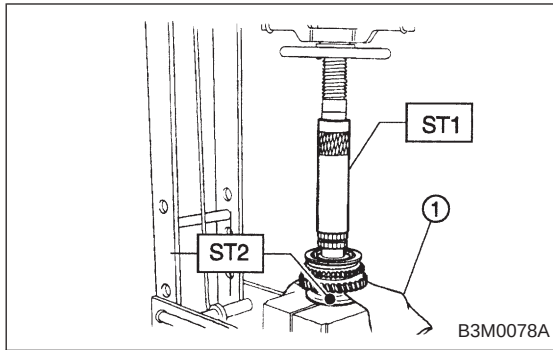
- Stake lock nut at two points.
- Using spring balancer, check that starting torque of roller bearing ② is 0.1 to 1.5 N·m (1 to 15 kg-cm, 0.9 to 13.0 in-lb).



4. DRIVEN GEAR ASSEMBLY (2500 cc MODEL)

Assemble a driven shaft and 1st driven gear that select for adjustment the proper radial clearance.

Driven shaft		1st driven gear
Part No.	Diameter A mm (in)	Part No.
32229AA150	49.959 — 49.966 (1.9669 — 1.9672)	32231AA290
32229AA140	49.967 — 49.975 (1.9672 — 1.9675)	32231AA280



- 1) Install sub gear to 1st driven gear.
- 2) Install 1st driven gear, 1st baulk ring, gear and hub assembly onto driven shaft.

NOTE:

Take care to install gear hub in proper direction.

- 3) Install 2nd driven gear bushing onto driven shaft using ST1, ST2 and press.

ST1 499277200 INSTALLER

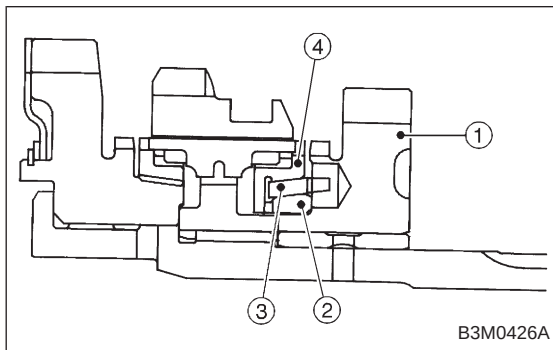
ST2 499587000 INSTALLER

CAUTION:

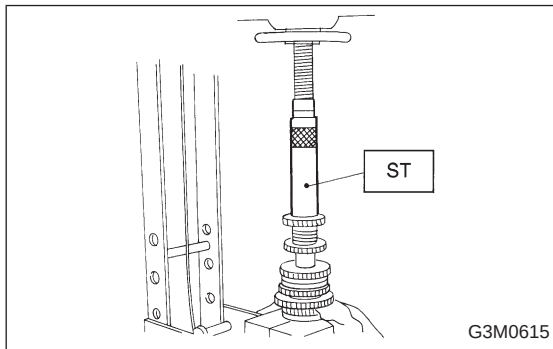
Attach a cloth ① to the end of driven shaft to prevent damage.

NOTE:

When press fitting, align oil holes of shaft and bush.



- 4) Install 2nd driven gear ①, inner baulk ring ②, synchro cone ③, outer baulk ring ④ and insert onto driven shaft.

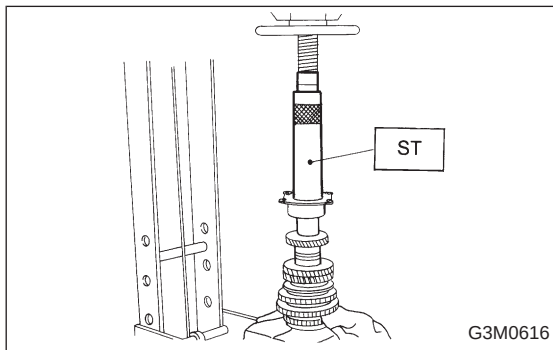


- 5) After installing key on driven shaft, install 3rd-4th driven gear using ST and press.

ST 499277200 INSTALLER

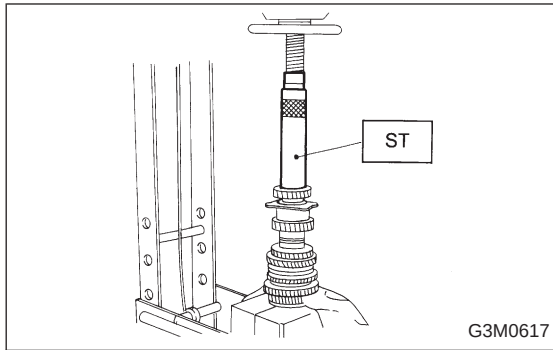
NOTE:

Align groove in baulk ring with insert.



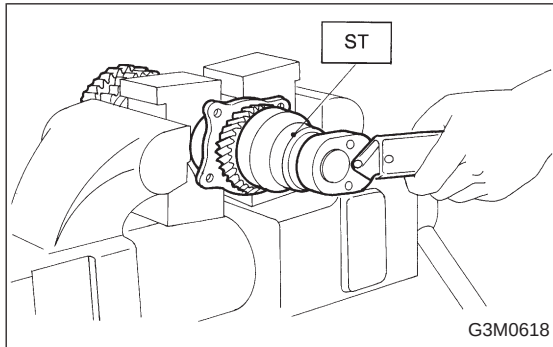
- 6) Install a set of roller bearings (42 x 74 x 40) onto the driven shaft using ST and press.

ST 499277200 INSTALLER



7) Position woodruff key in groove on the rear of driven shaft. Install 5th driven gear onto drive shaft using ST and press.

ST 499277200 INSTALLER

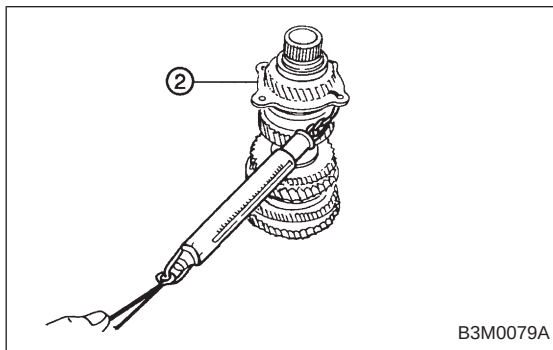


8) Install lock washer (42 x 53 x 2). Install lock nut (42 x 13) and tighten to the specified torque using ST.

ST 499987300 SOCKET WRENCH (50)

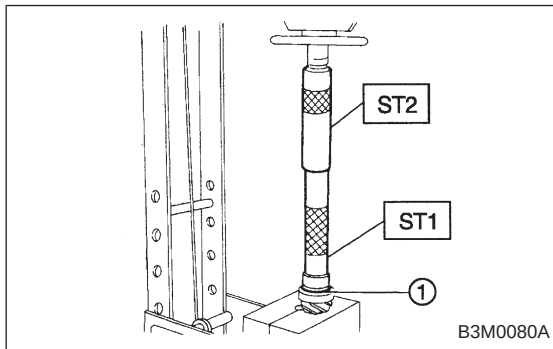
Tightening torque:

265±10 N·m (27±1 kg-m, 195±7 ft-lb)



NOTE:

- Stake lock nut at two points.
- Using spring balancer, check that starting torque of roller bearing ② is 0.1 to 1.5 N·m (1 to 15 kg-cm, 0.9 to 13.0 in-lb).

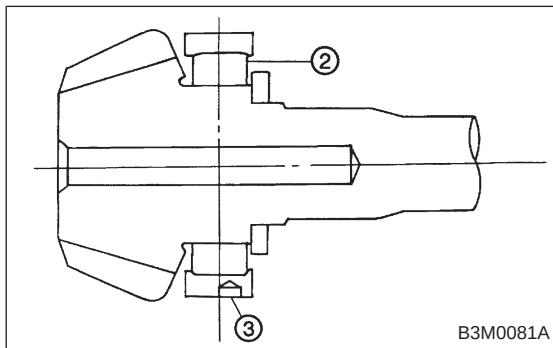


5. DRIVE PINION SHAFT

1) Install roller bearing onto drive pinion. Install washer ① (33 x 50 x 5) using ST1, ST2 and press.

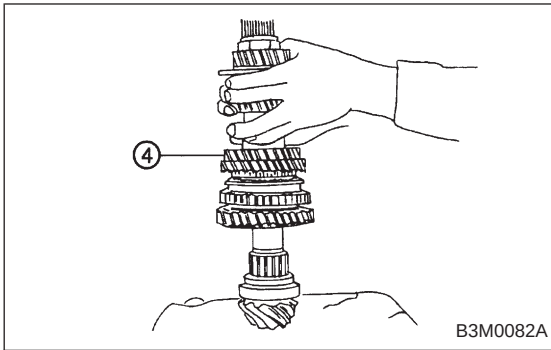
ST1 499277100 BUSH 1-2 INSTALLER

ST2 499277200 INSTALLER

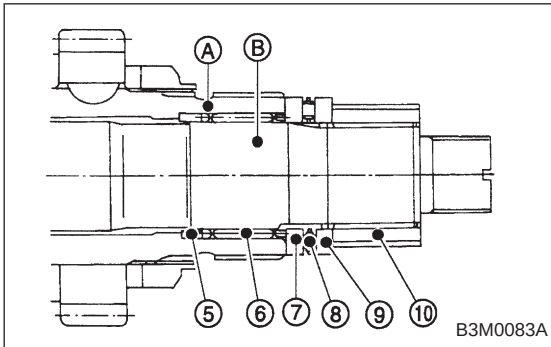


NOTE:

When installing roller bearing ②, note its directions (front and rear) because knock pin hole ③ in outer race is offset.



2) Install thrust bearing (33 x 50 x 3) and needle bearing (30 x 37 x 23). Install driven shaft assembly ④.



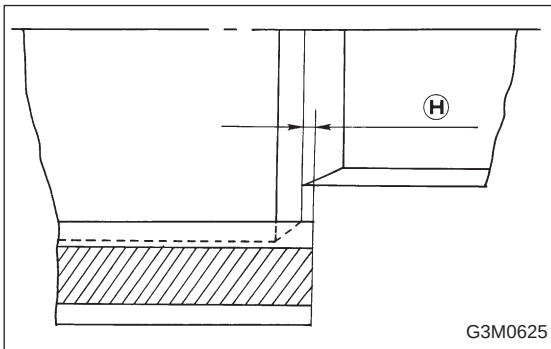
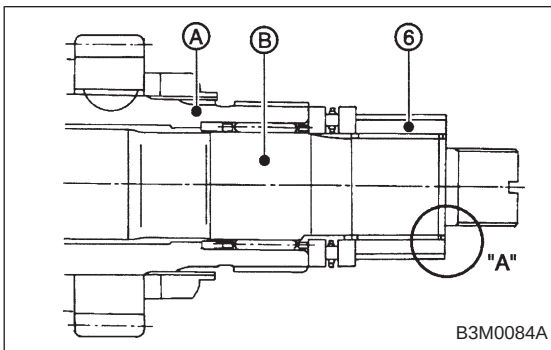
3) Install drive pinion collar ⑤, needle bearing ⑥ (25 x 30 x 20), adjusting washer No. 2 ⑦ (25 x 36 x 4), thrust bearing ⑧ (25 x 37.5 x 3), adjusting washer No. 1 ⑨ (25 x 36 x t) and differential bevel gear sleeve ⑩ in that order.

NOTE:

Be careful because spacer must be installed in proper direction.

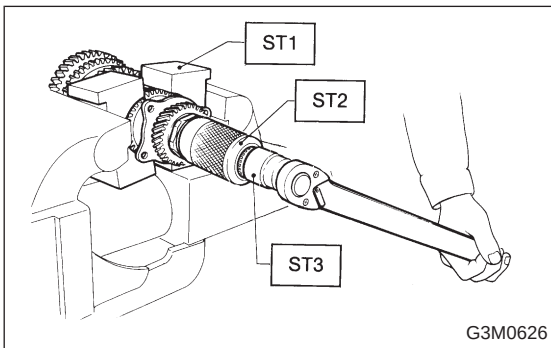
Ⓐ: Driven shaft

Ⓑ: Driven pinion shaft



6. ADJUSTMENT OF THRUST BEARING PRELOAD

1) After completing the preceding steps 1) through 3), select adjusting washer No. 2 so that dimension Ⓜ is zero through visual check. Position washer (18.3 x 30 x 4) and lock washer (18 x 30 x 2) and install lock nut (18 x 13.5).



2) Using ST1, ST2 and ST3, tighten lock nut to the specified torque.

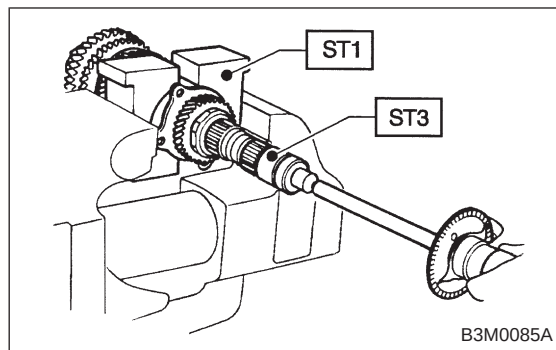
ST1 899884100 HOLDER

ST2 498427100 STOPPER

ST3 899988608 SOCKET WRENCH (27)

Tightening torque:

118±8 N·m (12±0.8 kg-m, 86.8±5.8 ft-lb)



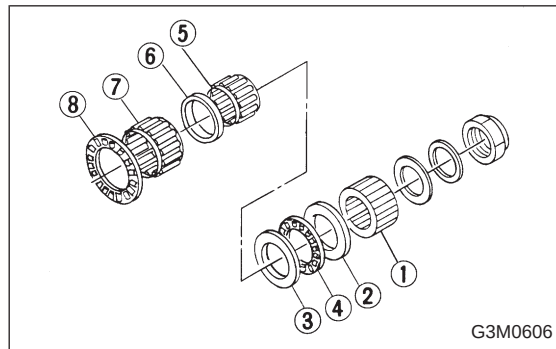
3) After removing ST2, measure starting torque using torque driver.

ST1 899884100 HOLDER

ST3 899988608 SOCKET WRENCH (27)

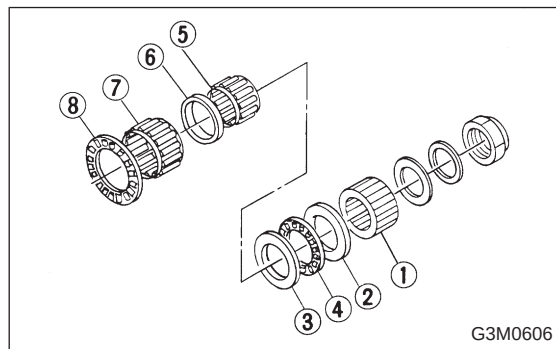
Starting torque:

$54 \pm 25 \text{ N}\cdot\text{m}$ ($5.5 \pm 2.5 \text{ kg}\cdot\text{m}$, $40 \pm 18 \text{ ft}\cdot\text{lb}$)



4) If starting torque is not within specified limit, select new adjusting washer No. 1 ② and recheck starting torque.

Adjusting washer No. 1	
Part No.	Thickness mm (in)
803025051	3.925 (0.1545)
803025052	3.950 (0.1555)
803025053	3.975 (0.1565)
803025054	4.000 (0.1575)
803025055	4.025 (0.1585)
803025056	4.050 (0.1594)
803025057	4.075 (0.1604)

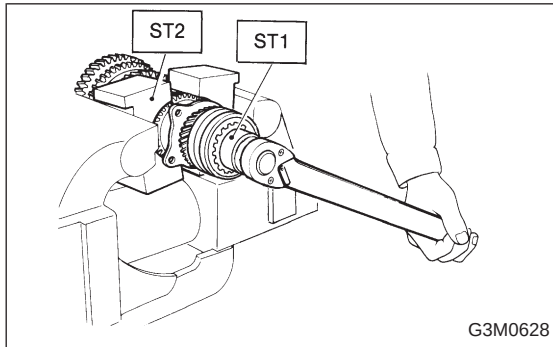


5) If specified starting torque range cannot be obtained when a No. 1 ② adjusting washer is used, then select a suitable No. 2 ③ adjusting washer from those listed in the following table. Repeat steps 1) through 4) to adjust starting torque.

Starting torque	Dimension H	Washer No. 2
Low	Small	Select thicker one.
High	Large	Select thinner one.

Adjusting washer No. 2	
Part No.	Thickness mm (in)
803025059	3.850 (0.1516)
803025054	4.000 (0.1575)
803025058	4.150 (0.1634)

- 6) Recheck that starting torque is within specified range, then clinch lock nut at four positions.



6. Drive Pinion Assembly (FWD Model)

A: DISASSEMBLY

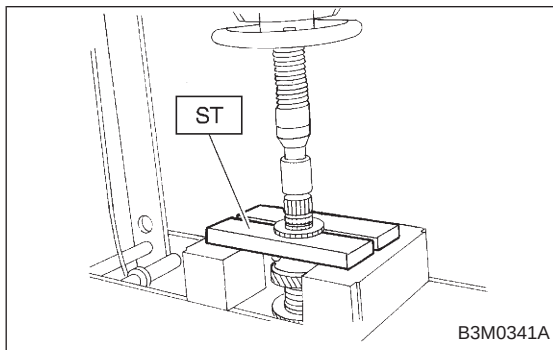
- 1) Loosen lock nut using ST1 and ST2.

ST1 499987100 or 499987003 or 899984103 SOCKET
WRENCH (35)

ST2 899884100 HOLDER

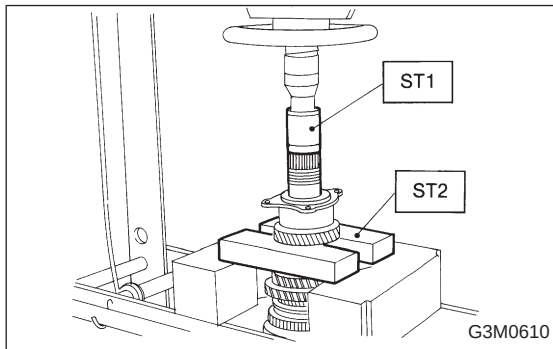
NOTE:

Remove caulking before taking off lock nut.



- 2) Remove 5th driven gear using a press.

ST 498077000 5TH DRIVEN GEAR REMOVER

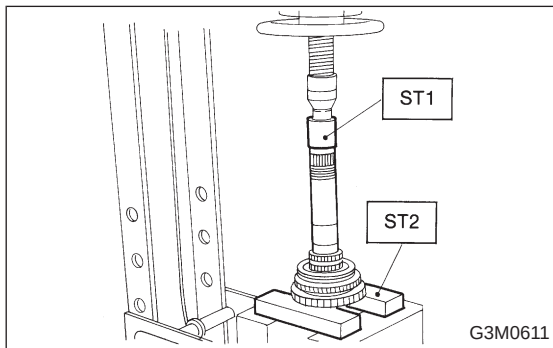


- 3) Remove woodruff key.

- 4) Remove roller bearing (29 x 74 x 38) and 3rd-4th driven gear using ST1 and ST2.

ST1 899714110 REMOVER

ST2 499757002 SNAP RING PRESS



- 5) Remove 2nd driven gear assembly.

- 6) Remove 3rd-4th driven gear key.

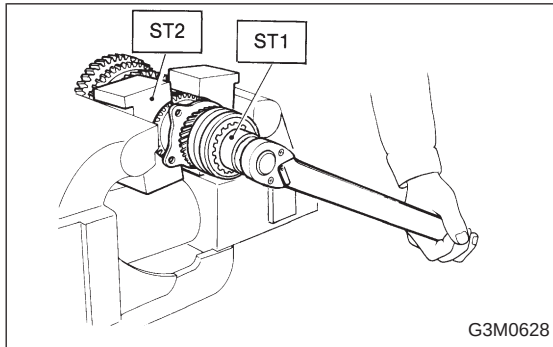
- 7) Remove 1st driven gear, 2nd gear bushing, and gear & hub assembly using ST1 and ST2.

Replace gear and hub if necessary. Do not attempt to disassemble if at all possible because they must engage at a specified point. If they have to be disassembled, mark the engaging point beforehand.

ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER

- 6) Recheck that starting torque is within specified range, then clinch lock nut at four positions.



6. Drive Pinion Assembly (FWD Model)

A: DISASSEMBLY

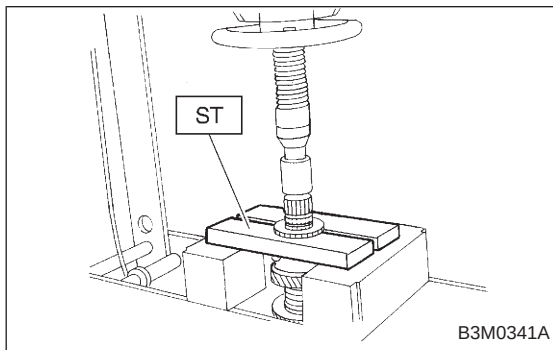
- 1) Loosen lock nut using ST1 and ST2.

ST1 499987100 or 499987003 or 899984103 SOCKET
WRENCH (35)

ST2 899884100 HOLDER

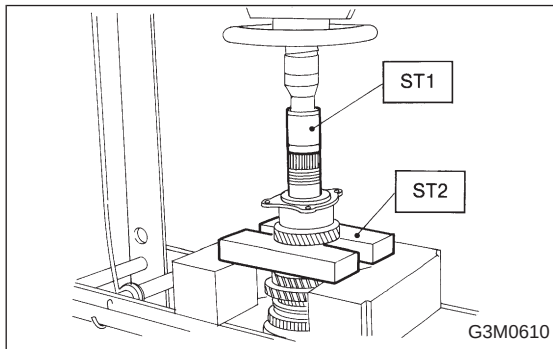
NOTE:

Remove caulking before taking off lock nut.



- 2) Remove 5th driven gear using a press.

ST 498077000 5TH DRIVEN GEAR REMOVER

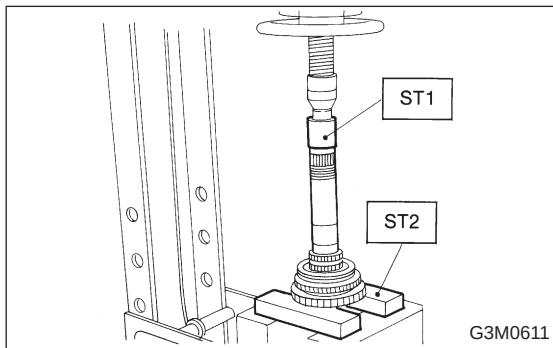


- 3) Remove woodruff key.

- 4) Remove roller bearing (29 x 74 x 38) and 3rd-4th driven gear using ST1 and ST2.

ST1 899714110 REMOVER

ST2 499757002 SNAP RING PRESS



- 5) Remove 2nd driven gear assembly.

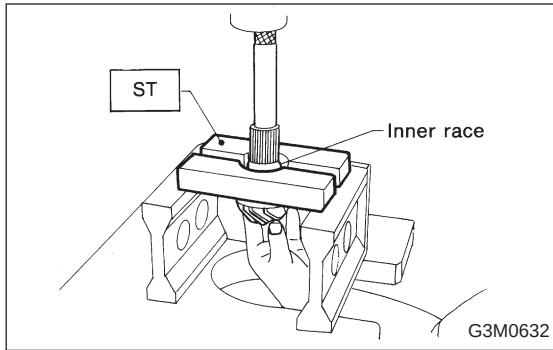
- 6) Remove 3rd-4th driven gear key.

- 7) Remove 1st driven gear, 2nd gear bushing, and gear & hub assembly using ST1 and ST2.

Replace gear and hub if necessary. Do not attempt to disassemble if at all possible because they must engage at a specified point. If they have to be disassembled, mark the engaging point beforehand.

ST1 499757002 SNAP RING PRESS

ST2 899714110 REMOVER

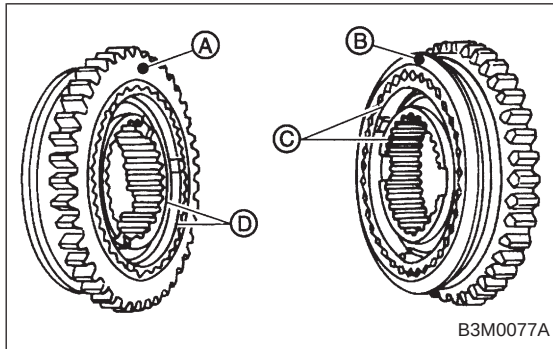


8) Remove 1st gear bushing, 1st driven gear thrust plate, and roller bearing (41 x 71 x 23) using ST and press.

ST 498517000 REPLACER

CAUTION:

Replace roller bearing (41 x 71 x 23) with a new one if this disassembly is performed.



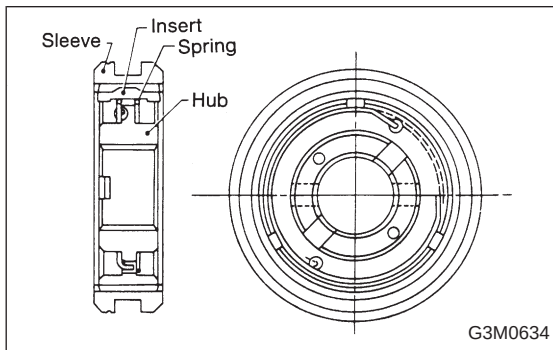
B: ASSEMBLY

1) Assemble gear and hub assembly.

- Ⓐ: 1st gear side
- Ⓑ: 2nd gear side
- Ⓒ: Flush surface
- Ⓓ: Stepped surface

NOTE:

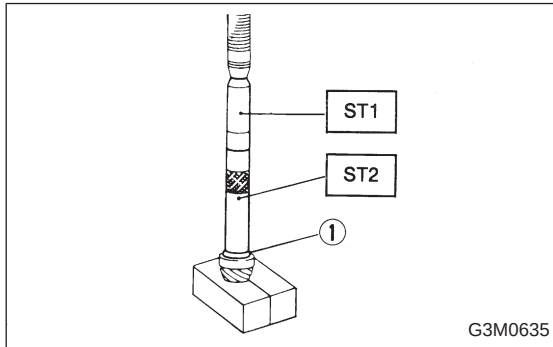
- Use new gear and hub assembly, if gear or hub have been replaced.
- Position open ends of springs 120° apart.



2) Assemble sleeve and hub assembly.

NOTE:

Make sure bent sections of springs on both sides are kept 180° apart and hooked at hub's holes.



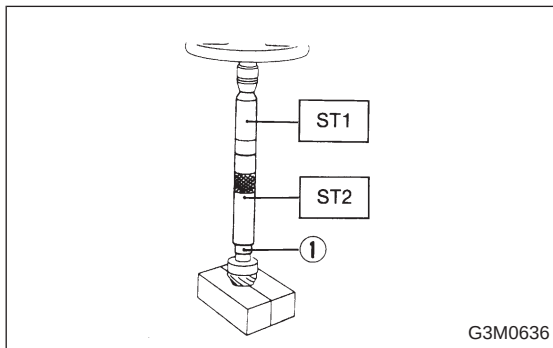
3) Drive roller bearing onto drive pinion shaft and 1st driven gear thrust washer ① using ST1 and ST2.

ST1 499877000 INSTALLER

ST2 499277100 INSTALLER

CAUTION:

Use new roller bearing, 1st gear thrust washer.



4) Install 1st-2nd driven gear bushing ① onto drive pinion shaft.

NOTE:

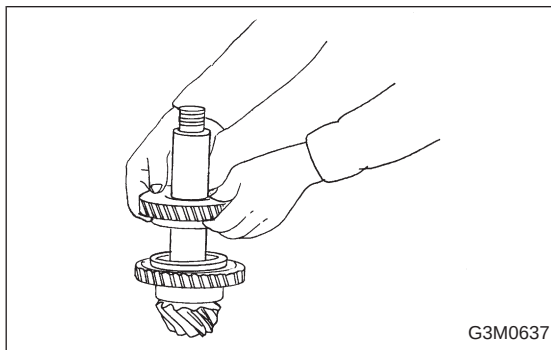
Bushing may be installed with either side up.

ST1 499877000 INSTALLER

ST2 499277100 INSTALLER

5) Measure outside diameter of 1st driven gear bushing to determine suitable 1st driven gear.

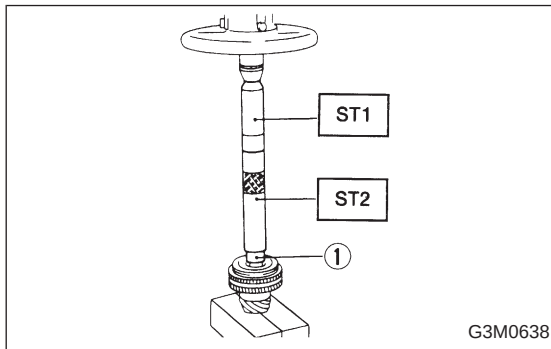
Bushing outside diameter mm (in)	1st driven gear
41.983 — 41.996 (1.6529 — 1.6534)	32231AA320
41.968 — 41.982 (1.6523 — 1.6528)	32231AA330
41.954 — 41.967 (1.6517 — 1.6522)	32231AA340



6) Install 1st driven gear, 1st-2nd bask ring and gear and hub assembly (already assembled in previous step) to drive pinion shaft by hand.

NOTE:

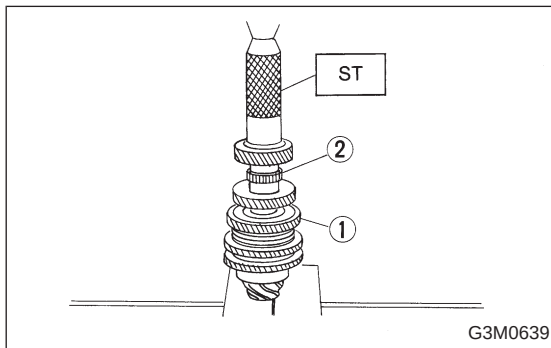
Align ring groove with insert.



7) Install 1st-2nd driven gear bushing ① to drive pinion shaft using ST1 and ST2.

ST1 499877000 INSTALLER

ST2 499277100 INSTALLER



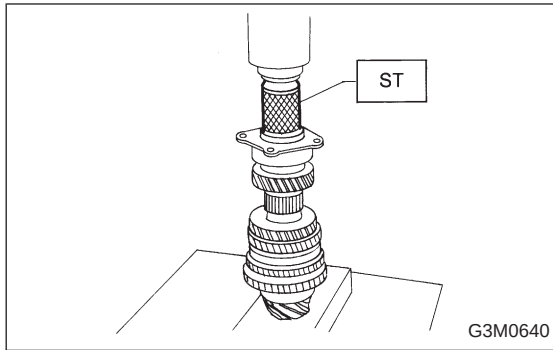
8) Install 2nd driven gear ① and 1st-2nd bask ring to drive pinion shaft by hand.

9) Install key into the groove on drive pinion shaft and install 3rd-4th driven gear.

NOTE:

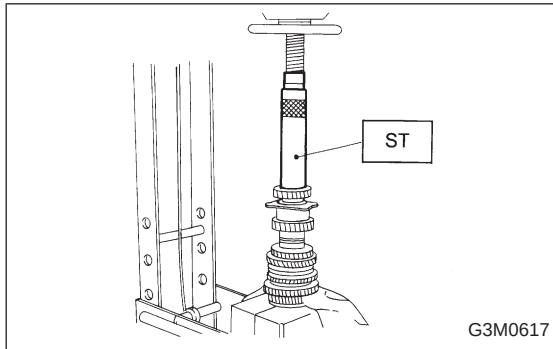
Ball bearing may be installed without using the tool. There should be no problem.

ST 499877000 INSTALLER



10) Install ball bearing (29 x 74 x 38) on drive pinion shaft with ST.

ST 499277100 INSTALLER

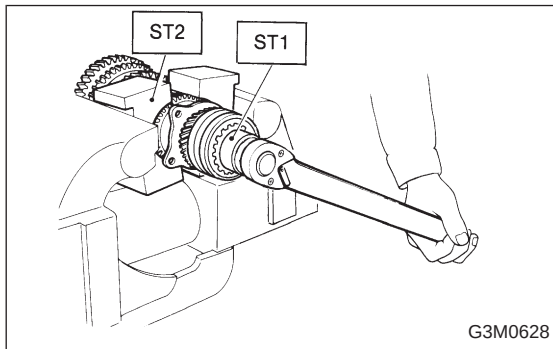
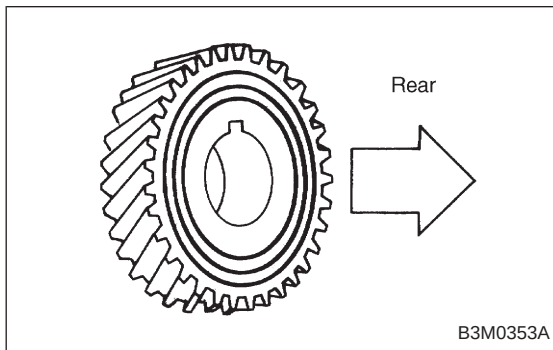


11) Position woodruff key in groove on the rear of drive pinion shaft. Install 5th driven gear onto drive shaft using ST and press.

ST 499277100 INSTALLER

CAUTION:

- Face 5th driven gear in the correct direction.
- Be careful not to dislocate woodruff key while installing 5th gear.



12) Install lock washer and tighten lock nut to the specified torque using ST1 and ST2.

ST1 499987100 or 499987003 or 899984103 SOCKET WRENCH (35)

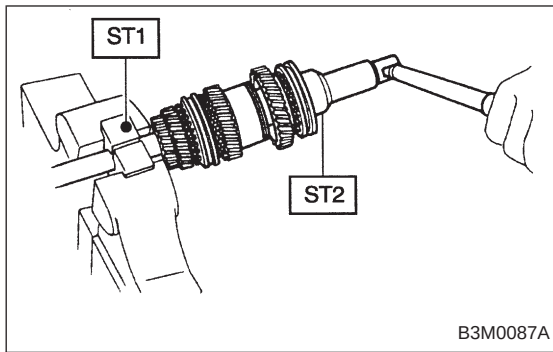
ST2 899884100 HOLDER

CAUTION:

- Discard old lock nuts, replace with new ones.
- Secure lock nut in four places.

Tightening torque:

112 — 124 N·m (11.4 — 12.6 kg·m, 82 — 91 ft·lb)



7. Main Shaft Assembly

A: DISASSEMBLY

1) Put vinyl tape around main shaft splines to protect oil seal from damage. Then pull out oil seal and needle bearing by hand.

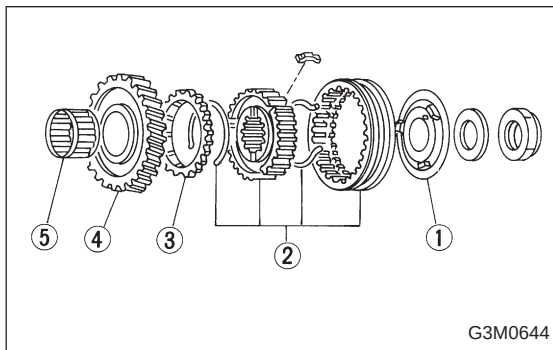
2) Remove lock nut from transmission main shaft assembly.

NOTE:

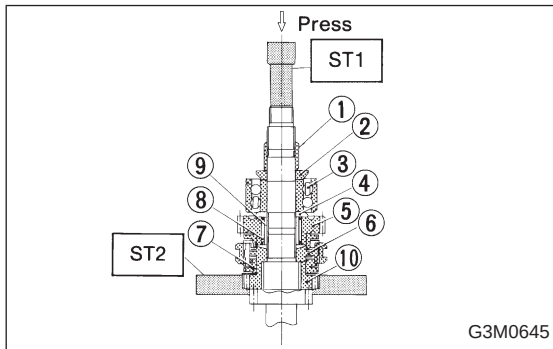
Remove caulking before taking off lock nut.

ST1 498937000 TRANSMISSION HOLDER

ST2 499987003 SOCKET WRENCH (35)



3) Remove insert stopper plate ①, sleeve and hub assembly No. 2, baulk ring ③, 5th drive gear ④, and needle bearing ⑤ (32 x 36 x 25.7).



4) Using ST1, ST2 and a press, remove:

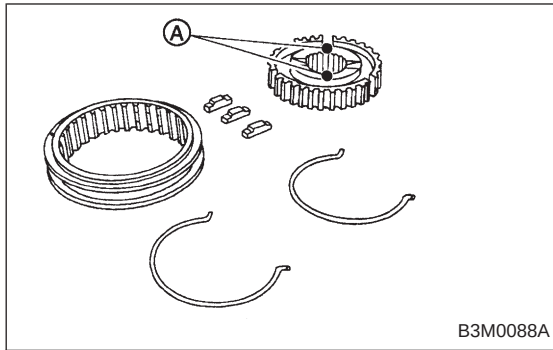
- 5th needle bearing inner race ①
- 5th gear thrust washer ②
- Ball bearing ③ (25.5 x 65 x 31)
- 4th gear thrust washer ④
- 4th drive gear ⑤
- Sleeve and hub assembly ⑥
- Baulk ring ⑦
- 4th needle bearing ⑧
- 4th needle bearing inner race ⑨
- 3rd drive gear ⑩

ST1 899864100 REMOVER

ST2 899714110 REMOVER

NOTE:

Replace sleeve and hub with new ones. Do not attempt to disassemble because they must engage at a specified point. If they should be disassembled, mark engagement point on splines beforehand.

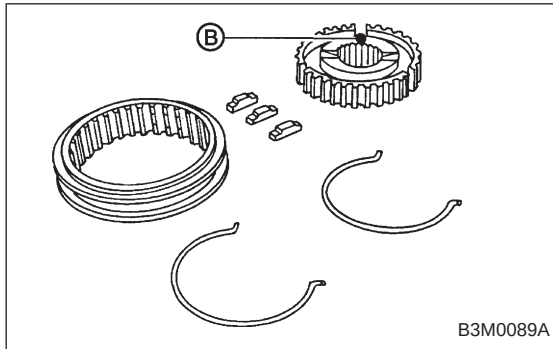
**B: ASSEMBLY**

1) Assemble sleeve and hub assembly for 3rd-4th and, 5th and high-low synchronizing.

NOTE:

Position open ends of spring 120° apart.

Ⓐ: Two holes for discrimination (3rd-4th hub)

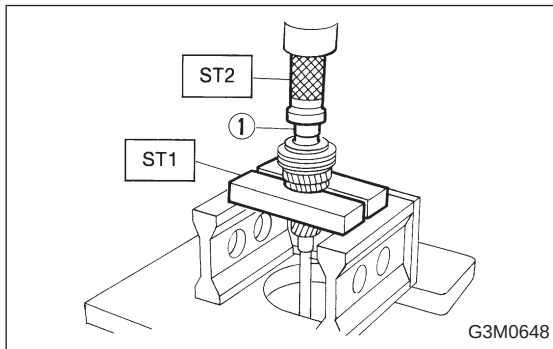


Ⓑ: One hole for discrimination (5th hub)

2) Install 3rd drive gear, baulk ring, and sleeve and hub assembly for 3rd-4th needle bearing (32 x 36 x 25.7) on transmission main shaft.

NOTE:

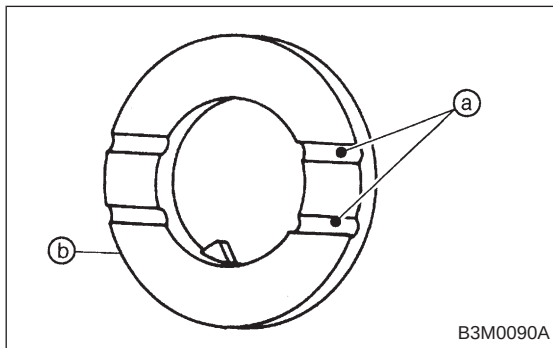
Align groove in baulk ring with shifting insert.



3) Install 4th needle bearing race ① onto transmission main shaft using ST1, ST2 and a press.

ST1 899714110 REMOVER

ST2 499877000 RACE 4-5 INSTALLER



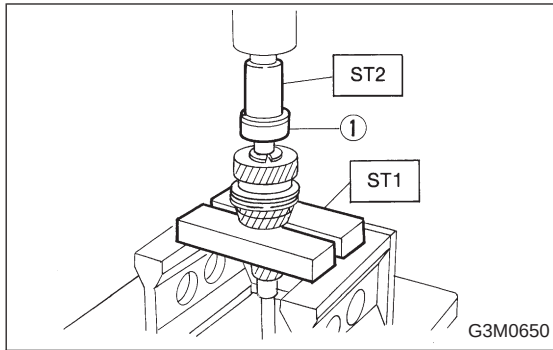
4) Install baulk ring, needle bearing (32 x 30 x 25.7), 4th drive gear and 4th gear thrust washer to transmission main shaft.

NOTE:

Face thrust washer in the correct direction.

Ⓐ: Groove

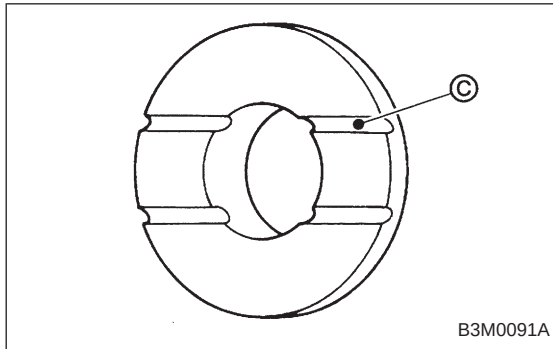
Ⓑ: 4th gear side



5) Drive ball bearing ① onto the rear section of transmission main shaft using ST1, ST2 and a press.

ST1 899714110 REMOVER

ST2 499877000 RACE 4-5 INSTALLER



6) Using the same tools as in step 5) above, install the following parts onto the rear section of transmission main shaft.

- 5th gear thrust washer

NOTE:

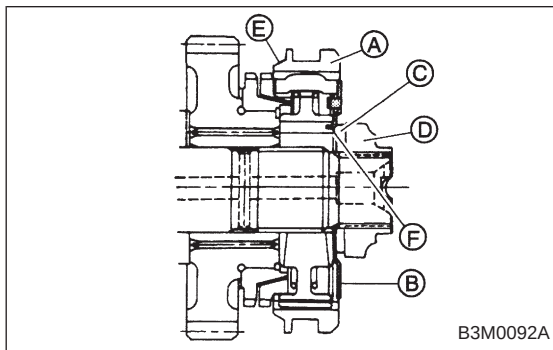
Face thrust washer in the correct direction.

Ⓒ: Face this surface to 5th gear side.

ST1 899714110 REMOVER

ST2 499877000 RACE 4-5 INSTALLER

- 5th needle bearing race



7) Install the following parts to the rear section of transmission main shaft.

- Needle bearing (32 x 36 x 25.7)
- 5th drive gear
- Baulk ring
- Sleeve ① and hub assembly
- Insert stopper plate ②
- Lock washer ③ (22 x 38 x 2)
- Tighten lock nuts ④ (22 x 13) to the specified torque using ST1 and ST2.

ST1 499987003 SOCKET WRENCH (35)

ST2 498937000 TRANSMISSION HOLDER

NOTE:

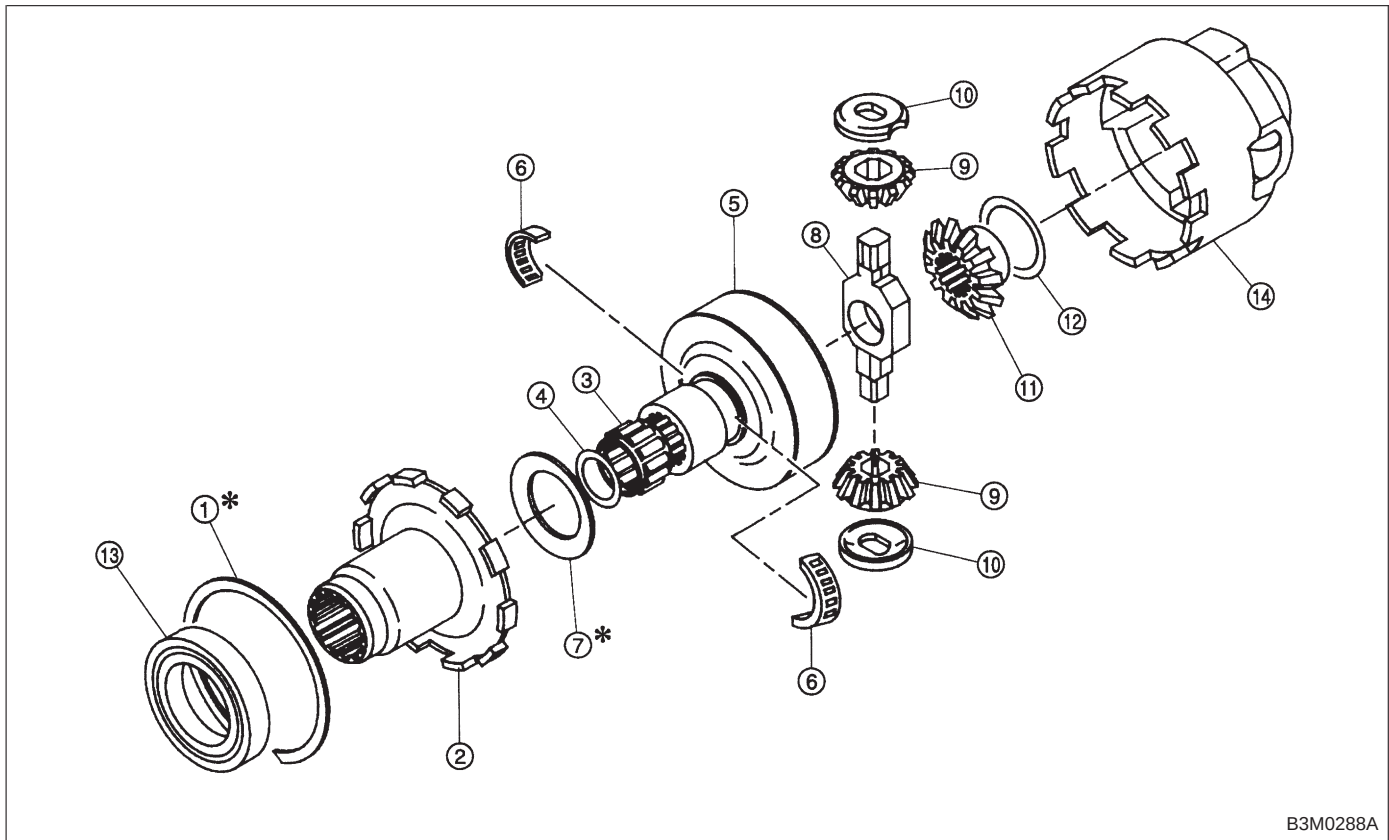
- Align groove ⑤ in baulk ring with shifting insert.
- Be sure to fit pawl ⑥ of insert stopper plate into 4 mm (0.16 in) dia. hole in the boss section of synchronizer hub.
- Secure lock nuts in two places after tightening.

Tightening torque:

118±6 N·m (12.0±0.6 kg-m, 86.8±4.3 ft-lb)

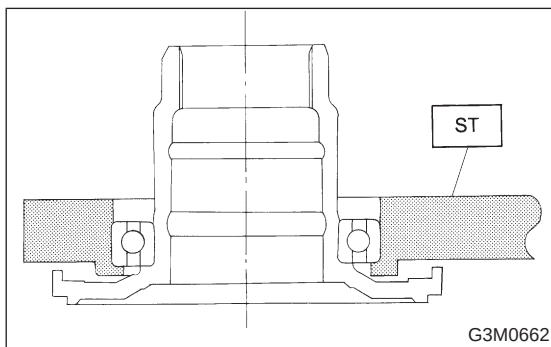
8. Center Differential (AWD Model)

A: DISASSEMBLY



B3M0288A

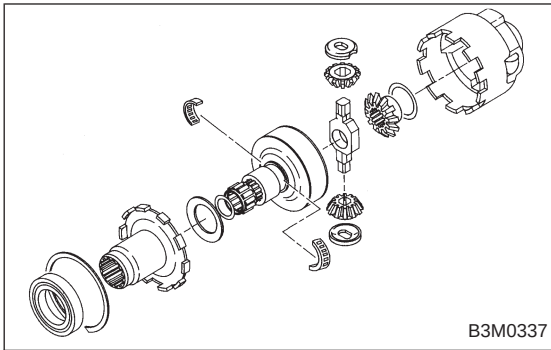
- 1) Remove snap ring ① (Inner-110) using flat bladed screwdriver.
- 2) Remove center differential cover ②.
- 3) Remove snap ring ④ and roller bearing ③.
- 4) Remove viscous coupling ⑤.
- 5) Remove needle bearings ⑥.
- 6) Remove adjusting washer ⑦ (45 x 62 x t).
- 7) Remove pinion shaft ⑧, bevel pinions ⑨ and retainers ⑩.
- 8) Remove side gear ⑪.
- 9) Remove thrust washer ⑫.



G3M0662

- 10) Remove ball bearing ⑬ using ST.
ST 498077300 CENTER DIFFERENTIAL BEARING REMOVER

CAUTION:
Do not reuse ball bearing.

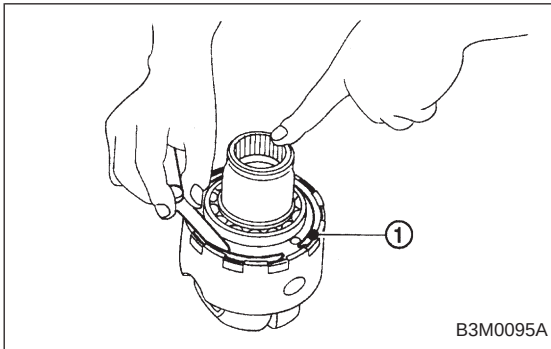
**B: ASSEMBLY**

Assembly is in the reverse order of disassembly.

Do the following:

- Install thrust washer with chamfered side of inner perimeter facing the side gear.
- Install adjusting washer with chamfered side of inner perimeter facing the viscous coupling using ST.

ST 499547300 INSTALLER SET



1) Selection of snap ring (Inner-110)

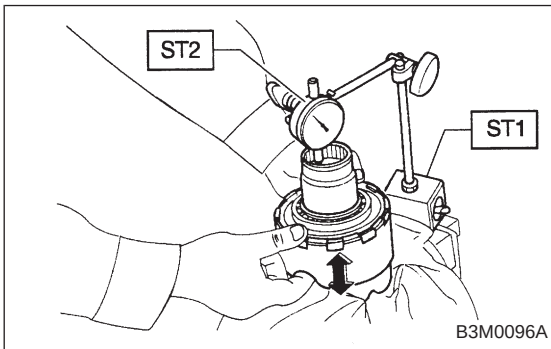
- (1) After assembling, using a thickness gauge measure clearance between snap ring ① and center differential case.

Clearance:

0 — 0.15 mm (0 — 0.0059 in)

- (2) If the measurement is not within the specification, select suitable snap ring.

Snap ring (Inner-110)	
Part No.	Thickness mm (in)
805100061	2.10 (0.0827)
805100062	2.21 (0.0870)
805100063	2.32 (0.0913)



2) Selection of adjusting washer (Backlash adjustment)

- (1) After assembling, set up a ST1 and ST2 to end of viscous coupling shaft. Move viscous coupling up and down, and measure backlash in the axial direction.

ST1 498247001 MAGNET BASE

ST2 498247100 DIAL GAUGE

Backlash:

0.62 — 0.86 mm (0.0244 — 0.0339 in)

- (2) If the measurement is not within the specification, select suitable washer.

Adjusting washer (45 x 62 x t)	
Part No.	Thickness mm (in)
803045041	1.60 (0.0630)
803045042	1.80 (0.0709)
803045043	2.00 (0.0787)
803045044	2.20 (0.0866)
803045045	2.40 (0.0945)

9. Front Differential

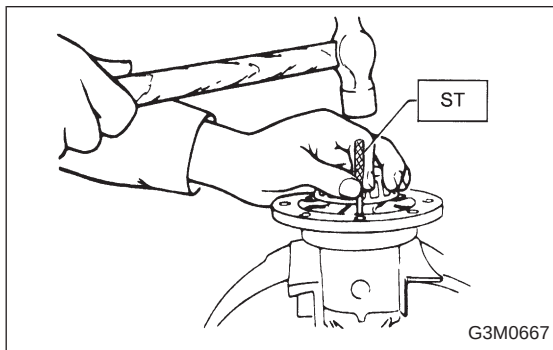
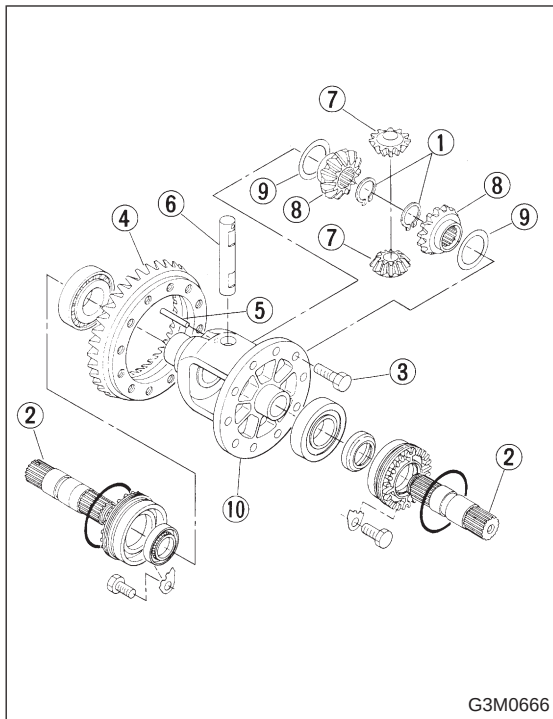
A: DISASSEMBLY

1) Remove right and left snap rings ① from differential, and then remove two axle drive shafts ②.

NOTE:

During reassembly, reinstall each axle drive shaft in the same place from which it was removed.

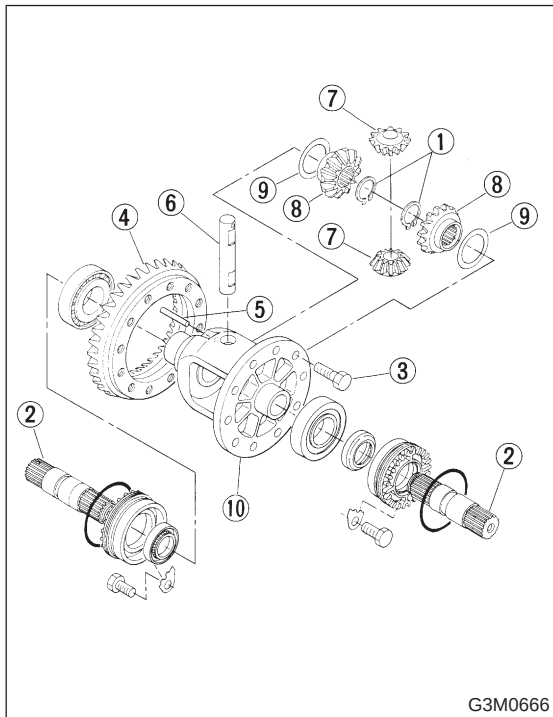
2) Loosen twelve bolts ③ and remove hypoid drive gear ④.

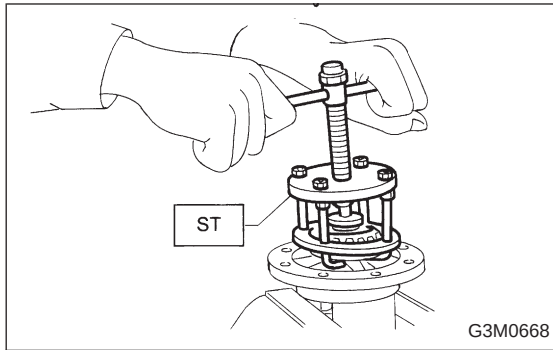


3) Drive out straight pin ⑤ from differential assembly toward hypoid driven gear.

ST 899904100 REMOVER

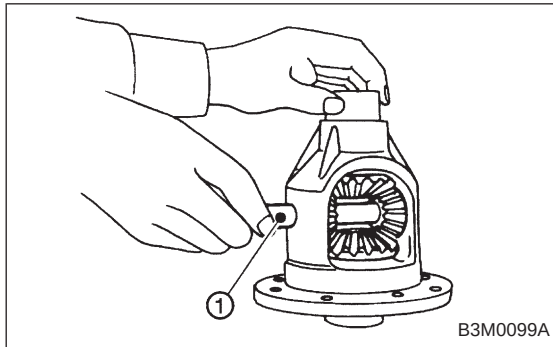
4) Pull out pinion shaft ⑥, and remove differential bevel pinion ⑦ and gear ⑧ and washer ⑨.





5) Remove roller bearing using ST.

ST 399527700 PULLER SET

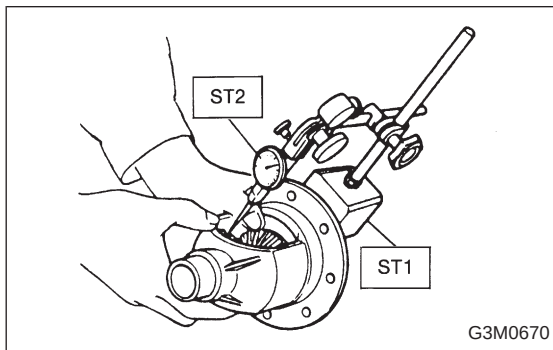


B: ASSEMBLY

1) Install bevel gear and bevel pinion together with washers, and insert pinion shaft ①.

NOTE:

Face the chamfered side of washer toward gear.



2) Measure backlash between bevel gear and pinion. If it is not within specifications, install a suitable washer to adjust it.

Standard backlash:

0.13 — 0.18 mm (0.0051 — 0.0071 in)

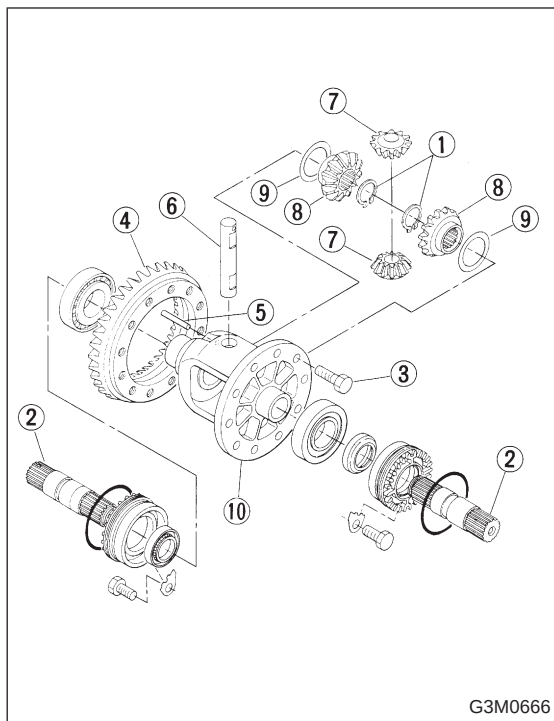
ST1 498247001 MAGNET BASE

ST2 498247100 DIAL GAUGE

NOTE:

Be sure the pinion gear tooth contacts adjacent gear teeth during measurement.

Washer (38.1 x 50 x t)	
Part No.	Thickness mm (in)
803038021	0.925 — 0.950 (0.0364 — 0.0374)
803038022	0.975 — 1.000 (0.0384 — 0.0394)
803038023	1.025 — 1.050 (0.0404 — 0.0413)

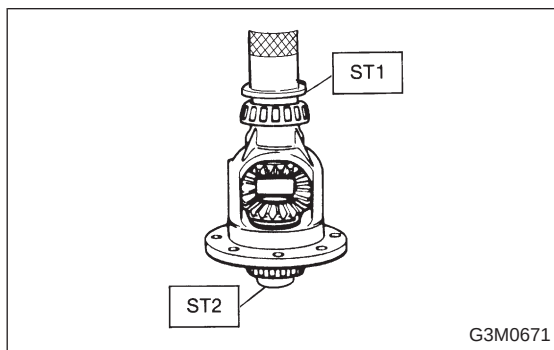


3) Align pinion shaft and differential case at their holes, and drive straight pin ⑤ into holes from the hypoid driven gear side, using ST.

ST 899904100 REMOVER

NOTE:

Lock straight pin after installing.



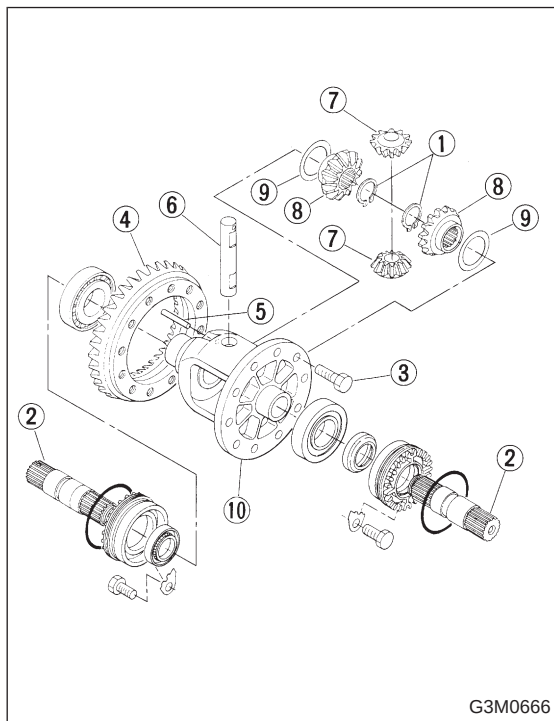
4) Install roller bearing (40 x 80 x 19.75) to differential case.

NOTE:

Be careful because roller bearing outer races are used as a set.

ST1 499277100 BUSH 1-2 INSTALLER

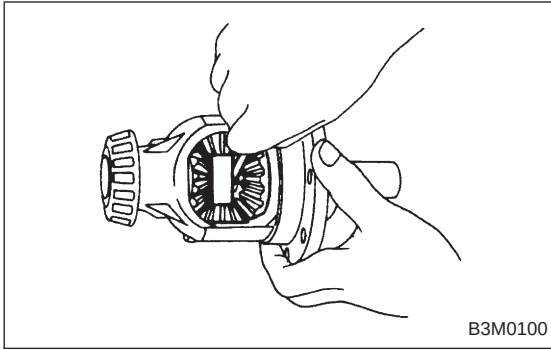
ST2 398497701 ADAPTER



5) Install hypoid driven gear ④ to differential case ⑩ using twelve bolts ③.

Tightening torque:

$62 \pm 5 \text{ N} \cdot \text{m}$ ($6.3 \pm 0.5 \text{ kg} \cdot \text{m}$, $45.6 \pm 3.6 \text{ ft} \cdot \text{lb}$)



6) Position drive axle shaft in differential case and hold it with outer snap ring (28). Using a thickness gauge, measure clearance between the shaft and case is within specifications.

Clearance:

0 — 0.2 mm (0 — 0.008 in)

If it is not within specifications, replace snap ring with a suitable one.

Snap ring (Outer-28)	
Part No.	Thickness mm (in)
805028011	1.05 (0.0413)
805028012	1.20 (0.0472)

1. Manual Transmission and Differential

Symptom and possible cause	Remedy
1. Gears are difficult to intermesh. The cause for difficulty in shifting gears can be classified into two kinds: one is malfunction of the gear shift system and the other is malfunction of the transmission. However, if the operation is heavy and engagement of the gears is difficult, defective clutch disengagement may also be responsible. Check whether the clutch is correctly functioning, before checking the gear shift system and transmission.	
(a) Worn, damaged or burred chamfer of internal spline of sleeve and reverse driven gear	Replace.
(b) Worn, damaged or burred chamfer of spline of gears	Replace.
(c) Worn or scratched bushings	Replace.
(d) Incorrect contact between synchronizer ring and gear cone or wear	Correct or replace.
2. Gear slips out. (1) Gear slips out when coasting on rough road. (2) Gear slips out during acceleration.	
(a) Defective pitching stopper adjustment	Adjust.
(b) Loose engine mounting bolts	Tighten or replace.
(c) Worn fork shifter, broken shifter fork rail spring	Replace.
(d) Worn or damaged ball bearing	Replace.
(e) Excessive clearance between splines of synchronizer hub and synchronizer sleeve	Replace.
(f) Worn tooth step of synchronizer hub (responsible for slip-out of 3rd gear)	Replace.
(g) Worn 1st driven gear, needle bearing and race	Replace.
(h) Worn 2nd driven gear, needle bearing and race	Replace.
(i) Worn 3rd drive gear and bushing	Replace.
(j) Worn 4th drive gear and bushing	Replace.
(k) Worn reverse idler gear and bushing	Replace.
3. Unusual noise comes from transmission. If an unusual noise is heard when the vehicle is parked with its engine idling and if the noise ceases when the clutch is disengaged, it may be considered that the noise comes from the transmission.	
(a) Insufficient or improper lubrication	Lubricate or replace with specified oil.
(b) Worn or damaged gears and bearings	Replace.
NOTE: If the trouble is only wear of the tooth surfaces, merely a high roaring noise will occur at high speeds, but if any part is broken, rhythmical knocking sound will be heard even at low speeds.	

Symptom and possible cause	Remedy
4. Broken differential (case, gear, bearing, etc.) Abnormal noise will develop and finally it will become impossible to continue to run due to broken pieces obstructing the gear revolution.	
(a) Insufficient or improper oil	Disassemble differential and replace broken components and at the same time check other components for any trouble, and replace if necessary.
(b) Use of vehicle under severe conditions such as excessive load and improper use of clutch	Readjust bearing preload and backlash and face contact of gears.
(c) Improper adjustment of taper roller bearing	Adjust.
(d) Improper adjustment of drive pinion and hypoid driven gear	Adjust.
(e) Excessive backlash due to worn differential side gear, washer or differential pinion	Add recommended oil to specified level. Do not use vehicle under severe operating conditions.
(f) Loose hypoid driven gear clamping bolts	Tighten.
5. Differential and hypoid gear noises Troubles of the differential and hypoid gear always appear as noise problems. Therefore noise is the first indication of the trouble. However noises from the engine, muffler, tire, exhaust gas, bearing, body, etc. are easily mistaken for the differential noise. Pay special attention to the hypoid gear noise because it is easily confused with other gear noises. There are the following four kinds of noises. (1) Gear noise when driving: If noise increases as vehicle speed increases it may be due to insufficient gear oil, incorrect gear engagement, damaged gears, etc. (2) Gear noise when coasting: Damaged gears due to maladjusted bearings and incorrect shim adjustment (3) Bearing noise when driving or when coasting: Cracked, broken or damaged bearings (4) Noise which mainly occurs when turning: Unusual noise from differential side gear, differential pinion, differential pinion shaft, etc.	
(a) Insufficient oil	Lubricate.
(b) Improper adjustment of hypoid driven gear and drive pinion	Check tooth contact.
(c) Worn teeth of hypoid driven gear and drive pinion	Replace as a set. Readjust bearing preload.
(d) Loose roller bearing	Readjust hypoid driven gear to drive pinion backlash and check tooth contact.
(e) Distorted hypoid driven gear or differential case	Replace.
(f) Worn washer and differential pinion shaft	Replace.